cSlope

Verification manual

Slope stability analyses compared with published results

This manual gives the factors of safety cSlope computes for 187 slope stability problems from the geotechnical literature and from the verification manuals of other slope stability programs, next to the values the sources publish and the range of the other programs' results.

Program version	cSlope 0.1.0-beta.2 (commit efce1f11)
Report generated	2026-11-14
Problems	187
cSlope analyses	362
Values published by the sources	214
Mean absolute deviation from other programs	0.32 % (301 results)
Mean absolute deviation from the literature	0.95 % (123 values)

Produced by packages/desktop/scripts/verification-report.ts from the problems in packages/engine/src/verification.

Contents

1 Introduction 3

2 Summary of results 4

2.1 Deviations from each reference 4

2.2 Distribution of the deviations 5

2.3 By method 5

2.4 Largest deviations 5

3 Verification problems 7

4 Strength reduction by finite elements 196

Appendix A Results of all problems 198

Section 3 has one page per problem, in the order of their ids; appendix A lists them all with their page numbers.

1 Introduction

Each problem is defined once, in cSlope's source code (packages/engine/src/verification/problems.ts). The automated tests run the same definitions on every change to the program, with a shorter search on the circular problems; this report runs them with the full search given below.

Analysis settings

Setting	Value
Slip surfaces	The surface the source gives, circular or not, where it gives one; otherwise a search for the critical circle, followed by non-circular optimization where the problem says so.
Number of slices	50, unless the problem states otherwise
Search	4000 entry and exit combinations, then 2000 more around the best surfaces
Convergence tolerance	0.001 on the factor of safety; 0.001 with non-circular optimization
Methods	Bishop simplified; Janbu simplified (uncorrected unless stated); Morgenstern-Price with a half-sine interslice function; Spencer, which is Morgenstern-Price with a constant interslice function
Pore pressure	From the piezometric line: the head is the vertical distance below the line (the usual convention). Problems described with "phreatic correction" use the vertical distance × $\cos^2\theta$, where θ is the inclination of the line.
Unit weight of water	9.807 kN/m³

Units

Problems published in imperial units without water are entered in those units (ft, psf, pcf); the factor of safety is dimensionless. Problems with water are converted to SI, because cSlope uses γ_w = 9.807 kN/m³ (62.4 pcf is 9.80 kN/m³, less than 0.1 % apart).

Source of the input data

The input data were transcribed from the publications and from the verification manuals of other slope stability programs, most of them commercial, and cross-checked against an independent open-source transcription of the same problems. For each problem the report lists the values the sources publish, with the method where the source names one; some are hand calculations, chart readings, trial wedges or finite element results rather than the method of slices. For each analysis it also gives the range of results that other programs publish in their verification manuals, without naming the programs. Problem ids are cSlope's own: the number is the slope, the letter a variant of it.

How to read the tables

In section 3 and in appendix A each row is one cSlope analysis (a method on a surface or a search): cSlope's factor of safety, each value the literature publishes for the same method with its source, and the range of the other programs' results. Dev. is the deviation, cSlope's value minus the reference in percent of the reference, as in the statistics of section 2; the reference of a published range is its lower end, that of the programs' range its midpoint. A finite element result between a published pair of bounds is marked "in bounds". A published value whose method cSlope does not have is listed with "–" in the Analysis column and is not compared. A value marked * is disputed: it is shown with the reason and left out of the statistics.

2 Summary of results

cSlope was run on 187 problems, with 362 analyses. Each result is compared with two kinds of reference. The first is the results that other slope stability programs publish for the same analysis in their verification manuals: 301 results, compared with the midpoint of the programs' range where more than one program gives a value. The second is the literature: 123 values from papers, books, design manuals and theory. The finite element factors of safety are compared with published finite element, finite difference, limit analysis and theoretical values (34 references). The deviation is cSlope's factor of safety minus the reference, in percent of the reference, so a negative mean deviation means that cSlope is lower on average.

2.1 Deviations from each reference

Table 1. Deviations of cSlope's factors of safety from each reference, in percent of the reference.

Reference	Results	Mean	Mean abs.	RMS	Largest	Within tolerance
Other programs	301	−0.20 %	0.32 %	0.45 %	2.01 %	329 of 329 checks
Literature	123	−0.13 %	0.95 %	1.53 %	6.59 %	85 of 85 checks
Finite elements (section 4)	34	−0.58 %	1.71 %	2.41 %	4.99 %	34 of 34 references

Against the other programs, 232 of the 301 results lie within 0.5 %, 289 within 1 % and 300 within 2 %; the largest deviation is −2.01 % (V038, Spencer, circles only). Against the literature the spread is wider: 85 of 123 values lie within 1 %, 103 within 2 % and 121 within 5 %, and the largest deviation is −6.59 % (V002, Spencer, non-circular). Many literature values are printed to two decimals, some are read from charts, some come from hand calculations or from another method such as finite elements or a trial wedge, and on some problems the sources differ from each other; section 2.4 lists the largest deviations.

"Within tolerance" counts the checks that the automated tests apply on every change to the program. Each check compares cSlope with one published value, so an analysis with values from three programs gives three checks, while the statistics use the midpoint of their range. A check agrees when cSlope is within 1 % of the reference on a given surface and 2 % for a search, plus half a unit of the reference's last printed digit, plus 1 % where the input or the reference was read from a figure, plus 1 % where the surface ends in a vertical or near-vertical scarp; 5 % against a finite element value. The finite element references have their own rule (section 4); 1 disputed reference is shown there and not counted. The 30 reliability indices and probabilities of failure of the probabilistic problems are compared with absolute tolerances: 30 of 30 agree. Section 3 and appendix A show every deviation but do not mark single results as passed or failed.

2.2 Distribution of the deviations

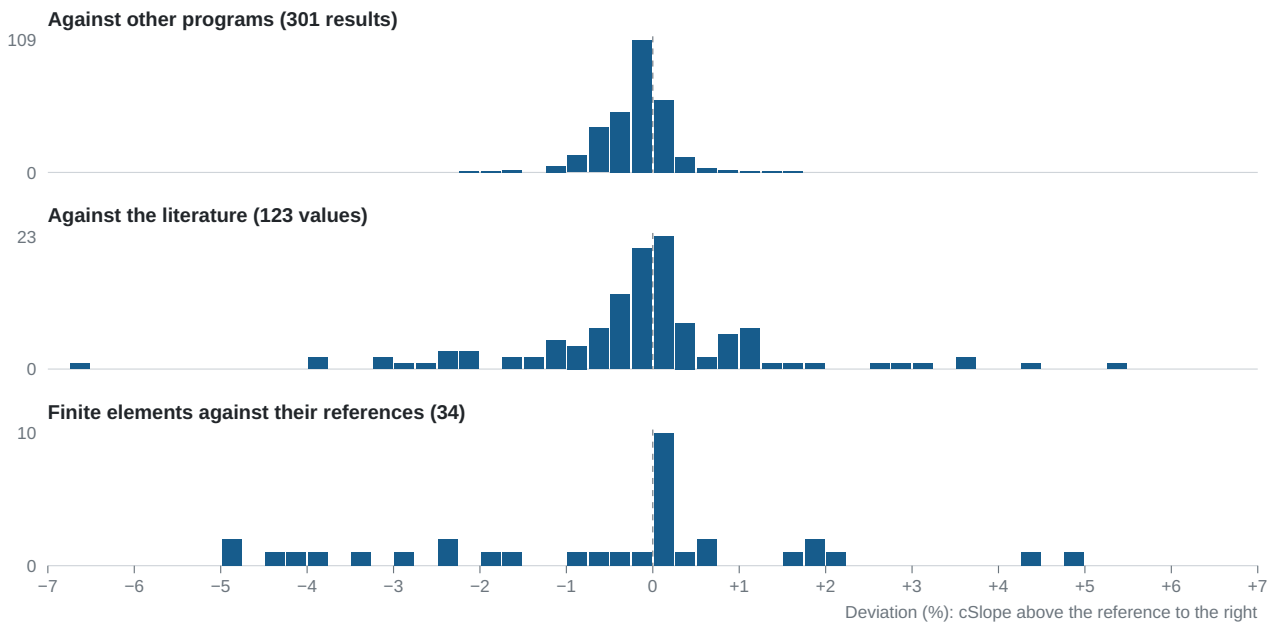


Figure 1. Number of results per 0.25 % of deviation. The dashed line is zero deviation.

2.3 By method

Table 2. Mean absolute and largest deviation by method, from the other programs and from the literature.

Method	Programs: n	Mean abs.	Largest	Literature: n	Mean abs.	Largest
Bishop simplified	116	0.27 %	0.97 %	44	0.86 %	3.90 %
Janbu simplified and corrected	39	0.25 %	1.81 %	2	1.59 %	2.97 %
Spencer and Morgenstern-Price	117	0.35 %	2.01 %	58	0.61 %	4.38 %
Non-circular optimization, all methods	29	0.45 %	1.69 %	19	2.12 %	6.59 %

The first three rows hold circles and the surfaces the sources give; the last holds the searches that optimize a circle into a non-circular surface, whatever the method. Non-circular searches depend on the search more than circular ones, and two programs need not end on the same surface.

2.4 Largest deviations

The largest deviations from each reference, with the notes of those problems. The problem id links to its section.

Table 3. The five largest deviations from other programs (reference: the midpoint of their range).

Problem	Analysis	Reference	Source	cSlope	Deviation
V038 (3.67)	Spencer, circles only	1.100	other programs	1.078	−2.01 %
V087 (3.162)	Janbu corrected	2.290	other programs	2.248	−1.81 %
V059b (3.99)	Spencer, non-circular	1.290	other programs	1.312	+1.69 %
V026 (3.48)	Spencer, non-circular	1.190	other programs	1.170	−1.66 %
V050b (3.86)	Spencer, non-circular	1.246	other programs	1.225	−1.65 %

Table 4. The five largest deviations from the literature.

Problem	Analysis	Reference	Source	cSlope	Deviation
V002 (3.2)	Spencer, non-circular	0.44	Kim et al. (2002)	0.411	−6.59 %
V002 (3.2)	Spencer, non-circular	0.39	Greco (1996)	0.411	+5.38 %
V065 (3.116)	Spencer	0.92	Leshchinsky & Han (2004)	0.960	+4.38 %
V002 (3.2)	Janbu corrected, non-circular	0.43	Arai & Tagyo (1985)	0.413	−3.95 %
V066e (3.121)	Bishop simplified	1.00	Leshchinsky & Han (2004)	0.961	−3.90 %

Table 5. The three largest deviations of the finite element analysis from its references.

Problem	Analysis	Reference	Source	cSlope	Deviation
V046b (3.79)	Strength reduction (FE)	1.13	another program's verification manual	1.186	+4.99 %
V053 (3.91)	Strength reduction (FE)	1.02	another program's verification manual	0.969	−4.97 %
V052a (3.88)	Strength reduction (FE)	1.88	another program's verification manual	1.789	−4.85 %

Notes of these problems

V038: Duncan & Wright's 1.16 is for γ = 125 pcf and their printed phreatic line, not this program input (100 pcf, a 9-point line); on their input cSlope gives 1.157 (V038b).

V059b: The paper gives its results in figures only; the comparison is with the other programs.

V050b: ACADS accepted 1.24–1.27; cSlope's search finds a surface along the weak layer at 1.228 (Spencer), unchanged at 50, 100 and 200 slices.

V065: Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

V066e: Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value. Here the reinforcement keeps the baseline strength of 10 kN/m, as in the program compared with, not the 11.6 kN/m designed for the surcharge, so the published 1.00 is not directly comparable.

V053: Chen & Shao give 1.009–1.025 with Spencer's assumption for a surface sliding along the weak seam (their test problem 3, Tables 5–6). Greco's Monte Carlo search finds a global minimum of 0.97 along the weak layer, entering the level ground far from the toe, and a local minimum of 1.08 on a nearly circular surface from the toe; the programs' block searches and cSlope give 1.01 on a commercial program's version of the geometry, 4 % above Greco's global minimum.

V087, V026, V002, V046b, V052a have no note: the table of each problem in section 3 gives every published value and the other programs' range for the same analysis.

3 Verification problems

3.1 V001: Arai & Tagyo (1985) example 1

Arai & Tagyo (1985), example 1.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 1.5:1 slope, 20 m high; critical-circle search.

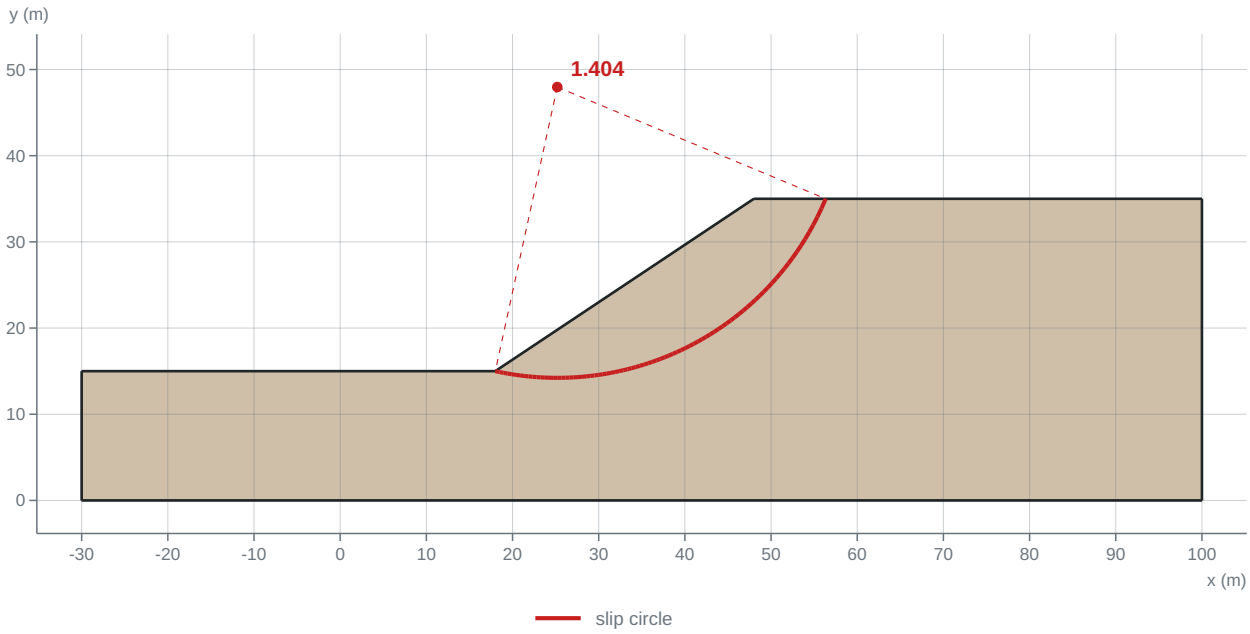


Figure 2. V001: critical circle (Bishop simplified, F = 1.404).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.82	c′ = 41.65	15.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.404	1.451	Arai & Tagyo (1985)	−3.21 %	1.409–1.417	−0.61 %
Morgenstern-Price (half-sine)	1.401				1.414	−0.90 %
Spencer, non-circular	1.384				1.386	−0.17 %
Janbu simplified, non-circular	1.252	1.265	Arai & Tagyo (1985)	−1.06 %		
Janbu corrected, non-circular	1.345	1.357	Arai & Tagyo (1985)	−0.91 %		
Strength reduction (FE)	1.402	1.40	another program's verification manual: finite-element strength reduction (verification manual, problem 10)	+0.17 %		

Arai & Tagyo's Bishop value 1.451 is 2–3 % above the other programs (1.409–1.417) and cSlope on the same slope; their non-circular Janbu values are within 1 % of cSlope's.

3.2 V002: Arai & Tagyo (1985) example 2, three layers

Arai & Tagyo (1985), example 2.

Critical circle search · Units: SI (m, kPa, kN/m³)

Slope with a weak middle layer between a stronger upper layer and a very strong base; critical-circle search.

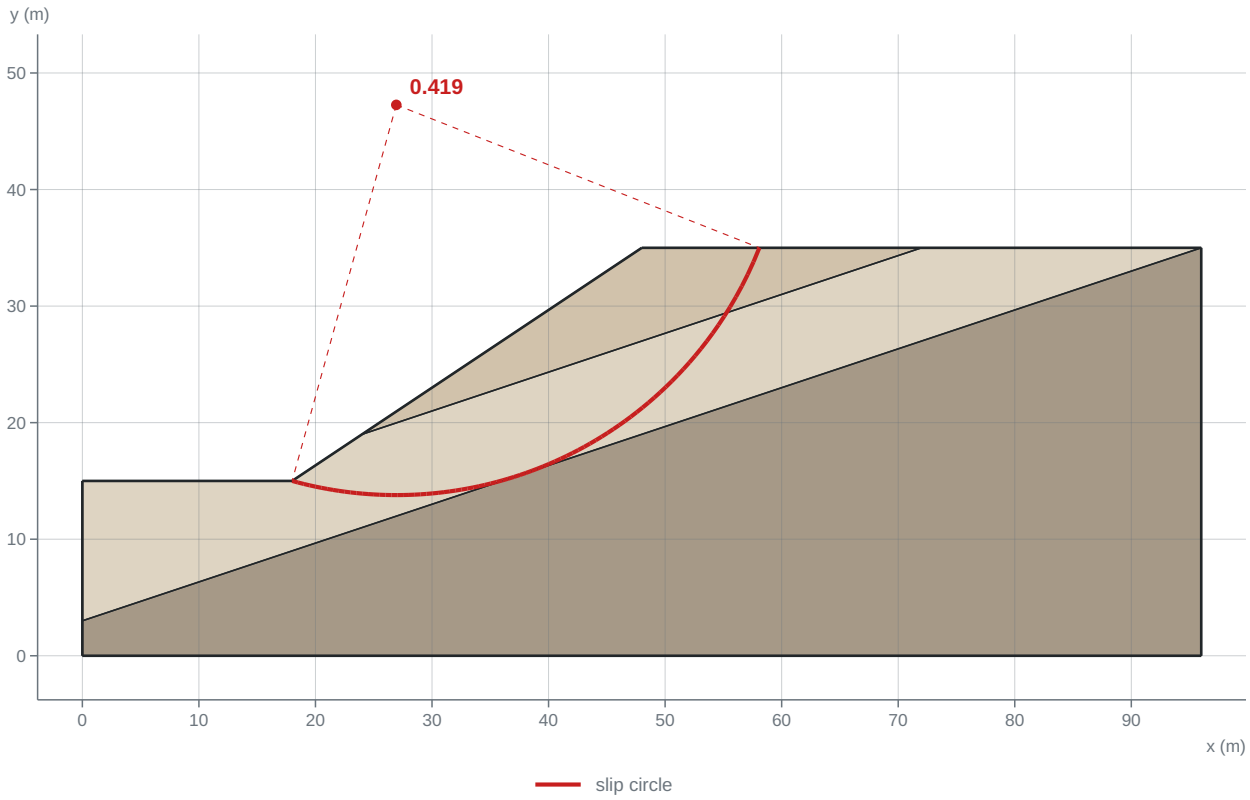


Figure 3. V002: critical circle (Bishop simplified, F = 0.419).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper	Mohr-Coulomb	18.82	c′ = 29.40	12.00°
Middle	Mohr-Coulomb	18.82	c′ = 9.80	5.00°
Lower	Mohr-Coulomb	18.82	c′ = 294.0	40.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	0.419	0.417	Arai & Tagyo (1985)	+0.55 %	0.420	−0.17 %
		0.43	Kim et al. (2002)	−2.49 %		
Spencer, non-circular	0.411	0.44	Kim et al. (2002)	−6.59 %	0.414	−0.72 %
		0.39	Greco (1996)	+5.38 %		
Janbu simplified, non-circular	0.392	0.405	Arai & Tagyo (1985)	−3.23 %		
Janbu corrected, non-circular	0.413	0.43	Arai & Tagyo (1985)	−3.95 %		
Strength reduction (FE)	0.425	0.40–0.45	Kim et al. (2002): limit analysis, lower and upper bound	in bounds		

3.3 V003: Arai & Tagyo (1985) example 3, water table

Arai & Tagyo (1985), Soils and Foundations 25(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous slope with a water table; critical-circle search.

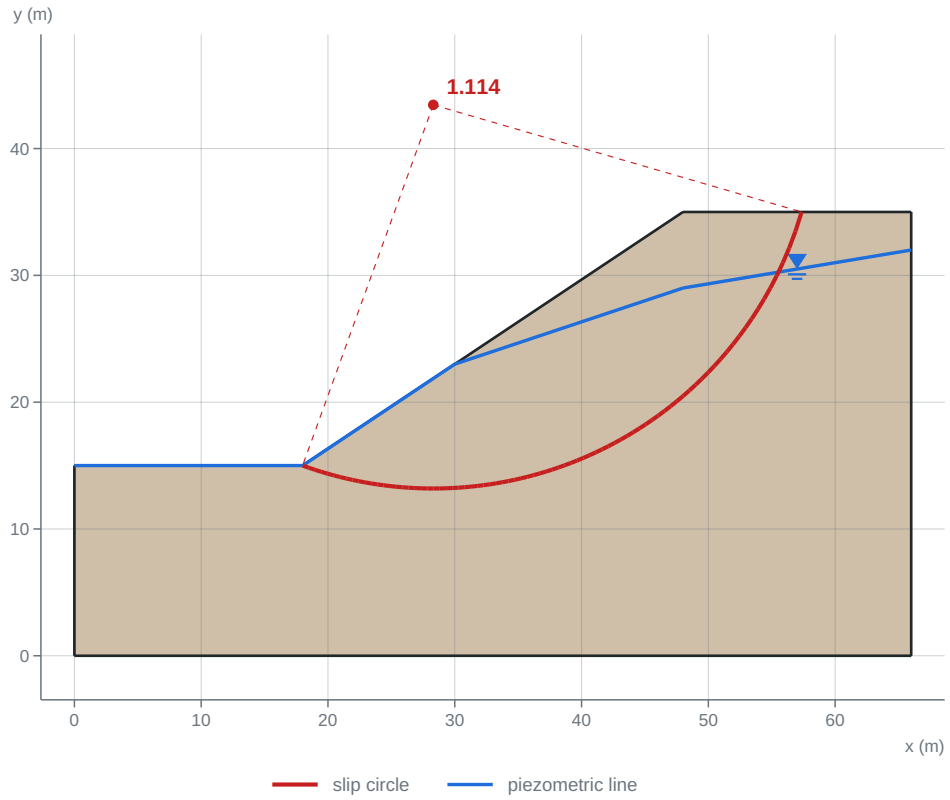


Figure 4. V003: critical circle (Bishop simplified, F = 1.114).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
A&T ex3 soil	Mohr-Coulomb	18.82	c′ = 41.65	15.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.114	1.138	Arai & Tagyo (1985)	−2.11 %	1.112–1.118	−0.09 %
Spencer	1.115				1.118	−0.30 %
Janbu corrected	1.130				1.131	−0.05 %
Janbu simplified, non-circular	0.966	0.995	Arai & Tagyo (1985)	−2.91 %		
Janbu corrected, non-circular	1.047	1.071	Arai & Tagyo (1985)	−2.23 %		
Strength reduction (FE)	1.107	1.09	another program's verification manual: finite-element strength reduction (verification manual, problem 12)	+1.55 %		

Arai & Tagyo's Bishop value 1.138 is about 2 % above the other programs (1.112–1.118 with Bishop).

3.4 V004a: 4:1 slope, Mohr-Coulomb fit

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

12 m high 4:1 slope with the same soil as V008a; critical-circle search.

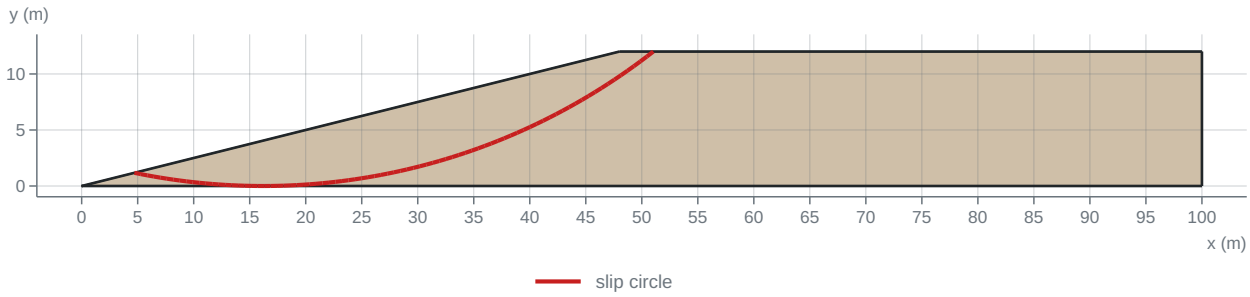


Figure 5. V004a: critical circle (Spencer, F = 2.802).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c′ = 11.64	24.70°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	2.802				2.794	+0.30 %
Janbu simplified	2.677				2.662	+0.55 %
–	–	2.66	Baker (2003): Mohr-Coulomb strength			

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.5 V004b: Baker (2003) ex. 2 slope, Mohr-Coulomb iteration parameters

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V004a 4:1 slope with Baker's iterated Mohr-Coulomb parameters (c′ = 2.439 kPa, φ′ = 30.392°); critical-circle search.

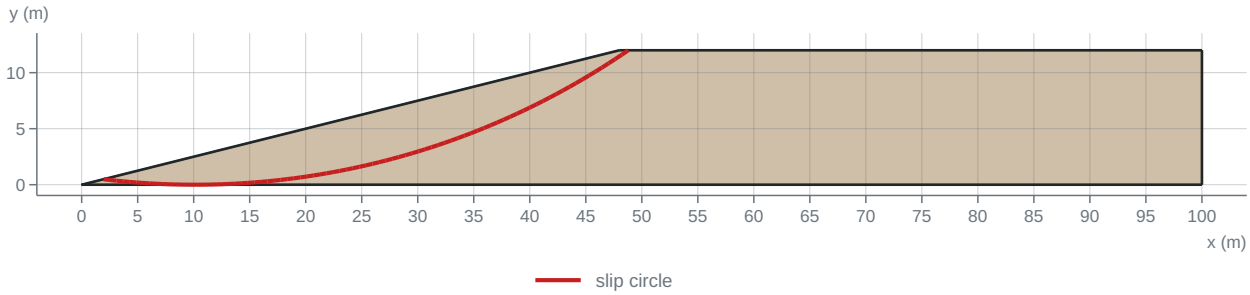


Figure 6. V004b: critical circle (Janbu simplified, F = 2.609).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c′ = 2.44	30.39°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu simplified	2.609				2.610	−0.04 %
Spencer	2.667				2.696	−1.09 %
–	–	2.64	Baker (2003): non-linear strength envelope			
–	–	2.66	Baker (2003): Mohr-Coulomb strength			

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.6 V005: Baker (1993) dam, end of construction, infinite-slope face

Baker, R., Frydman, S. & Talesnick, M. (1993). Slope stability analysis for undrained loading conditions. Int. J. Numer. Anal. Meth. Geomech. 17, 15–43.

Critical circle search · Units: SI (m, kPa, kN/m³)

Compacted-clay embankment (c′ = 6.5 kPa, φ′ = 40°) on a deep natural-clay foundation (c′ = 0, φ′ = 32°) with a 6 m crest and a straight 4H:1V downstream face, whose natural-clay part governs; dry; circular search. Theoretical (infinite slope) F = tan 32°/tan 14.04° = 2.50.

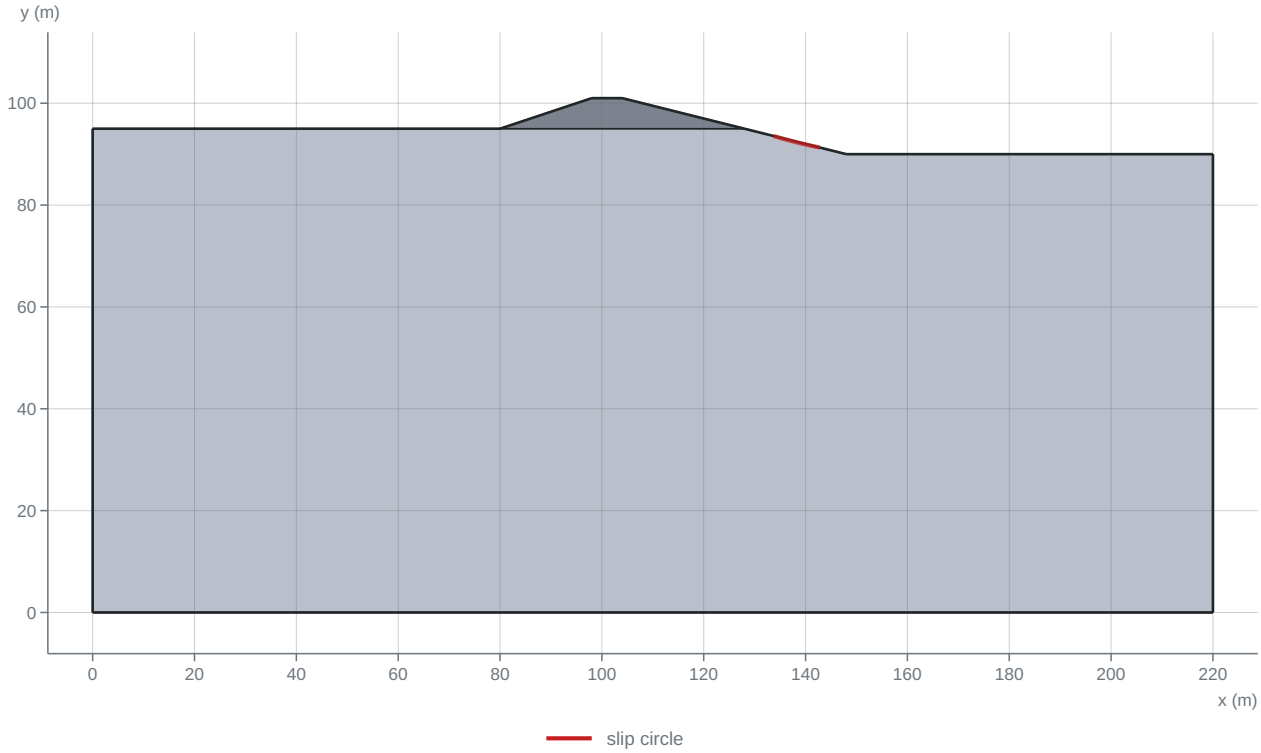


Figure 7. V005: critical circle (Bishop simplified, F = 2.502).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Compacted clay	Mohr-Coulomb	18.00	c′ = 6.50	40.00°
Natural clay	Mohr-Coulomb	18.00	c′ = 0.00	32.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.502	2.50	Theory: infinite slope	+0.07 %		
Spencer	2.502					
Spencer, non-circular	2.500	2.41	Baker, Frydman & Talesnick (1993): Spencer, non-circular (dynamic programming)	+3.75 %		

3.7 V006: Baker & Leshchinsky (2001) planar surface

Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Homogeneous 10 m slope on the critical plane from the toe to the crest at 49.6°.

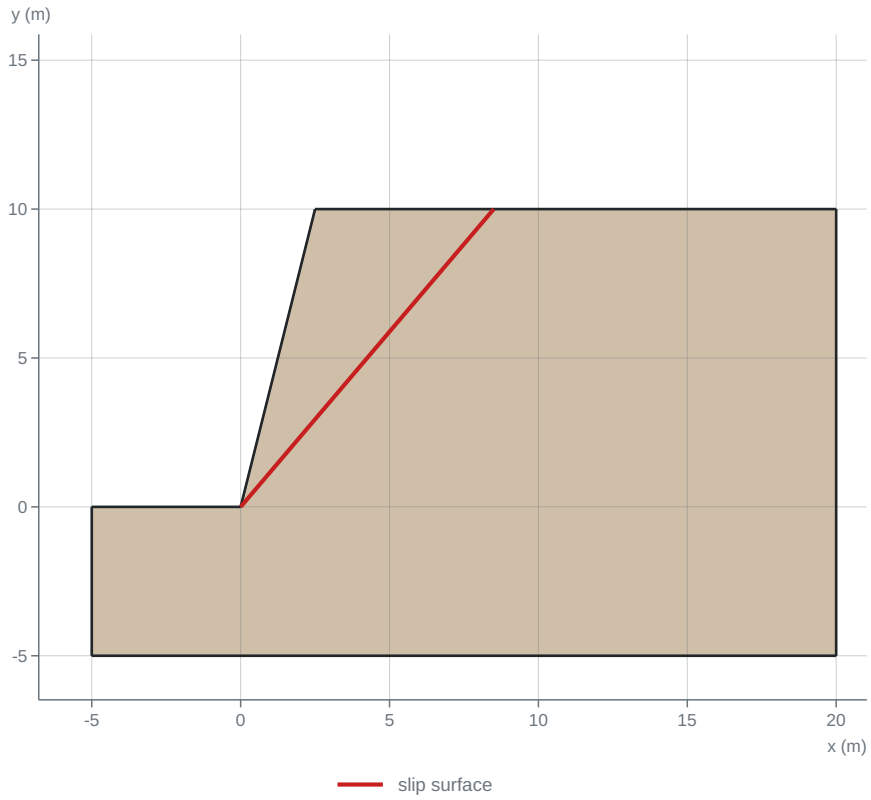


Figure 8. V006: analyzed surface (Spencer, F = 1.352).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c′ = 30.00	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.352				1.351	+0.07 %
Janbu simplified	1.352				1.351	+0.07 %
–	–	1.35	Baker & Leshchinsky (2001): Culmann (planar wedge)			

3.8 V007a: Baker (2003) ex. 1, Mohr-Coulomb, Janbu simplified on the reference circle

Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V008a slope (Mohr-Coulomb fit c′ = 11.64 kPa, φ′ = 24.7°) with Janbu simplified (uncorrected) on the reference critical circle (Fig. 44.3).

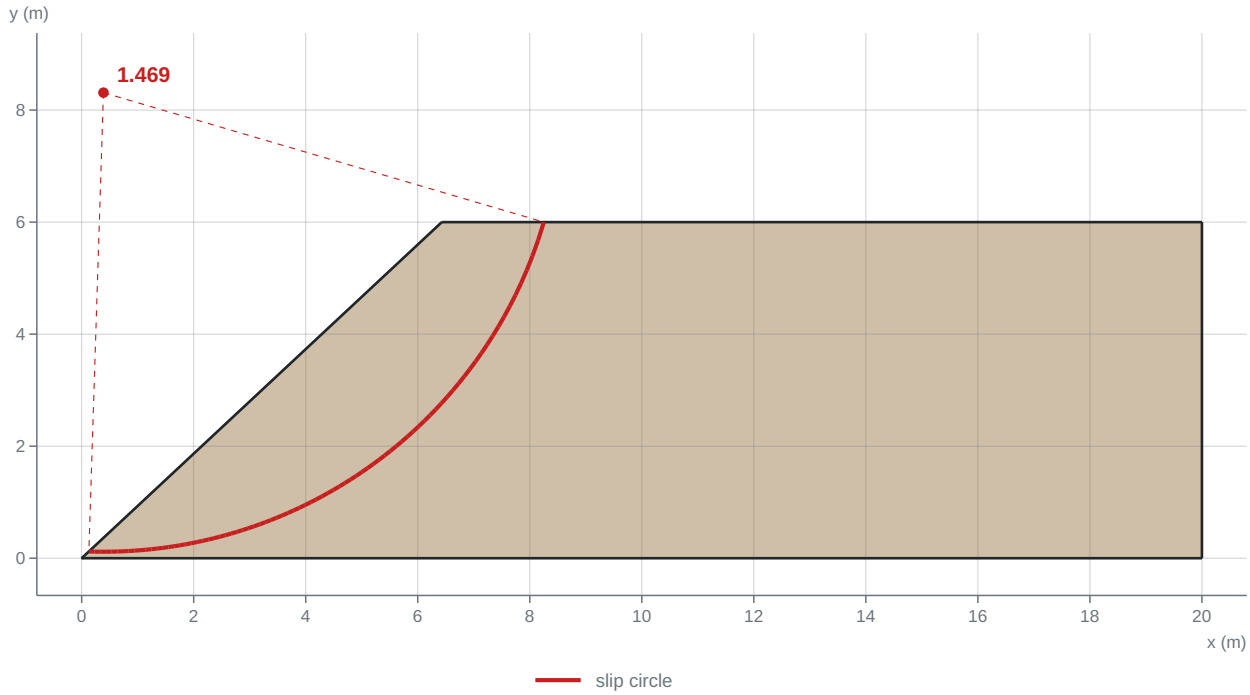


Figure 9. V007a: analyzed circle (Janbu simplified, F = 1.469).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c′ = 11.64	24.70°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu simplified	1.469				1.469	−0.03 %
–	–	1.50	Baker (2003): Mohr-Coulomb strength			

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.9 V007b: Baker (2003) ex. 1 slope, Mohr-Coulomb iteration parameters

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V008a slope with Baker's Mohr-Coulomb parameters from his iterative linearisation (c′ = 0.39 kPa, φ′ = 38.6°); critical-circle search.

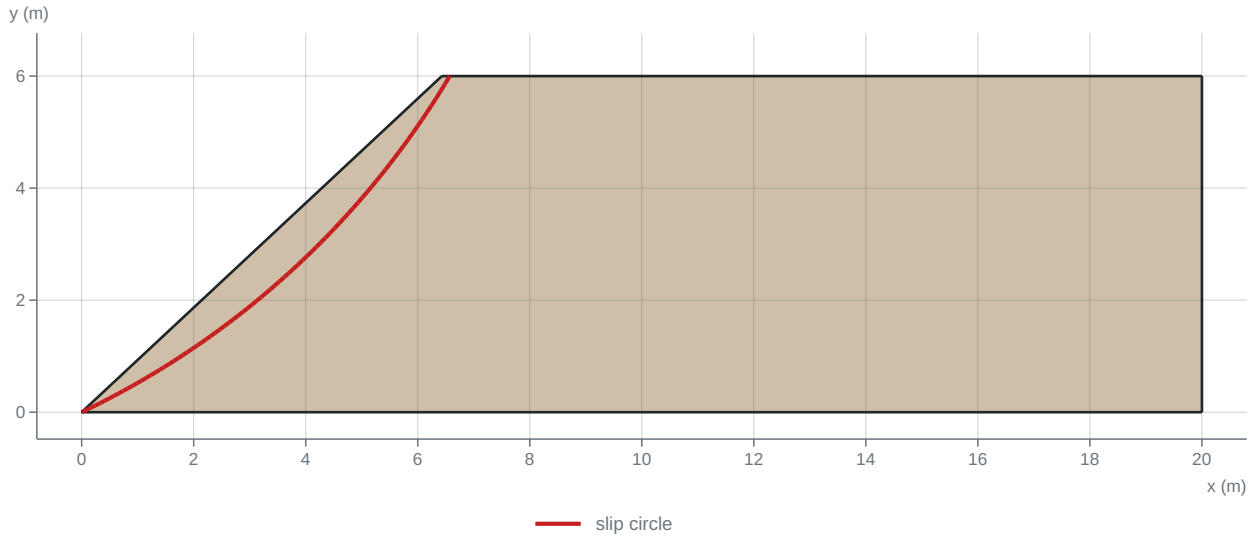


Figure 10. V007b: critical circle (Janbu simplified, F = 0.955).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c′ = 0.39	38.60°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu simplified	0.955				0.957	−0.17 %
Spencer	0.978				0.981	−0.35 %
–	–	0.97	Baker (2003): non-linear strength envelope			
–	–	1.50	Baker (2003): Mohr-Coulomb strength			

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387); 1.50 is his Mohr-Coulomb value for the same slope.

3.10 V007c: Baker (2003) ex. 3, Janbu simplified

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V008b London clay slope (c′ = 6 kPa, φ′ = 32°) with Janbu simplified (uncorrected); critical-circle search.

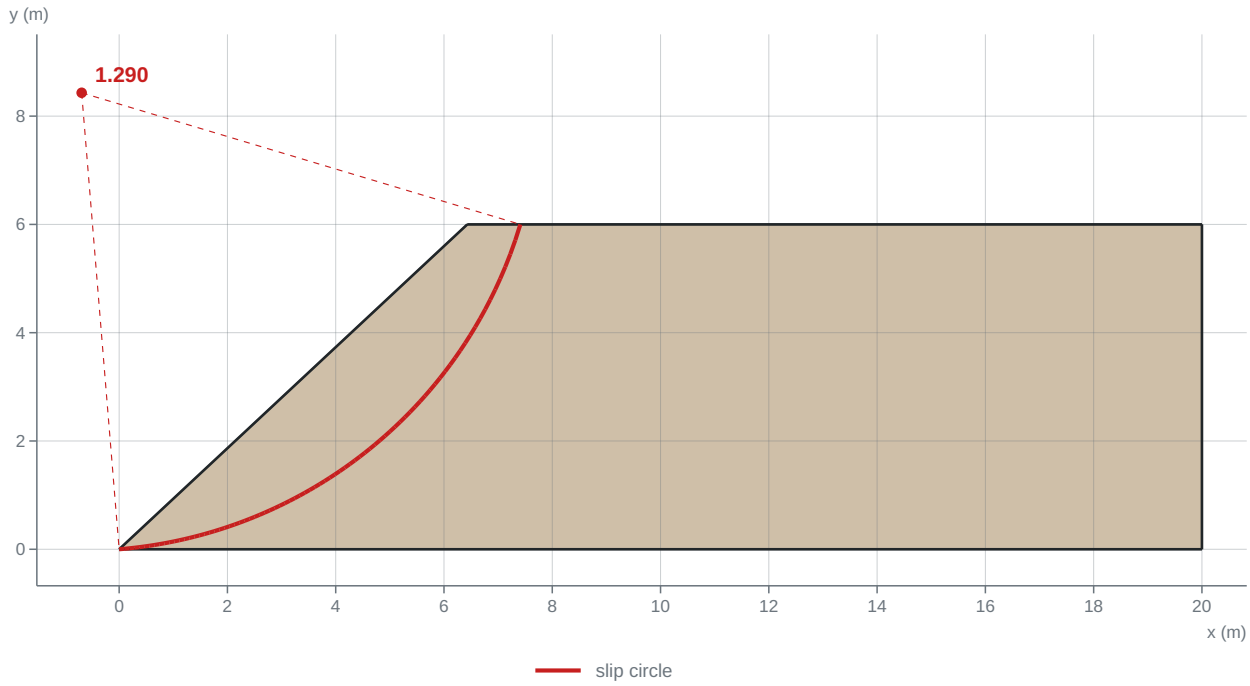


Figure 11. V007c: critical circle (Janbu simplified, F = 1.290).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c′ = 6.00	32.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu simplified	1.290				1.291	−0.05 %
–	–	1.35	Baker (2003): Mohr-Coulomb strength			

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.11 V008a: Baker (2003) example 1, Mohr-Coulomb fit

Baker (2003), example 1.

Critical circle search · Units: SI (m, kPa, kN/m³)

6 m high slope at about 43° in soil with a linear Mohr-Coulomb fit to a curved envelope; critical-circle search.

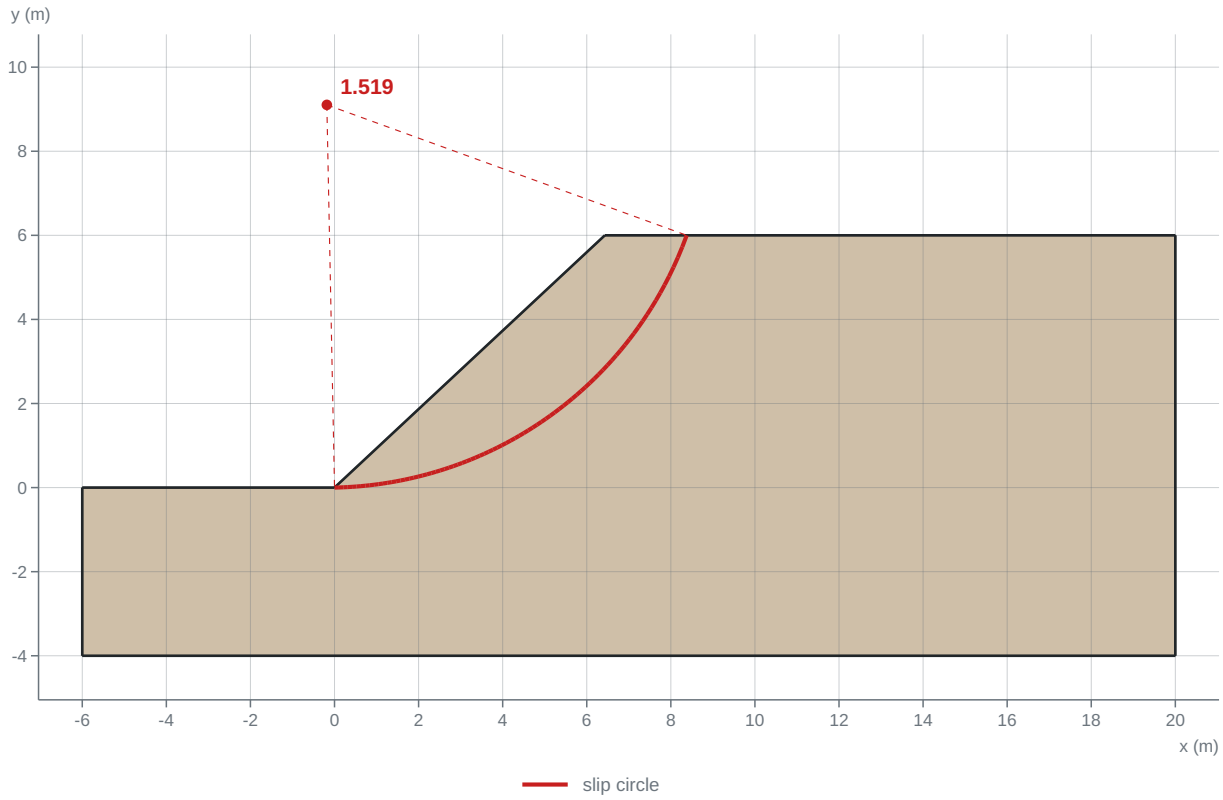


Figure 12. V008a: critical circle (Spencer, F = 1.519).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c′ = 11.64	24.70°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.519				1.536	−1.11 %
–	–	1.50	Baker (2003): Mohr-Coulomb strength			

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.12 V008b: Baker (2003) London clay slope, Mohr-Coulomb fit

Baker (2003) example 3.

Critical circle search · Units: SI (m, kPa, kN/m³)

43° slope, 6 m high, with the Mohr-Coulomb envelope fitted to London clay data (c′ = 6 kPa, φ′ = 32°); critical-circle search.

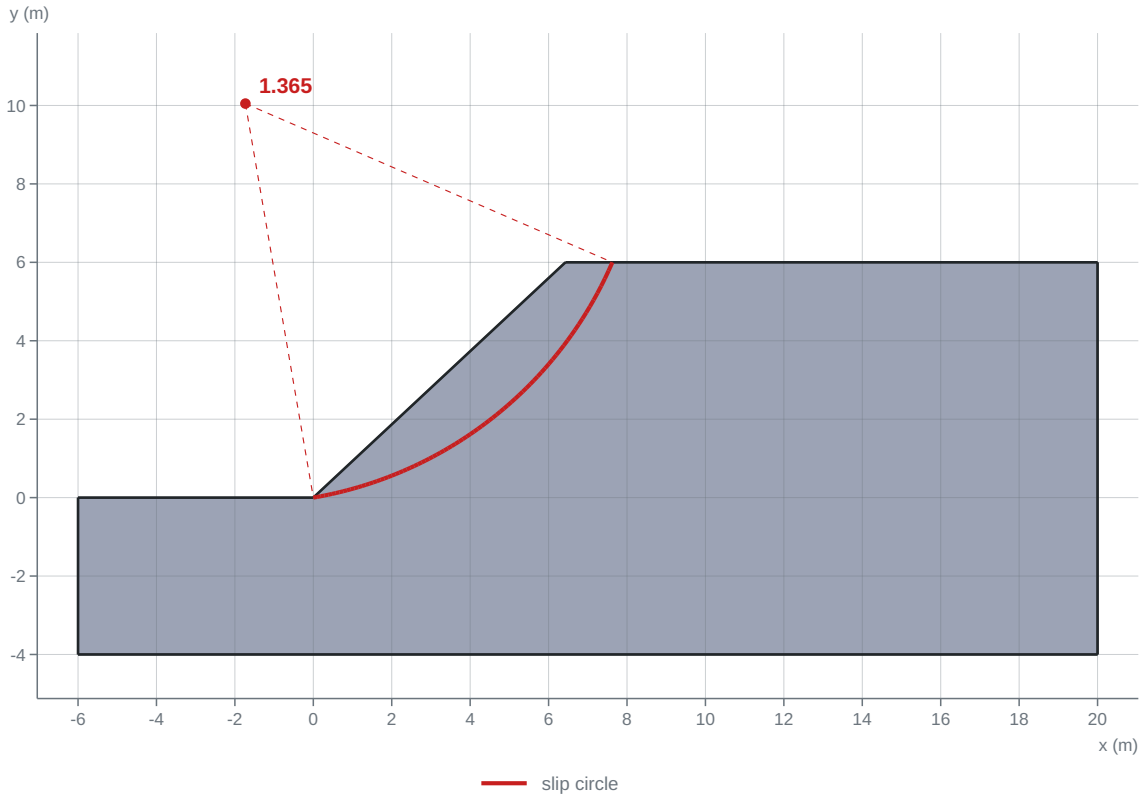


Figure 13. V008b: critical circle (Spencer, F = 1.365).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
London clay	Mohr-Coulomb	18.00	c′ = 6.00	32.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.365				1.366	−0.09 %
–	–	1.35	Baker (2003): Mohr-Coulomb strength			

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.13 V009a: Baker & Leshchinsky clay-core dam, reservoir and crest crack

Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.

Fixed circle · Units: SI (m, kPa, kN/m³)

Granular dam with a clay core (a hexagon reaching the base) on a hard base, half-full reservoir standing on the face (free water), phreatic surface through the core and a 5 m dry tension crack at the crest; a commercial program's critical circle.

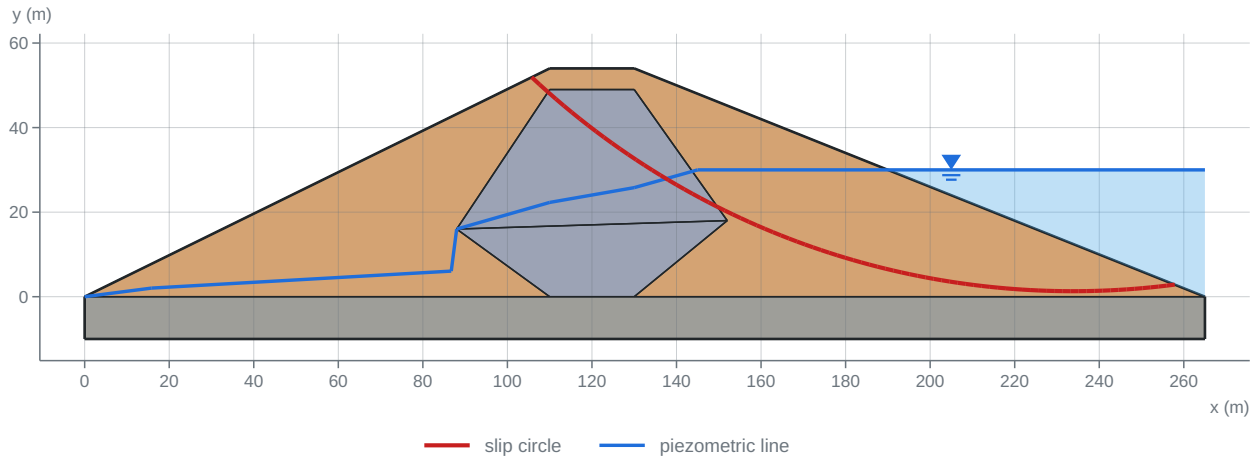


Figure 14. V009a: analyzed circle (Spencer, F = 1.927).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular fill	Mohr-Coulomb	21.50	c′ = 0.00	40.00°
Clay core	Mohr-Coulomb	20.00	c′ = 20.00	20.00°
Hard base	Mohr-Coulomb	24.00	c′ = 200.0	45.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.927				1.925	+0.08 %

The published 1.91 belongs to Baker & Leshchinsky's own non-circular surface (V009b), not to this circle.

3.14 V009b: Baker & Leshchinsky dam, Baker's non-circular surface

Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

As V009a, on the non-circular surface published by Baker & Leshchinsky.

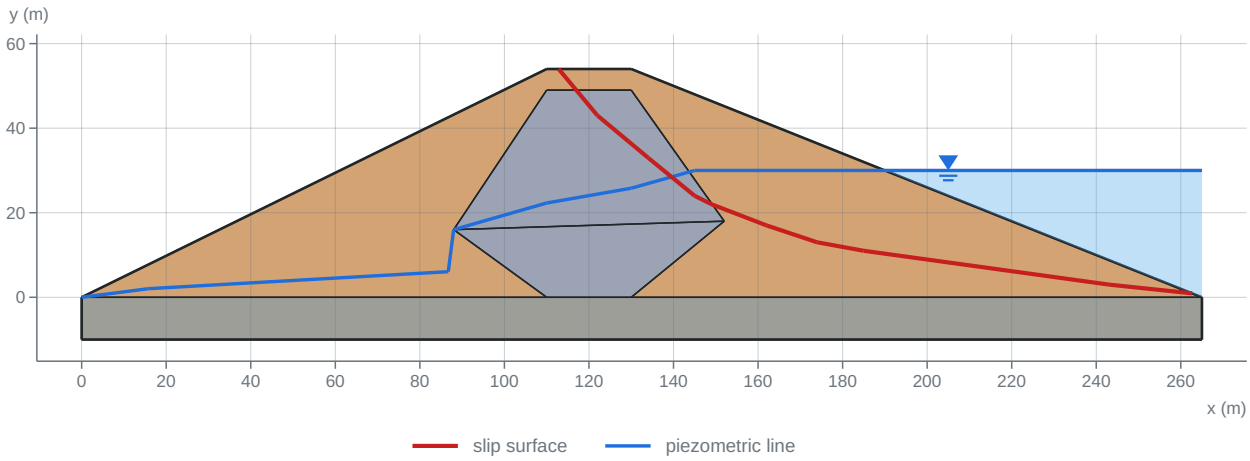


Figure 15. V009b: analyzed surface (Spencer, F = 1.882).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular fill	Mohr-Coulomb	21.50	c′ = 0.00	40.00°
Clay core	Mohr-Coulomb	20.00	c′ = 20.00	20.00°
Hard base	Mohr-Coulomb	24.00	c′ = 200.0	45.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.882	1.91	Baker & Leshchinsky	−1.47 %		

3.15 V009c: Baker & Leshchinsky clay-core dam, critical surface search

Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.

Circle search with non-circular optimization · Units: SI (m, kPa, kN/m³)

The V009a clay-core dam (reservoir on the face, 5 m dry crest crack) with a circular search and non-circular optimization.

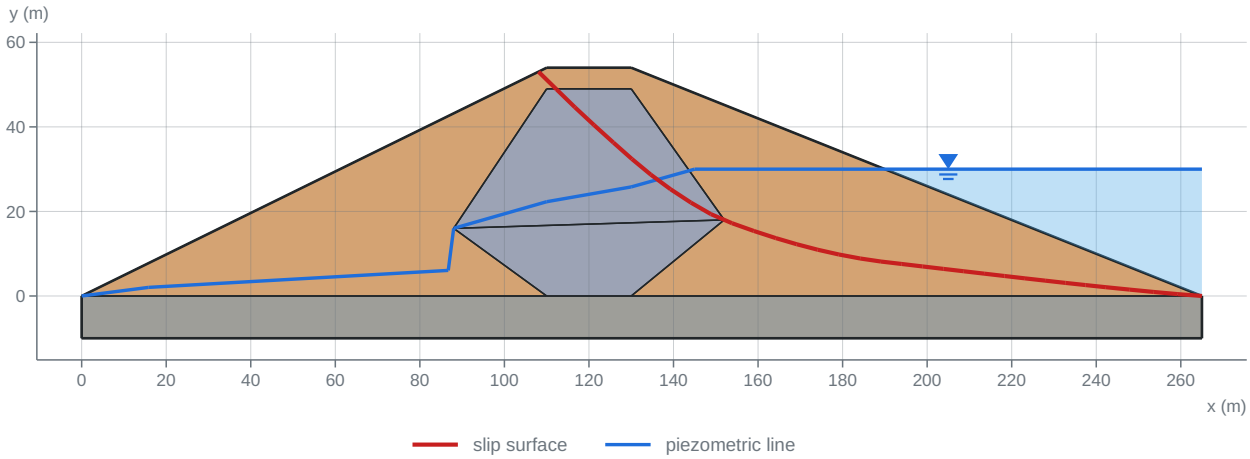


Figure 16. V009c: critical surface (Spencer, non-circular, F = 1.867).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular fill	Mohr-Coulomb	21.50	c′ = 0.00	40.00°
Clay core	Mohr-Coulomb	20.00	c′ = 20.00	20.00°
Hard base	Mohr-Coulomb	24.00	c′ = 200.0	45.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer, non-circular	1.867	1.91	Baker & Leshchinsky (2001)	−2.25 %	1.868	−0.05 %

Baker & Leshchinsky's 1.91 is for their own surface (V009b); searches find lower surfaces.

3.16 V010: Borges & Cardoso embankment 2 (geosynthetic)

Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.

Fixed circle · Units: SI (m, kPa, kN/m³)

5 m embankment (φ′ = 35°) on four soft clays with Su increasing with depth (Borges & Cardoso Table 6, p. 408: clay 3 from 16 to 18.4 kPa, clay 4 from 18.4 to 55.1 kPa), a 200 kN/m geosynthetic at y = 1 m (axial, active, not anchored, interface friction 33.7° on both faces); a commercial program's critical circle.

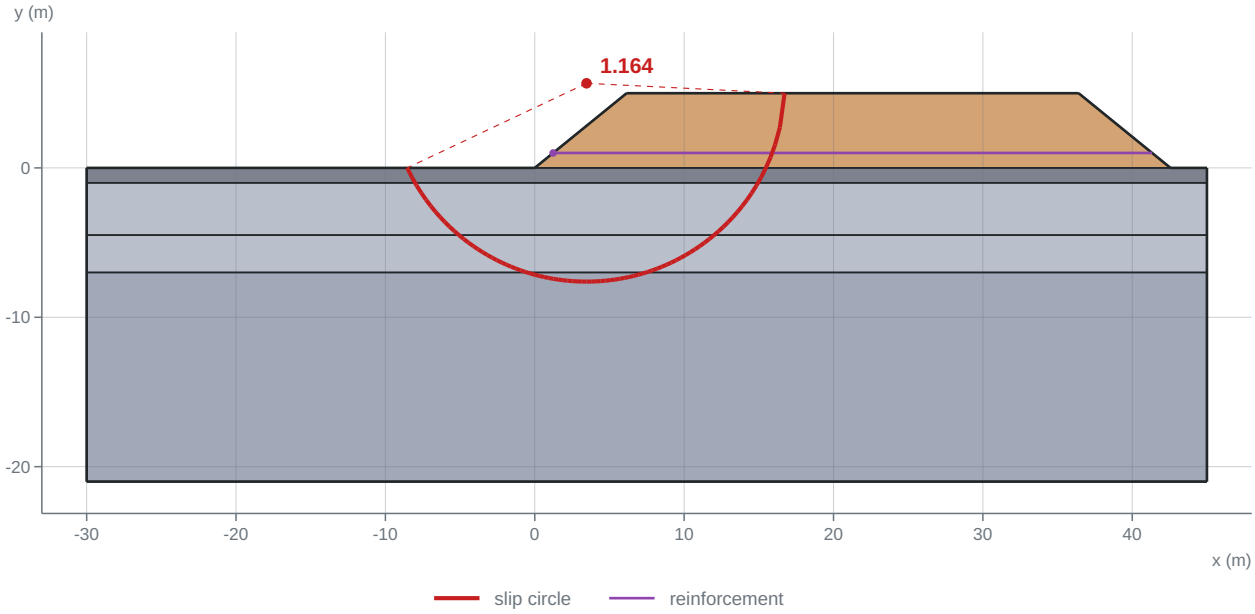


Figure 17. V010: analyzed circle (Morgenstern-Price (half-sine), F = 1.164).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	20.00	c′ = 0.00	35.00°
Clay 1	Undrained (φ = 0)	17.00	Su = 33.00	0°
Clay 2	Undrained (φ = 0)	17.00	Su = 16.00	0°
Clay 3	Undrained (φ = 0)	17.00	Su = 16.00 at y = -4.50, +0.95 kPa/m below	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 18.38 at y = -7.00, +2.63 kPa/m below	0°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	1.164				1.171	−0.56 %
Bishop simplified	1.166				1.170	−0.36 %

3.17 V010b: Borges & Cardoso embankment 2 (geosynthetic), the paper's circle B

Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421; case 2, Table 11, Fig. 14.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V010 embankment as Borges & Cardoso analyze it: the 200 kN/m geosynthetic at 0.9 m height (p. 407) with its full horizontal force in the resisting moment (passive), on their circle B (3.0, 8.0, R = 15.04).

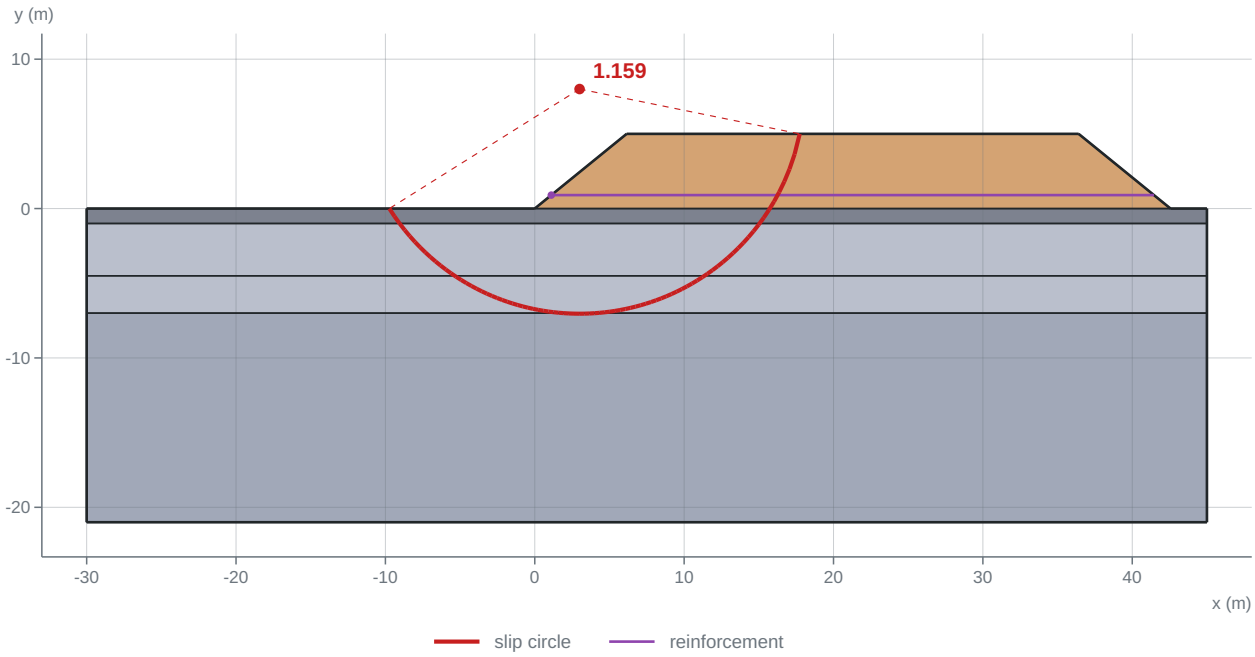


Figure 18. V010b: analyzed circle (Bishop simplified, F = 1.159).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	20.00	c′ = 0.00	35.00°
Clay 1	Undrained (φ = 0)	17.00	Su = 33.00	0°
Clay 2	Undrained (φ = 0)	17.00	Su = 16.00	0°
Clay 3	Undrained (φ = 0)	17.00	Su = 16.00 at y = -4.50, +0.95 kPa/m below	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 18.38 at y = -7.00, +2.63 kPa/m below	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.159	1.15	Borges & Cardoso	+0.79 %		
–	–	1.19	Borges & Cardoso (2002): authors' moment method, circle A			
–	–	1.22	Borges & Cardoso (2002): finite elements, circle A			

Borges & Cardoso use their own moment-equilibrium method of slices on circles (Kaniraj & Abdullah 1993 with their improvements); simplified Bishop is the nearest of cSlope's methods. Circle B's coordinates and the geosynthetic level are printed in the paper (Table 11, p. 407).

3.18 V011a: Borges & Cardoso embankment 3, H = 7 m, circle A

Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.

Fixed circle · Units: SI (m, kPa, kN/m³)

Granular fill (two stages) on five soft clay layers with a T = 200 kN/m geosynthetic at y = 0.9 m (axial, passive as in a commercial program, pullout over 1.3 m from each end); Bishop on circle A of the paper.

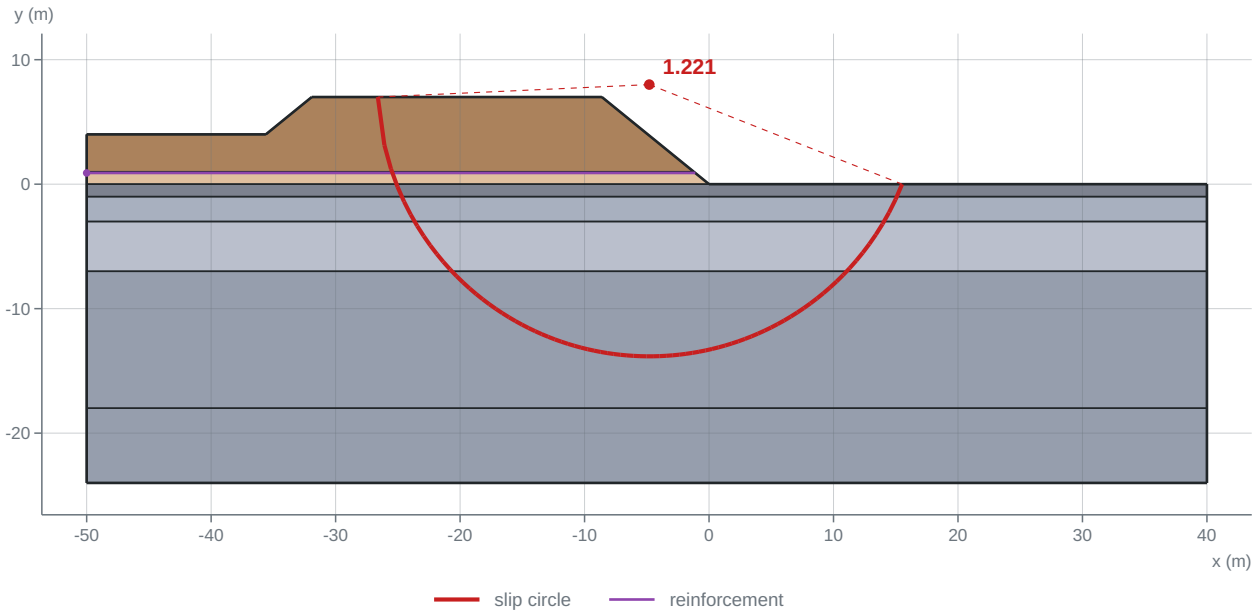


Figure 19. V011a: analyzed circle (Bishop simplified, F = 1.221).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper embankment	Mohr-Coulomb	21.90	c′ = 0.00	35.00°
Lower embankment	Mohr-Coulomb	17.20	c′ = 0.00	33.00°
Clay 1	Undrained (φ = 0)	18.00	Su = 43.00	0°
Clay 2	Undrained (φ = 0)	16.60	Su = 31.00	0°
Clay 3	Undrained (φ = 0)	13.50	Su = 30.00	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 32.00	0°
Clay 5	Undrained (φ = 0)	17.50	Su = 32.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.221				1.230	−0.72 %
–	–	1.25	Borges & Cardoso (2002)			

Borges & Cardoso print two decimals and do not name the method; the other programs differ from them by up to 2.5 %.

3.19 V011b: Borges & Cardoso embankment 3, H = 7 m, circle B

Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V011a on circle B.

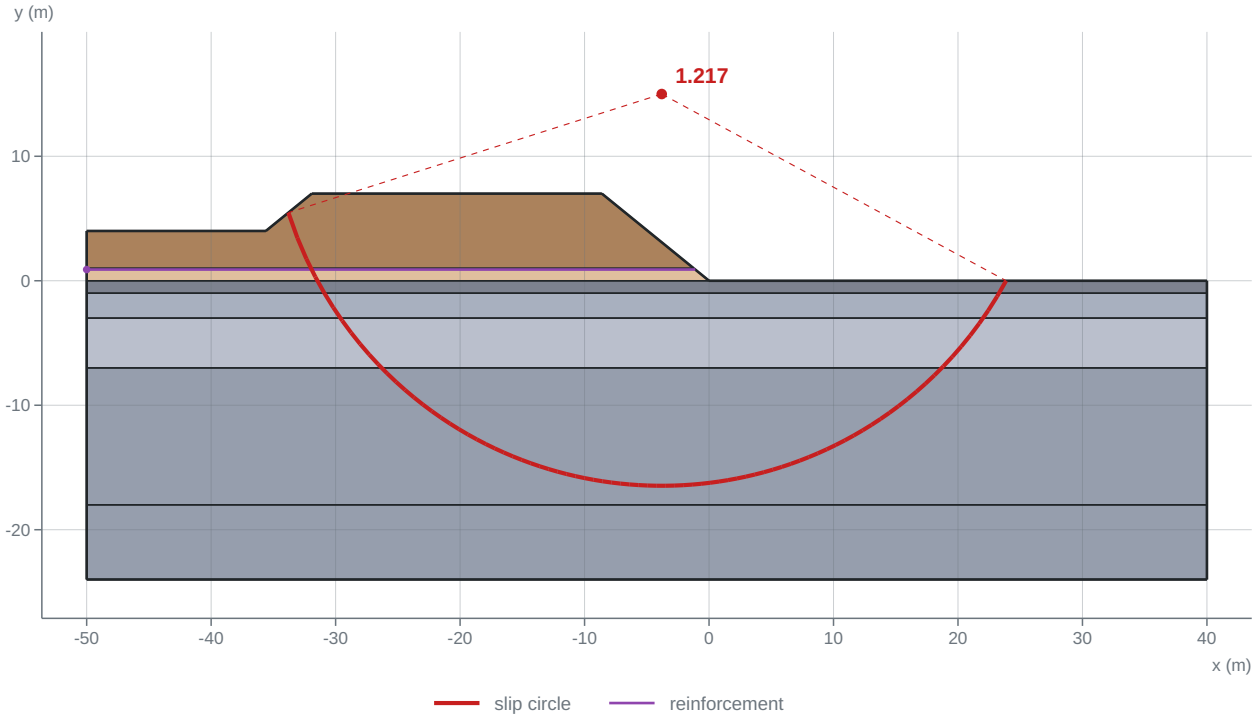


Figure 20. V011b: analyzed circle (Bishop simplified, F = 1.217).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper embankment	Mohr-Coulomb	21.90	c′ = 0.00	35.00°
Lower embankment	Mohr-Coulomb	17.20	c′ = 0.00	33.00°
Clay 1	Undrained (φ = 0)	18.00	Su = 43.00	0°
Clay 2	Undrained (φ = 0)	16.60	Su = 31.00	0°
Clay 3	Undrained (φ = 0)	13.50	Su = 30.00	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 32.00	0°
Clay 5	Undrained (φ = 0)	17.50	Su = 32.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.217				1.220	−0.27 %
–	–	1.19	Borges & Cardoso (2002)			

Borges & Cardoso print two decimals and do not name the method; the other programs differ from them by up to 2.5 %.

3.20 V012: Borges & Cardoso embankment 3, H = 8.75 m, circle C

Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.

Fixed circle · Units: SI (m, kPa, kN/m³)

The second fill stage (8.75 m crest) on circle C.

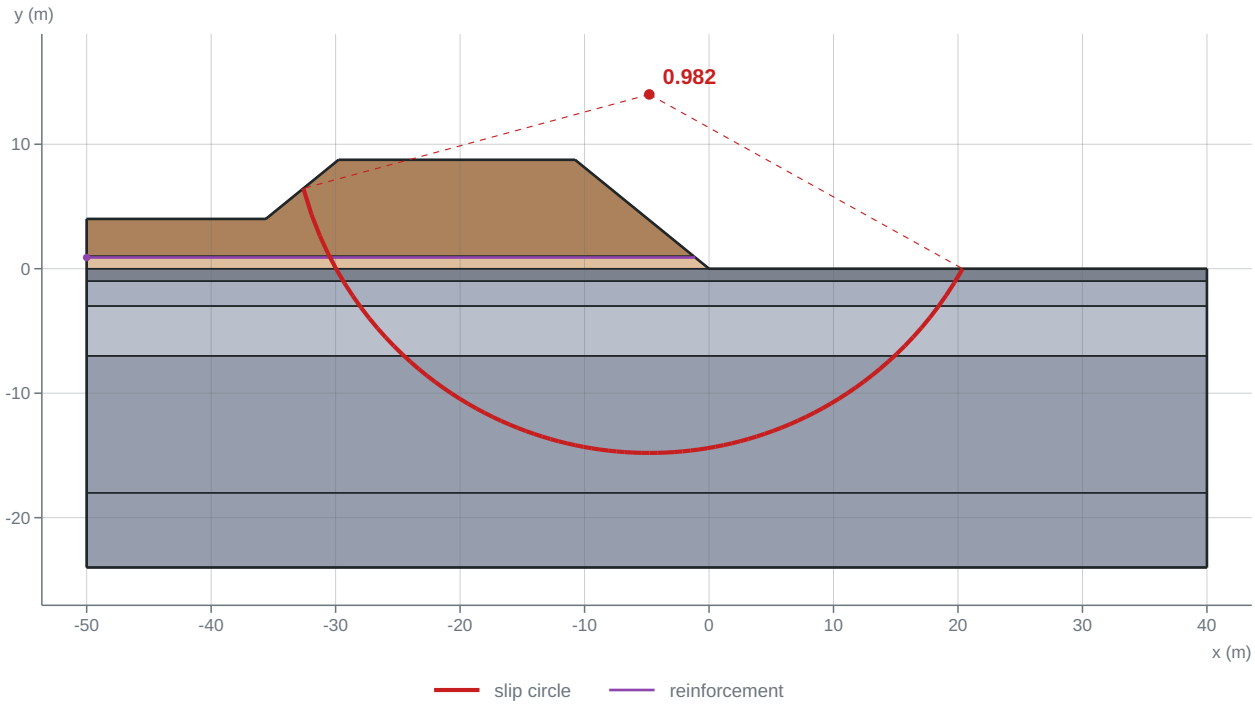


Figure 21. V012: analyzed circle (Bishop simplified, F = 0.982).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper embankment	Mohr-Coulomb	21.90	c′ = 0.00	35.00°
Lower embankment	Mohr-Coulomb	17.20	c′ = 0.00	33.00°
Clay 1	Undrained (φ = 0)	18.00	Su = 43.00	0°
Clay 2	Undrained (φ = 0)	16.60	Su = 31.00	0°
Clay 3	Undrained (φ = 0)	13.50	Su = 30.00	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 32.00	0°
Clay 5	Undrained (φ = 0)	17.50	Su = 32.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	0.982				0.980	+0.19 %
Strength reduction (FE)	0.928	0.95	another program's verification manual: finite-element strength reduction (verification manual, problem 24)	−2.35 %		
–	–	0.99	Borges & Cardoso (2002)			

Borges & Cardoso print two decimals and do not name the method.

3.21 V013a: Cai & Ugai (2000) slope, no pile (see V013b, V013c, V013d and V013e for the pile rows)

Cai & Ugai (2000), Soils and Foundations 40(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 1:1.5 slope, 10 m high, without piles; critical-circle search.

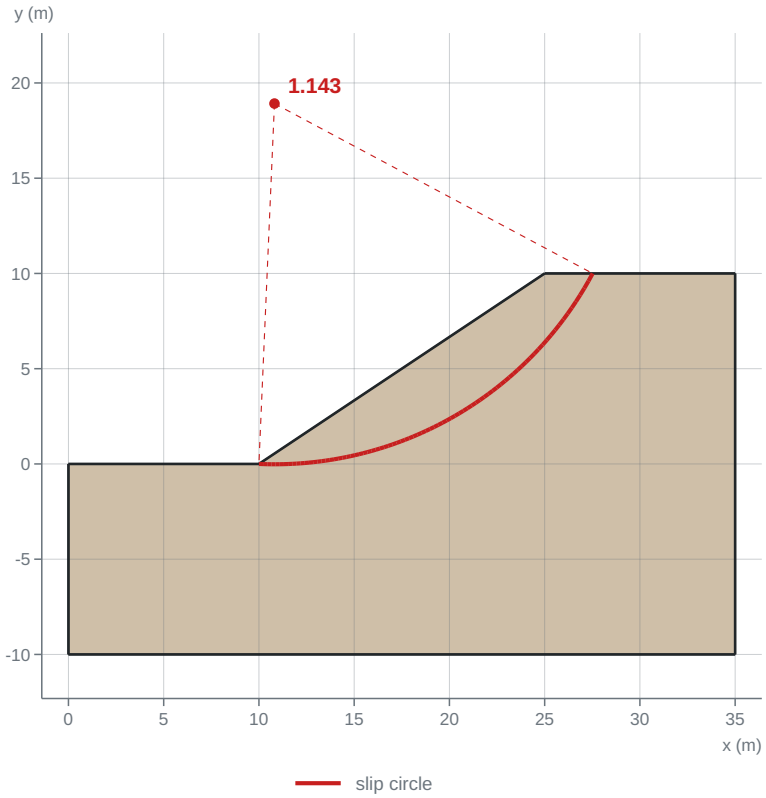


Figure 22. V013a: critical circle (Bishop simplified, F = 1.143).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c′ = 10.00	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.143	1.13	Cai & Ugai	+1.15 %	1.140	+0.26 %
Strength reduction (FE)	1.133	1.14	Cai & Ugai (2000): finite-element strength reduction (3D, without piles)	−0.64 %		

3.22 V013b: Cai & Ugai slope, pile row D1/D = 2

Cai & Ugai (2000), Soils and Foundations 40(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

A row of 0.8 m piles at mid-slope, center-to-center 1.6 m, with the Ito & Matsui force (passive); Bishop search.

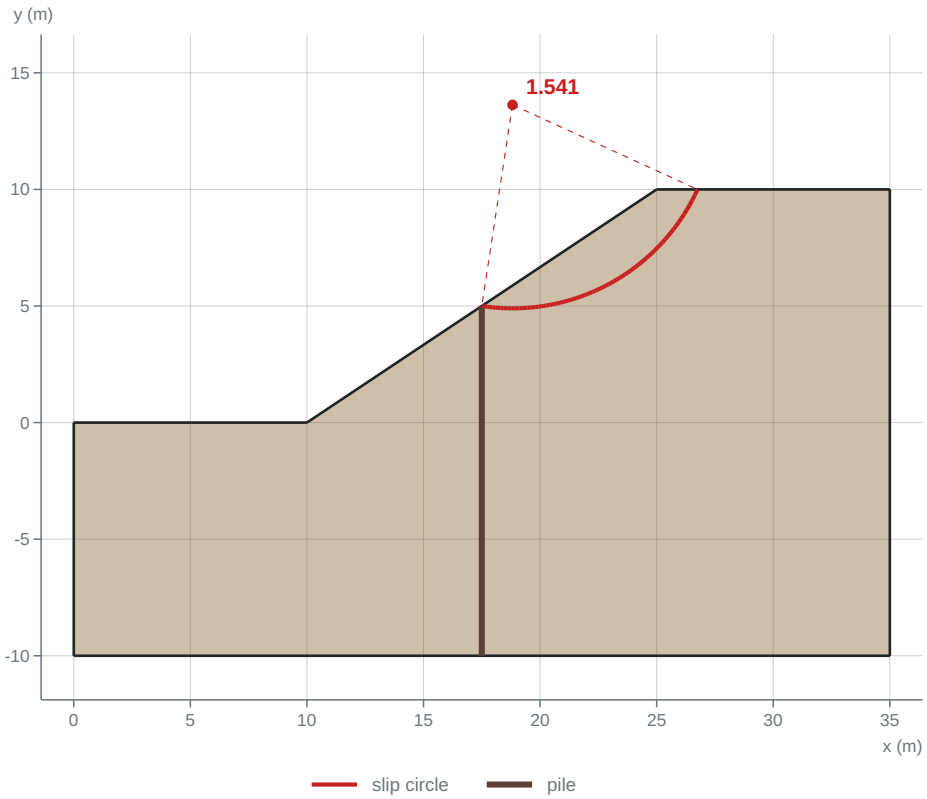


Figure 23. V013b: critical circle (Bishop simplified, F = 1.541).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c′ = 10.00	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.541				1.540	+0.07 %
–	–	1.54	Cai & Ugai (2000)			

Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.

3.23 V013c: Cai & Ugai slope, pile row D1/D = 3

Cai & Ugai (2000), Soils and Foundations 40(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

As V013b at 2.4 m spacing.

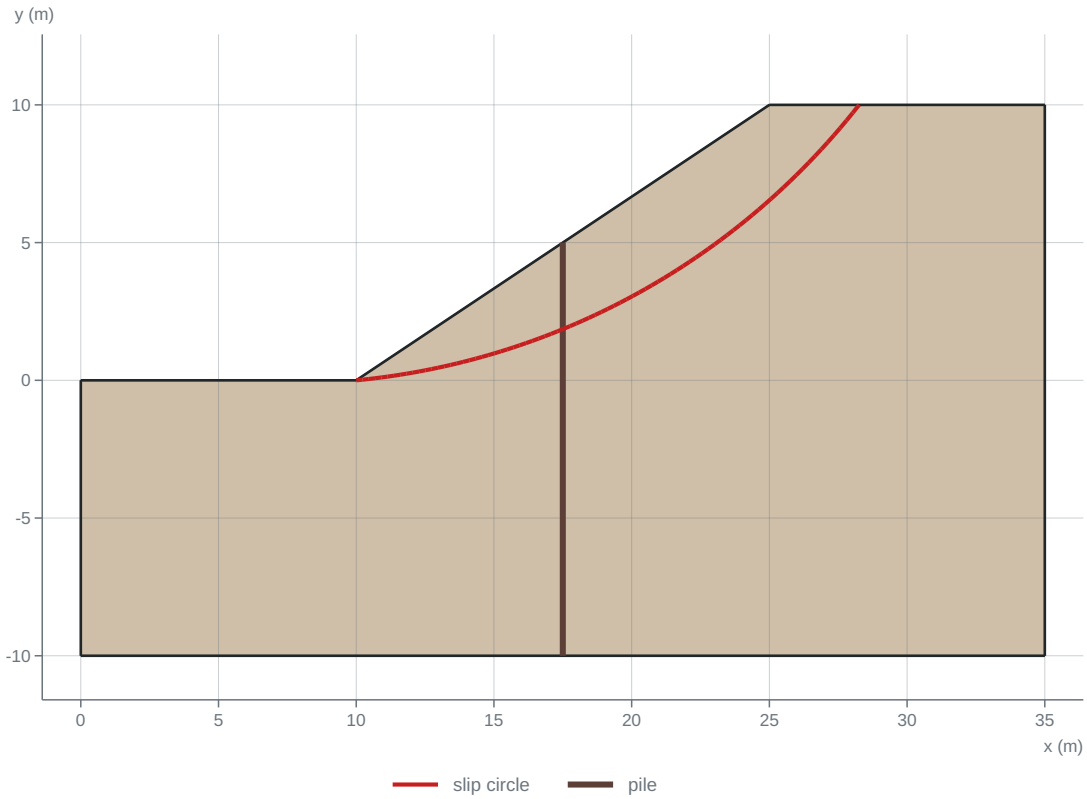


Figure 24. V013c: critical circle (Bishop simplified, F = 1.437).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c′ = 10.00	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.437				1.430	+0.51 %
–	–	1.37	Cai & Ugai (2000)			

Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.

3.24 V013d: Cai & Ugai slope, pile row D1/D = 4

Cai & Ugai (2000), Soils and Foundations 40(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

As V013b at 3.2 m spacing.

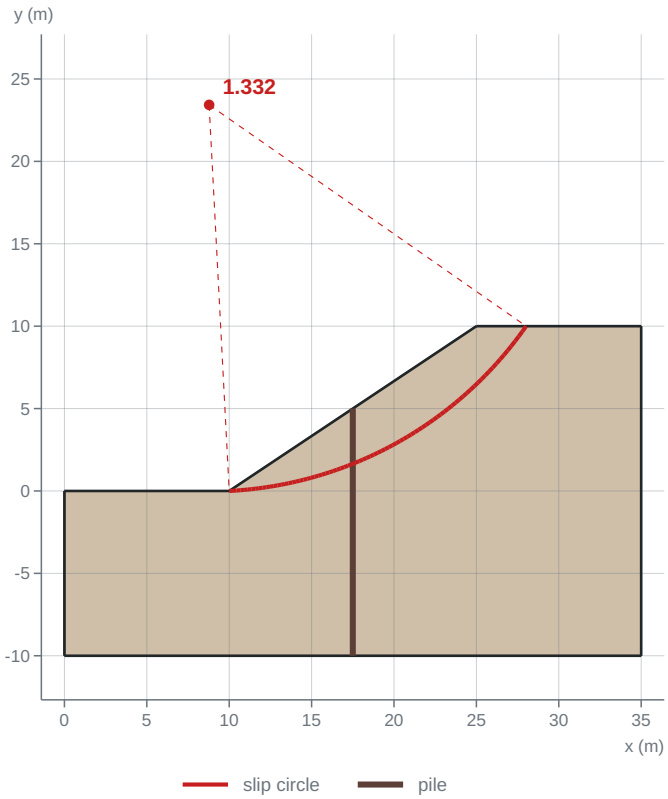


Figure 25. V013d: critical circle (Bishop simplified, F = 1.332).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c′ = 10.00	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.332				1.330	+0.12 %
–	–	1.31	Cai & Ugai (2000)			

Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.

3.25 V013e: Cai & Ugai slope, pile row D1/D = 6

Cai & Ugai (2000), Soils and Foundations 40(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

As V013b at 4.8 m spacing.

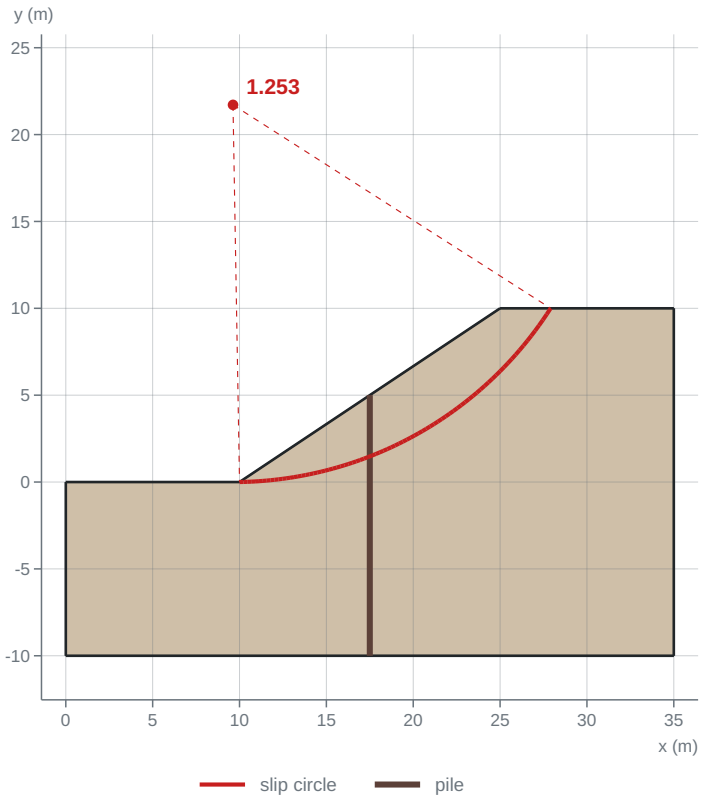


Figure 26. V013e: critical circle (Bishop simplified, F = 1.253).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c′ = 10.00	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.253				1.250	+0.24 %
–	–	1.25	Cai & Ugai (2000)			

Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.

3.26 V014: Nailed wall, 14 rows, deep wedge

Caltrans soil nail reference manual.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Nail-reinforced wall with 14 horizontal rows (per-row length, capacity and bond; fully anchored at the wall, tangent, passive) on the printed bilinear wedge. Converted to SI.

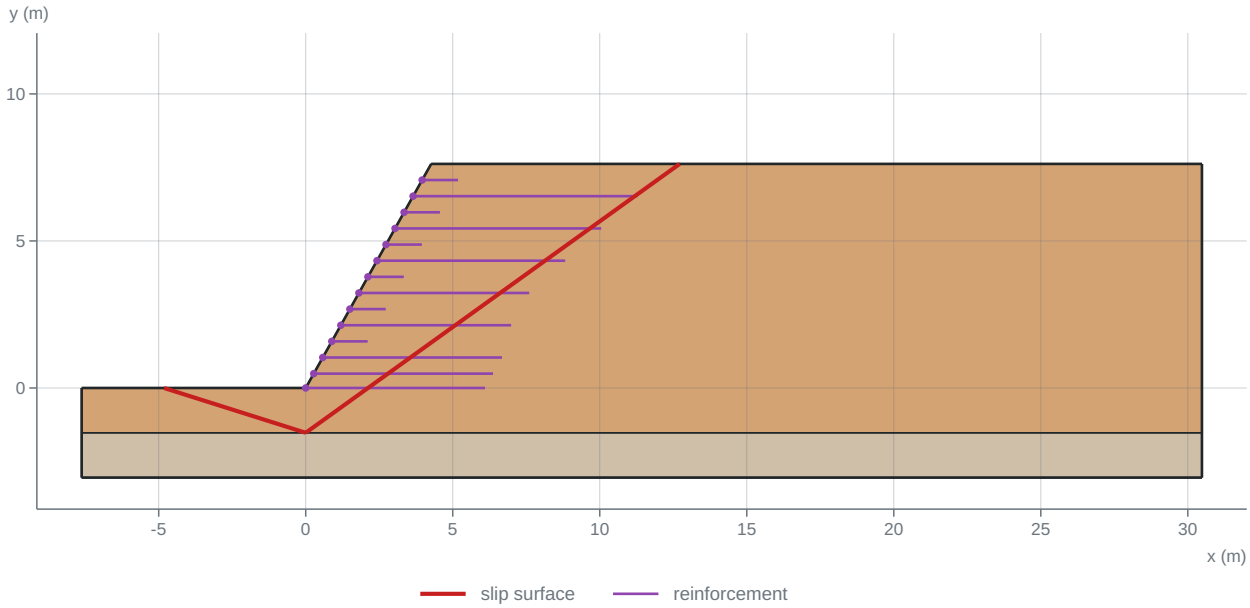


Figure 27. V014: analyzed surface (Janbu corrected, F = 1.448).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	19.64	c′ = 0.00	32.00°
Layer 2	Mohr-Coulomb	20.11	c′ = 23.94	35.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	1.448				1.448–1.460	−0.40 %
Spencer	1.575				1.576	−0.03 %

The reference results come from design programs quoted in the soil nail manual; no independent value is published.

3.27 V015: Soldier-pile tieback wall

Caltrans soil nail reference manual.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Two-layer slope cut by a soldier-pile wall with two grouted tieback rows (8 ft spacing); the soldier pile is a shear force of 5900 lb/ft where the surface passes, as a commercial program models it; the manual's bilinear wedge from the wall toe. Converted to SI.

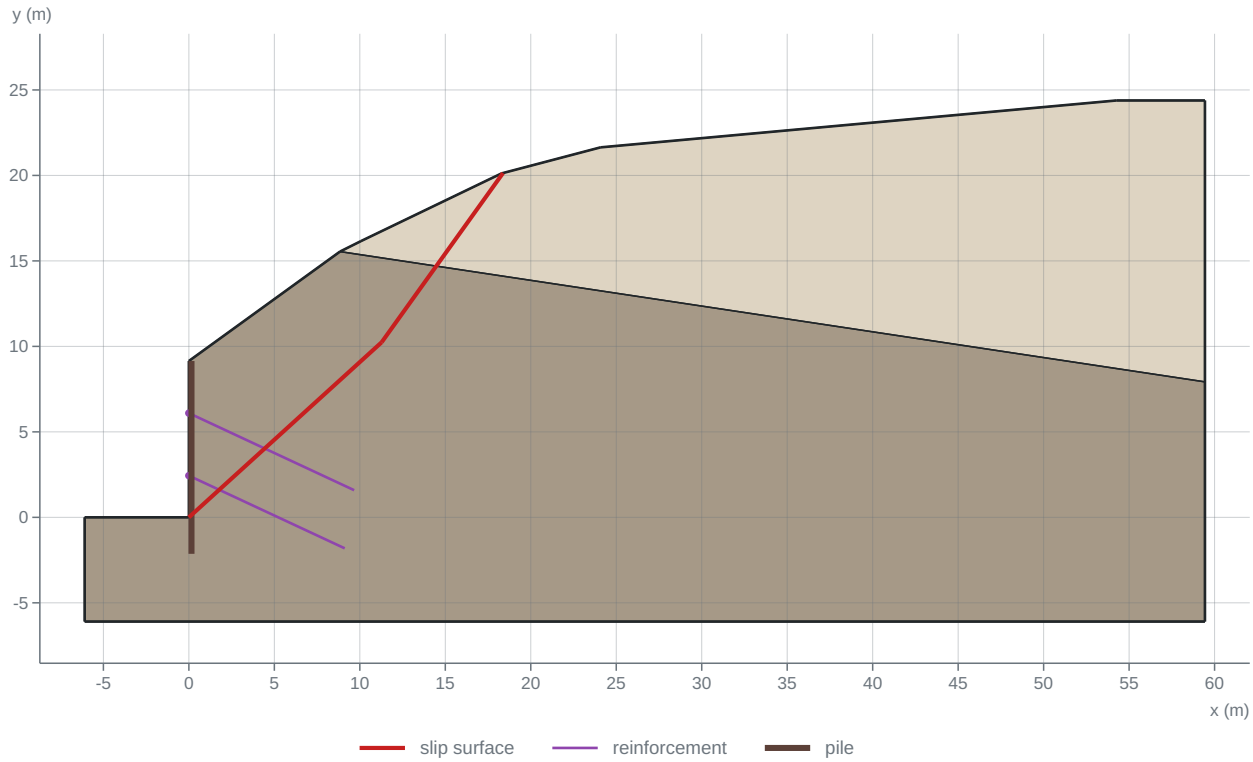


Figure 28. V015: analyzed surface (Janbu corrected, F = 1.476).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	18.85	c′ = 28.73	24.00°
Layer 2	Mohr-Coulomb	20.42	c′ = 14.36	34.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	1.476				1.479	−0.18 %
Janbu simplified	1.442				1.446	−0.29 %

The reference results come from design programs quoted in the soil nail manual; no independent value is published.

3.28 V016a: Stepped gabion wall, equivalent cohesion

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

4 m gabion wall of 1 m cubes with the steps facing outwards on a sloping two-layer profile, dry; mesh as an equivalent cohesion (59.7 kPa); the reference circle.

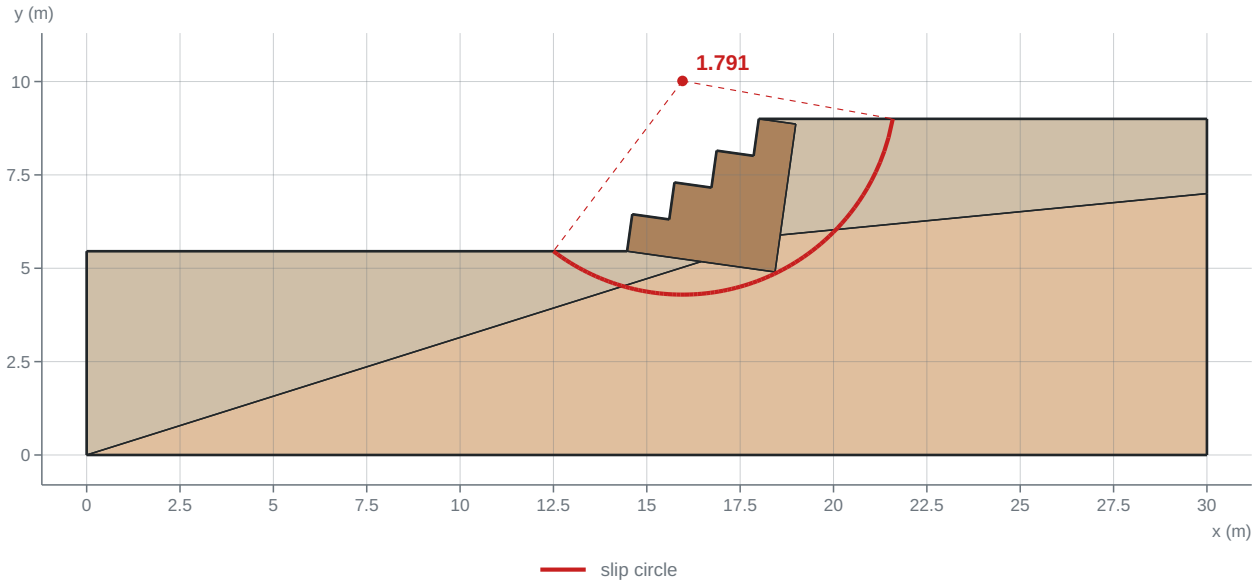


Figure 29. V016a: analyzed circle (Bishop simplified, F = 1.791).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c′ = 5.00	30.00°
Soil #2	Mohr-Coulomb	20.00	c′ = 0.00	25.00°
Gabion wall	Mohr-Coulomb	23.00	c′ = 59.70	42.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.791				1.787–1.790	+0.15 %
Spencer	1.798				1.791	+0.40 %

The paper gives its results in figures only; the comparison is with the other programs.

3.29 V016b: Stepped gabion wall, mesh as geosynthetic supports

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V016a with the mesh as geosynthetic supports (T = 100 kN/m) at the course interfaces.

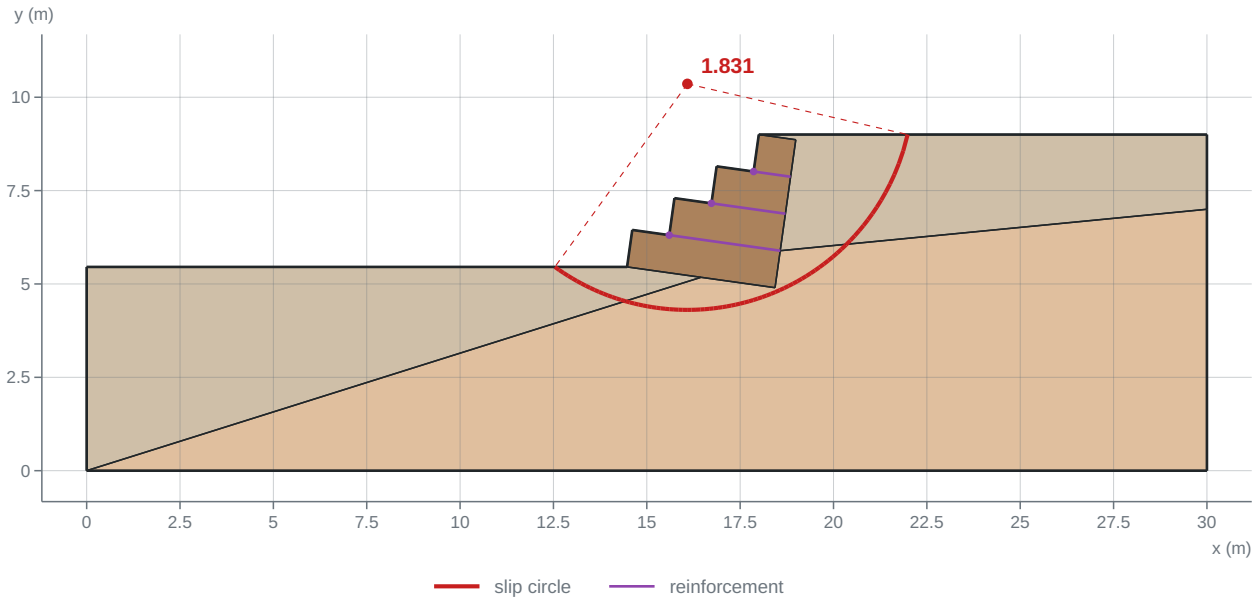


Figure 30. V016b: analyzed circle (Bishop simplified, F = 1.831).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c′ = 5.00	30.00°
Soil #2	Mohr-Coulomb	20.00	c′ = 0.00	25.00°
Gabion wall	Mohr-Coulomb	23.00	c′ = 0.00	42.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.831				1.830–1.835	−0.09 %
Spencer	1.836				1.839	−0.17 %

The paper gives its results in figures only; the comparison is with the other programs.

3.30 V016c: Gabion wall with weak joint layers

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

The V016 wall with 6 cm weak bands at the course interfaces (φ = 37.8°, c = 20.4 kPa from the joint tensile strength); the V016a reference circle, which passes below the wall and crosses no joint.

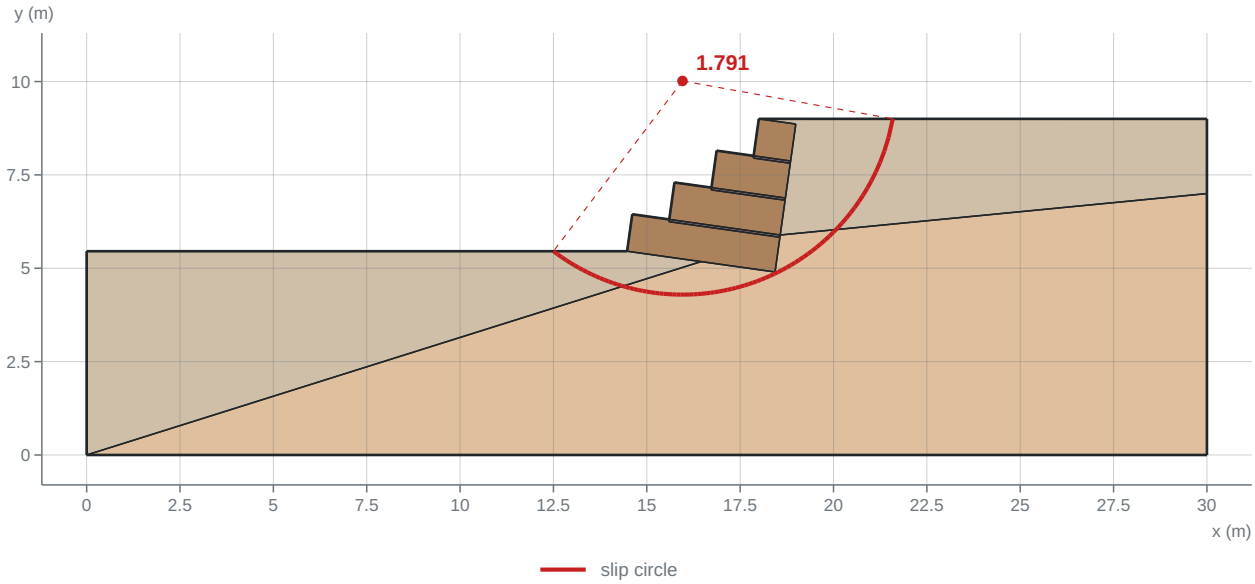


Figure 31. V016c: analyzed circle (Bishop simplified, F = 1.791).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c′ = 5.00	30.00°
Soil #2	Mohr-Coulomb	20.00	c′ = 0.00	25.00°
Gabion wall	Mohr-Coulomb	23.00	c′ = 59.70	42.00°
Weak joint	Mohr-Coulomb	23.00	c′ = 20.40	37.80°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.791				1.790	+0.08 %

The paper gives its results in figures only; the comparison is with the other programs.

3.31 V017a: Ontario gabion wall (Cao et al. 2016), equivalent cohesion

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

6 m battered gabion wall retaining a 30° backfill with a 12 kPa crest surcharge and a water table in the retained slope; the gabion mesh as an equivalent cohesion (100 kPa); the reference circle. Courses from the base 4, 3, 2.5, 2, 1.5 and 1 m wide.

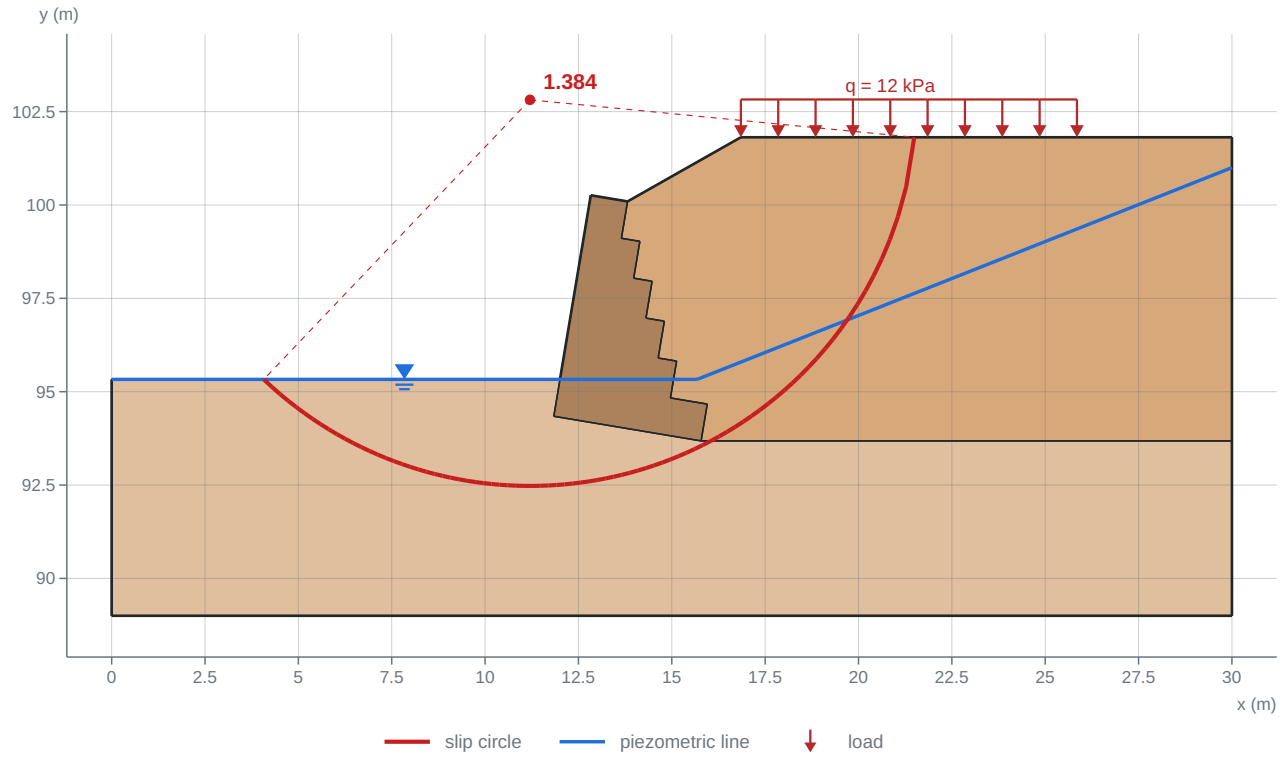


Figure 32. V017a: analyzed circle (Bishop simplified, F = 1.384).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c′ = 0.00	32.00°
Soil #2	Mohr-Coulomb	20.00	c′ = 0.00	30.00°
Gabion wall	Mohr-Coulomb	20.00	c′ = 100.0	45.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.384				1.373	+0.82 %
Spencer	1.400				1.386	+1.02 %

The paper gives its results in figures only; the comparison is with the other programs.

3.32 V017b: Ontario gabion wall (Cao et al. 2016), mesh as geosynthetic supports

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V017a with the steel mesh as geosynthetic supports at every course interface instead of the equivalent cohesion.

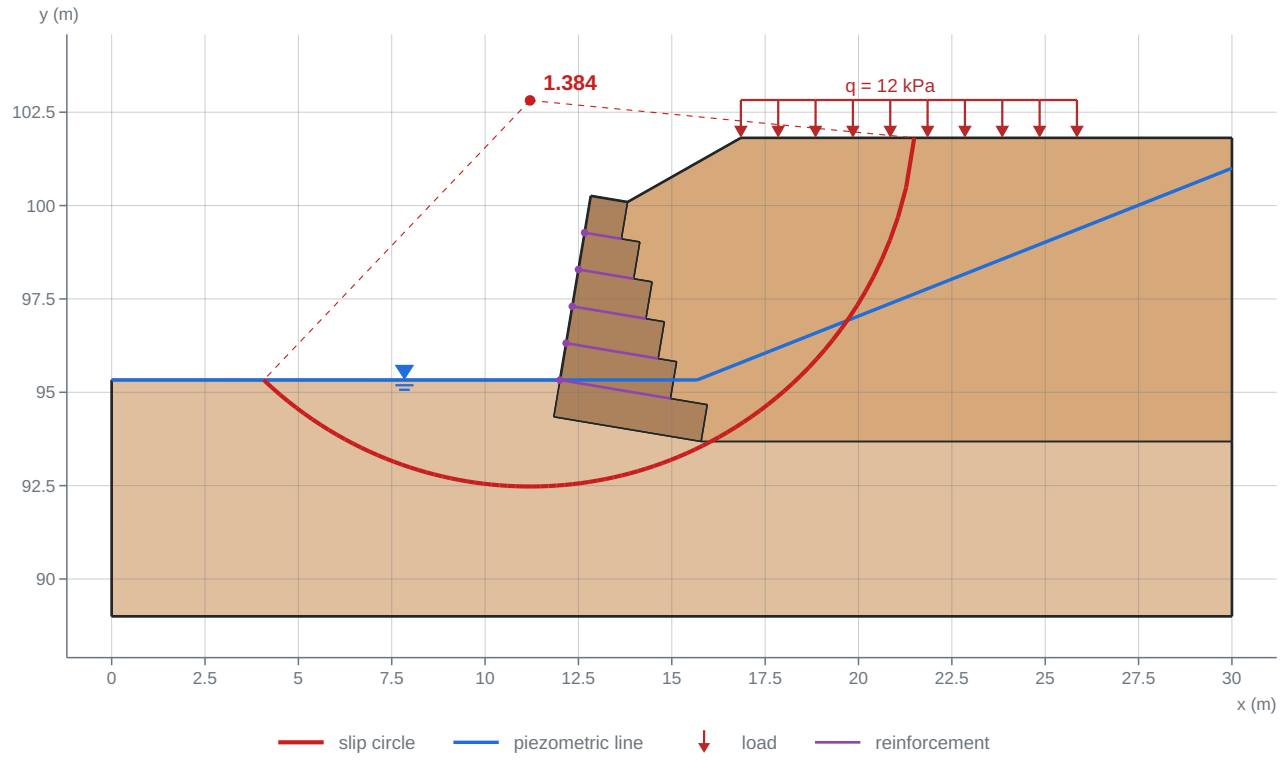


Figure 33. V017b: analyzed circle (Bishop simplified, F = 1.384).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c′ = 0.00	32.00°
Soil #2	Mohr-Coulomb	20.00	c′ = 0.00	30.00°
Gabion wall	Mohr-Coulomb	20.00	c′ = 0.00	45.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.384				1.378	+0.46 %

The paper gives its results in figures only; the comparison is with the other programs.

3.33 V018: Chen & Shao (1988) frictionless 60° slope, Prandtl mechanism

Chen, Z. & Shao, C. (1988). Evaluation of minimum factor of safety in slope stability analysis. Can. Geotech. J. 25(4), 735–748.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Weightless, frictionless 10 m slope at 60° under the critical strip load (149.31 kPa) on the Prandtl surface (wedge, fan and passive wedge); theoretical F = 1.0.

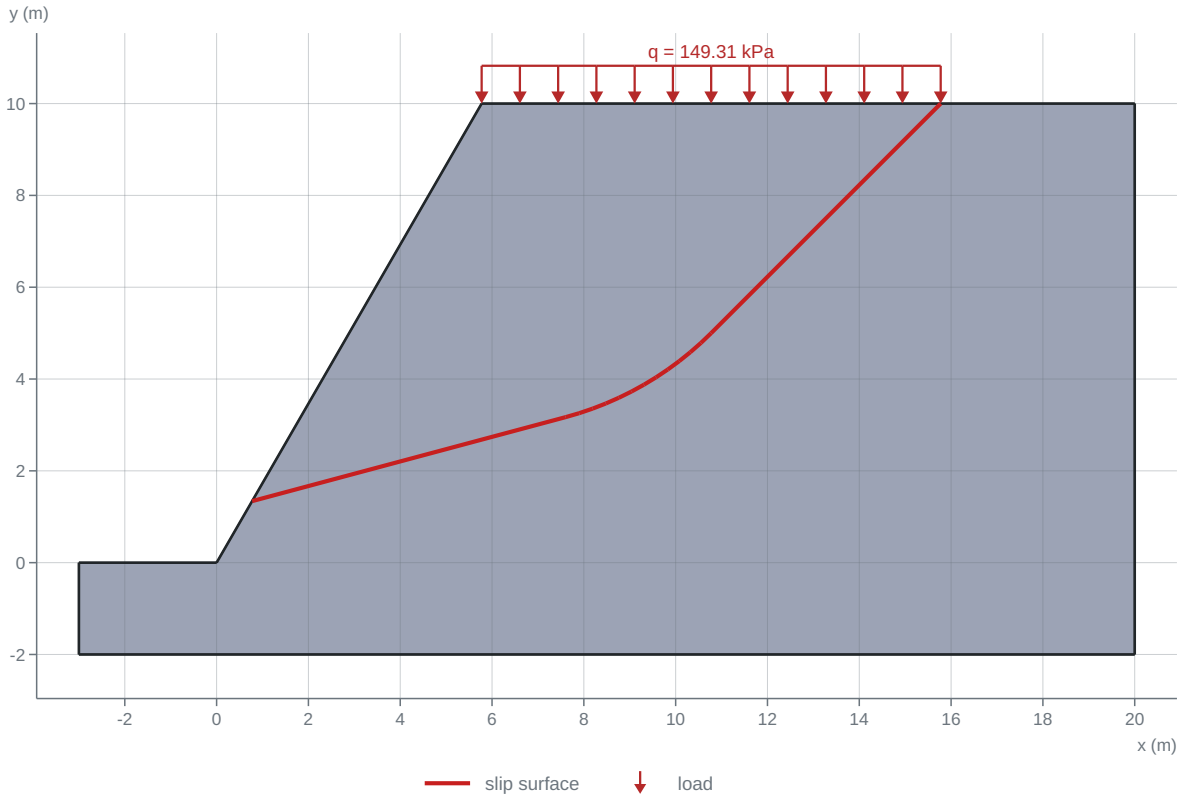


Figure 34. V018: analyzed surface (Spencer, F = 1.052).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	Mohr-Coulomb	0.00	c′ = 49.00	0.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.052	1.05	Chen & Shao (1988)	+0.19 %	1.051	+0.10 %
Morgenstern-Price (specified f(x))	1.008				1.009–1.017	−0.51 %
Strength reduction (FE)	1.000	1.00	Chen & Shao (1988): plasticity theory (Prandtl mechanism)	0.00 %		

3.34 V019: Cheng et al. 11 m homogeneous slope

Cheng, Y.M., Lansivaara, T. & Wei, W.B. (2007). Two-dimensional slope stability analysis by limit equilibrium and strength reduction methods. Computers and Geotechnics 34(3), 137–150.

Critical circle search · Units: SI (m, kPa, kN/m³)

11 m homogeneous slope (c = 10 kPa, φ = 30°, γ = 20 kN/m³); circular search.

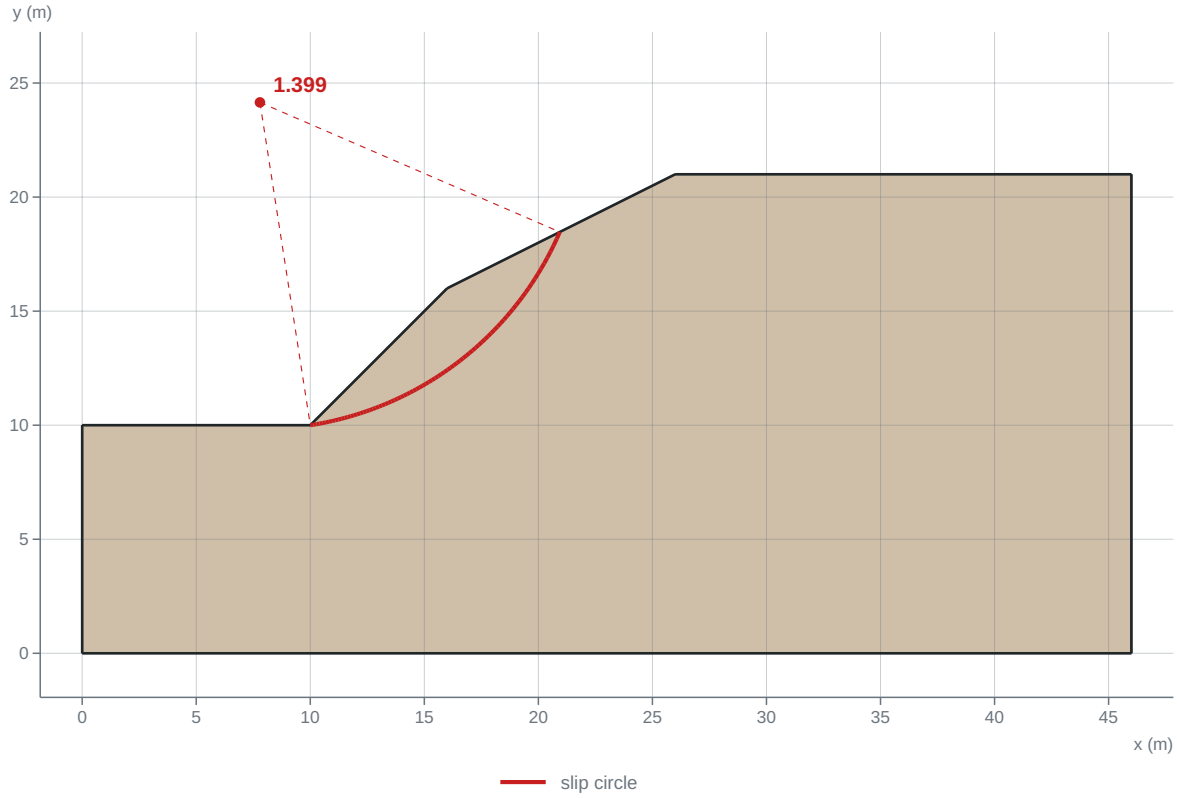


Figure 35. V019: critical circle (Spencer (40 slices, vertical pore-pressure head), F = 1.399).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Mohr-Coulomb	20.00	c′ = 10.00	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer (40 slices, vertical pore-pressure head)	1.399				1.380–1.398	+0.71 %
Spencer (vertical pore-pressure head)	1.399	1.383	Cheng et al.: Spencer	+1.13 %		
Bishop simplified	1.401				1.401	+0.01 %
Strength reduction (FE)	1.404	1.47	Cheng et al. (2007): finite-element strength reduction	−4.46 %		

3.35 V020: Cheng et al. benched slope, global minimum

Cheng, Y.M., Lansivaara, T. & Wei, W.B. (2007). Two-dimensional slope stability analysis by limit equilibrium and strength reduction methods. Computers and Geotechnics 34(3), 137–150.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous benched slope; the global minimum on the mid-lower face by circular search.

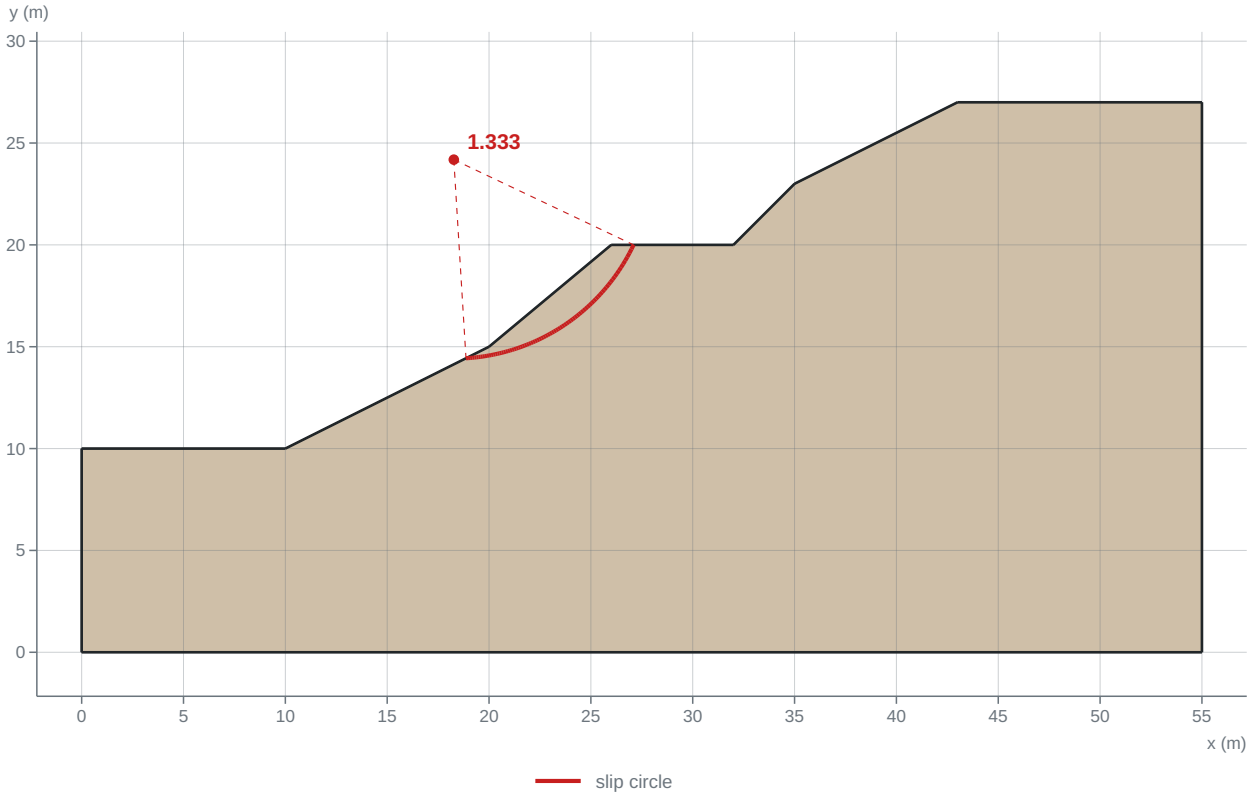


Figure 36. V020: critical circle (Spencer, F = 1.333).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Mohr-Coulomb	20.00	c′ = 5.00	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.333	1.33	Cheng et al.	+0.23 %	1.336–1.338	−0.29 %
Bishop simplified	1.337				1.342	−0.34 %
Strength reduction (FE)	1.353	1.327	Cheng et al. (2007): finite-element strength reduction	+1.95 %		

3.36 V021a: Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 2

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 1, the Congress Street cut as a commercial program models it (Su = 55, 43, 56 kPa under a strengthless sand cap, digitized piezometric line), on the critical circle tangent to the base of clay 2.

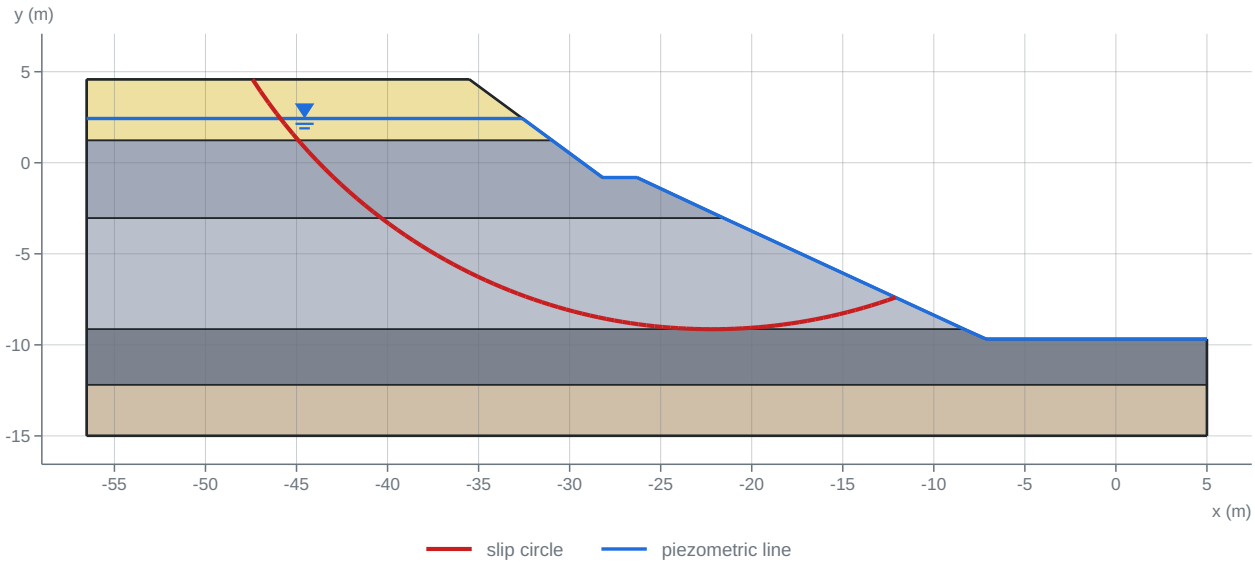


Figure 37. V021a: analyzed circle (Morgenstern-Price (half-sine), F = 1.129).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 55.00	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 43.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 56.00	0°
Bed	Mohr-Coulomb	22.00	c′ = 200.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	1.129				1.132	−0.29 %
Bishop simplified	1.129	1.1178	Chowdhury & Xu	+0.98 %		
Bishop simplified (60 slices)	1.129				1.128	+0.04 %

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
Pf, share F < 1 (Monte Carlo)	–	0.246	0.239
β lognormal (Monte Carlo)	–	0.650	0.673

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
Pf, share F < 1 (Monte Carlo)	–	0.253	0.245

3.37 V021b: Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 3

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V021a on the critical circle tangent to the base of clay 3.

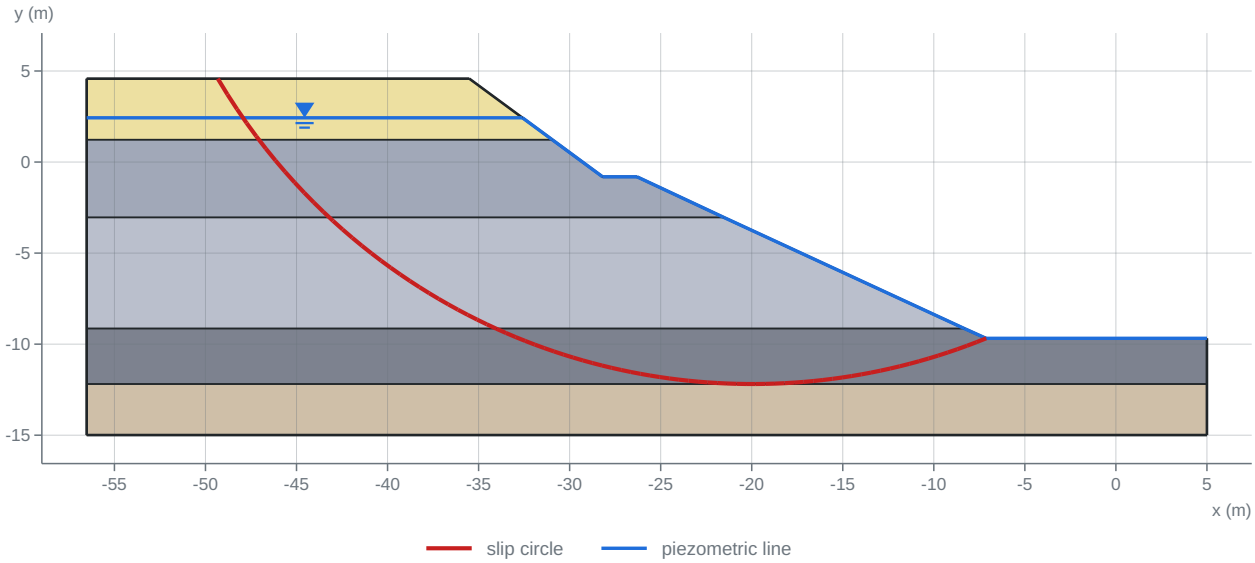


Figure 38. V021b: analyzed circle (Morgenstern-Price (half-sine), F = 1.110).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 55.00	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 43.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 56.00	0°
Bed	Mohr-Coulomb	22.00	c′ = 200.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	1.110				1.116	−0.52 %
Bishop simplified	1.110	1.1088	Chowdhury & Xu	+0.13 %		
Bishop simplified (60 slices)	1.110				1.109	+0.10 %

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
Pf, share F < 1 (Monte Carlo)	–	0.279	0.281
β lognormal (Monte Carlo)	–	0.537	0.555

3.38 V021c: Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 2

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 2 (Su = 68.1, 39.3, 50.8 kPa), circle tangent to the base of clay 2.

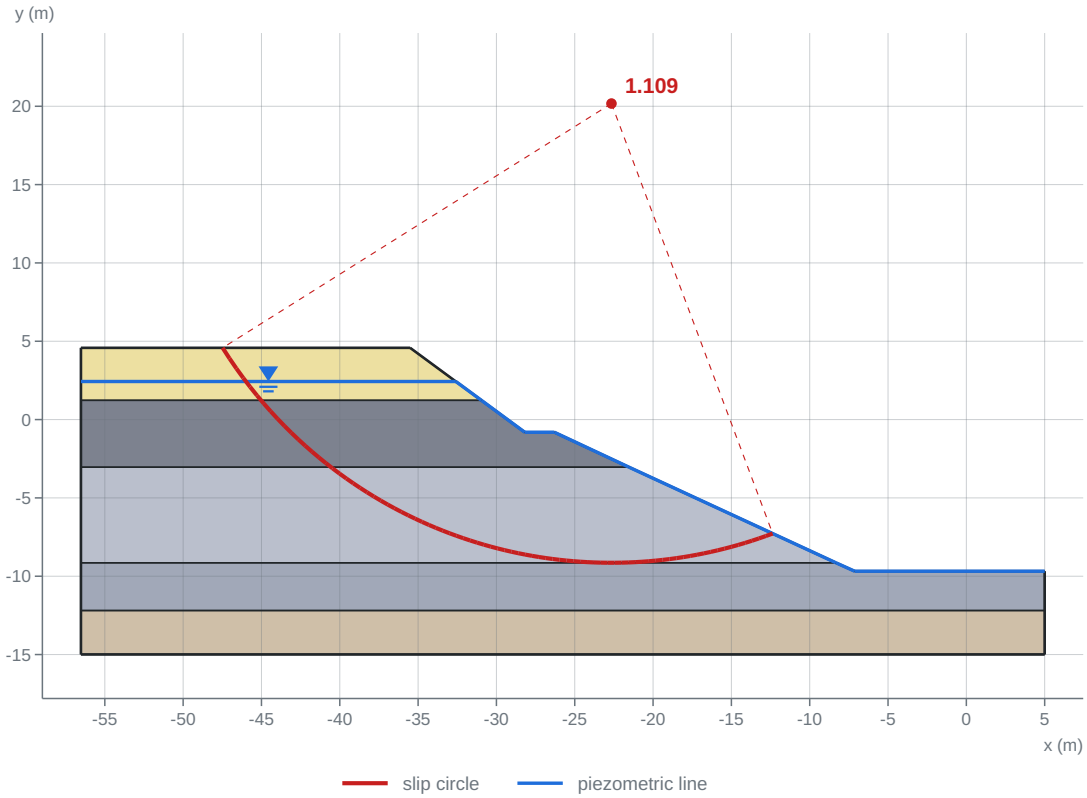


Figure 39. V021c: analyzed circle (Morgenstern-Price (half-sine), F = 1.109).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 68.10	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 39.30	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 50.80	0°
Bed	Mohr-Coulomb	22.00	c′ = 200.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	1.109				1.110	−0.11 %
Bishop simplified	1.109	1.1096	Chowdhury & Xu	−0.07 %	1.108	+0.07 %

3.39 V021d: Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 3

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 2, circle tangent to the base of clay 3.

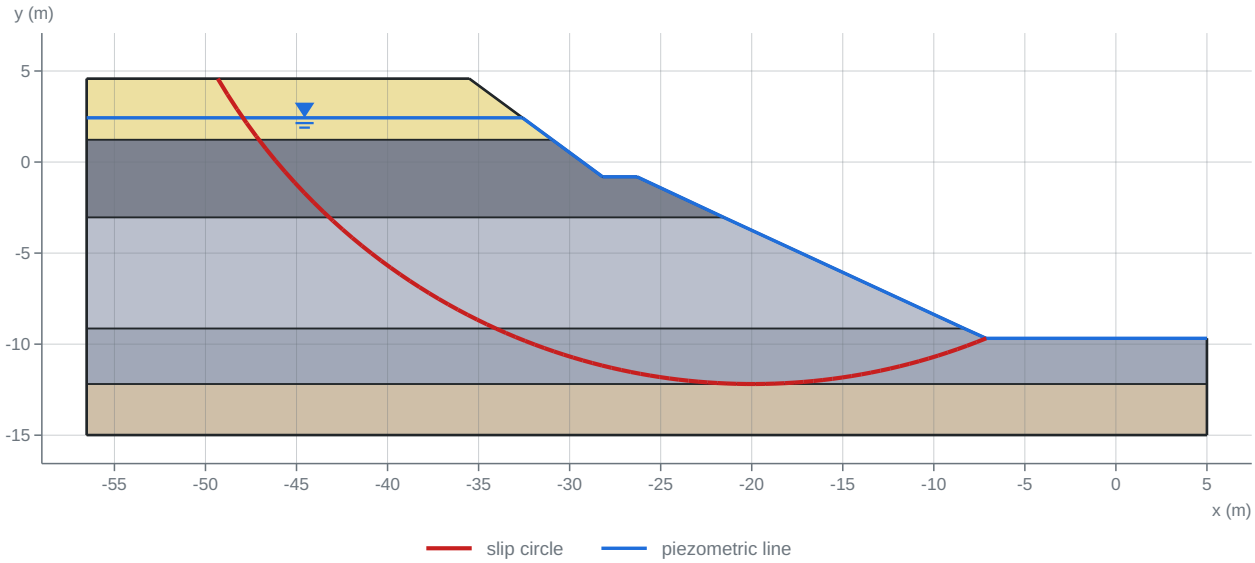


Figure 40. V021d: analyzed circle (Morgenstern-Price (half-sine), F = 1.058).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 68.10	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 39.30	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 50.80	0°
Bed	Mohr-Coulomb	22.00	c′ = 200.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	1.058				1.063	−0.43 %
Bishop simplified	1.058	1.0639	Chowdhury & Xu	−0.52 %	1.058	+0.04 %

3.40 V021e: Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 2

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 3 (Su = 136, 80, 102 kPa), circle tangent to the base of clay 2.

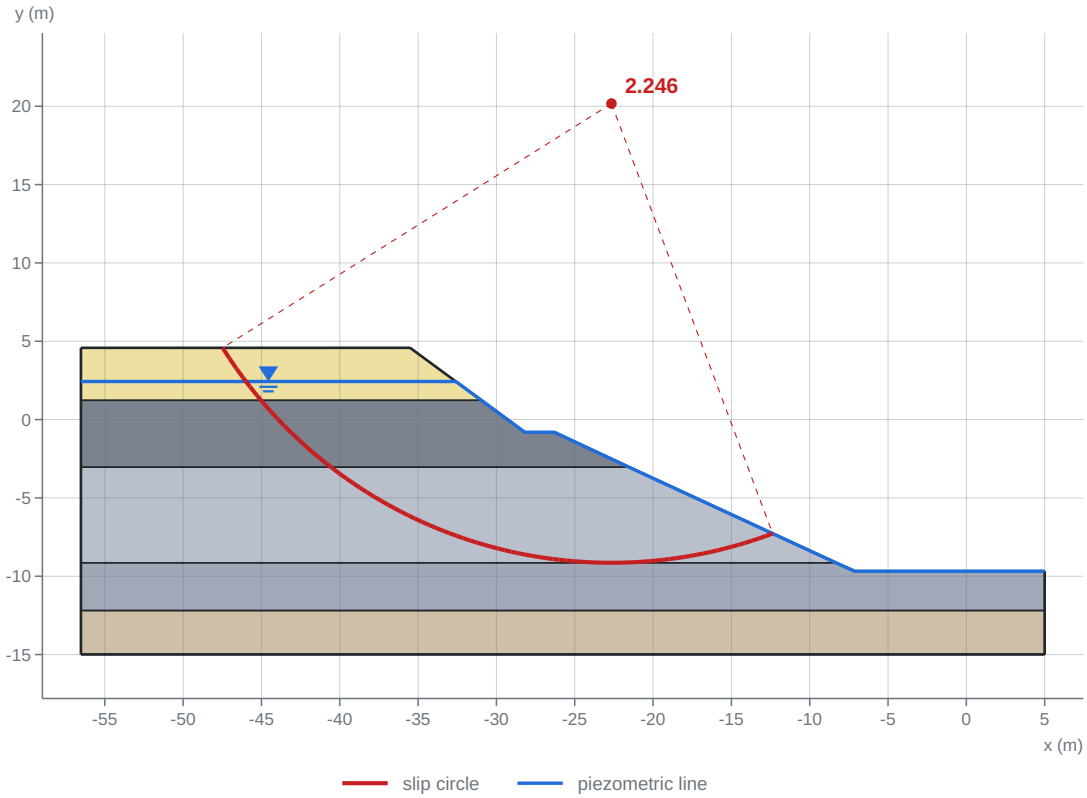


Figure 41. V021e: analyzed circle (Morgenstern-Price (half-sine), F = 2.246).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 136.0	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 80.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 102.0	0°
Bed	Mohr-Coulomb	22.00	c′ = 200.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	2.246				2.248	−0.10 %
Bishop simplified	2.246	2.2343	Chowdhury & Xu	+0.51 %	2.245	+0.03 %

3.41 V021f: Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 3

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 3, circle tangent to the base of clay 3.

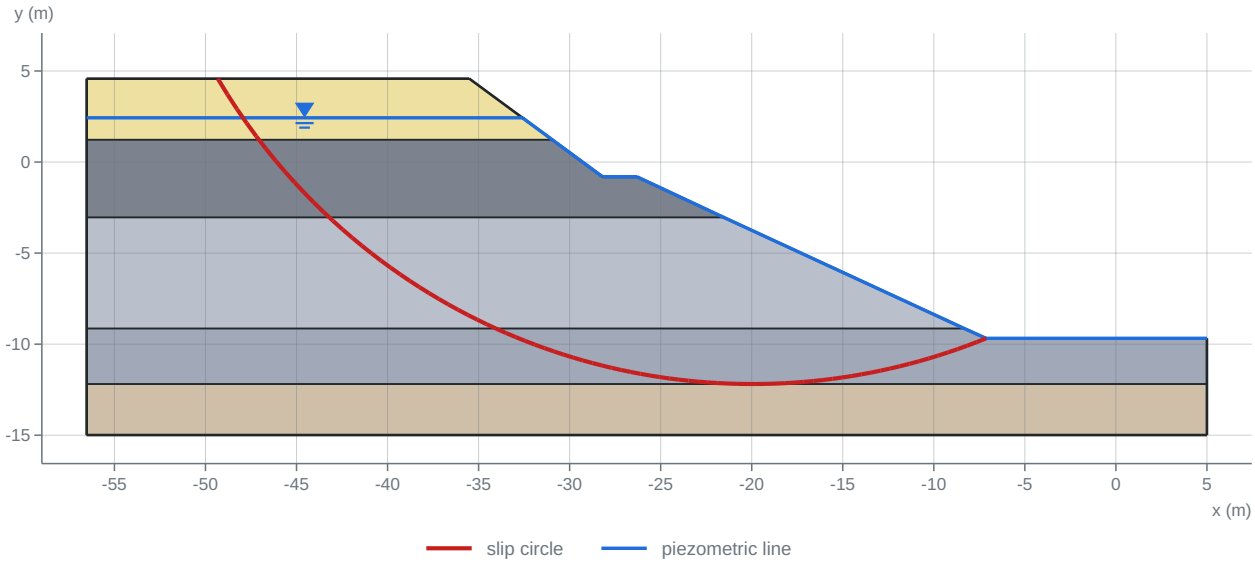


Figure 42. V021f: analyzed circle (Morgenstern-Price (half-sine), F = 2.129).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 136.0	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 80.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 102.0	0°
Bed	Mohr-Coulomb	22.00	c′ = 200.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	2.129				2.138	−0.43 %
Bishop simplified	2.129	2.1396	Chowdhury & Xu	−0.50 %	2.128	+0.04 %

3.42 V021g: Chowdhury & Xu (1995) example 4, Congress Street cut, circle tangent to the base of clay 2

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 4 (c-φ clays with the digitized piezometric line), circle tangent to the base of clay 2.

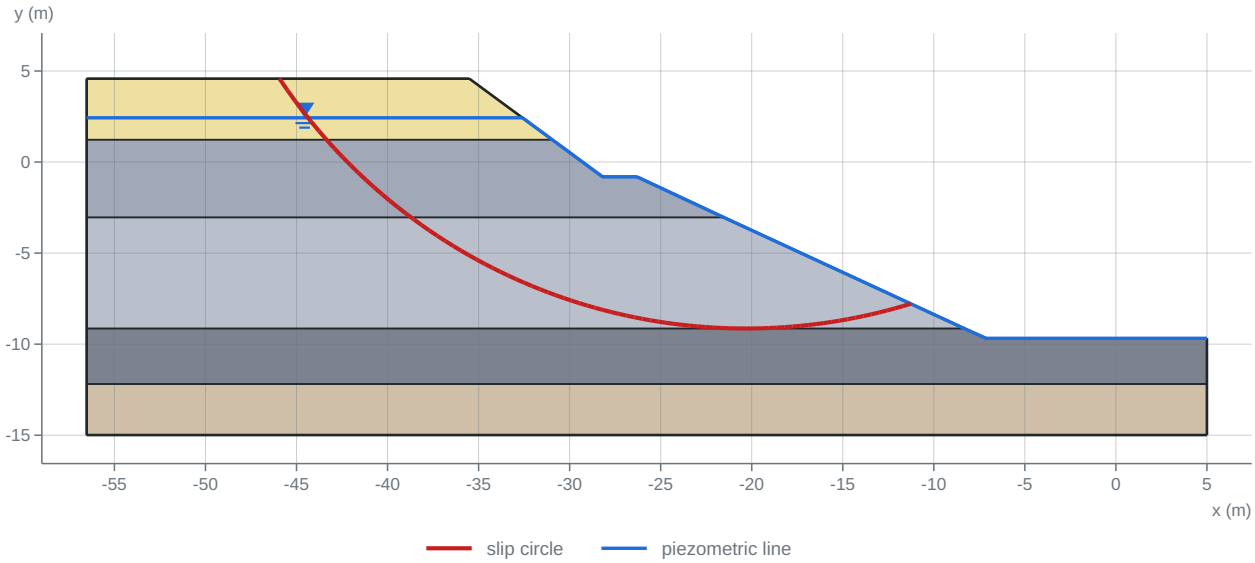


Figure 43. V021g: analyzed circle (Morgenstern-Price (half-sine), F = 1.429).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Mohr-Coulomb	17.00	c′ = 55.00	5.00°
Clay 2	Mohr-Coulomb	22.00	c′ = 43.00	7.00°
Clay 3	Mohr-Coulomb	22.00	c′ = 56.00	8.00°
Bed	Mohr-Coulomb	22.00	c′ = 200.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	1.429				1.422	+0.47 %
Bishop simplified	1.429	1.4239	Chowdhury & Xu	+0.33 %		

3.43 V022a: Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y = 15

Chowdhury & Xu (1995).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 5: embankment fill (c′ = 10 kPa, φ′ = 12°) on soft clay (Su = 40 kPa), the geometry as a commercial program's verification manual draws it; circle tangent to the fill–clay interface.

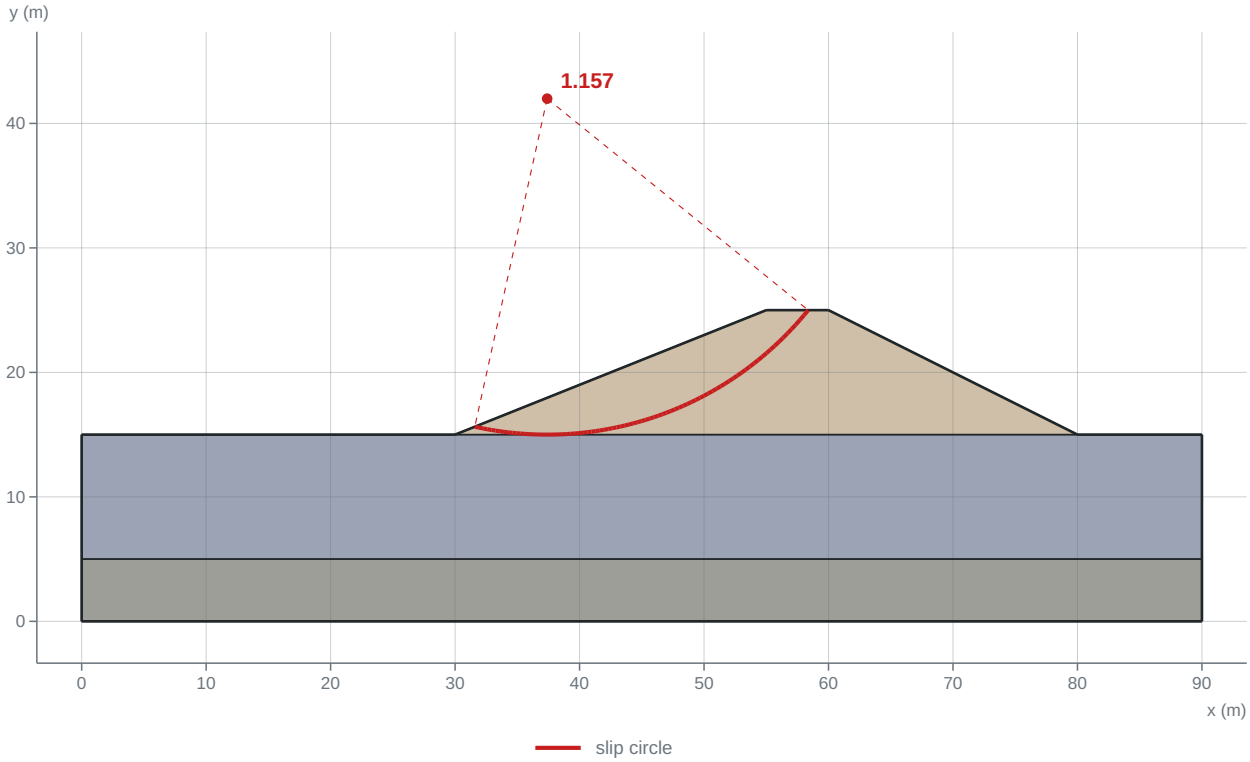


Figure 44. V022a: analyzed circle (Morgenstern-Price (half-sine), F = 1.157).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c′ = 10.00	12.00°
Layer 2	Undrained (φ = 0)	18.00	Su = 40.00	0°
Hard layer	Undrained (φ = 0)	20.00	Su = 1000.0	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	1.157				1.158	−0.06 %
Bishop simplified	1.159	1.1625	Chowdhury & Xu	−0.33 %		
Bishop simplified (60 slices)	1.159				1.160	−0.12 %

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
Pf, share F < 1 (Monte Carlo)	–	0.206–0.212	0.206
β lognormal (Monte Carlo)	–	0.799–0.800	0.812
β lognormal (Taylor series)	–	0.787	0.789

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
Pf, share F < 1 (Monte Carlo)	–	0.210	0.208

3.44 V022b: Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y = 5

Chowdhury & Xu (1995).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V022a with the circle tangent to the base of the soft clay.

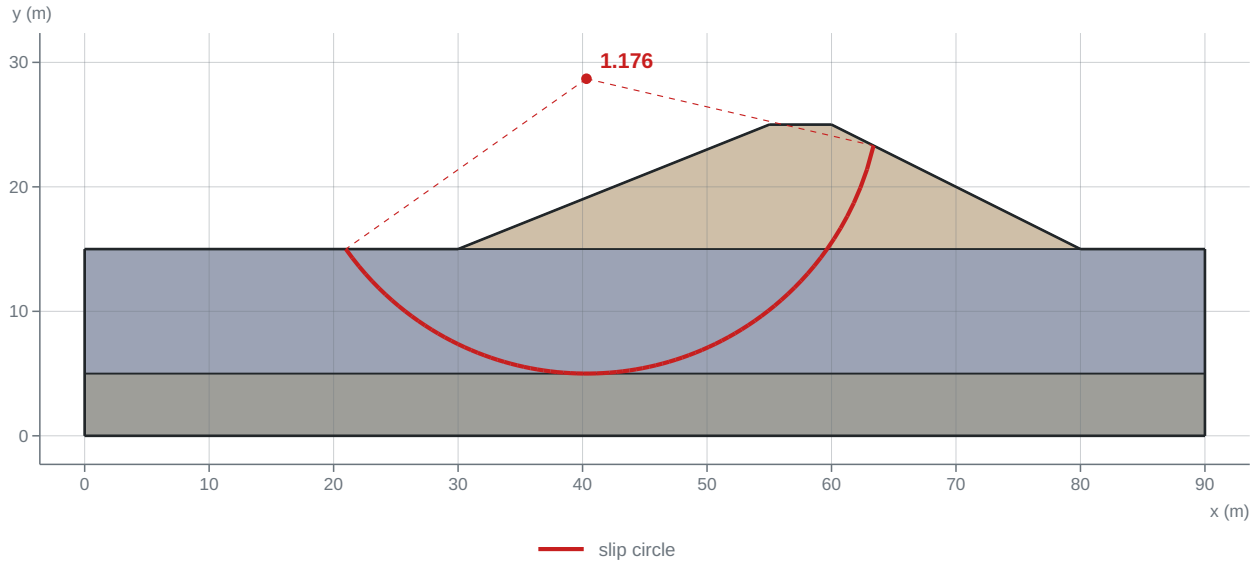


Figure 45. V022b: analyzed circle (Morgenstern-Price (half-sine), F = 1.176).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c′ = 10.00	12.00°
Layer 2	Undrained (φ = 0)	18.00	Su = 40.00	0°
Hard layer	Undrained (φ = 0)	20.00	Su = 1000.0	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	1.176				1.178	−0.20 %
Bishop simplified	1.178	1.1479	Chowdhury & Xu (1995)	+2.62 %	1.185	−0.59 %

Chowdhury & Xu's 1.1479 is about 2.5 % below the other programs on this family of circles.

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
Pf, share F < 1 (Monte Carlo)	–	0.199–0.209	0.211
β lognormal (Monte Carlo)	–	0.789–0.820	0.793
β lognormal (Taylor series)	–	0.798	0.806

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
Pf, share F < 1 (Monte Carlo)	–	0.214	0.215

3.45 V023: LASH terminal trench, Duncan's Taylor series example

Duncan (2000), J. Geotech. Geoenviron. Eng. 126(4).

Fixed circle · Units: SI (m, kPa, kN/m³)

Underwater trench in San Francisco Bay Mud: Su = 100 psf at el. −20 ft increasing 9.8 psf/ft, γ = 100 pcf, water at el. 0. Duncan's surface as a circle, Spencer; σγ = 3.3 pcf and σ of the Su gradient 1.2 psf/ft (a commercial program's Table 29.2). Converted to SI.

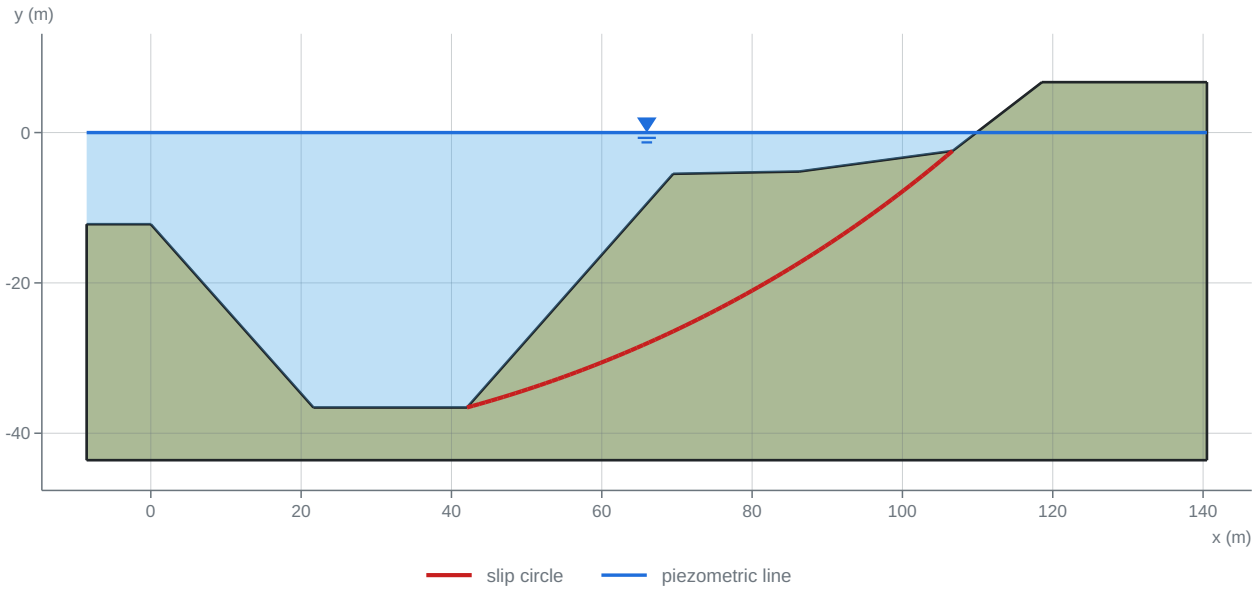


Figure 46. V023: analyzed circle (Spencer, F = 1.138).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
San Francisco Bay Mud	Undrained (φ = 0)	15.71	Su = 4.79 at y = -6.10, +1.54 kPa/m below	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.138				1.145	−0.65 %
–	–	1.17	Duncan (2000)			

Duncan (2000) prints two decimals; the other programs give 1.145 on the same input.

Probabilistic analysis on the same surface: 2 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Taylor series)	0.900 Duncan (2000)	–	0.889
Pf lognormal (Taylor series)	0.180 Duncan (2000)	–	0.187

3.46 V024: Dam on a layered foundation with artesian uplift, piezometric line (D&W fig. 6.39)

Duncan & Wright, Soil Strength and Slope Stability, figs. 6.39–6.40.

Fixed circle · Units: SI (m, kPa, kN/m³)

Symmetric dam with a narrow clay core on 30 ft of clay over permeable sand; a commercial program's piezometric line from the underseepage; the circle tangent to el. 197 ft. Converted to SI.

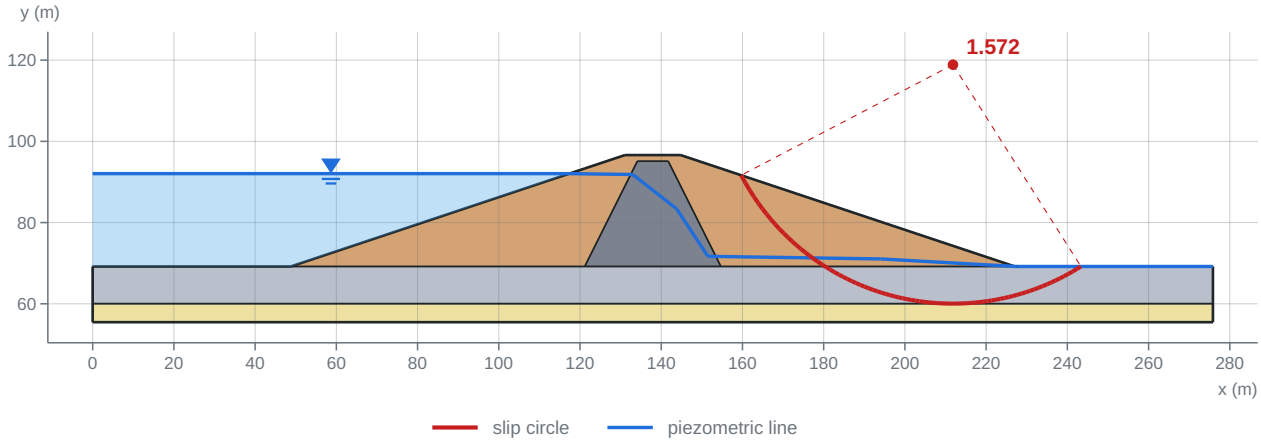


Figure 47. V024: analyzed circle (Bishop simplified, F = 1.572).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Outer shell	Mohr-Coulomb	19.64	c′ = 0.00	34.00°
Clay core	Mohr-Coulomb	19.16	c′ = 4.79	26.00°
Foundation clay	Mohr-Coulomb	19.32	c′ = 0.00	24.00°
Foundation sand	Mohr-Coulomb	19.95	c′ = 0.00	32.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.572	1.57	Duncan & Wright	+0.15 %	1.563–1.572	+0.31 %
Spencer	1.563				1.557	+0.37 %

3.47 V025: Duncan & Wright fig. 6.38, piezometric line

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.38.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 2:1 slope with a nearly flat piezometric line; critical-circle search. Converted to SI; the water line was traced from a figure.

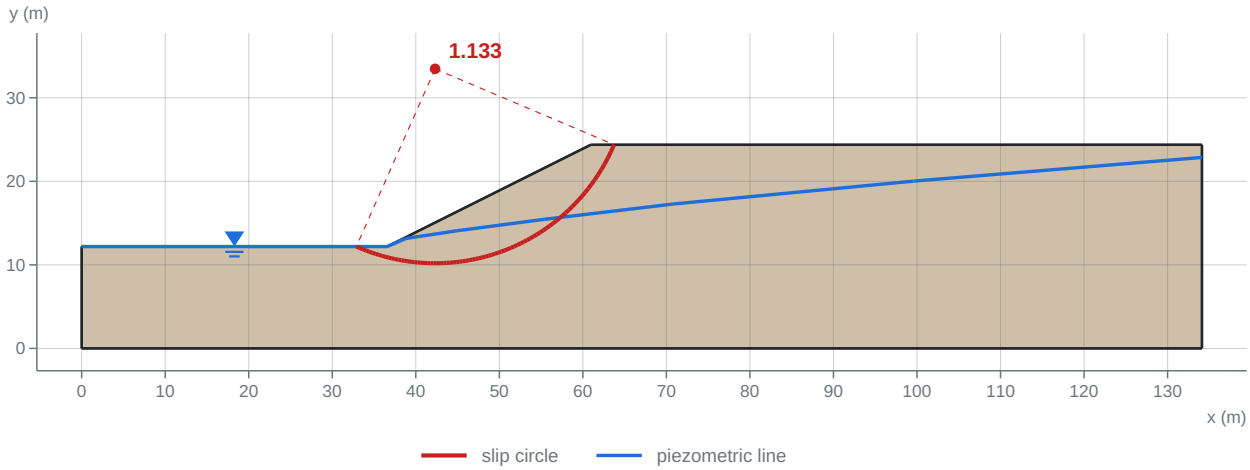


Figure 48. V025: critical circle (Bishop simplified, F = 1.133).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	19.64	c′ = 9.58	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.133				1.142	−0.81 %
Spencer	1.133				1.142	−0.80 %
–	–	1.141	Duncan & Wright			

3.48 V026: Duncan & Wright fig. 7.12, embankment on clay

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.12.

Critical circle search · Units: imperial (ft, psf, pcf)

Sand embankment on a thick undrained clay foundation; critical-circle search.

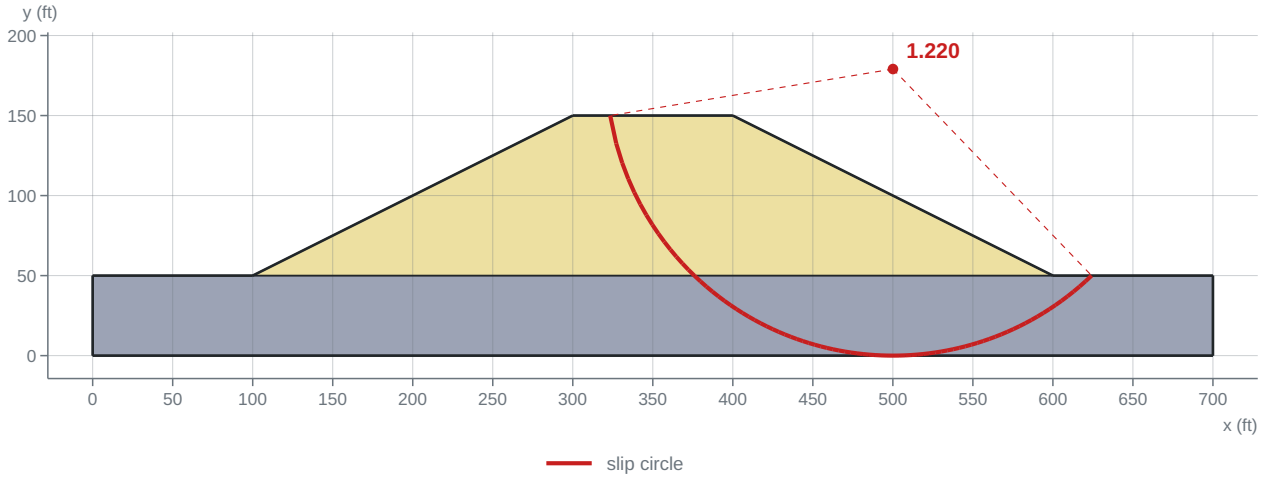


Figure 49. V026: critical circle (Bishop simplified, F = 1.220).

Material	Model	γ (pcf)	Strength (psf)	φ
Sand	Mohr-Coulomb	140.0	c′ = 0.00	40.00°
Clay	Mohr-Coulomb	140.0	c′ = 2500.0	0.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.220	1.22	Duncan & Wright	+0.01 %	1.228	−0.64 %
Spencer	1.195	1.19	Duncan & Wright	+0.45 %		
Spencer, non-circular	1.170				1.190	−1.66 %
Strength reduction (FE)	1.186	1.19	Duncan, Wright & Brandon (2014): finite-element strength reduction	−0.30 %		

3.49 V027: Duncan & Wright fig. 14.3, undrained clay

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.3.

Critical circle search · Units: imperial (ft, psf, pcf)

Homogeneous undrained clay slope (φ = 0); critical-circle search.

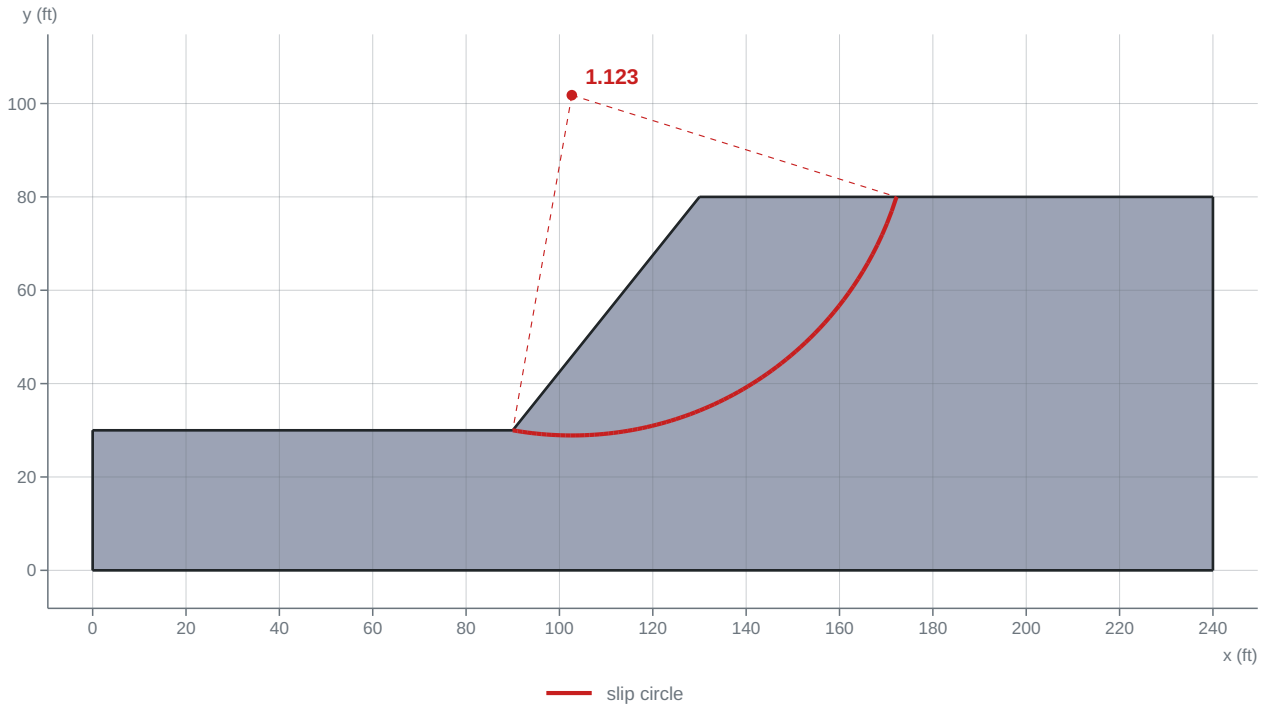


Figure 50. V027: critical circle (Bishop simplified, F = 1.123).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay	Mohr-Coulomb	100.0	c′ = 1000.0	0.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.123				1.126	−0.24 %
Spencer	1.136	1.124	Duncan & Wright	+1.03 %		

3.50 V028: Duncan & Wright fig. 14.3, undrained clay on a 46.5 ft foundation

Verification manual of a commercial slope stability program.

Critical circle search · Units: imperial (ft, psf, pcf)

The V027 φ = 0 clay slope with a 46.5 ft thick foundation; critical-circle search (the minimum of the toe and base-tangent circles of Tables 78.4–78.5).

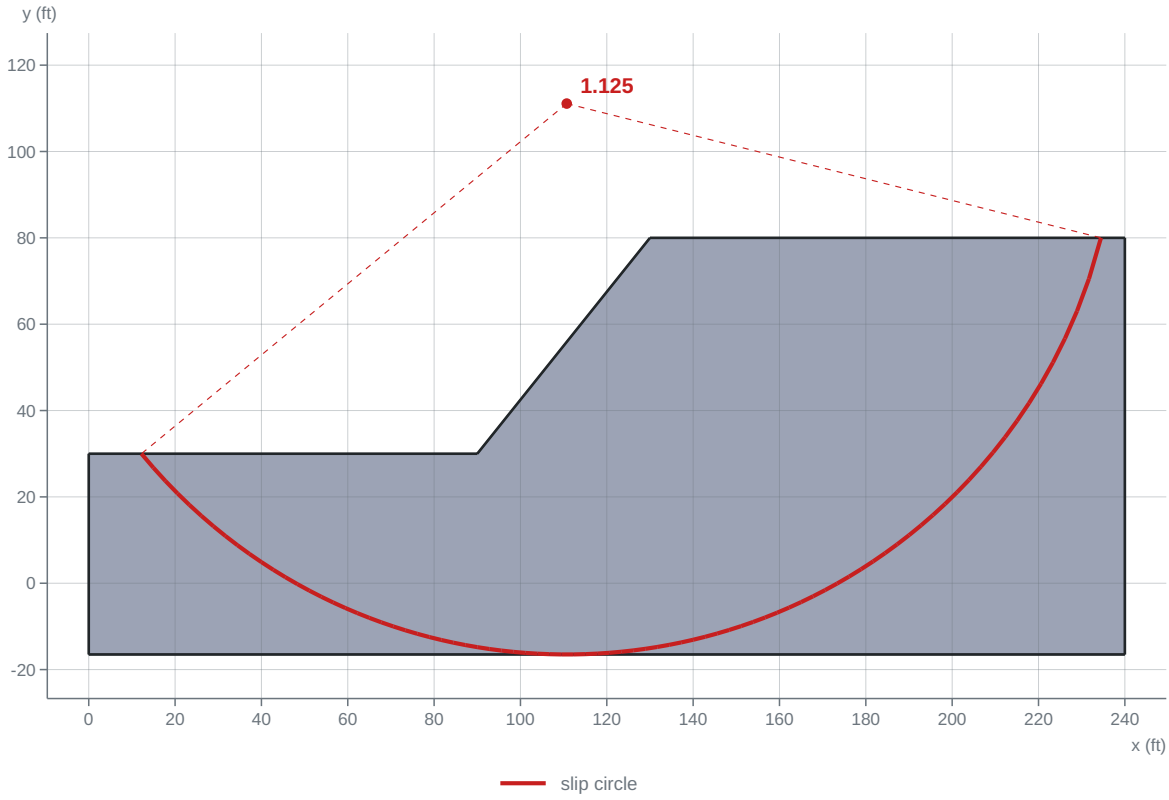


Figure 51. V028: critical circle (Bishop simplified, F = 1.125).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay	Mohr-Coulomb	100.0	c′ = 1000.0	0.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.125				1.126	−0.12 %
Spencer	1.124	1.124	Duncan & Wright	+0.03 %		

3.51 V029: Duncan & Wright fig. 14.3, undrained clay on a 60 ft foundation

Verification manual of a commercial slope stability program.

Critical circle search · Units: imperial (ft, psf, pcf)

As V028 with a 60 ft thick foundation (Tables 78.6–78.7).

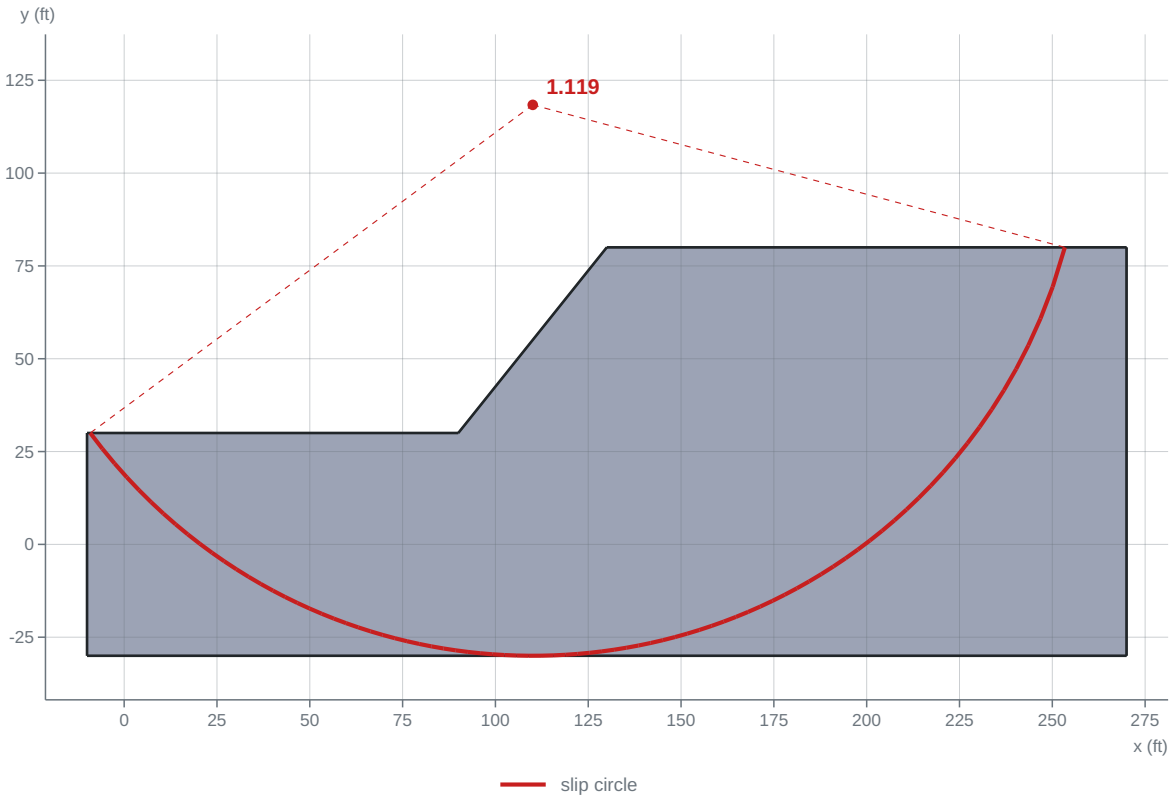


Figure 52. V029: critical circle (Bishop simplified, F = 1.119).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay	Mohr-Coulomb	100.0	c′ = 1000.0	0.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.119				1.125	−0.50 %
Spencer	1.119	1.119	Duncan & Wright	+0.04 %		

3.52 V030: Duncan & Wright fig. 14.4

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.4.

Critical circle search · Units: imperial (ft, psf, pcf)

Cohesionless fill on undrained clay; critical-circle search.

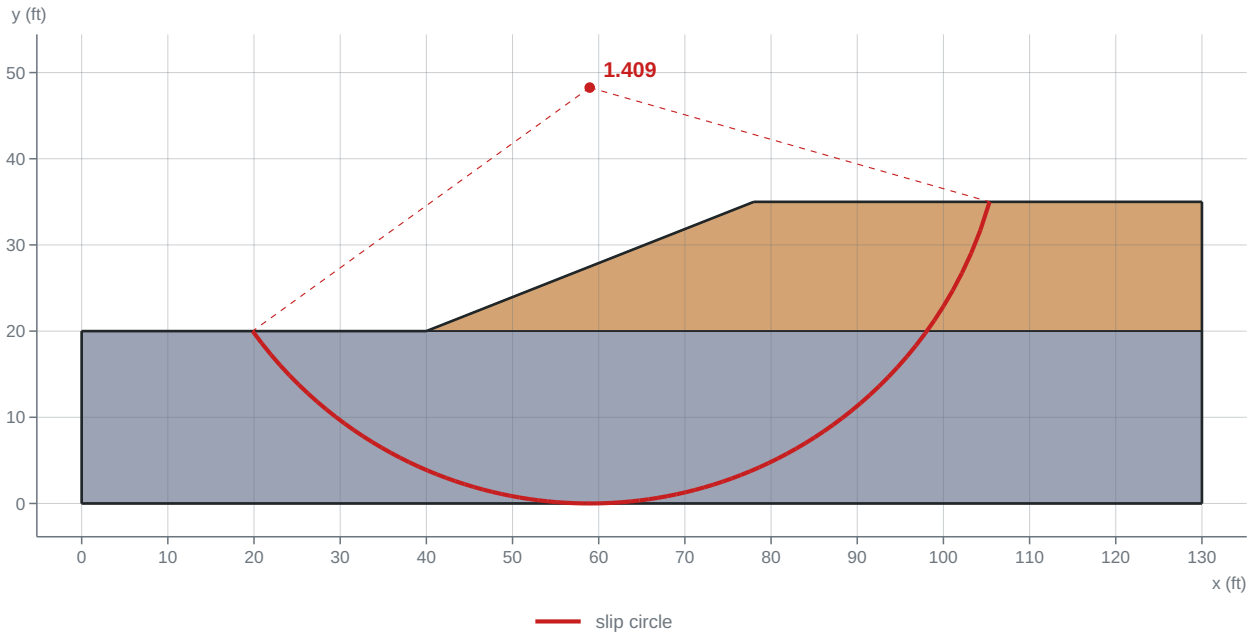


Figure 53. V030: critical circle (Bishop simplified, F = 1.409).

Material	Model	γ (pcf)	Strength (psf)	φ
Fill	Mohr-Coulomb	120.0	c′ = 0.00	30.00°
Clay	Mohr-Coulomb	120.0	c′ = 450.0	0.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.409				1.412	−0.21 %
Spencer	1.400	1.40	Duncan & Wright	−0.02 %		

3.53 V031a: Duncan & Wright fig. 14.5, R = 87 ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.5.

Fixed circle · Units: imperial (ft, psf, pcf)

Embankment over alternating cohesive and frictional layers; circle of radius 87 ft.

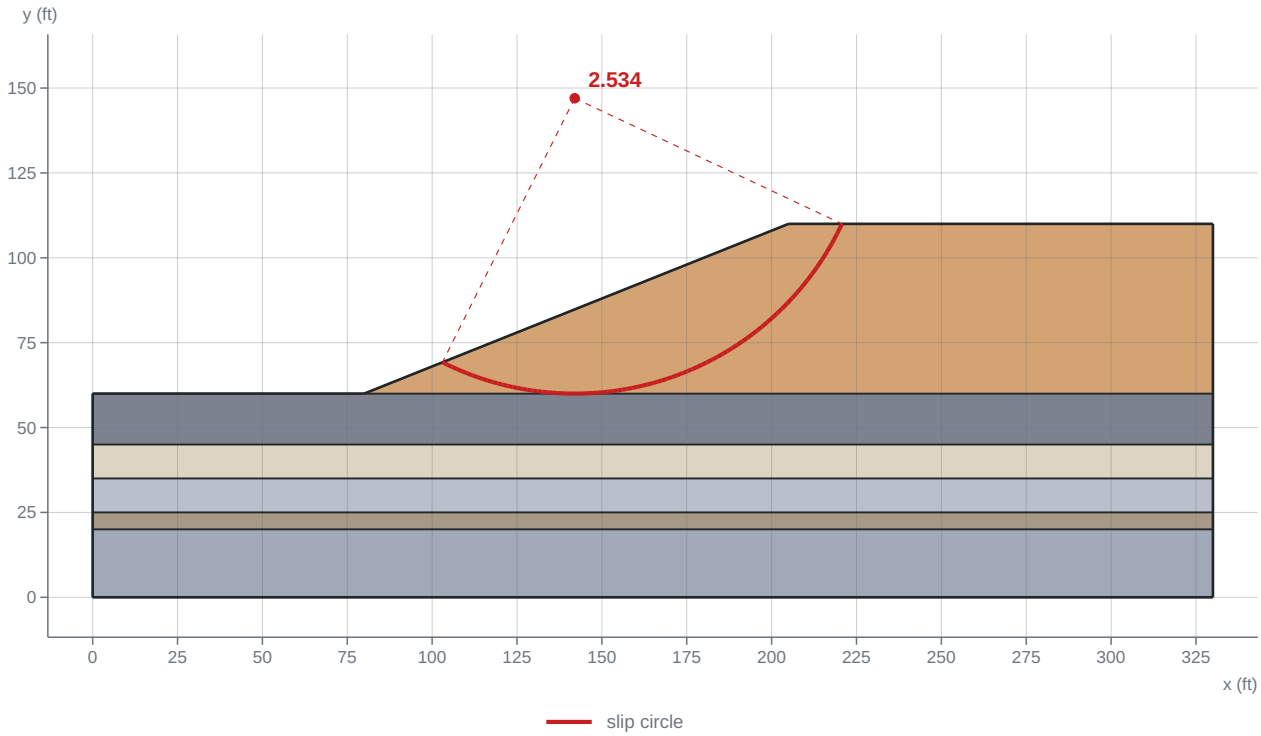


Figure 54. V031a: analyzed circle (Bishop simplified, F = 2.534).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	120.0	c′ = 1.00	35.00°
L60	Mohr-Coulomb	110.0	c′ = 950.0	0.00°
L45	Mohr-Coulomb	122.0	c′ = 1.00	32.00°
L35	Mohr-Coulomb	98.00	c′ = 500.0	0.00°
L25	Mohr-Coulomb	131.0	c′ = 1.00	37.00°
L20	Mohr-Coulomb	103.0	c′ = 600.0	0.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.534				2.549	−0.59 %
Spencer	2.531	2.56	Duncan & Wright	−1.14 %	2.545	−0.55 %

3.54 V031b: Duncan & Wright fig. 14.5, R = 102 ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.5.

Fixed circle · Units: imperial (ft, psf, pcf)

As V031a with a deeper circle of radius 102 ft, reaching the weak clay layers.

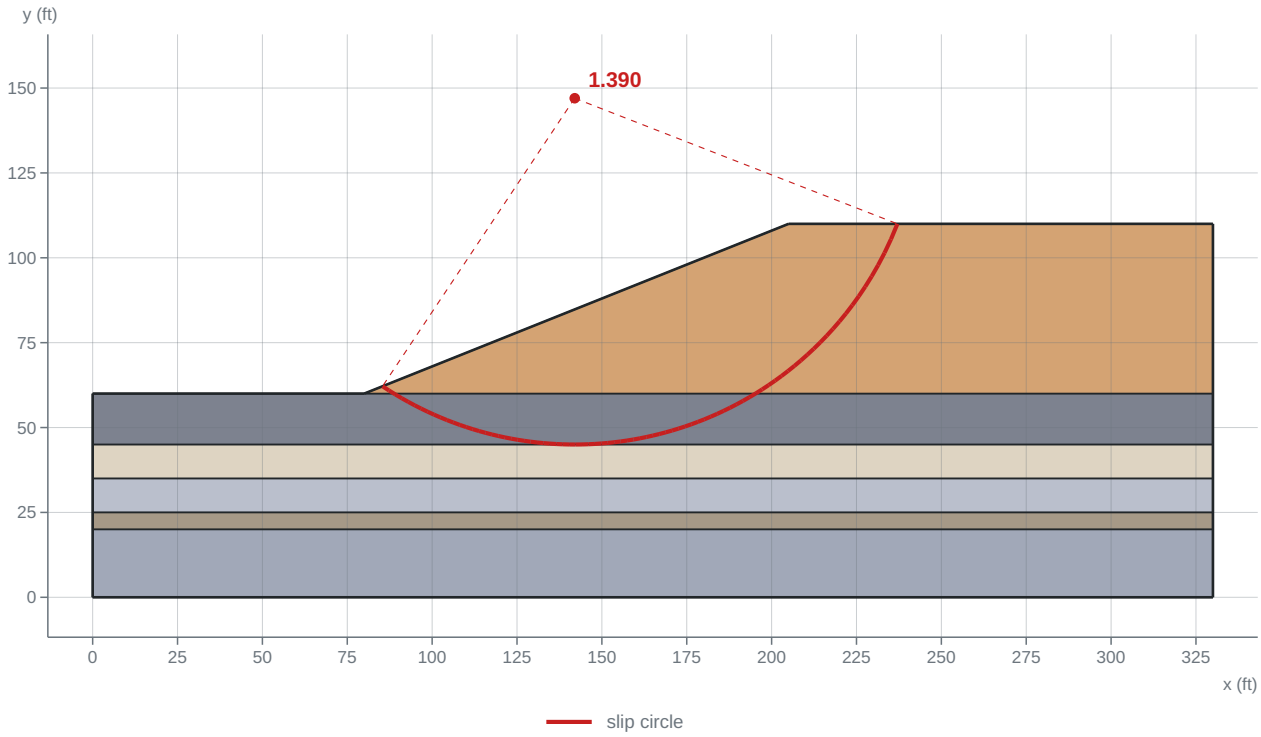


Figure 55. V031b: analyzed circle (Bishop simplified, F = 1.390).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	120.0	c′ = 1.00	35.00°
L60	Mohr-Coulomb	110.0	c′ = 950.0	0.00°
L45	Mohr-Coulomb	122.0	c′ = 1.00	32.00°
L35	Mohr-Coulomb	98.00	c′ = 500.0	0.00°
L25	Mohr-Coulomb	131.0	c′ = 1.00	37.00°
L20	Mohr-Coulomb	103.0	c′ = 600.0	0.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.389				1.398	−0.61 %
Spencer	1.352	1.35	Duncan & Wright	+0.16 %	1.359	−0.51 %

3.55 V032a: Duncan & Wright fig. 14.7, deep circle

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.7.

Critical circle search · Units: imperial (ft, psf, pcf)

Cohesionless embankment on an undrained clay foundation. Entry is limited to the crest and exit to the toe flat to find the published deep circle.

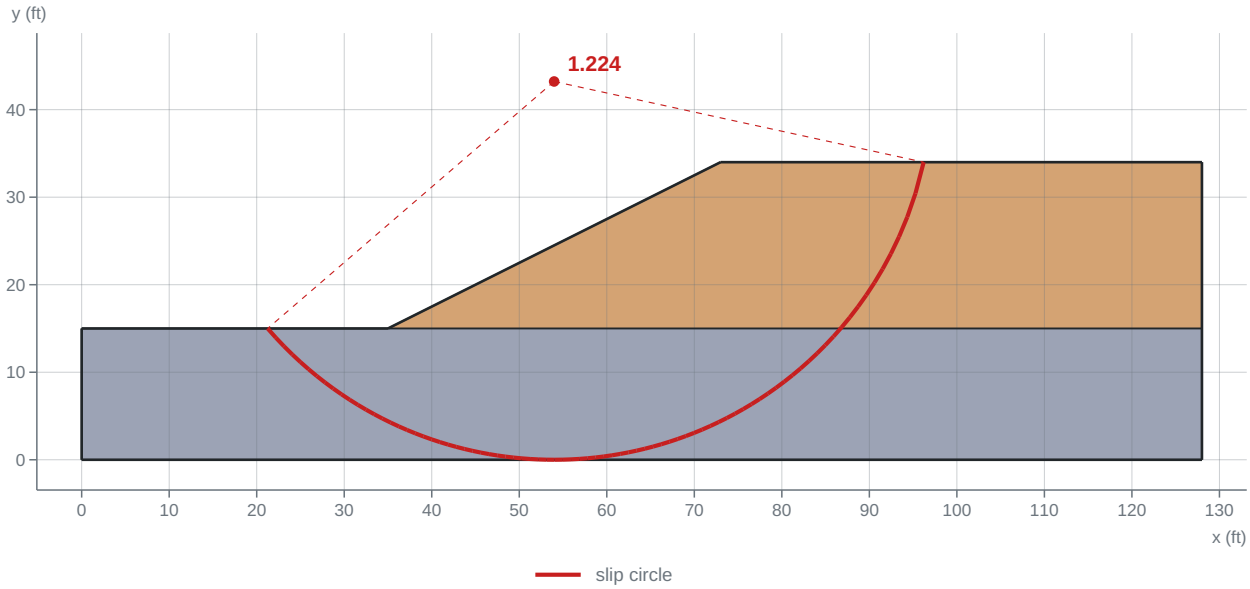


Figure 56. V032a: critical circle (Bishop simplified, F = 1.224).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	124.0	c′ = 0.00	30.00°
Foundation	Mohr-Coulomb	98.00	c′ = 500.0	0.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.224				1.230	−0.46 %
Spencer	1.206	1.21	Duncan & Wright	−0.36 %	1.209	−0.27 %

3.56 V032b: Duncan & Wright fig. 14.7, infinite-slope mechanism

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.7. Verification manual of a commercial slope stability program.

Critical circle search · Units: imperial (ft, psf, pcf)

The V032a embankment without search limits: the critical surface is the shallow infinite-slope sliver in the cohesionless fill (tan 30° / tan 26.6° = 1.155).

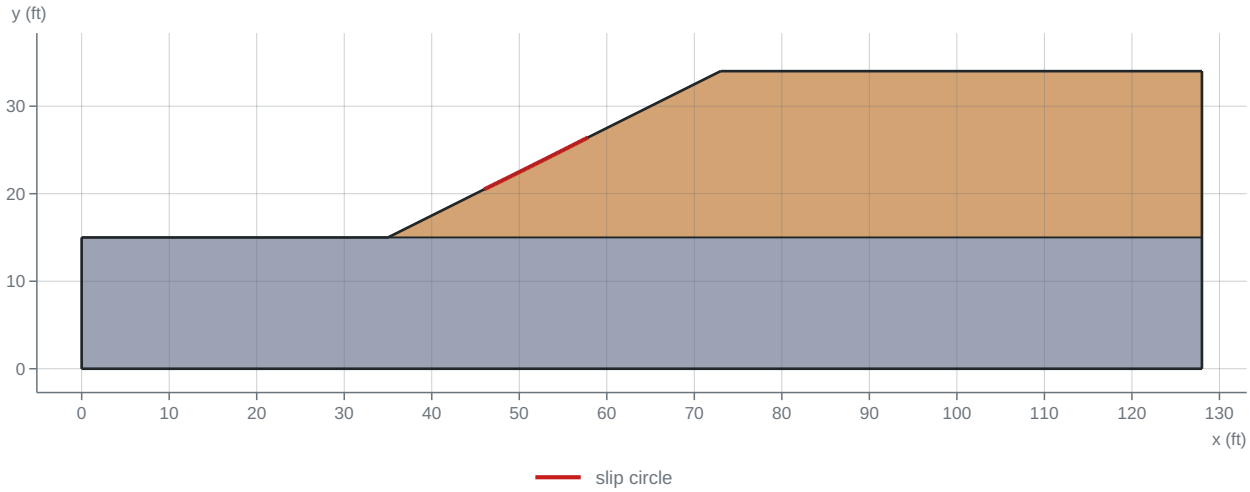


Figure 57. V032b: critical circle (Bishop simplified, F = 1.155).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	124.0	c′ = 0.00	30.00°
Foundation	Mohr-Coulomb	98.00	c′ = 500.0	0.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.155				1.155	−0.03 %
Spencer	1.155	1.15	Duncan & Wright	+0.41 %	1.155	−0.03 %

3.57 V033: Duncan & Wright fig. 14.20a, water table

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20a.

Critical circle search · Units: SI (m, kPa, kN/m³)

Cohesive-frictional fill over sand with a piezometric line; critical-circle search. Converted to SI.

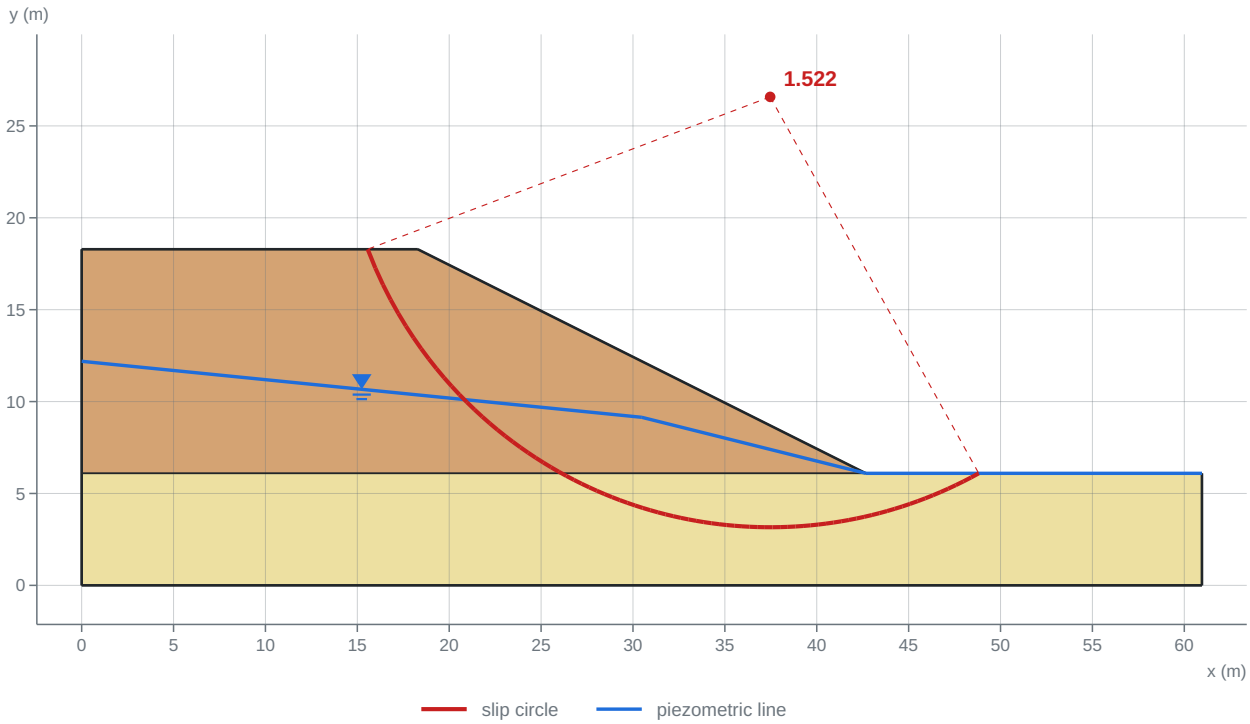


Figure 58. V033: critical circle (Bishop simplified, F = 1.522).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Mohr-Coulomb	19.64	c′ = 28.73	25.00°
Sand	Mohr-Coulomb	20.74	c′ = 0.00	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.522				1.533	−0.73 %
Spencer	1.534	1.529	Duncan & Wright	+0.30 %		
Spencer, non-circular	1.522				1.534	−0.78 %

3.58 V034a: Duncan & Wright fig. 14.20b, Su = 200 + 15·depth

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20b.

Critical circle search · Units: imperial (ft, psf, pcf)

Granular fill over clay with undrained strength increasing linearly with depth; critical-circle search.

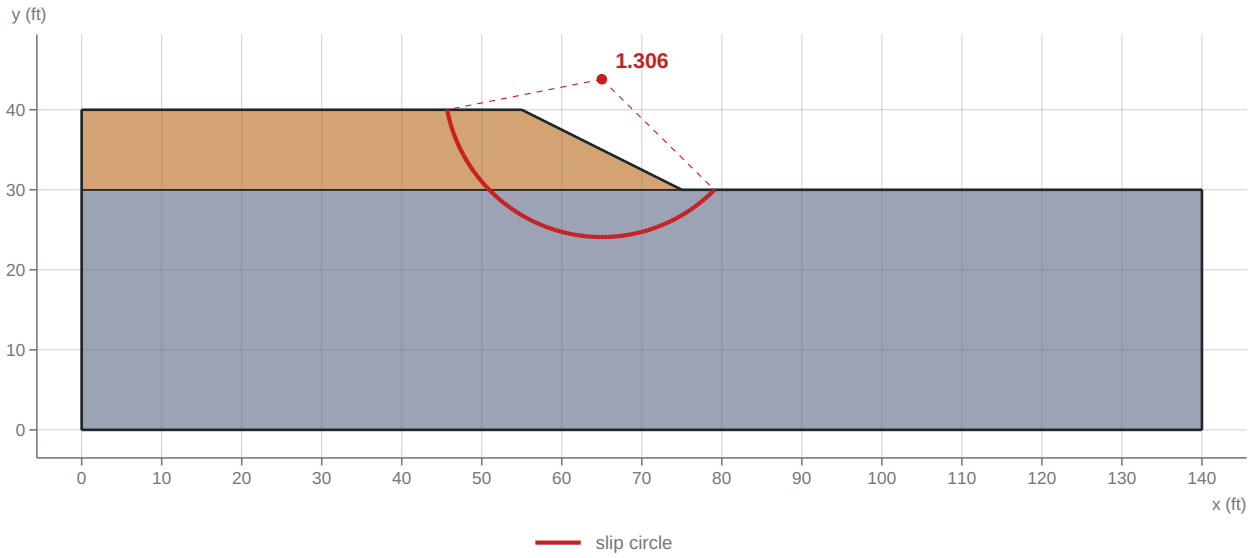


Figure 59. V034a: critical circle (Bishop simplified, F = 1.306).

Material	Model	γ (pcf)	Strength (psf)	φ
Fill	Mohr-Coulomb	123.0	c′ = 0.00	36.00°
Clay	Undrained (φ = 0)	97.00	Su = 200.0 at y = 30.00, +15.00 psf/ft below	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.306				1.313	−0.51 %
Spencer	1.277	1.276	Duncan & Wright	+0.06 %		
Spencer, non-circular	1.261				1.262	−0.09 %

3.59 V034b: Duncan & Wright fig. 14.20b, Su = 300 psf

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20b.

Critical circle search · Units: imperial (ft, psf, pcf)

As V034a with a constant undrained strength of 300 psf.

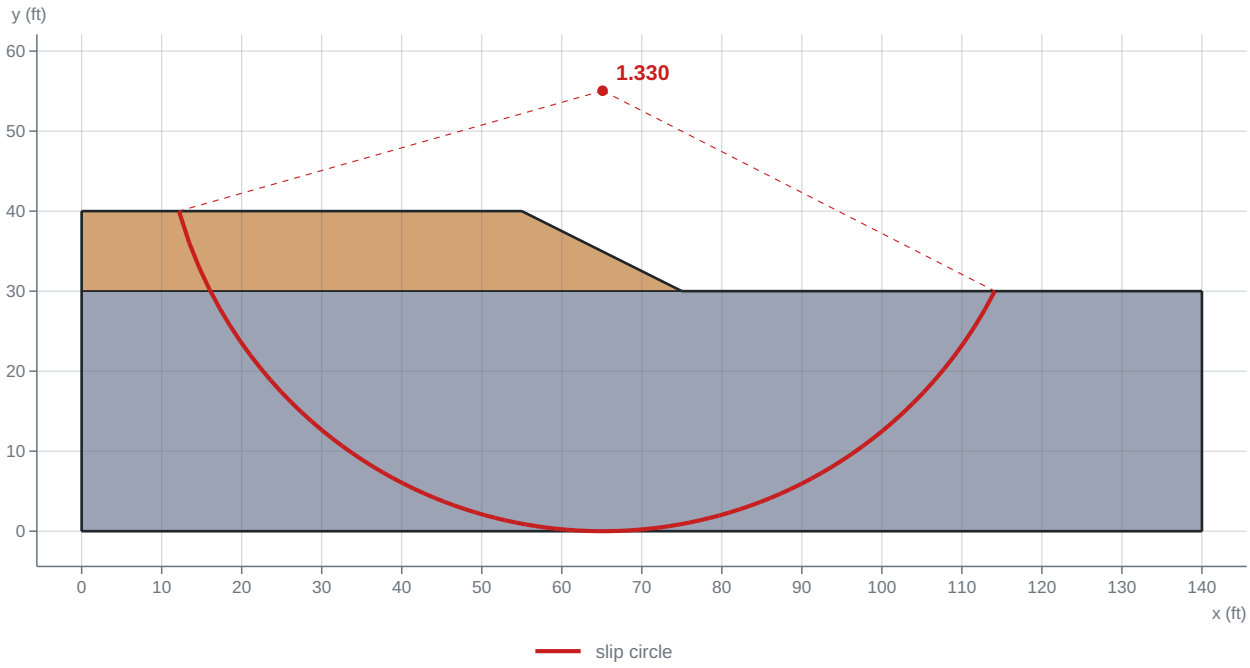


Figure 60. V034b: critical circle (Bishop simplified, F = 1.330).

Material	Model	γ (pcf)	Strength (psf)	φ
Fill	Mohr-Coulomb	123.0	c′ = 0.00	36.00°
Clay	Undrained (φ = 0)	97.00	Su = 300.0	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.330				1.335	−0.35 %
Spencer	1.328	1.328	Duncan & Wright	+0.03 %		
Spencer, non-circular	1.168				1.182	−1.18 %

3.60 V035a: Duncan & Wright tieback, active

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.34.

Fixed circle · Units: SI (m, kPa, kN/m³)

Homogeneous slope with a 9000 lb/ft tieback applied horizontally at mid-height as an active force, on the reference printed GLE circle. Converted to SI.

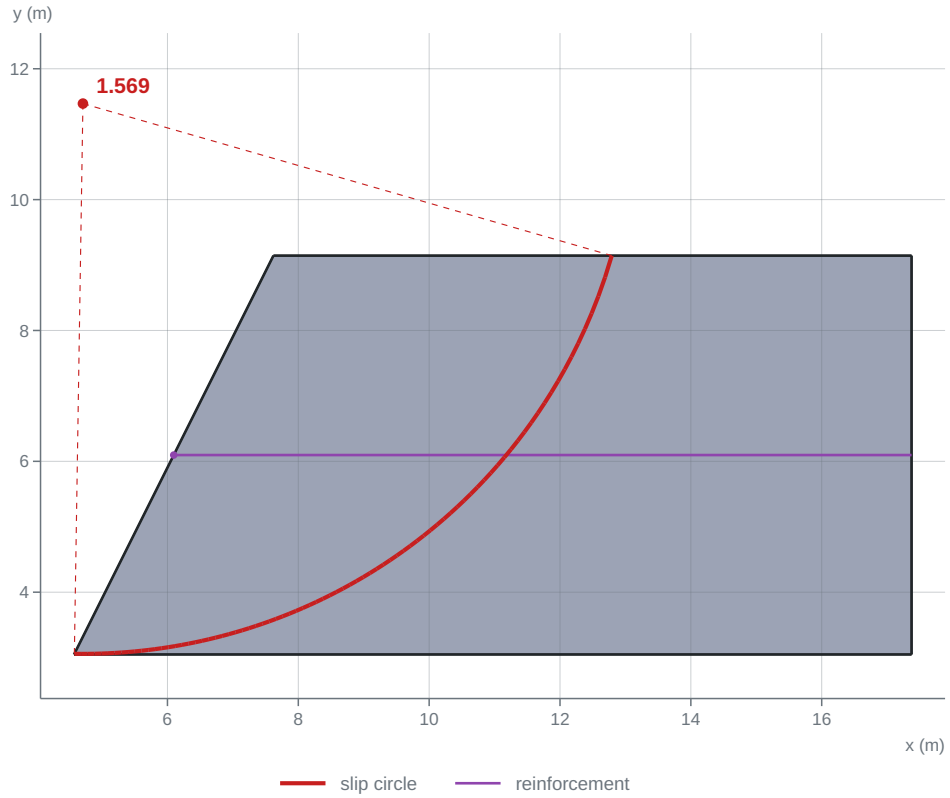


Figure 61. V035a: analyzed circle (Morgenstern-Price (half-sine), F = 1.569).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
saturated clay	Undrained (φ = 0)	15.39	Su = 16.76	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	1.569				1.575	−0.40 %
–	–	1.51	Duncan & Wright			

Duncan & Wright's 1.51 is about 4 % below the other programs on this surface.

3.61 V035b: Duncan & Wright tieback, passive

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.34.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V035a with the tieback applied as a passive force (divided by F), on the reference printed Bishop circle.

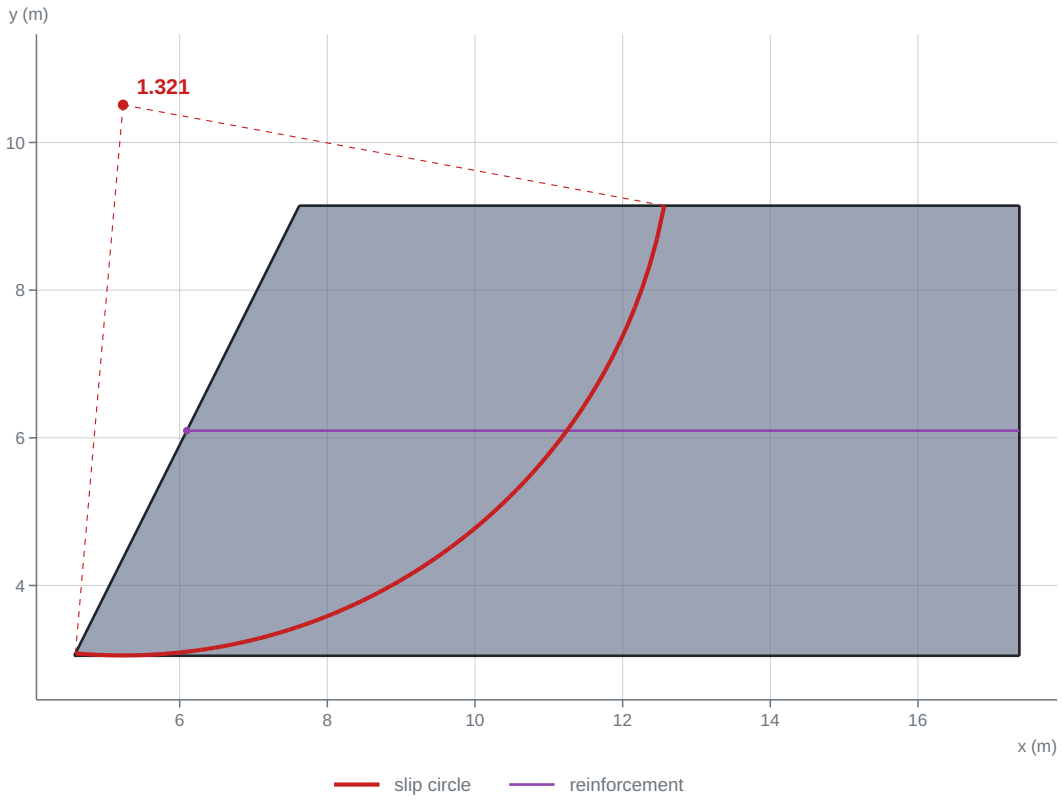


Figure 62. V035b: analyzed circle (Bishop simplified, F = 1.321).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
saturated clay	Undrained (φ = 0)	15.39	Su = 16.76	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.321	1.32	Duncan & Wright	+0.07 %	1.324	−0.23 %

3.62 V036a: Embankment on an undrained foundation, cz = 0 (D&W fig. 15.9)

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

Embankment (c′ = 0, φ′ = 35°, γ = 125 pcf) on a 20 ft undrained foundation, cu = 300 psf, constant; circular search. Converted to SI.

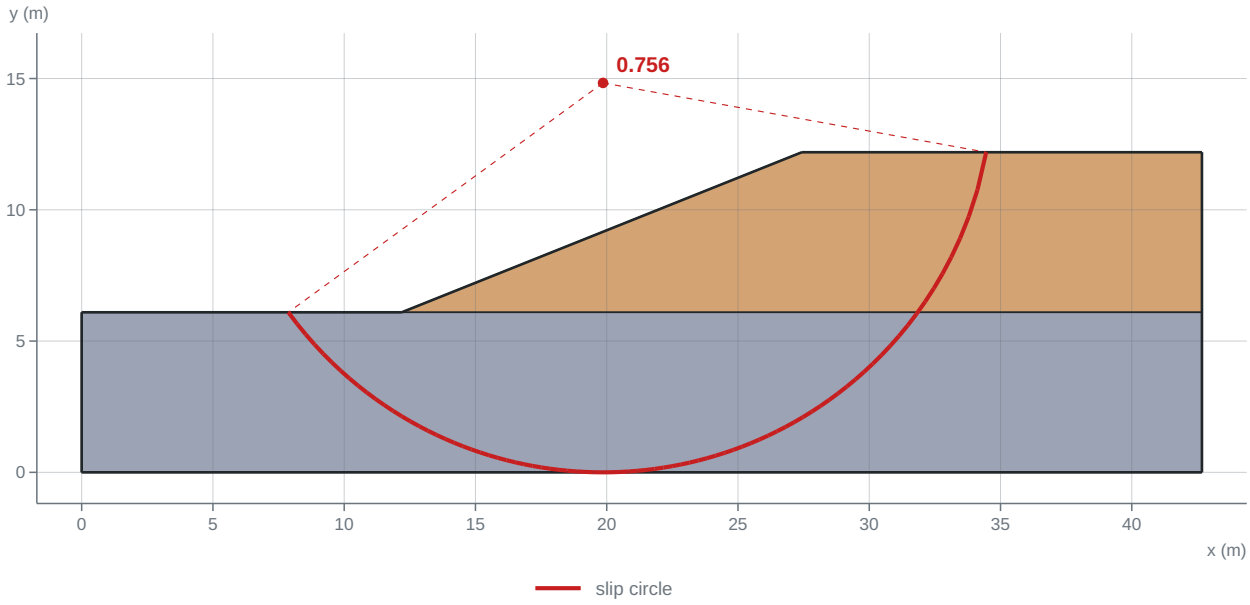


Figure 63. V036a: critical circle (Bishop simplified, F = 0.756).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c′ = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	0.756				0.761	−0.60 %
Spencer	0.751	0.75	Duncan & Wright	+0.20 %	0.756	−0.60 %
Spencer, non-circular	0.737				0.740	−0.43 %

3.63 V036b: Embankment on an undrained foundation, cz = 5 psf/ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V036a with cu increasing 5 psf/ft with depth.

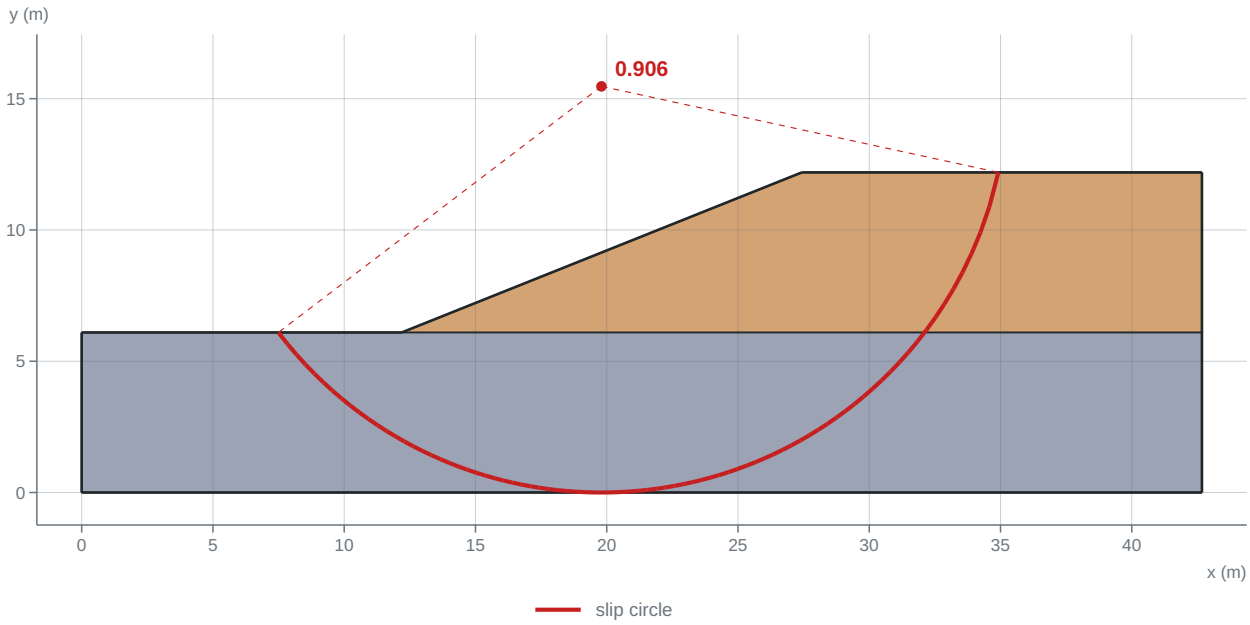


Figure 64. V036b: critical circle (Bishop simplified, F = 0.906).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c′ = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36 at y = 6.10, +0.79 kPa/m below	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	0.906				0.909	−0.35 %
Spencer	0.898	0.90	Duncan & Wright	−0.23 %	0.898	−0.01 %

3.64 V036c: Embankment on an undrained foundation, cz = 10 psf/ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V036a with cu increasing 10 psf/ft with depth.

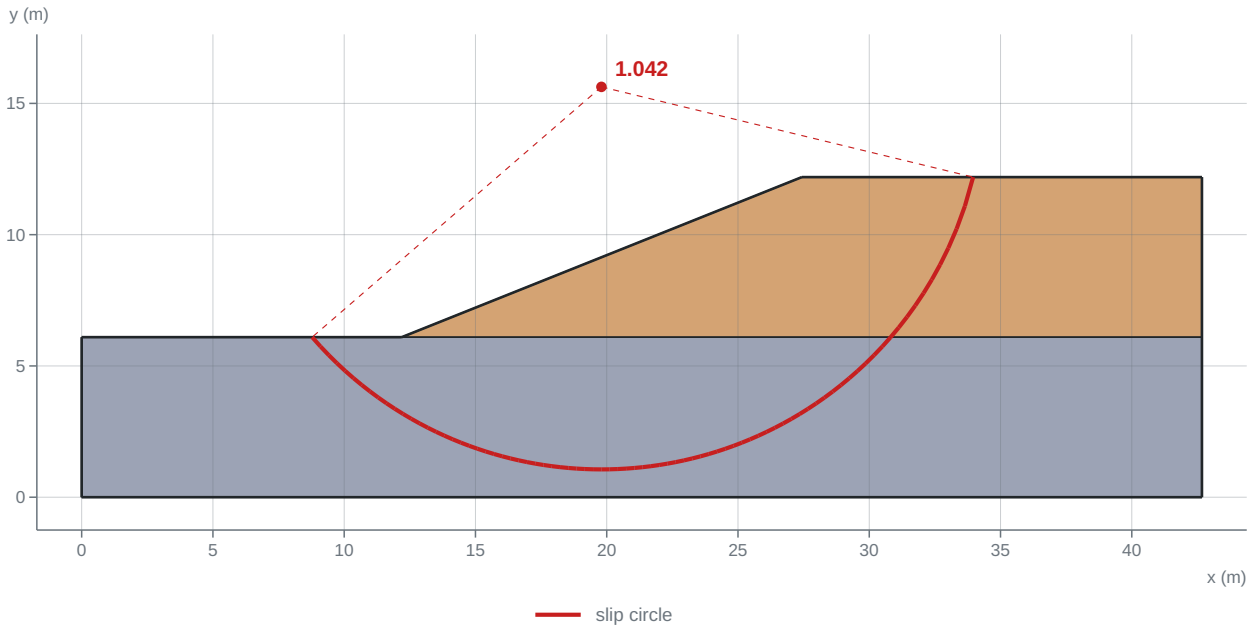


Figure 65. V036c: critical circle (Bishop simplified, F = 1.042).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c′ = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36 at y = 6.10, +1.57 kPa/m below	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.042				1.045	−0.25 %
Spencer	1.029	1.03	Duncan & Wright	−0.13 %	1.032	−0.32 %
Spencer, non-circular	1.016				1.018	−0.15 %

3.65 V036d: Embankment on an undrained foundation, cz = 15 psf/ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V036a with cu increasing 15 psf/ft with depth.

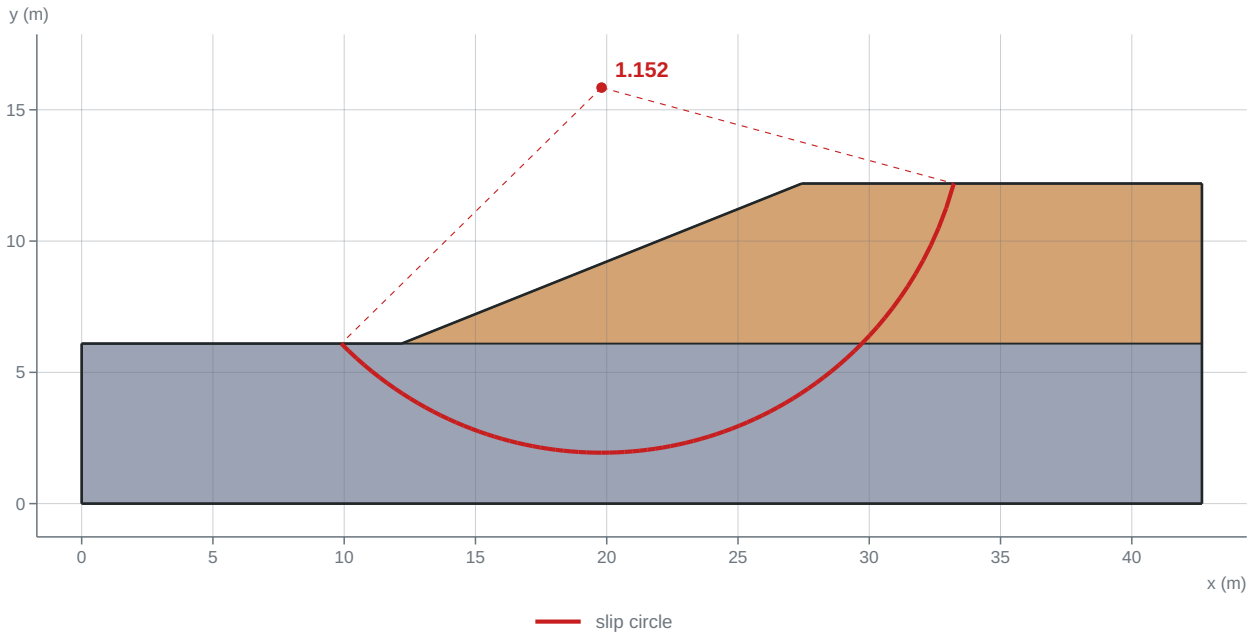


Figure 66. V036d: critical circle (Bishop simplified, F = 1.152).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c′ = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36 at y = 6.10, +2.36 kPa/m below	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.152				1.154	−0.17 %
Spencer	1.132	1.13	Duncan & Wright	+0.14 %	1.134	−0.21 %
Spencer, non-circular	1.119				1.116	+0.30 %

3.66 V037: Geogrid-reinforced fill on rock (D&W fig. 7.28)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.28.

Critical circle search · Units: SI (m, kPa, kN/m³)

24 ft fill slope (φ′ = 37°) on a strong rock foundation with five 800 lb/ft geogrids, 20 ft long, every 4 ft; circular search for surfaces through the toe. Converted to SI.

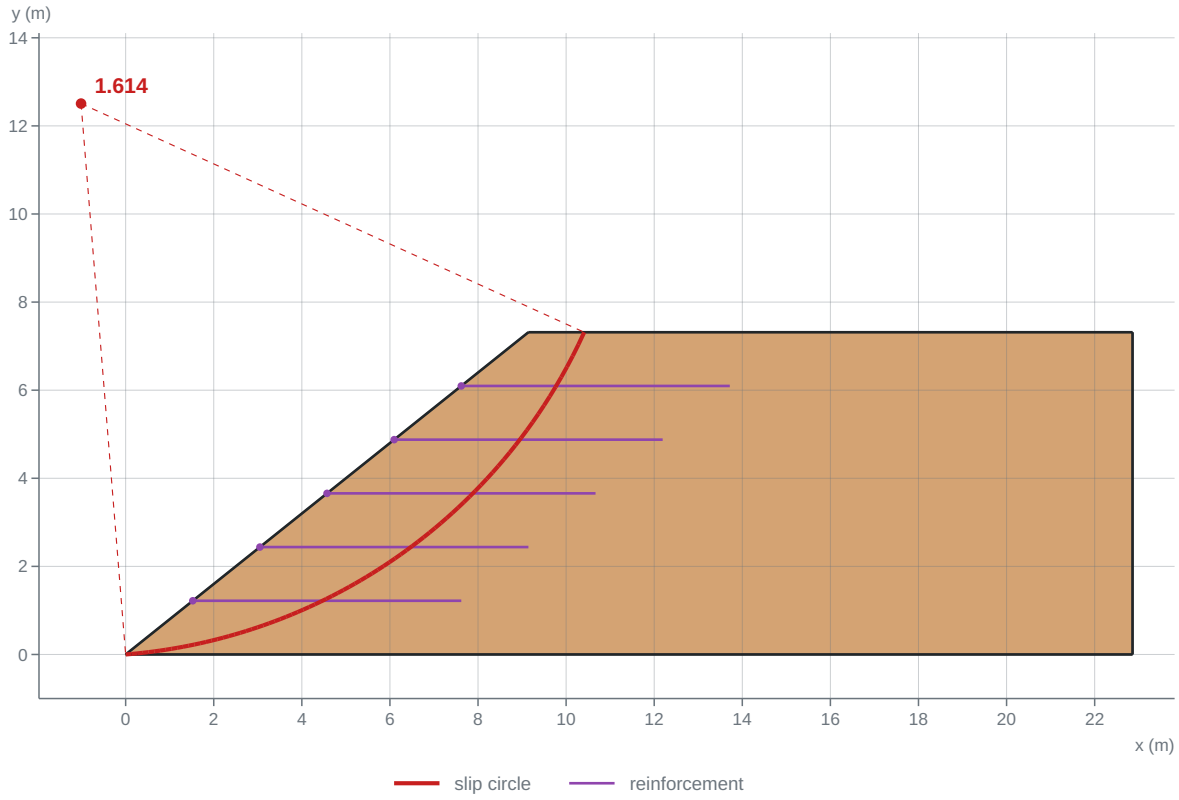


Figure 67. V037: critical circle (Bishop simplified, F = 1.614).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
fill	Mohr-Coulomb	20.42	c′ = 0.00	37.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.614				1.629	−0.95 %
Spencer	1.607	1.62	Duncan, Wright & Brandon (2014)	−0.77 %	1.620	−0.77 %

3.67 V038: Homogeneous dam with a piezometric line (D&W fig. 7.19)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.19.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous embankment (c′ = 100 psf, φ′ = 30°, γ = 100 pcf) with the pool at el. 40 ft and a commercial program's piezometric line; critical-circle search. Converted to SI.

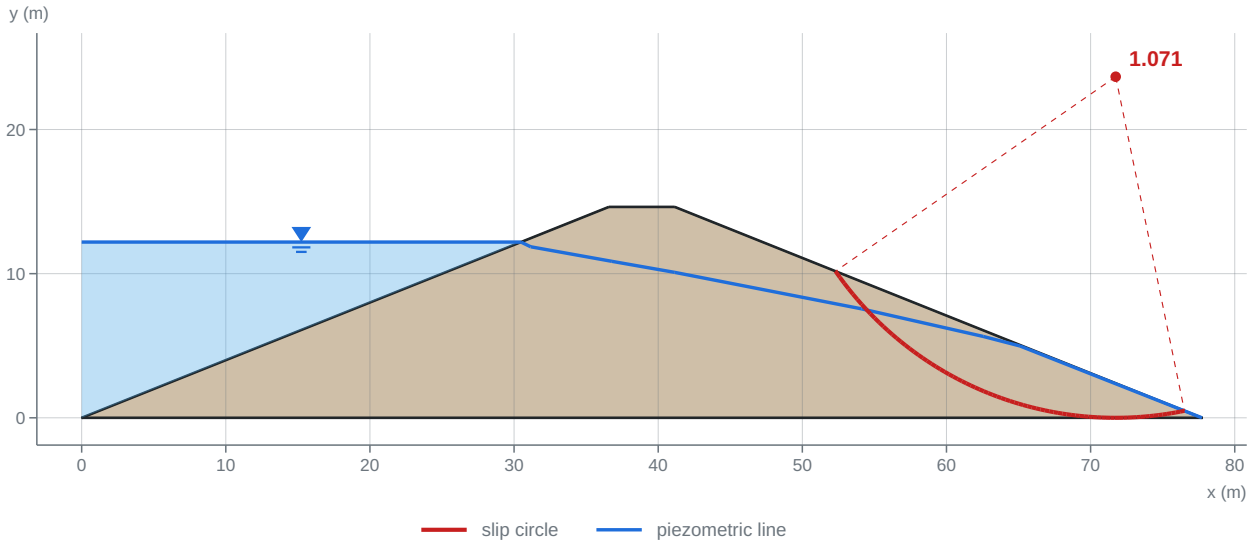


Figure 68. V038: critical circle (Bishop simplified, circles only, F = 1.071).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	15.71	c′ = 4.79	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified, circles only	1.071				1.070–1.090	−0.87 %
Spencer, circles only	1.078				1.100	−2.01 %
–	–	1.16	Duncan & Wright			

Duncan & Wright's 1.16 is for γ = 125 pcf and their printed phreatic line, not this program input (100 pcf, a 9-point line); on their input cSlope gives 1.157 (V038b).

3.68 V038b: Homogeneous dam with a piezometric line, Duncan & Wright's input

Duncan, J.M., Wright, S.G. & Brandon, T.L. (2014). Soil Strength and Slope Stability, 2nd ed., Wiley; §7.7, Figs. 7.30–7.36, Table 7.8.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V038 dam with the unit weight Duncan, Wright & Brandon's slice calculations use (125 pcf) and their printed phreatic line (30 points, Fig. 7.31); critical-circle search. Converted to SI.

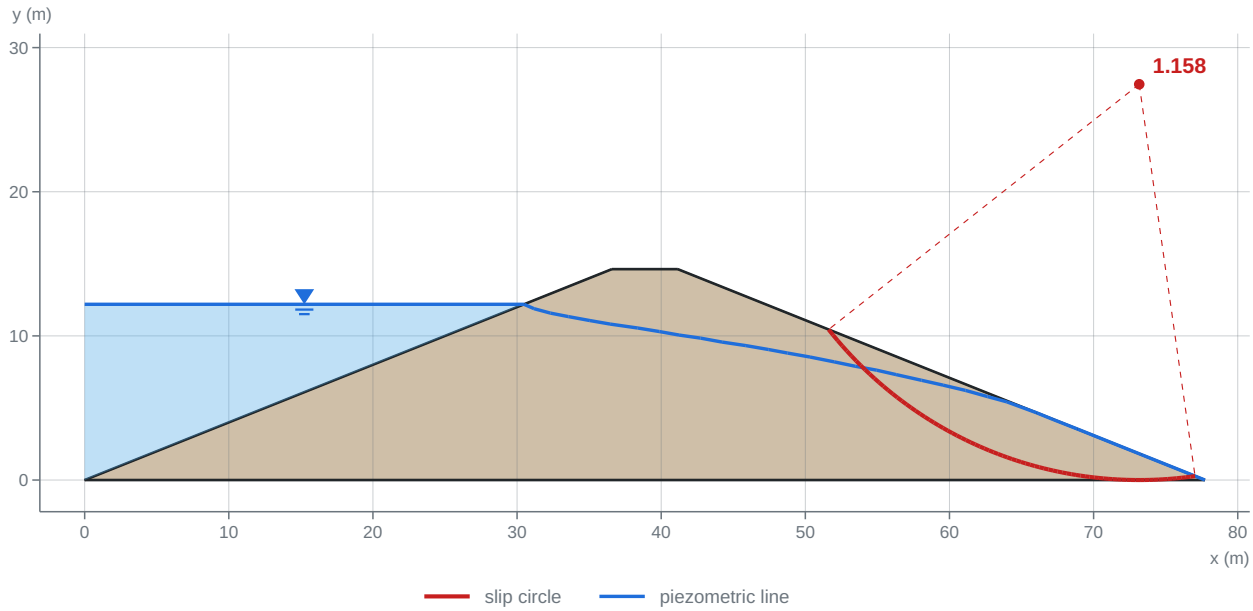


Figure 69. V038b: critical circle (Spencer, F = 1.158).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	19.64	c′ = 4.79	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.158	1.16	Duncan & Wright	−0.17 %		
		1.19	Duncan, Wright & Brandon (2014): Spencer, finite-element pore pressures	−2.69 %		
Strength reduction (FE)	1.073	1.07	Duncan, Wright & Brandon (2014): finite-element strength reduction, piezometric line	+0.33 %		

The book contradicts itself on the unit weight: its figures say 100 pcf, its hand calculations and slice weights use 125 pcf. With 125 pcf and the book's line cSlope reproduces the 1.16 that all three of the book's programs give (Table 7.8); with 100 pcf it gives 1.03.

3.69 V039: James Bay dyke (Duncan & Wright fig. 7.16)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.16.

Critical circle search · Units: SI (m, kPa, kN/m³)

Granular fill embankment with a wide berm (c′ = 0, φ′ = 30°) on a 4 m clay crust (Su = 41), 8 m marine clay (34.5) and 6.5 m lacustrine clay (31.2); circular search. The critical circle is deep and tangent to the base of the lacustrine clay.

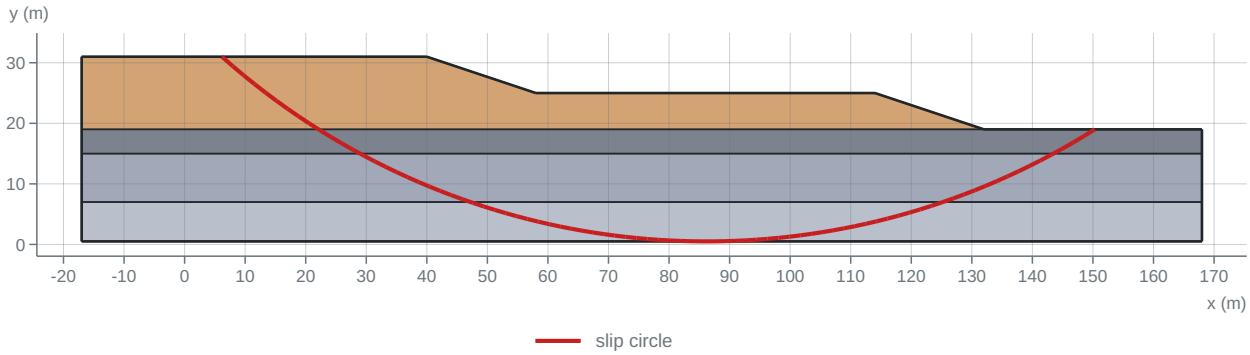


Figure 70. V039: critical circle (Spencer, circles only, F = 1.447).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Mohr-Coulomb	20.00	c′ = 0.00	30.00°
Clay crust	Undrained (φ = 0)	20.00	Su = 41.00	0°
Marine clay	Undrained (φ = 0)	18.80	Su = 34.50	0°
Lacustrine clay	Undrained (φ = 0)	20.30	Su = 31.20	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer, circles only	1.447	1.45	Duncan & Wright: Spencer	−0.19 %		
Spencer	1.166	1.17	Duncan & Wright: Spencer, composite circle, composite surface	−0.32 %		
Spencer, non-circular	1.160	1.16	Duncan & Wright	+0.03 %	1.167	−0.57 %
Strength reduction (FE)	1.240	1.26	Duncan, Wright & Brandon (2014): finite-element strength reduction	−1.56 %		

3.70 V040: James Bay dyke, reference model (deterministic)

El-Ramly et al. (2002); Duncan & Wright fig. 7.16.

Critical circle search · Units: SI (m, kPa, kN/m³)

The James Bay dyke as a commercial program models it (crest 36 m, berm at 30 m, clay crust Su = 43 kPa, till below y = 5.5); critical-circle search, deterministic.

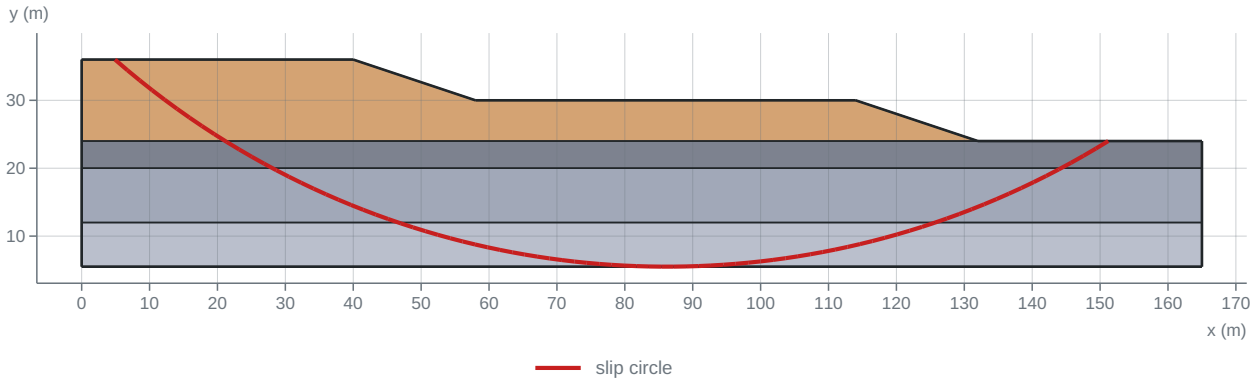


Figure 71. V040: critical circle (Bishop simplified, circles only, F = 1.459).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	20.00	c′ = 0.00	30.00°
Clay crust	Undrained (φ = 0)	18.80	Su = 43.00	0°
Marine clay	Undrained (φ = 0)	18.80	Su = 34.50	0°
Lacustrine clay	Undrained (φ = 0)	20.30	Su = 31.20	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified, circles only	1.460	1.46	El-Ramly et al. (2002): Bishop simplified	−0.03 %	1.461	−0.10 %

3.71 V041a: Submerged slope, pool 30 ft above the crest (D&W fig. 6.27)

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.27.

Critical circle search · Units: SI (m, kPa, kN/m³)

Fully submerged homogeneous slope (c = 100 psf, φ = 20°, γ = 128 pcf) with the pool 30 ft above the crest: the extra water weight and pore pressure cancel. Converted to SI.

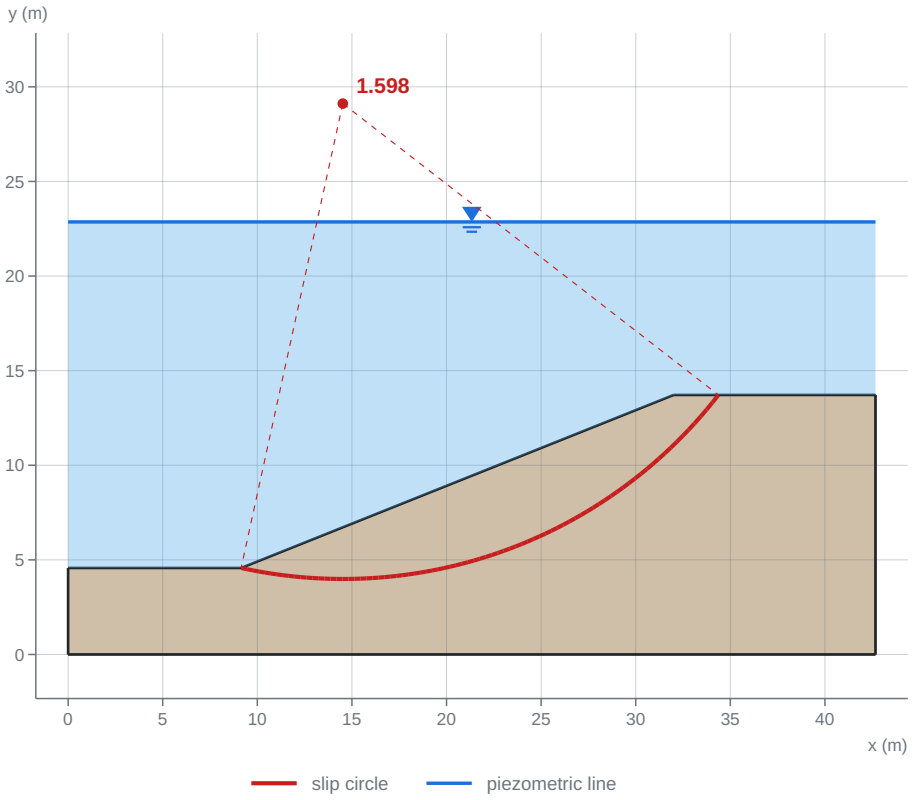


Figure 72. V041a: critical circle (Bishop simplified, F = 1.598).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	20.11	c′ = 4.79	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.598				1.603	−0.29 %
Spencer	1.596	1.60	Duncan & Wright	−0.28 %	1.599	−0.21 %
Spencer, non-circular	1.591				1.592	−0.04 %

3.72 V041b: Submerged slope, pool 60 ft above the crest (D&W fig. 6.27)

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.27.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V041a with the pool 60 ft above the crest: the same factor of safety.

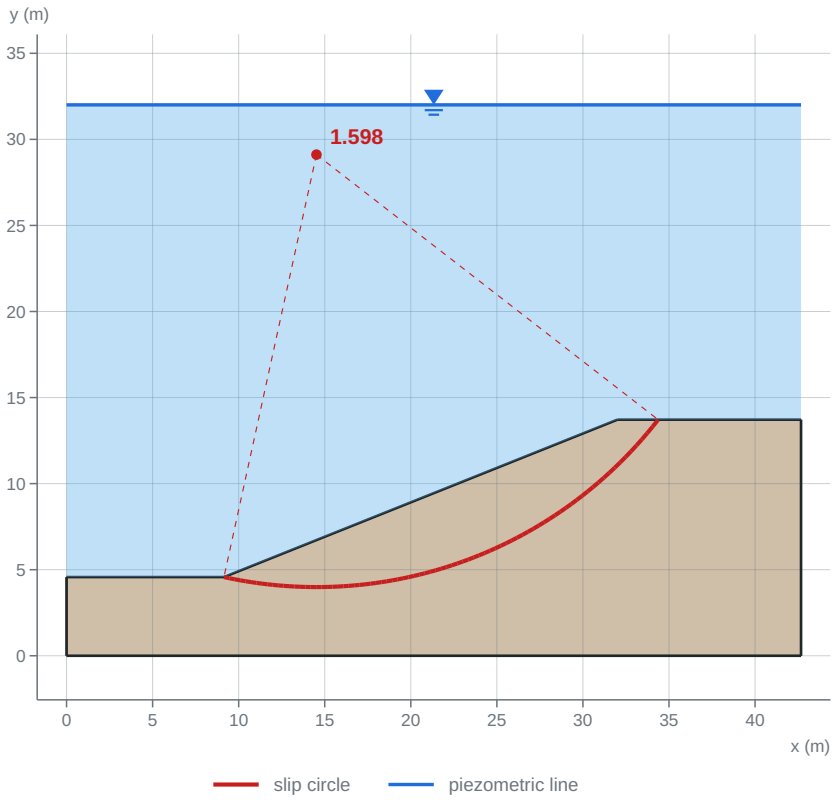


Figure 73. V041b: critical circle (Bishop simplified, F = 1.598).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	20.11	c′ = 4.79	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.598				1.603	−0.29 %
Spencer	1.595	1.60	Duncan & Wright	−0.31 %	1.599	−0.24 %
Spencer, non-circular	1.588				1.590	−0.13 %

3.73 V042: Thick-core dam with a piezometric line (D&W fig. 7.24)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.24.

Fixed circle · Units: SI (m, kPa, kN/m³)

Symmetric dam with a thick clay core (φ′ = 20°) in a shell (φ′ = 38°), pond at el. 315 ft, a commercial program's piezometric line; the deep circle. Converted to SI.

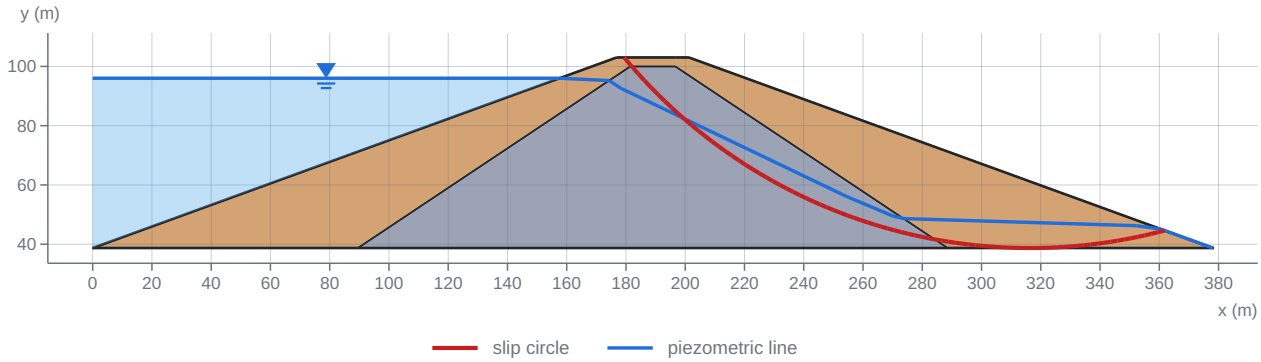


Figure 74. V042: analyzed circle (Bishop simplified, F = 1.582).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Shell	Mohr-Coulomb	21.99	c′ = 0.00	38.00°
Core	Mohr-Coulomb	18.85	c′ = 0.00	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.582				1.584	−0.11 %
Spencer	1.649				1.648	+0.08 %
–	–	1.67	Duncan & Wright			

3.74 V043: Syncrude tailings dyke (El-Ramly et al.), composite circle

El-Ramly, H., Morgenstern, N.R. & Cruden, D.M. (2003). Probabilistic stability analysis of a tailings dyke on presheared clay-shale. Can. Geotech. J. 40(1), 192–208.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Cohesionless tailings sand, glacio-fluvial sands and tills (φ = 34° here; El-Ramly et al. give the sandy till 35.7°) on a presheared clay-shale (φ = 7.5°) with a water table; the reference circle (327.5, 394, R = 124) reaches below the model base, so the surface follows the base there (composite).

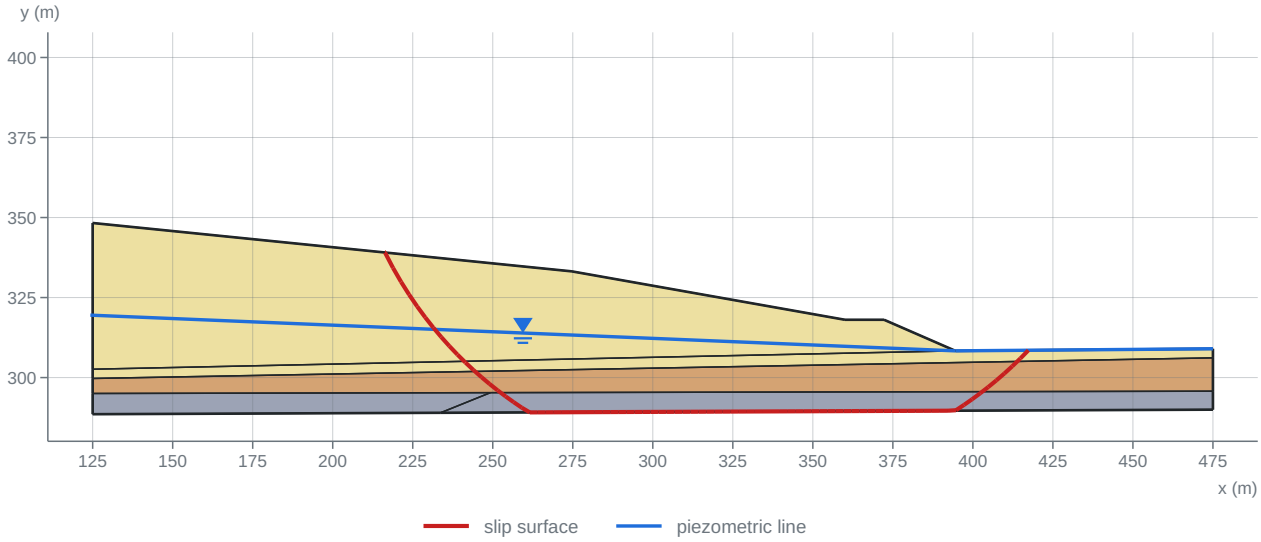


Figure 75. V043: analyzed surface (Bishop simplified, F = 1.320).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Tailing sand (TS)	Mohr-Coulomb	20.00	c′ = 0.00	34.00°
Glacio-fluvial sand (Pf4)	Mohr-Coulomb	17.00	c′ = 0.00	34.00°
Sandy till (Pgs)	Mohr-Coulomb	17.00	c′ = 0.00	34.00°
Clayey till (Pgc)	Mohr-Coulomb	17.00	c′ = 0.00	7.50°
Disturbed clay-shale (Kca)	Mohr-Coulomb	17.00	c′ = 0.00	7.50°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.320	1.31	El-Ramly et al.	+0.78 %	1.320	+0.02 %

3.75 V044: Fredlund & Krahn (1977), dry

Fredlund, D.G. & Krahn, J. (1977). Comparison of slope stability methods of analysis. Can. Geotech. J. 14(3).

Fixed circle · Units: imperial (ft, psf, pcf)

Homogeneous 2:1 slope, 40 ft high, on a firm base: Fredlund & Krahn's comparison problem, on the published circle and with no pore pressure.

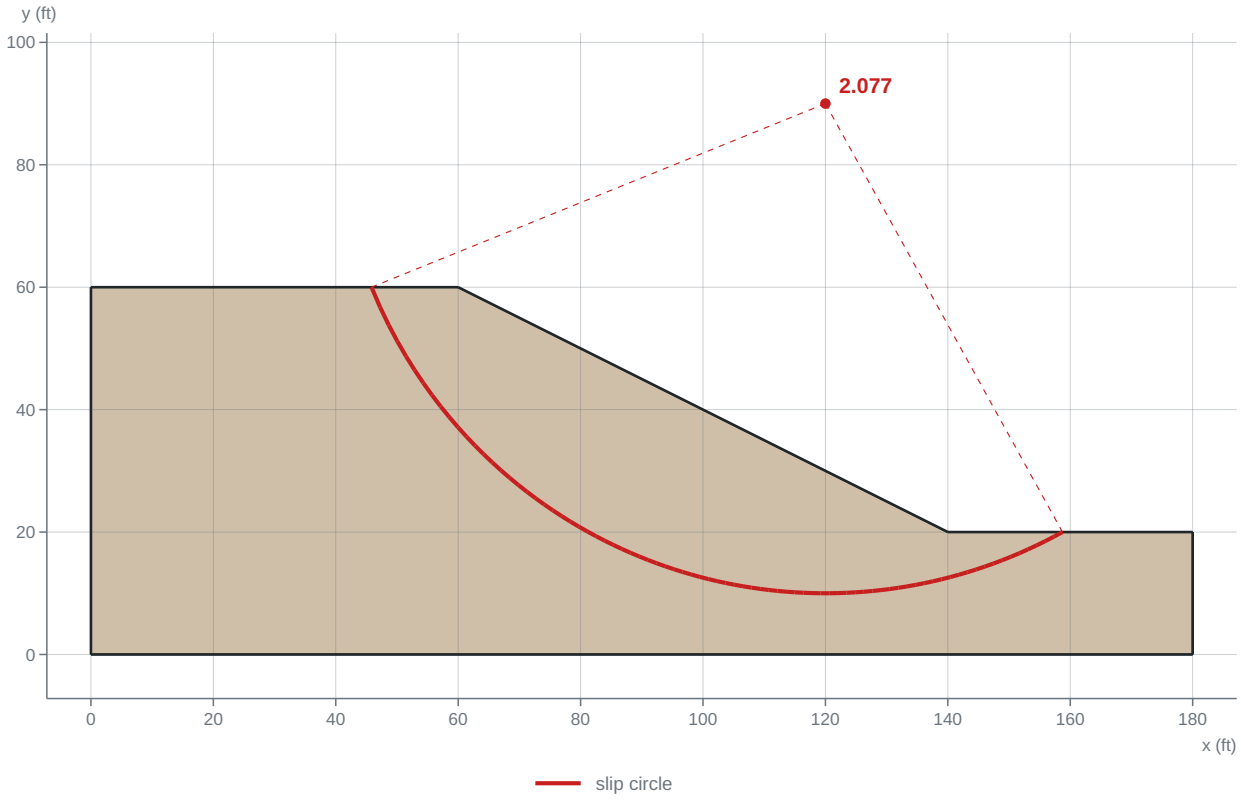


Figure 76. V044: analyzed circle (Bishop simplified, F = 2.077).

Material	Model	γ (pcf)	Strength (psf)	φ
Soil	Mohr-Coulomb	120.0	c′ = 600.0	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.077	2.08	Fredlund & Krahn	−0.16 %		
Morgenstern-Price (half-sine)	2.072	2.076	Fredlund & Krahn	−0.17 %		
Spencer	2.073	2.073	Fredlund & Krahn	−0.00 %		
Strength reduction (FE)	1.980	2.00	another program's verification manual: finite-element strength reduction (verification manual, problem 17)	−0.99 %		

3.76 V044b: Fredlund & Krahn (1977), r_u = 0.25

Fredlund, D.G. & Krahn, J. (1977). Comparison of slope stability methods of analysis. Can. Geotech. J. 14(3), 429–439.

Fixed circle · Units: imperial (ft, psf, pcf)

The V044 slope and circle with a pore pressure ratio r_u = 0.25 (Fredlund & Krahn's case 3).

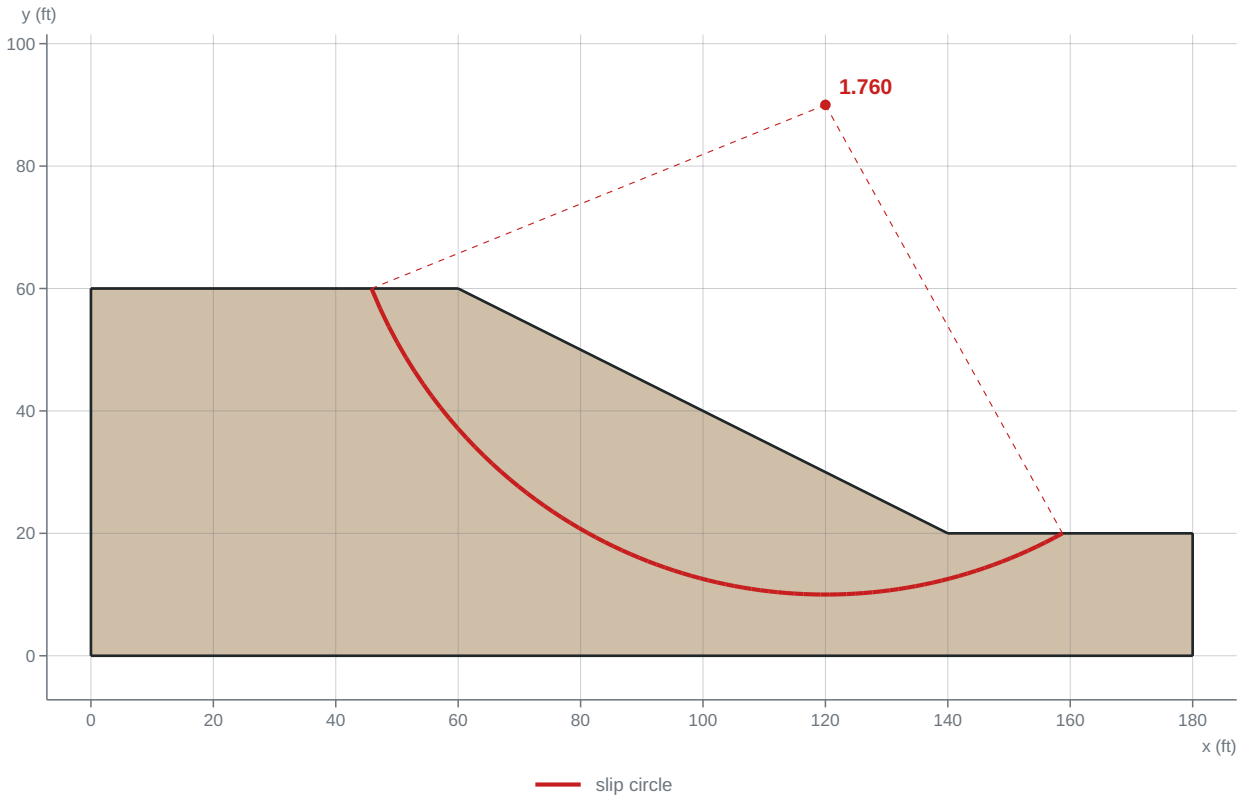


Figure 77. V044b: analyzed circle (Bishop simplified, F = 1.760).

Material	Model	γ (pcf)	Strength (psf)	φ
Soil	Mohr-Coulomb	120.0	c′ = 600.0	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.760	1.766	Fredlund & Krahn	−0.33 %		
Morgenstern-Price (half-sine)	1.758	1.764	Fredlund & Krahn	−0.36 %		
Spencer	1.758	1.761	Fredlund & Krahn	−0.15 %		

3.77 V045: Fredlund & Krahn (1977), with water table

Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3).

Fixed circle · Units: SI (m, kPa, kN/m³)

The Fredlund & Krahn slope with a piezometric line, analyzed on the published circle. Converted to SI.

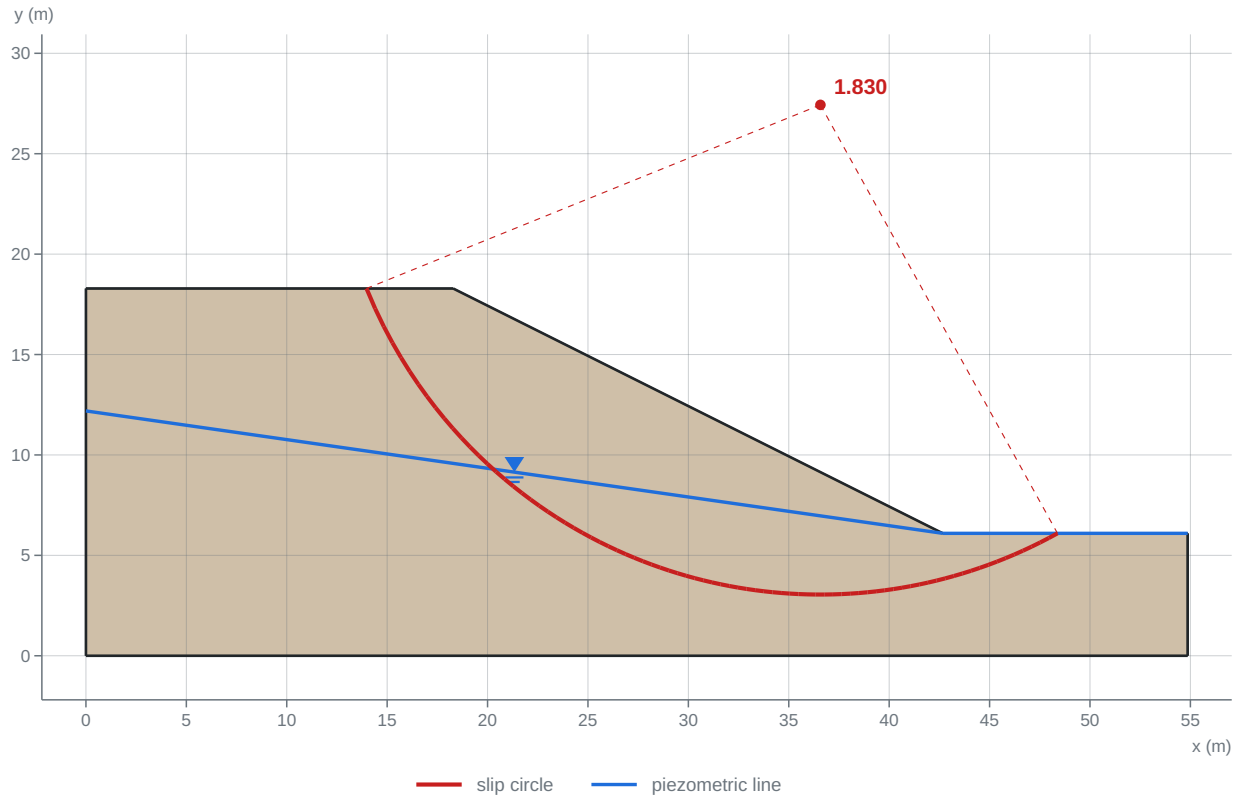


Figure 78. V045: analyzed circle (Bishop simplified, F = 1.830).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.85	c′ = 28.73	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.830	1.834	Fredlund & Krahn	−0.22 %		
Morgenstern-Price (half-sine)	1.828	1.832	Fredlund & Krahn	−0.23 %		
Spencer	1.828	1.83	Fredlund & Krahn	−0.09 %		
Strength reduction (FE)	1.781	1.78	another program's verification manual: finite-element strength reduction (verification manual, problem 17)	+0.05 %		

3.78 V046a: Fredlund & Krahn slope on a weak seam, composite surface

Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3).

Specified non-circular surface · Units: imperial (ft, psf, pcf)

The Fredlund & Krahn slope with a 1 ft weak seam (c′ = 0, φ′ = 10°) on bedrock at el. 15 (their case 2); the published circle (120, 90, R = 80) reaches below the seam, so the surface follows the arc down to the bedrock, runs along the seam and climbs back on the arc. Dry.

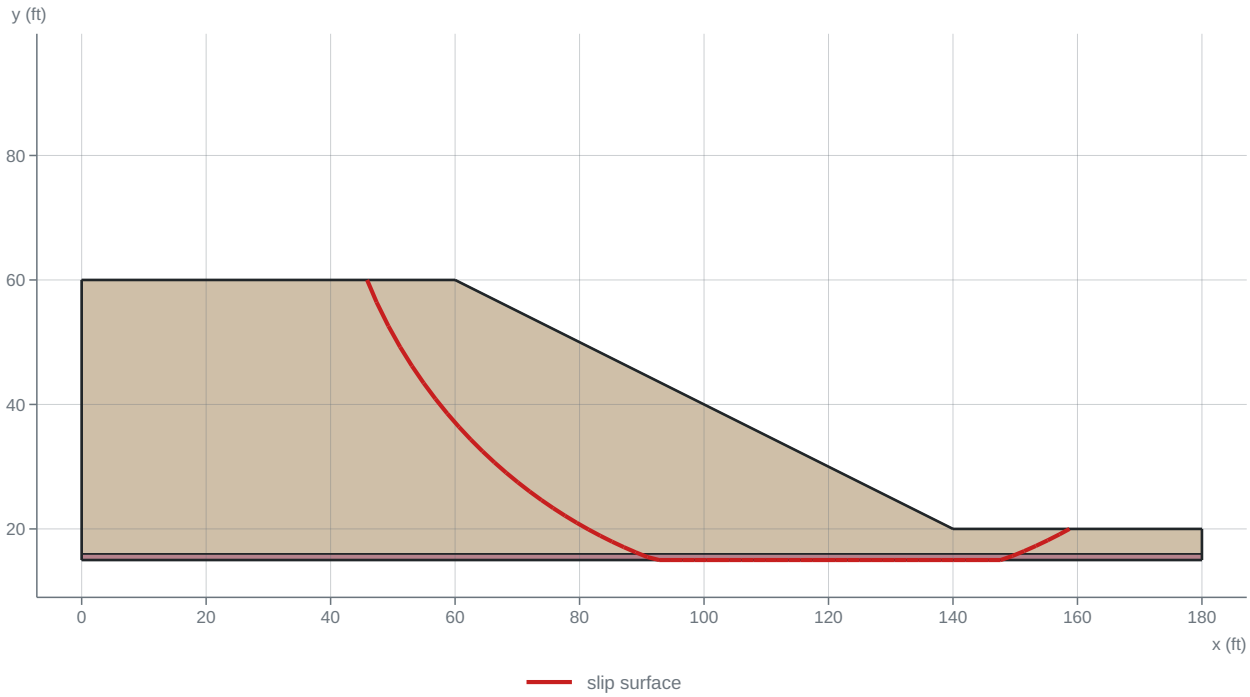


Figure 79. V046a: analyzed surface (Bishop simplified, F = 1.380).

Material	Model	γ (pcf)	Strength (psf)	φ
Upper soil	Mohr-Coulomb	120.0	c′ = 600.0	20.00°
Weak seam	Mohr-Coulomb	120.0	c′ = 0.00	10.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.380	1.377	Fredlund & Krahn	+0.25 %	1.382	−0.12 %
Spencer	1.379	1.373	Fredlund & Krahn	+0.47 %	1.382	−0.19 %
Morgenstern-Price (half-sine)	1.370	1.37	Fredlund & Krahn	−0.01 %		
Strength reduction (FE)	1.337	1.34	another program's verification manual: finite-element strength reduction (verification manual, problem 18)	−0.25 %		

3.79 V046b: Fredlund & Krahn slope on a weak seam, with a piezometric line

Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

As V046a with the piezometric line of V045 (from (0, 40) ft to the toe) for both materials: Fredlund & Krahn's case 6. Converted to SI.

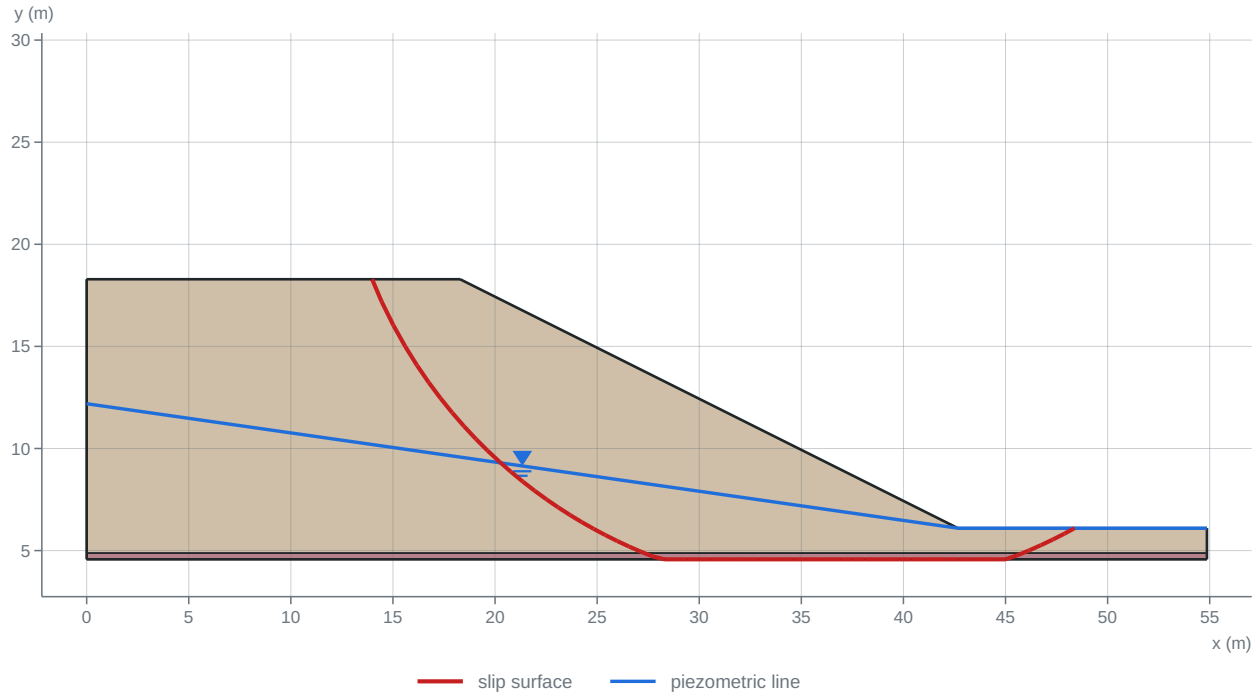


Figure 80. V046b: analyzed surface (Bishop simplified, F = 1.241).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper soil	Mohr-Coulomb	18.85	c′ = 28.73	20.00°
Weak seam	Mohr-Coulomb	18.85	c′ = 0.00	10.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.241	1.248	Fredlund & Krahn	−0.57 %		
Spencer	1.242	1.245	Fredlund & Krahn	−0.27 %		
Morgenstern-Price (half-sine)	1.235					
Strength reduction (FE)	1.186	1.13	another program's verification manual: finite-element strength reduction (verification manual, problem 18)	+4.99 %		

3.80 V046c: Fredlund & Krahn slope on a weak seam, r_u = 0.25

Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3), 429–439.

Specified non-circular surface · Units: imperial (ft, psf, pcf)

As V046a (weak seam at el. 15–16 ft, composite surface) with a pore pressure ratio r_u = 0.25 in both materials: Fredlund & Krahn's case 4.

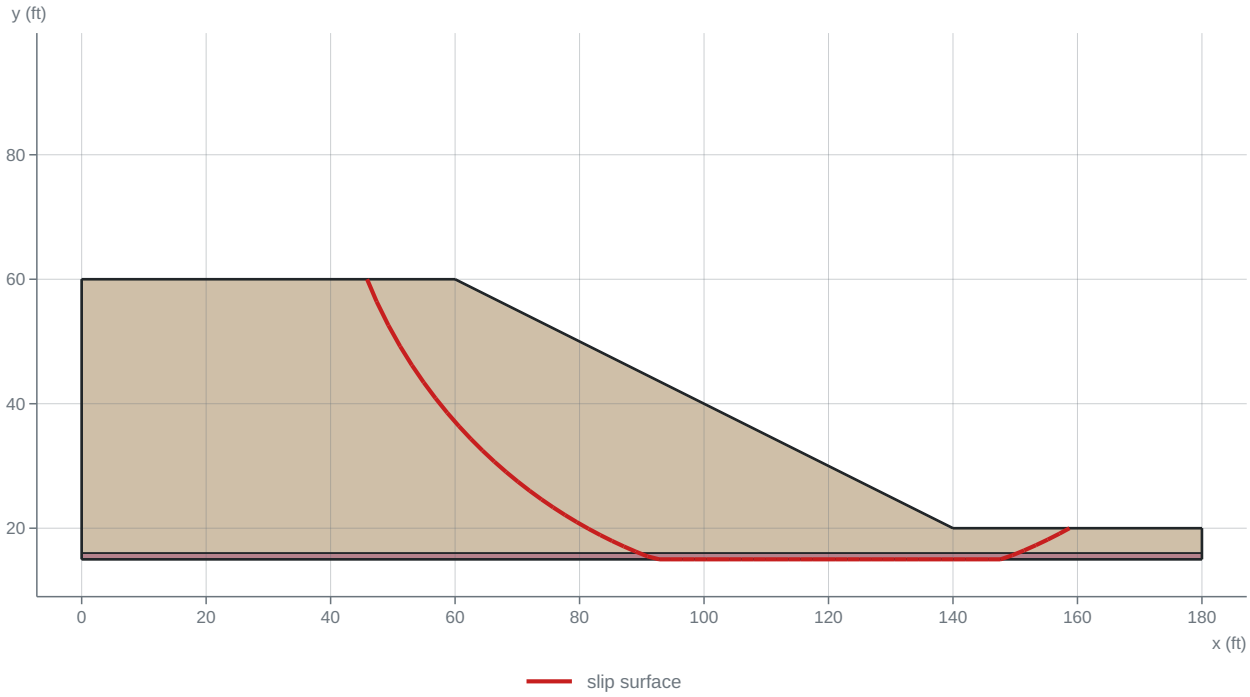


Figure 81. V046c: analyzed surface (Bishop simplified, F = 1.122).

Material	Model	γ (pcf)	Strength (psf)	φ
Upper soil	Mohr-Coulomb	120.0	c′ = 600.0	20.00°
Weak seam	Mohr-Coulomb	120.0	c′ = 0.00	10.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.122	1.124	Fredlund & Krahn	−0.19 %	1.124	−0.19 %
Spencer	1.122				1.124	−0.16 %
Morgenstern-Price (half-sine)	1.113	1.118	Fredlund & Krahn	−0.49 %		

3.81 V047a: ACADS 1(a), homogeneous slope, dry

Giam & Donald (1989).

Critical circle search · Units: SI (m, kPa, kN/m³)

ACADS 1(a): homogeneous 2:1 slope, 10 m high (c′ = 3 kPa, φ′ = 19.6°, γ = 20 kN/m³), dry; critical-circle search.

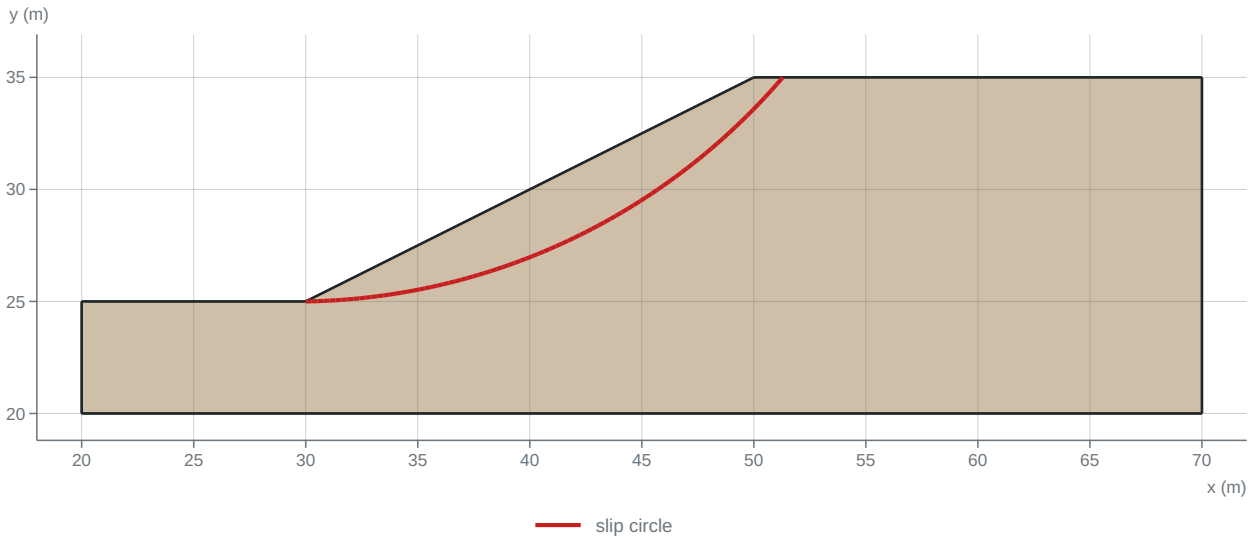


Figure 82. V047a: critical circle (Bishop simplified, F = 0.985).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c′ = 3.00	19.60°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	0.985				0.987–0.992	−0.44 %
Spencer	0.984	1.00	ACADS	−1.59 %	0.986	−0.19 %
Janbu corrected	0.986				0.990	−0.44 %
Morgenstern-Price (half-sine)	0.984				0.991	−0.70 %
Strength reduction (FE)	0.956	0.99	another program's verification manual: finite-element strength reduction (verification manual, problem 1)	−3.44 %		

3.82 V047b: ACADS tension crack, reference crack depth 3.6 m (results correspond to a dry crack)

Giam & Donald (1989), ACADS problem 1(b).

Critical circle search · Units: SI (m, kPa, kN/m³)

ACADS 1(b) as a commercial program models it: c′ = 32 kPa, φ′ = 10°, a 3.6 m tension crack (dry); critical-circle search.

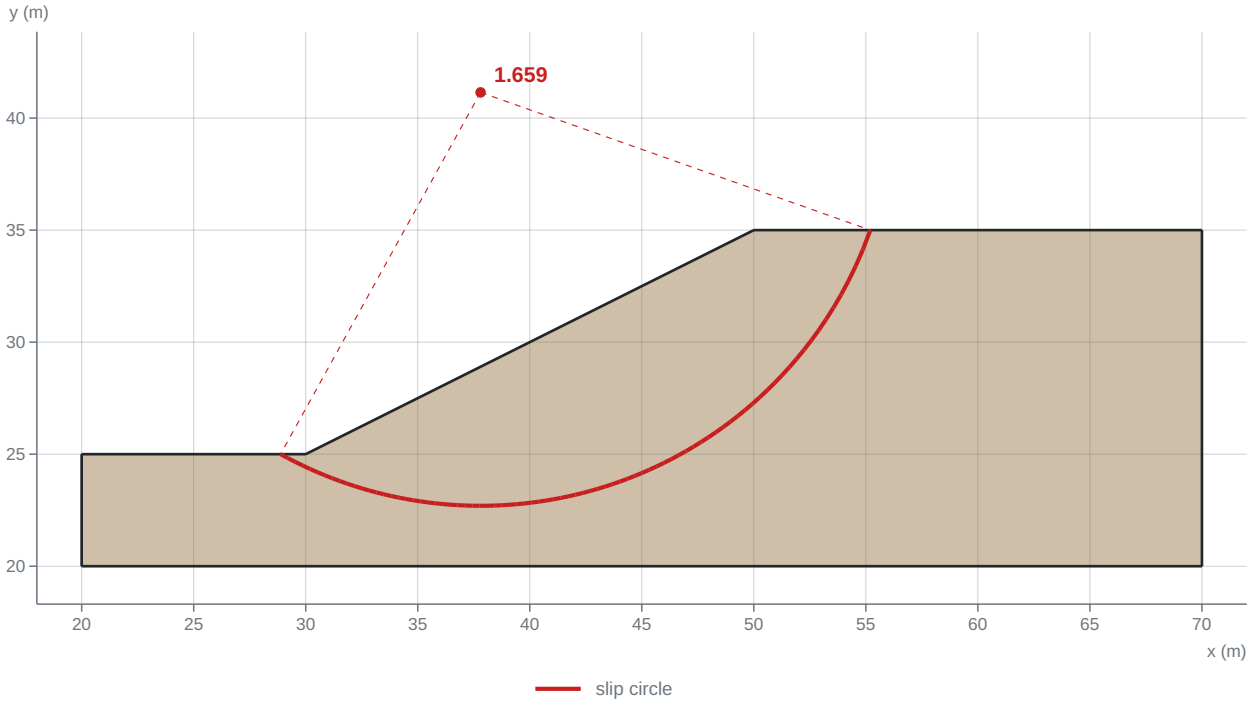


Figure 83. V047b: critical circle (Bishop simplified, F = 1.659).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c′ = 32.00	10.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.659				1.664	−0.30 %
Morgenstern-Price (half-sine)	1.656				1.660	−0.25 %
–	–	1.65–1.70	Giam & Donald (1989), ACADS			

3.83 V048: ACADS 1(b), water-filled tension crack

Giam & Donald (1989), ACADS problem 1(b).

Critical circle search · Units: SI (m, kPa, kN/m³)

The ACADS 1(a) slope (10 m high, 2:1) with c′ = 32 kPa, φ′ = 10° and a water-filled tension crack 4 m deep, the crack zone of a commercial program's Fig. 2.1 (35 → 31 m; the formula 2c/(γ√Ka) gives 3.814 m); critical-circle search.

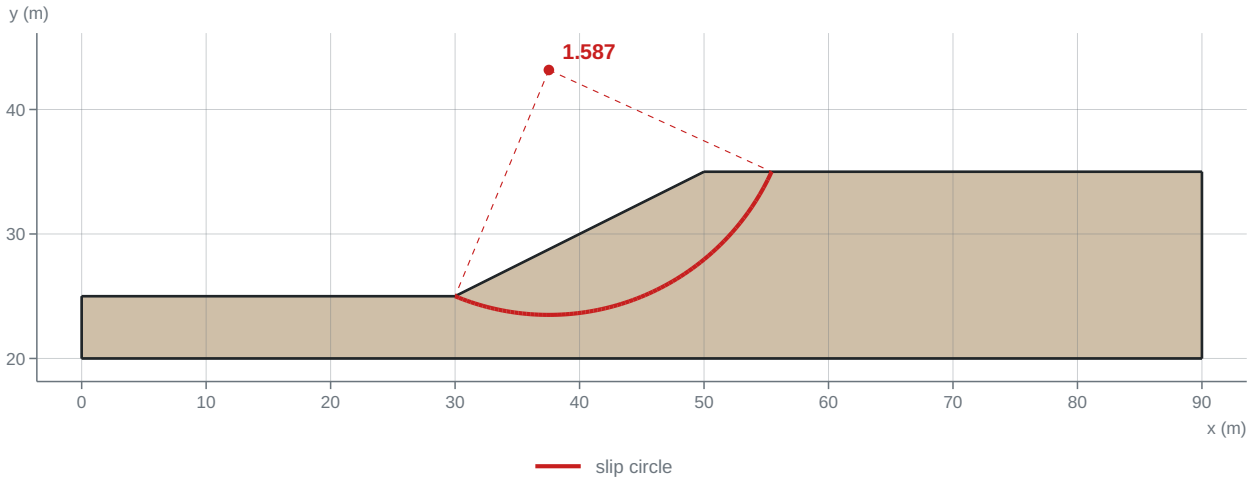


Figure 84. V048: critical circle (Bishop simplified, F = 1.587).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c′ = 32.00	10.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.587				1.596	−0.57 %
Spencer	1.583				1.592	−0.60 %
Janbu corrected	1.475				1.489	−0.93 %
–	–	1.65	Giam & Donald (1989), ACADS			

The ACADS reference 1.65 corresponds to a dry tension crack; with the crack filled with water the programs give about 1.59–1.60.

3.84 V049: ACADS 1(c), three-layer slope

Giam & Donald (1989), ACADS slope stability programs review.

Critical circle search · Units: SI (m, kPa, kN/m³)

Non-homogeneous three-layer slope (ACADS 1c); critical-circle search.

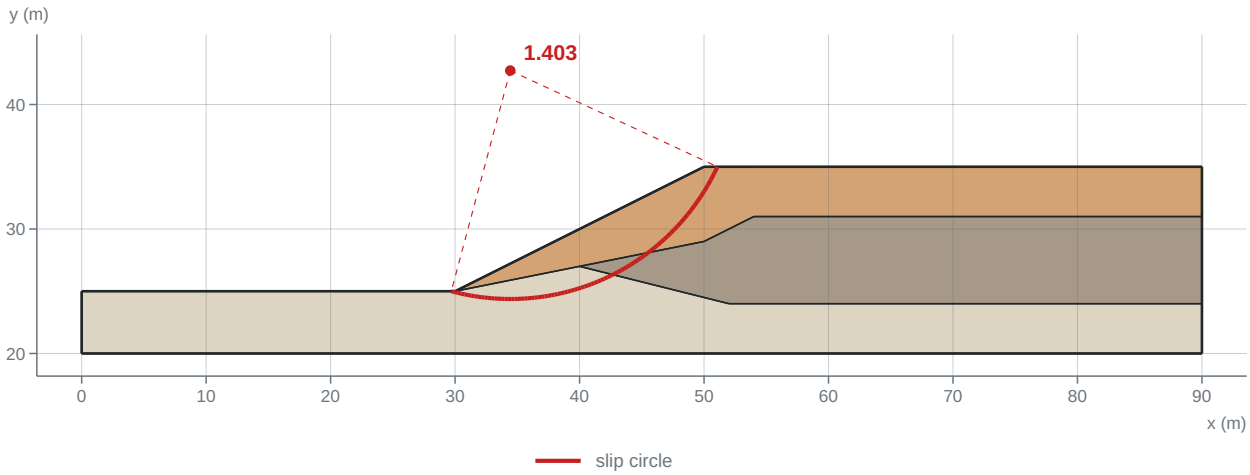


Figure 85. V049: critical circle (Bishop simplified, F = 1.403).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	19.50	c′ = 0.00	38.00°
Soil #2	Mohr-Coulomb	19.50	c′ = 5.30	23.00°
Soil #3	Mohr-Coulomb	19.50	c′ = 7.20	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.403				1.403–1.405	−0.06 %
Spencer	1.372				1.375	−0.19 %
Janbu corrected	1.356				1.357	−0.10 %
Morgenstern-Price (half-sine)	1.372				1.382	−0.73 %
Strength reduction (FE)	1.335	1.36	another program's verification manual: finite-element strength reduction (verification manual, problem 2)	−1.82 %		
–	–	1.39	Giam & Donald (1989), ACADS			

3.85 V050a: ACADS 3(b), weak layer, specified non-circular surface

Giam & Donald (1989), ACADS problem 3(b).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Slope with a 0.5 m weak layer (c′ = 0, φ′ = 10°) and a water table at its base, analyzed on the published four-point non-circular surface that runs along the weak layer.

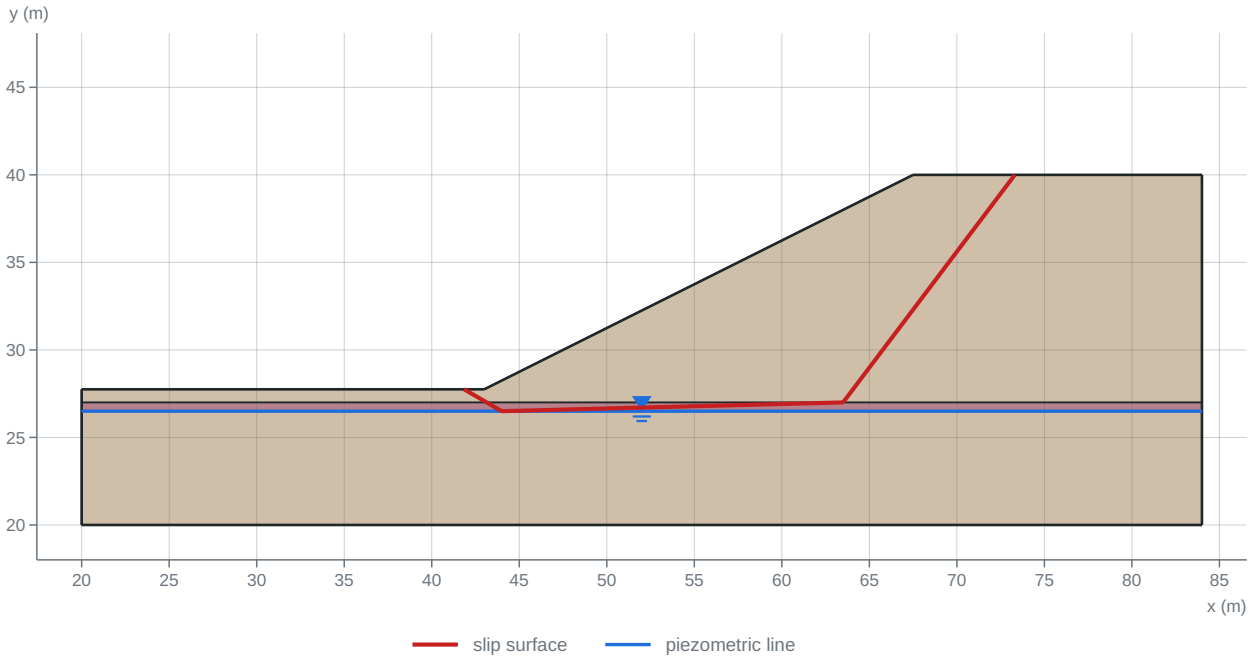


Figure 86. V050a: analyzed surface (Janbu corrected, F = 1.294).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.84	c′ = 28.50	20.00°
Weak layer	Mohr-Coulomb	18.84	c′ = 0.00	10.00°
Soil 1 below	Mohr-Coulomb	18.84	c′ = 28.50	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	1.294				1.294	−0.02 %
Spencer	1.276				1.277	−0.06 %
Morgenstern-Price (half-sine)	1.260				1.261	−0.04 %
–	–	1.34	Giam & Donald (1989), ACADS			

The ACADS referee value comes from the 1989 survey; programs today give 1.26–1.29 on the specified surface.

3.86 V050b: ACADS 3(a), weak layer, non-circular search

Giam & Donald (1989), ACADS problem 3(a).

Circle search with non-circular optimization · Units: SI (m, kPa, kN/m³)

As V050a, with a circular search followed by non-circular optimization; the critical surface slides along the weak layer with a back scarp to the crest.

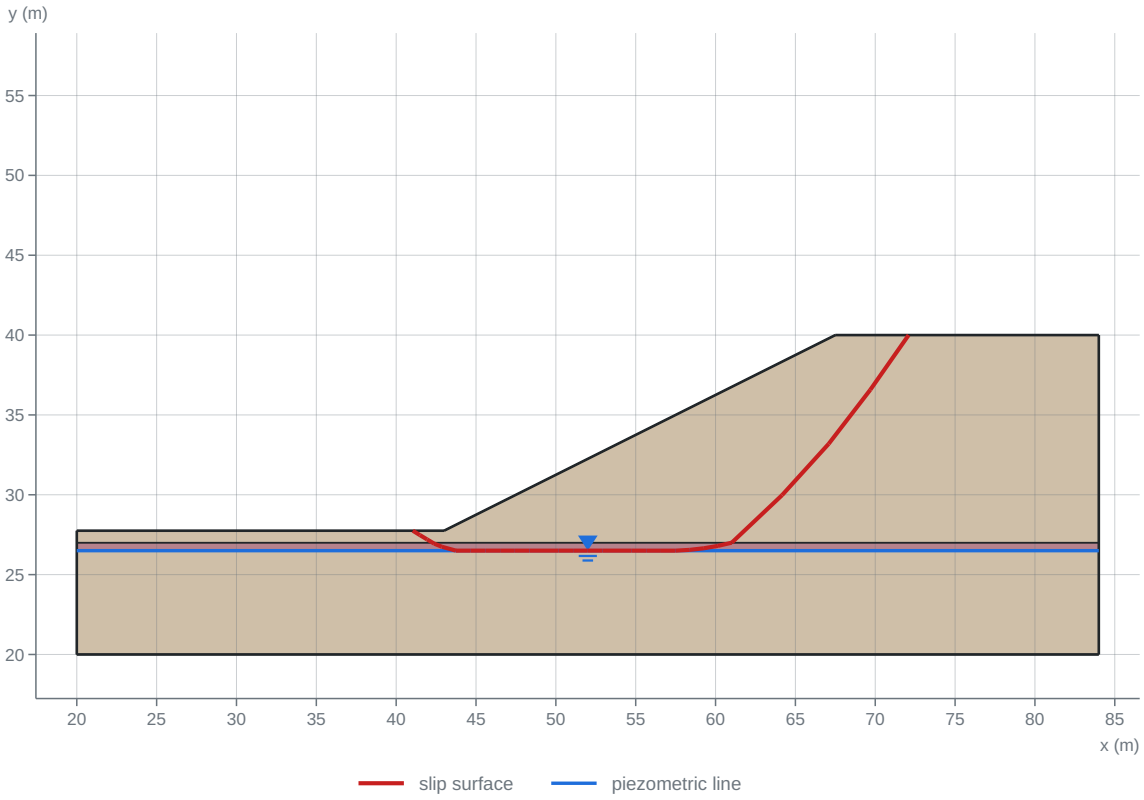


Figure 87. V050b: critical surface (Spencer, non-circular, F = 1.225).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.84	c′ = 28.50	20.00°
Weak layer	Mohr-Coulomb	18.84	c′ = 0.00	10.00°
Soil 1 below	Mohr-Coulomb	18.84	c′ = 28.50	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer, non-circular	1.225				1.246	−1.65 %
Strength reduction (FE)	1.283	1.26	another program's verification manual: finite-element strength reduction (verification manual, problem 5)	+1.84 %		
–	–	1.24–1.27	Giam & Donald (1989), ACADS			

ACADS accepted 1.24–1.27; cSlope's search finds a surface along the weak layer at 1.228 (Spencer), unchanged at 50, 100 and 200 slices.

3.87 V051: ACADS 4, weak layer, water table and surcharges

Giam & Donald (1989), ACADS slope stability programs review.

Circle search with non-circular optimization · Units: SI (m, kPa, kN/m³)

Weak-layer slope with a piezometric line and two surcharge strips (20 kPa on the bench, 20–40 kPa ramp on the crest); non-circular search. The published results are widely spread, so the problem tests the search.

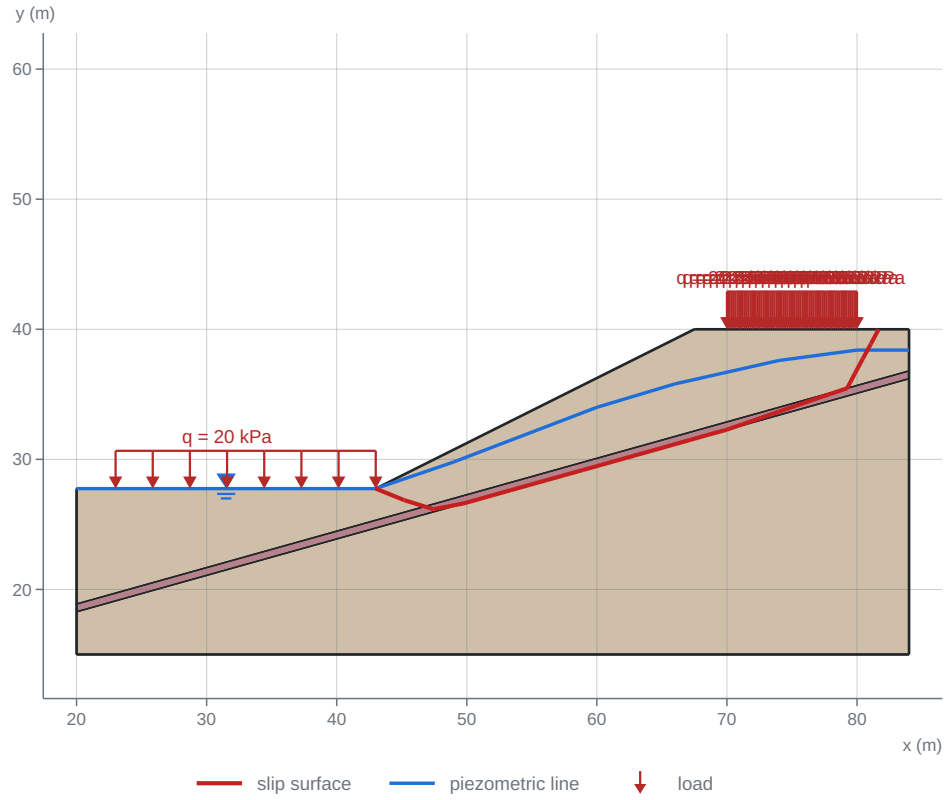


Figure 88. V051: critical surface (Spencer, non-circular, F = 0.714).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.84	c′ = 28.50	20.00°
Weak Layer	Mohr-Coulomb	18.84	c′ = 0.00	10.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer, non-circular	0.714				0.707	+0.99 %
–	–	0.78	Giam & Donald (1989), ACADS			

The ACADS referee value 0.78 comes from the 1989 survey, where the participating programs averaged 0.81; optimized non-circular searches today give about 0.68–0.71.

3.88 V052a: Talbingo Dam, end of construction

Giam & Donald (1989), ACADS slope stability programs review.

Critical circle search · Units: SI (m, kPa, kN/m³)

ACADS 2(a): Talbingo Dam, four zones, critical-circle search. The minimum is a shallow slide parallel to the steeper upstream face.

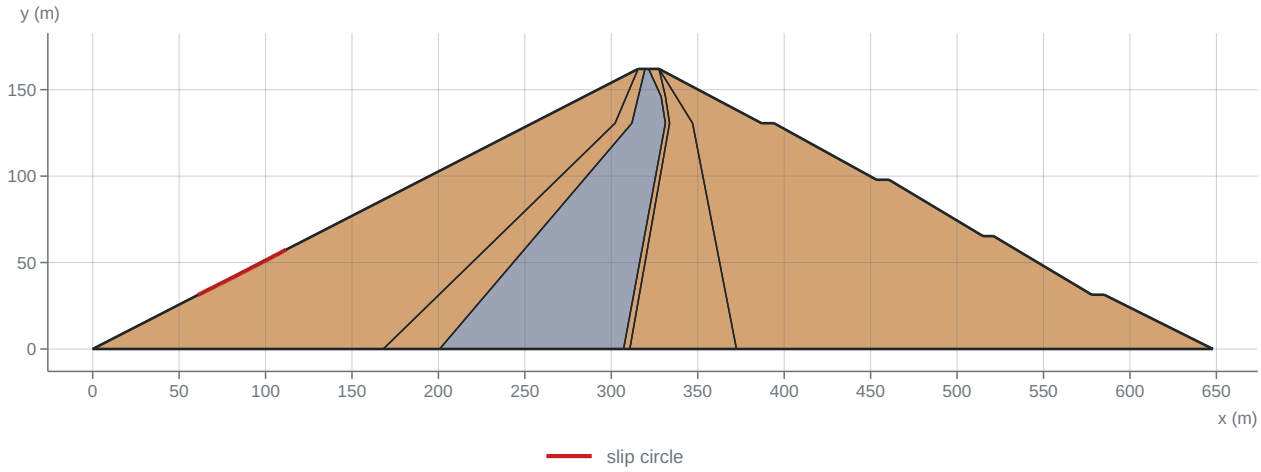


Figure 89. V052a: critical circle (Bishop simplified, F = 1.948).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Rockfill	Mohr-Coulomb	20.40	c′ = 0.00	45.00°
Transitions	Mohr-Coulomb	20.40	c′ = 0.00	45.00°
Filter	Mohr-Coulomb	20.40	c′ = 0.00	45.00°
Core	Mohr-Coulomb	18.10	c′ = 85.00	23.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.948				1.948–1.951	−0.09 %
Spencer	1.948				1.948	−0.02 %
Janbu corrected	1.950				1.949	+0.06 %
Strength reduction (FE)	1.789	1.88	another program's verification manual: finite-element strength reduction (verification manual, problem 4)	−4.85 %		
–	–	1.95	Giam & Donald (1989), ACADS			

3.89 V052b: Talbingo Dam, specified circle

Giam & Donald (1989), ACADS slope stability programs review.

Fixed circle · Units: SI (m, kPa, kN/m³)

ACADS 2(b): Talbingo Dam on the circle (100.3, 291.0, R = 278.8).

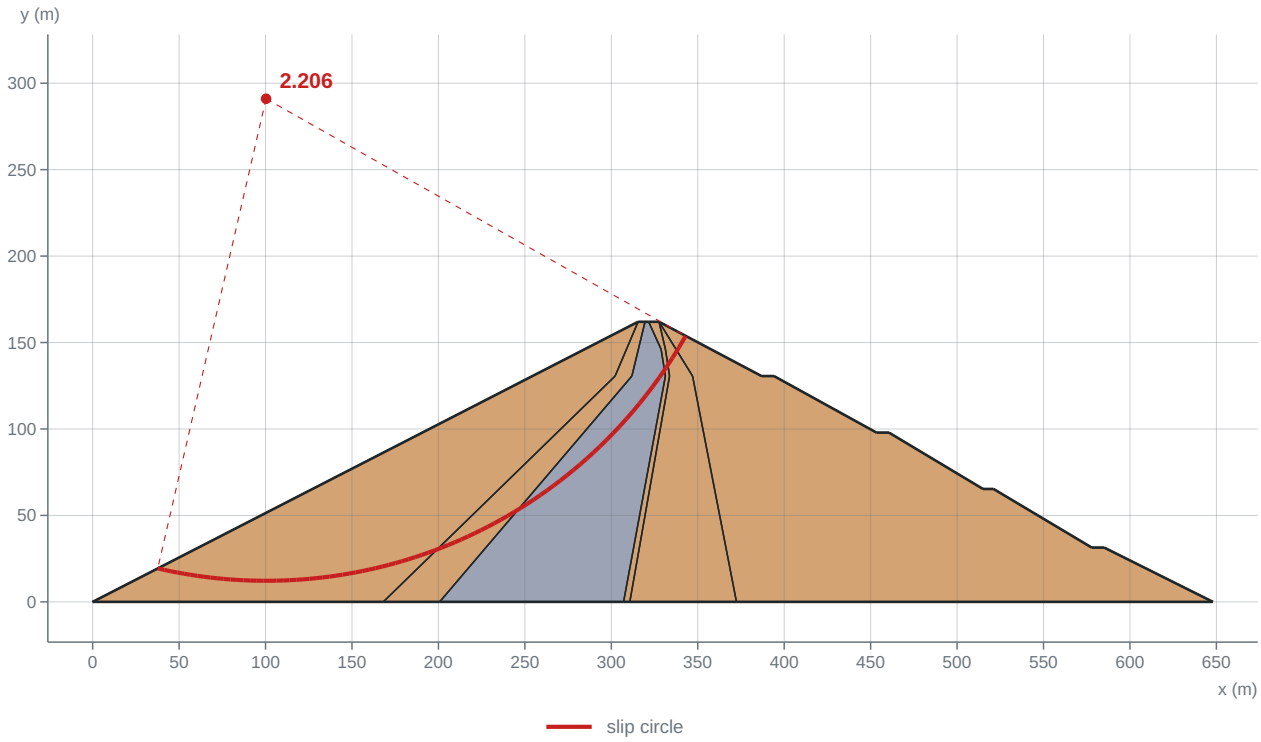


Figure 90. V052b: analyzed circle (Bishop simplified, F = 2.206).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Rockfill	Mohr-Coulomb	20.40	c′ = 0.00	45.00°
Transitions	Mohr-Coulomb	20.40	c′ = 0.00	45.00°
Filter	Mohr-Coulomb	20.40	c′ = 0.00	45.00°
Core	Mohr-Coulomb	18.10	c′ = 85.00	23.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.206				2.208	−0.07 %
Spencer	2.290				2.292	−0.08 %
Janbu corrected	2.073				2.073	−0.00 %
Morgenstern-Price (half-sine)	2.299				2.299	+0.01 %
–	–	2.29	Giam & Donald (1989), ACADS			

3.90 V099: ACADS 5, excavated slope with seepage

Giam & Donald (1989), ACADS problem 5.

Finite elements only · Units: SI (m, kPa, kN/m³)

1:2 cut, 10 m deep, below an initially horizontal ground with the water table 2 m down; long-term steady seepage towards the excavation, whose floor is under 1 m of water. Checked by finite elements only: the limit equilibrium values come from a flow net that the input cannot reproduce (a straight water table gives a different pore pressure field).

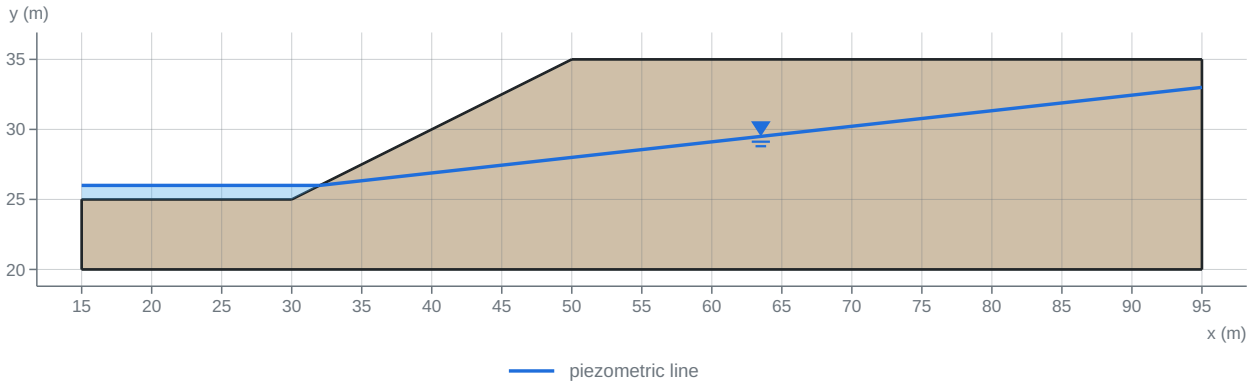


Figure 91. V099: geometry and water.

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c′ = 11.00	28.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Strength reduction (FE)	1.543	1.48	another program's verification manual: finite-element strength reduction with finite element seepage (verification manual, problem 7)	+4.28 %		

3.91 V053: Greco (1996) ex. 5 / Chen & Shao weak-layer slope, reference geometry

Greco (1996) example 5; Chen & Shao (1988).

Circle search with non-circular optimization · Units: SI (m, kPa, kN/m³)

Greco's weak-seam slope (V054) with a commercial program's geometry and water table; circular search with non-circular optimization, vertical pore-pressure head.

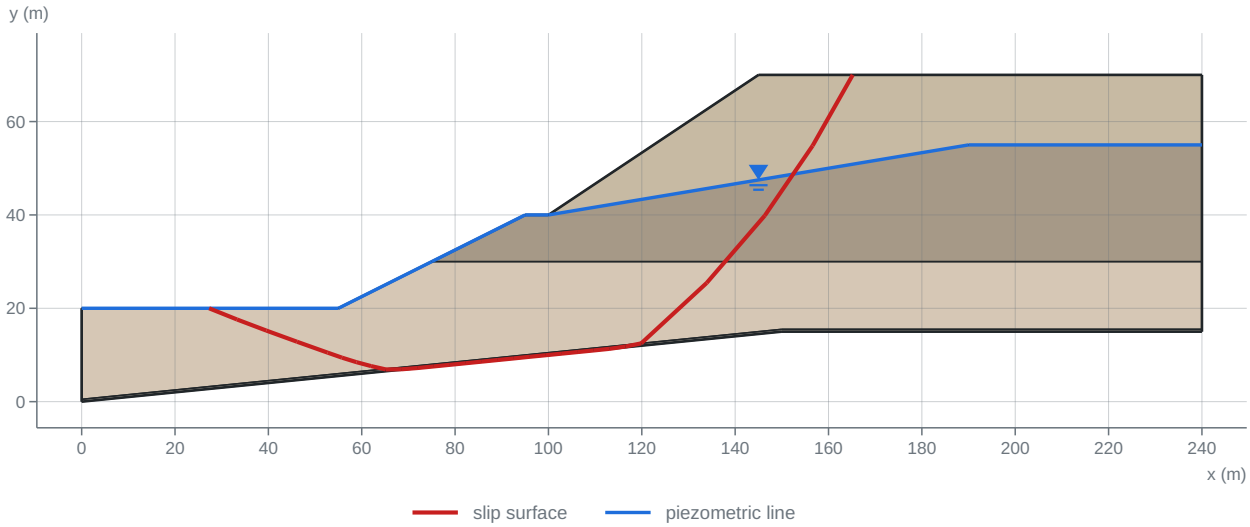


Figure 92. V053: critical surface (Spencer, non-circular, F = 1.006).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c′ = 9.80	35.00°
Layer 2	Mohr-Coulomb	19.00	c′ = 58.80	25.00°
Layer 3	Mohr-Coulomb	21.50	c′ = 19.80	30.00°
Layer 4	Mohr-Coulomb	21.50	c′ = 9.80	16.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer, non-circular	1.006	1.009–1.025	Chen & Shao (1988): Spencer, non-circular (test problem 3)	−0.28 %	1.010	−0.38 %
		0.97	Greco (1996): Spencer, non-circular (global minimum along the weak layer)	+3.73 %		
Spencer	1.092	1.08	Greco (1996): Spencer, local minimum (nearly circular surface from the toe)	+1.08 %		
Strength reduction (FE)	0.969	1.02	another program's verification manual: finite-element strength reduction (verification manual, problem 16)	−4.97 %		

Chen & Shao give 1.009–1.025 with Spencer's assumption for a surface sliding along the weak seam (their test problem 3, Tables 5–6). Greco's Monte Carlo search finds a global minimum of 0.97 along the weak layer, entering the level ground far from the toe, and a local minimum of 1.08 on a nearly circular surface from the toe; the programs' block searches and cSlope give 1.01 on a commercial program's version of the geometry, 4 % above Greco's global minimum.

3.92 V054: Greco (1996) example 5 / Chen & Shao (1988)

Greco (1996), example 5; Chen & Shao (1988).

Critical circle search · Units: SI (m, kPa, kN/m³)

Four layers including a weak seam, with a water table; critical-circle search, vertical pore-pressure head.

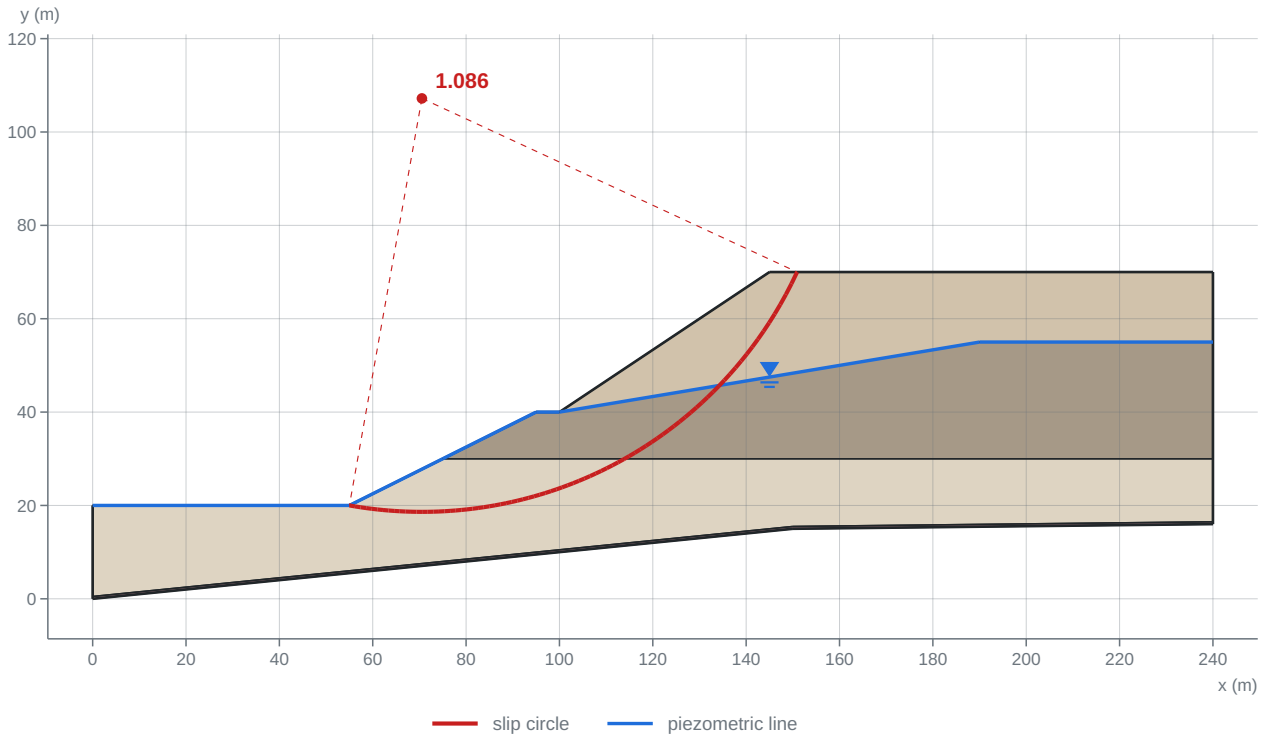


Figure 93. V054: critical circle (Bishop simplified, F = 1.086).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c′ = 9.80	35.00°
Layer 2	Mohr-Coulomb	19.00	c′ = 58.80	25.00°
Layer 3	Mohr-Coulomb	21.50	c′ = 19.80	30.00°
Seam	Mohr-Coulomb	21.50	c′ = 9.80	16.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.086				1.087	−0.09 %
Spencer	1.092	1.08	Greco (1996): Spencer, local minimum (nearly circular surface from the toe)	+1.08 %	1.093	−0.12 %

Greco's 1.08 is the local minimum of his search, a nearly circular surface from the toe; his global minimum, 0.97 along the weak layer, is compared in V053.

3.93 V055: Griffiths & Lane example 2, slope with a foundation layer

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

The 2:1 slope (c/γH = 0.05, φ = 20°) on a foundation of the same soil; unconstrained circular search finds the toe circle. Converted to SI.

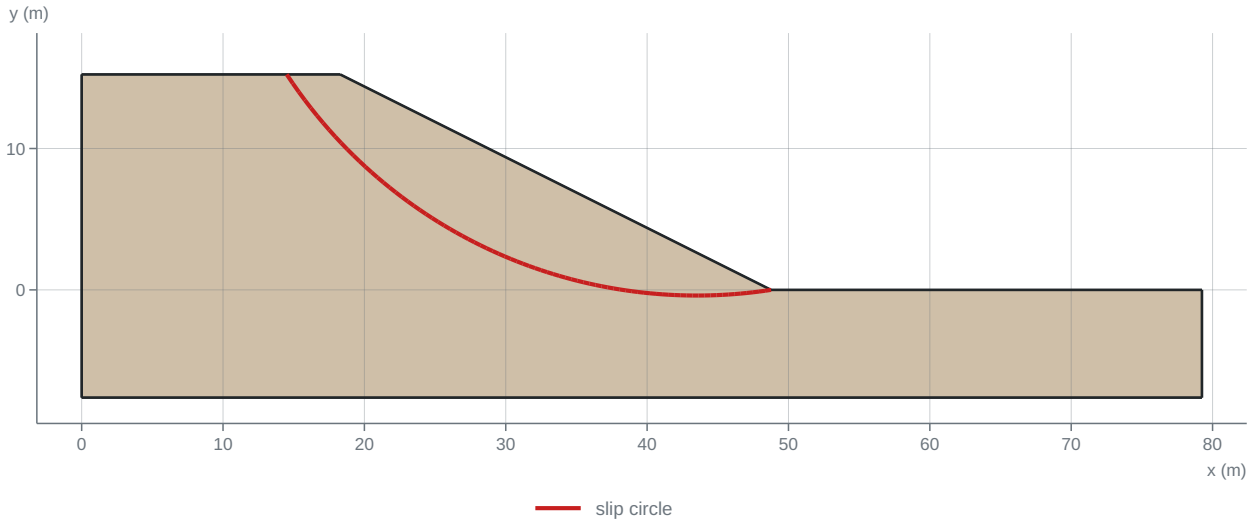


Figure 94. V055: critical circle (Spencer, F = 1.366).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Mohr-Coulomb	19.64	c′ = 14.96	20.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.367	1.40	Griffiths & Lane: finite elements	−2.39 %	1.366	+0.04 %
Strength reduction (FE)	1.342	1.40	Griffiths & Lane (1999): finite elements	−4.11 %		

3.94 V056: Griffiths & Lane example 3, thin weak layer, cu2/cu1 = 0.2

Griffiths & Lane (1999), Géotechnique 49(3).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Undrained clay slope with a thin weak layer at a fifth of the strength, on the three-line wedge through the layer (the paper's Janbu mechanism). Converted to SI.

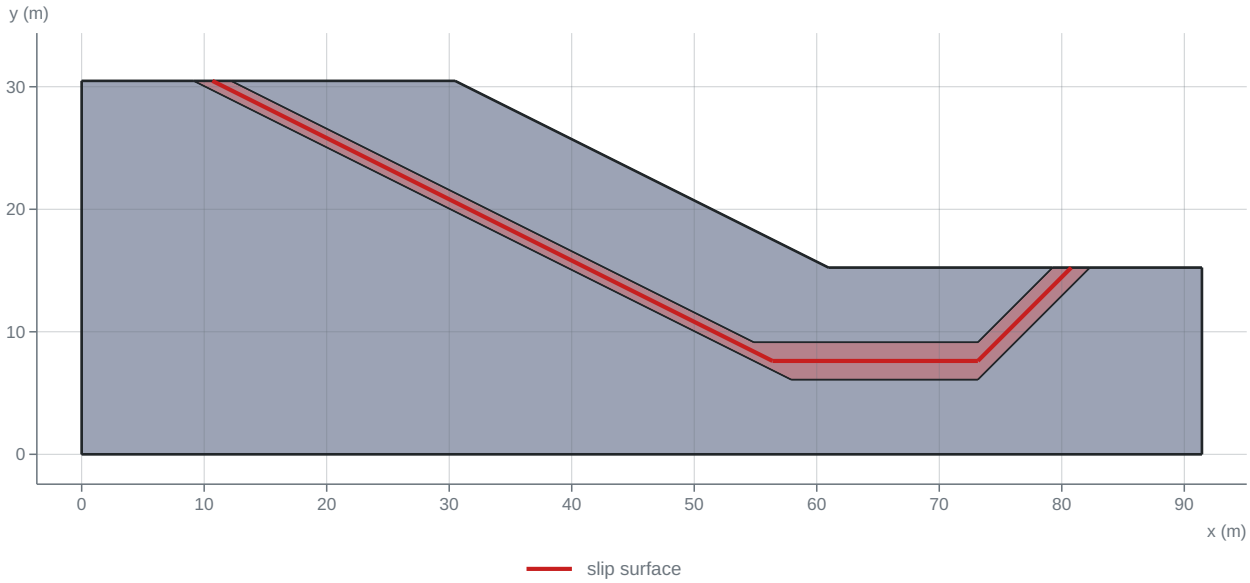


Figure 95. V056: analyzed surface (Janbu corrected, F = 0.499).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Surrounding clay	Undrained (φ = 0)	19.64	Su = 74.81	0°
Thin weak layer	Undrained (φ = 0)	19.64	Su = 14.96	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	0.499	0.485	Griffiths & Lane: Janbu wedge	+2.97 %		
Strength reduction (FE)	0.509	0.60	Griffiths & Lane (1999): finite elements (read from fig. 7)	*−15.10 % **		

** Disputed, not counted in section 2: Griffiths & Lane (1999)'s 0.60. Griffiths & Lane's 0.60 is read from fig. 7 on their mesh of 60 × 20 eight-node elements of 0.1H (fig. 8), one element across the inclined part of the weak layer, with failure as non-convergence in 1000 viscoplastic iterations. It disagrees with the paper's own Janbu wedge for the same mechanism (0.485) and with the limit analysis bounds, 0.488–0.498 (exact for φu = 0; cSlope's lower and upper bound programs on refined meshes): cSlope's strength reduction falls with the mesh, 0.64 on a uniform 0.2H mesh, 0.59 on 0.16H, 0.545 on 0.1H, 0.51 refined and 0.504 on a 0.5 m mesh, so the paper's value measures its mesh rather than cSlope. Not counted in the agreement or the error statistics.*

3.95 V057a: Griffiths & Lane example 4, cu2/cu1 = 1 (deep base circle)

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

Undrained clay slope over a foundation of the same strength; the deep base circle. Converted to SI.

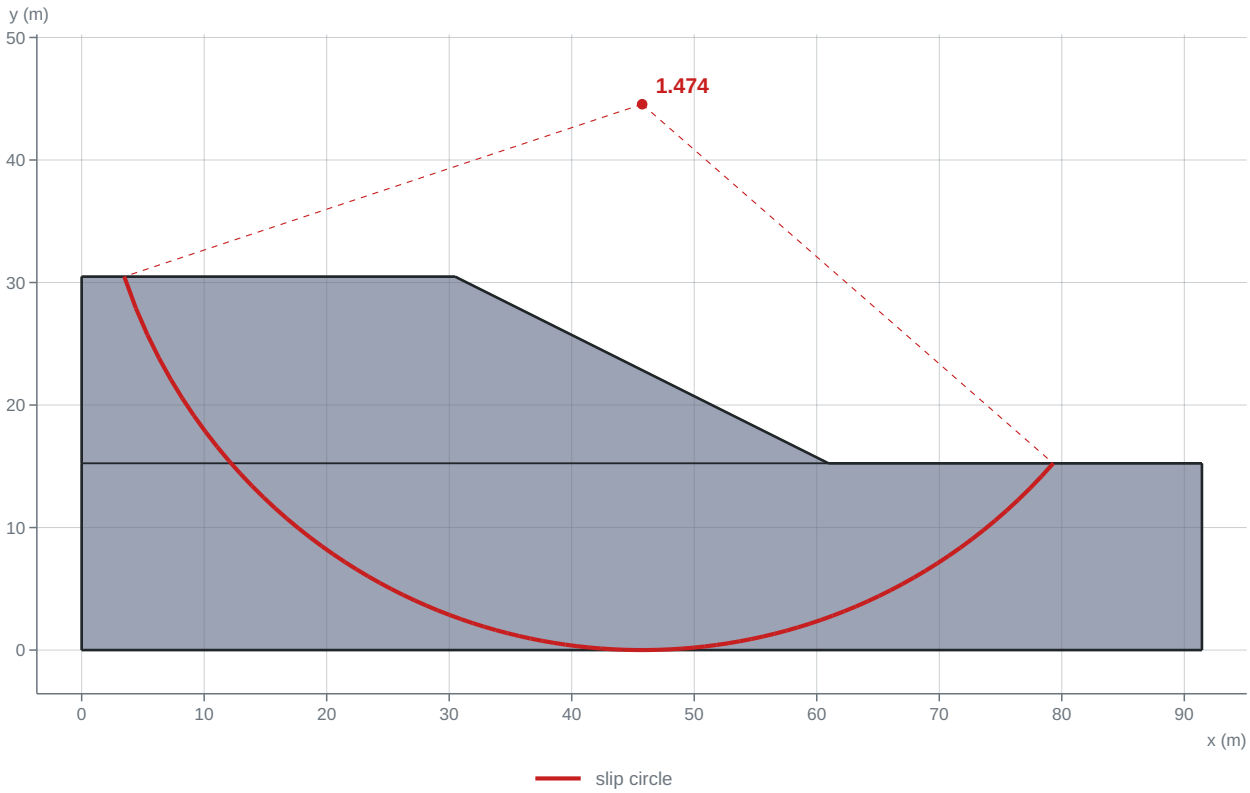


Figure 96. V057a: critical circle (Spencer, F = 1.474).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Slope clay	Undrained (φ = 0)	19.64	Su = 74.81	0°
Foundation clay	Undrained (φ = 0)	19.64	Su = 74.81	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.474	1.46	Griffiths & Lane: limit-equilibrium chart	+0.93 %	1.468	+0.38 %
Strength reduction (FE)	1.451	1.45	Griffiths & Lane (1999): finite elements	+0.10 %		
–	–	1.47	Griffiths & Lane (1999): stability chart (Taylor)			

3.96 V057b: Griffiths & Lane example 4, cu2/cu1 = 2 (toe circle)

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

The same slope over a foundation twice as strong; the shallow toe circle. Converted to SI.

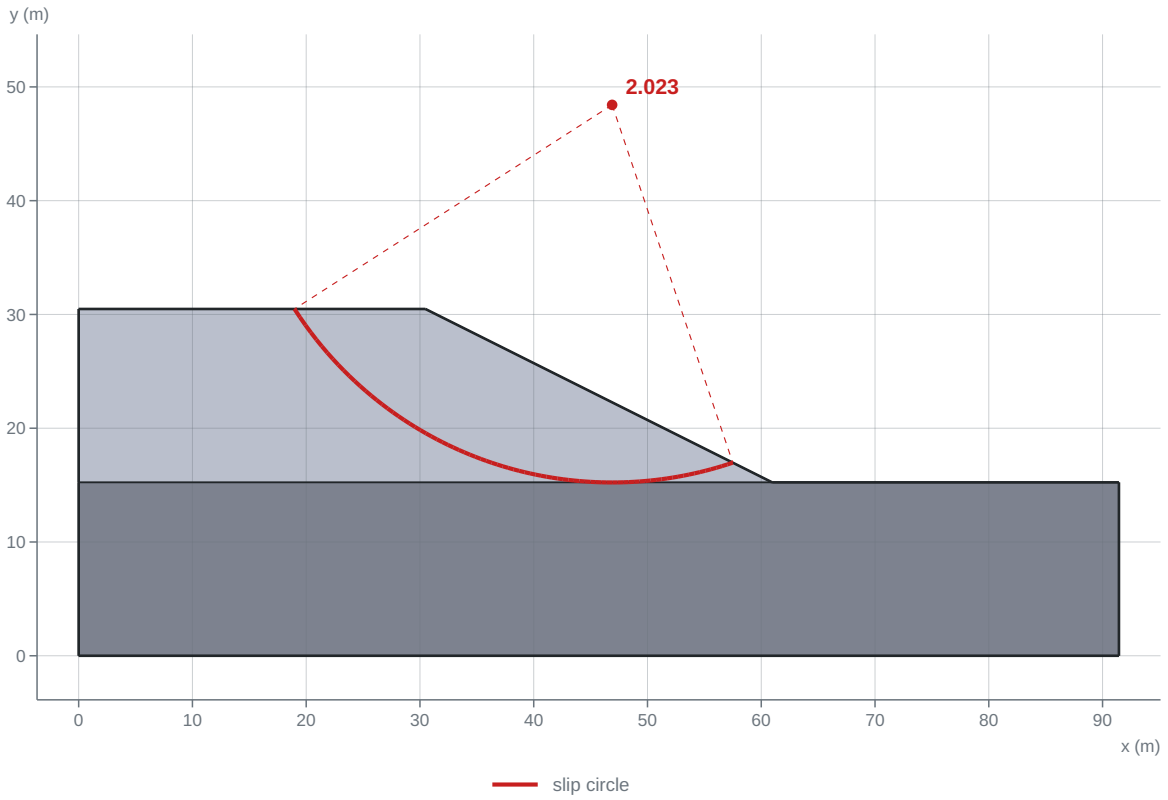


Figure 97. V057b: critical circle (Spencer, F = 2.023).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Slope clay	Undrained (φ = 0)	19.64	Su = 74.81	0°
Foundation clay	Undrained (φ = 0)	19.64	Su = 149.6	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	2.023	2.04	Griffiths & Lane: limit-equilibrium chart	−0.84 %	2.022	+0.04 %
Strength reduction (FE)	2.042	2.03	Griffiths & Lane (1999): finite elements	+0.60 %		
–	–	2.10	Griffiths & Lane (1999): stability chart (Taylor)			

3.97 V058: Griffiths & Lane example 6, earth dam with full reservoir

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

Two-sided earth dam with the reservoir full and a piezometric line; circular search. Converted to SI.

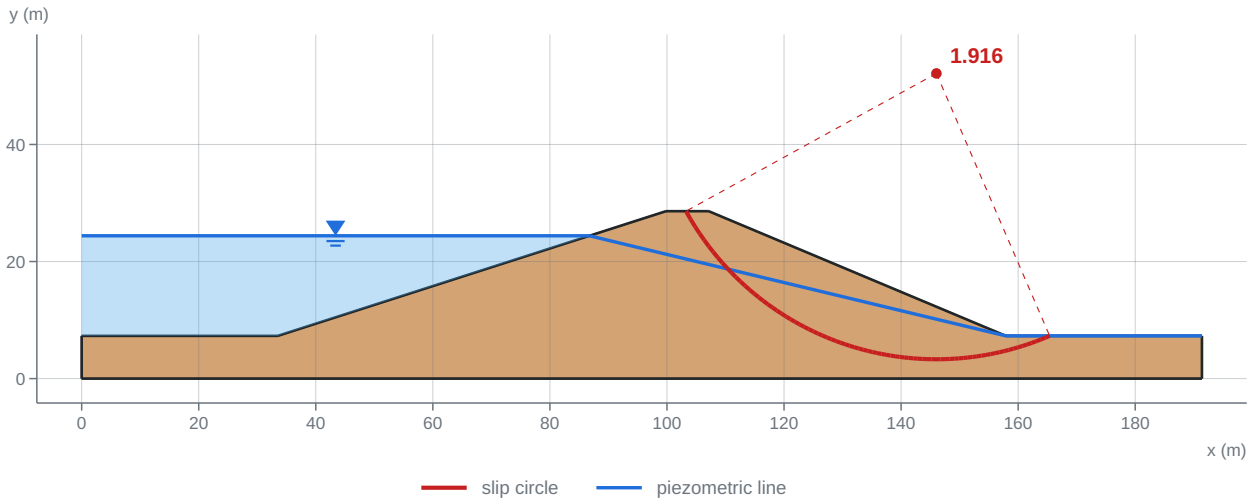


Figure 98. V058: critical circle (Spencer, F = 1.916).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.20	c′ = 13.80	37.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.917	1.90	Griffiths & Lane: finite elements	+0.87 %	1.915	+0.08 %
Strength reduction (FE)	1.854	1.90	Griffiths & Lane (1999): finite elements	−2.44 %		

3.98 V059a: Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.4

Guo & Griffiths (2020), Can. Geotech. J. 57.

Circle search with non-circular optimization · Units: SI (m, kPa, kN/m³)

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 84 kPa), γ = 20, firm base; optimized non-circular search. The deep mechanism governs.

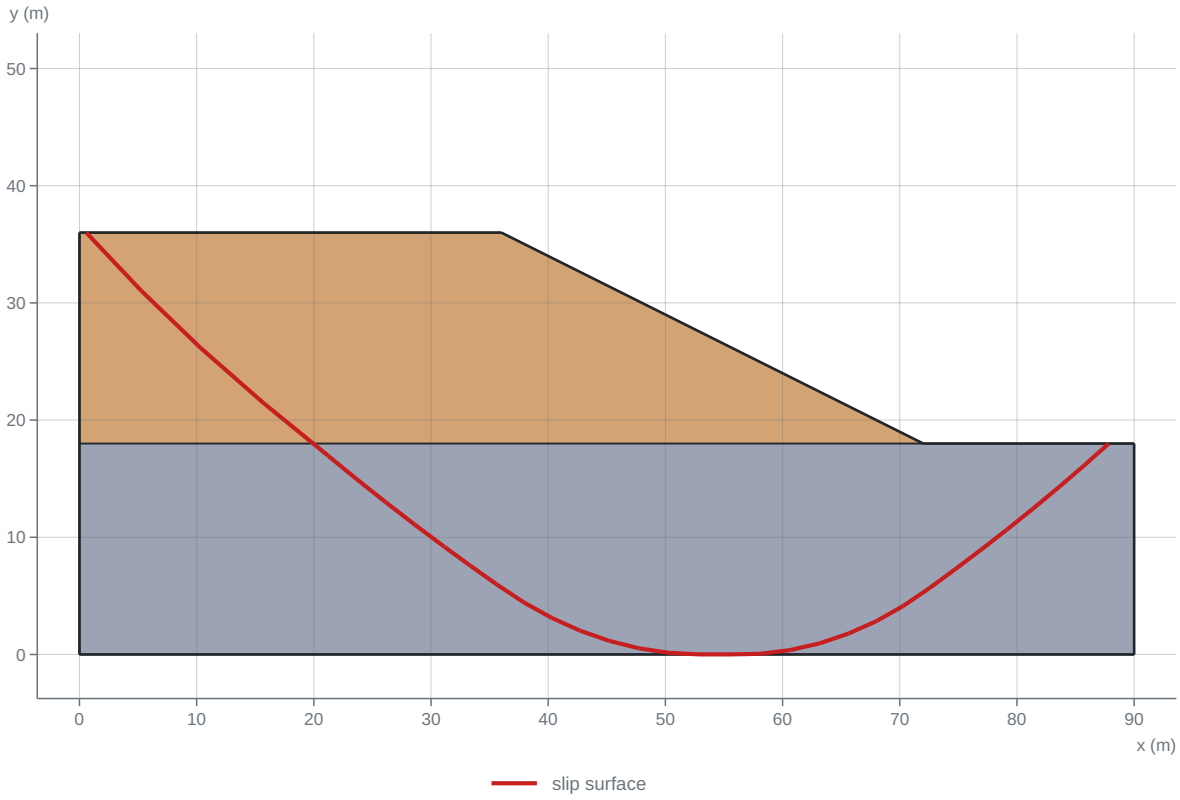


Figure 99. V059a: critical surface (Spencer, non-circular, F = 1.214).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Undrained (φ = 0)	20.00	Su = 60.00	0°
Foundation	Undrained (φ = 0)	20.00	Su = 84.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer, non-circular	1.214				1.215	−0.11 %
Strength reduction (FE)	1.281	1.28	Guo & Griffiths (2020): finite elements (read from fig. 2)	+0.08 %		

The paper gives its results in figures only; the comparison is with the other programs.

3.99 V059b: Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.5

Guo & Griffiths (2020), Can. Geotech. J. 57.

Circle search with non-circular optimization · Units: SI (m, kPa, kN/m³)

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 90 kPa), γ = 20, firm base; optimized non-circular search. The deep mechanism governs.

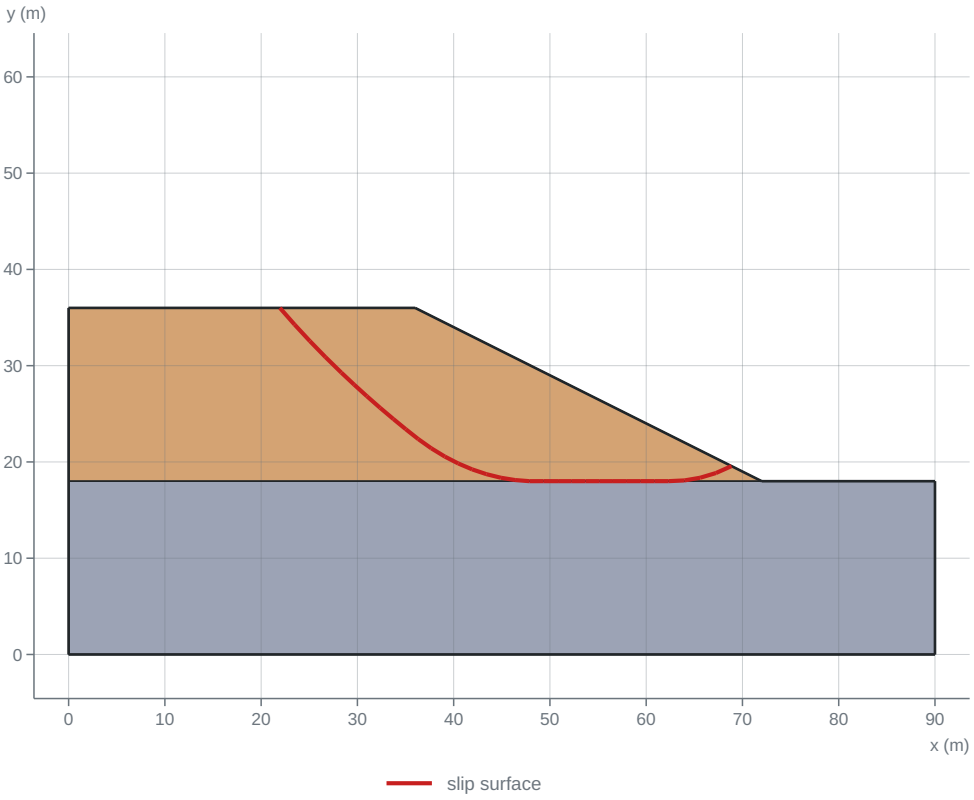


Figure 100. V059b: critical surface (Spencer, non-circular, F = 1.312).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Undrained (φ = 0)	20.00	Su = 60.00	0°
Foundation	Undrained (φ = 0)	20.00	Su = 90.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer, non-circular	1.312				1.290	+1.69 %
Strength reduction (FE)	1.353	1.35	Guo & Griffiths (2020): finite elements	+0.19 %		

The paper gives its results in figures only; the comparison is with the other programs.

3.100 V059c: Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.6

Guo & Griffiths (2020), Can. Geotech. J. 57.

Circle search with non-circular optimization · Units: SI (m, kPa, kN/m³)

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 96 kPa), γ = 20, firm base; optimized non-circular search. The shallow mechanism governs.

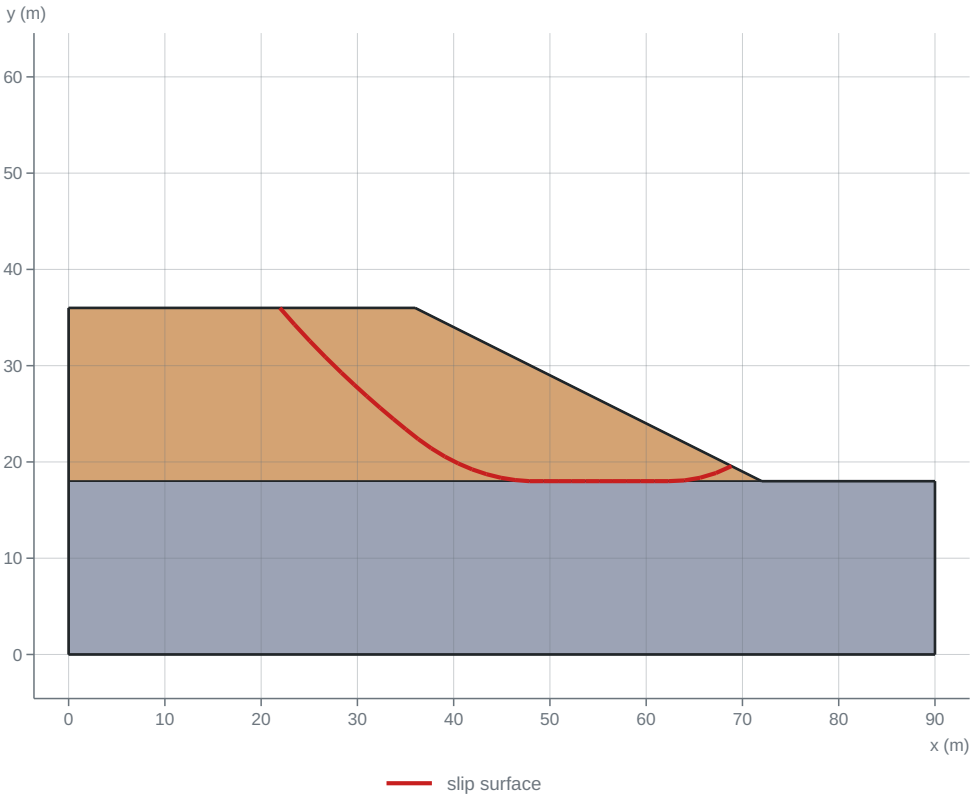


Figure 101. V059c: critical surface (Spencer, non-circular, F = 1.312).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Undrained (φ = 0)	20.00	Su = 60.00	0°
Foundation	Undrained (φ = 0)	20.00	Su = 96.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer, non-circular	1.312				1.315	−0.24 %
Strength reduction (FE)	1.362	1.36	Guo & Griffiths (2020): finite elements (read from fig. 2)	+0.12 %		

The paper gives its results in figures only; the comparison is with the other programs.

3.101 V060a: Cannon Dam end of construction (Hassan & Wolff), critical circle

Hassan & Wolff (1999), J. Geotech. Geoenviron. Eng. 125(4).

Critical circle search · Units: SI (m, kPa, kN/m³)

Two clay fills with a sand-filter strip under the crest and a spoil-covered downstream toe; Bishop search for the minimum factor of safety at the mean strengths. The reliability index on the minimum-β surface is V060b.

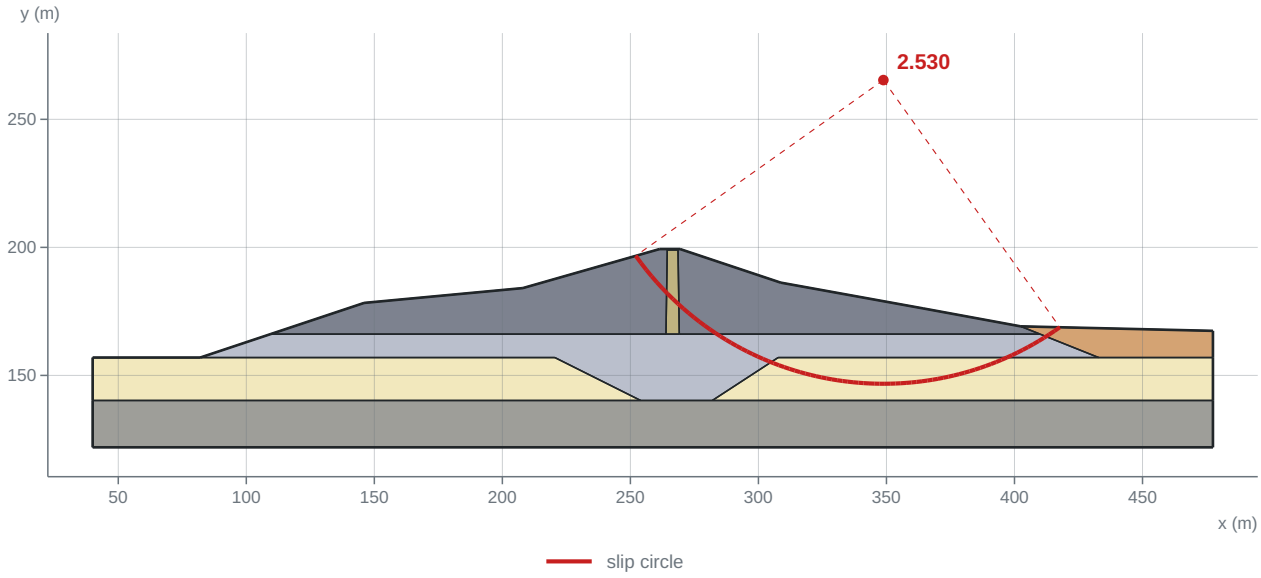


Figure 102. V060a: critical circle (Bishop simplified, F = 2.530).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II clay fill	Mohr-Coulomb	22.00	c′ = 143.6	15.00°
Sand filter	Mohr-Coulomb	22.00	c′ = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c′ = 5.00	35.00°
Phase I clay fill	Mohr-Coulomb	22.00	c′ = 117.8	8.50°
Foundation sand	Mohr-Coulomb	20.00	c′ = 5.00	18.00°
Limestone	Mohr-Coulomb	23.00	c′ = 1000.0	45.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.530				2.529	+0.05 %

Hassan & Wolff give 2.753 on a fixed circle (V062a), not a search result.

3.102 V060b: Cannon Dam end of construction (Hassan & Wolff), reliability on the minimum-β circle

Hassan & Wolff (1999), J. Geotech. Geoenviron. Eng. 125(4).

Fixed circle · Units: SI (m, kPa, kN/m³)

The V060a dam on the circle of minimum reliability index: Phase I fill φ = 8.5° ± 8.5°, c = 117.8 ± 58.9 kPa; Phase II fill c = 143.6 ± 79 kPa, φ = 15° ± 9°, treated as independent normal variables (another program's Taylor series has no correlation term); Bishop.

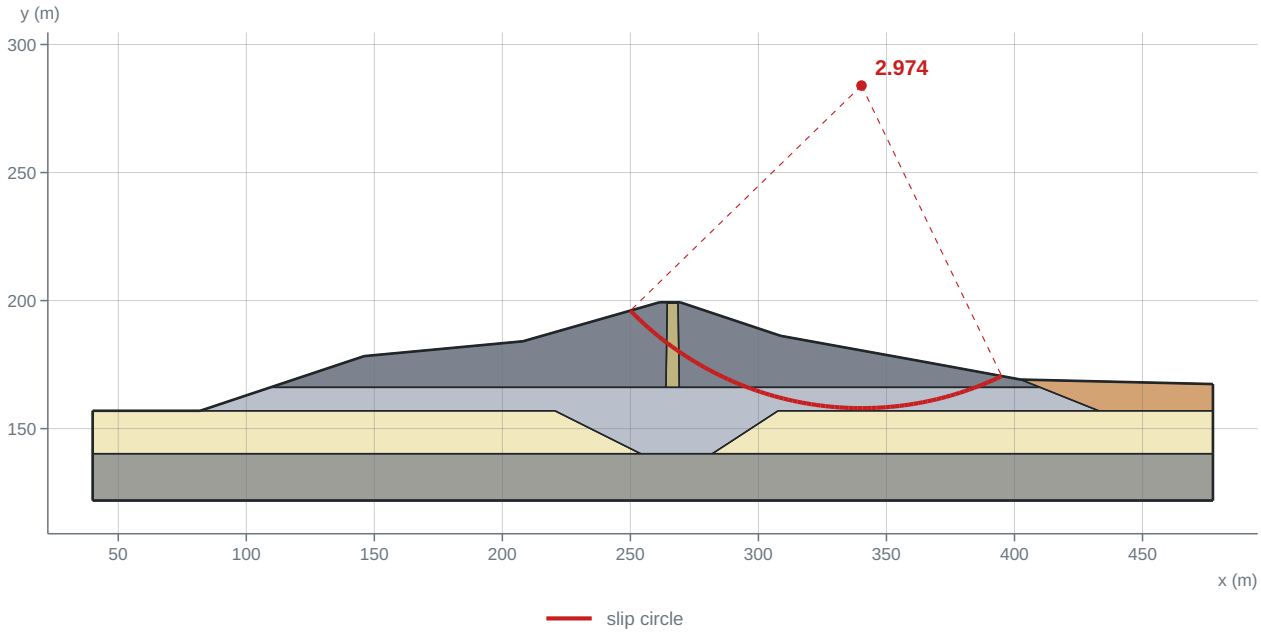


Figure 103. V060b: analyzed circle (Bishop simplified, F = 2.974).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II clay fill	Mohr-Coulomb	22.00	c′ = 143.6	15.00°
Sand filter	Mohr-Coulomb	22.00	c′ = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c′ = 5.00	35.00°
Phase I clay fill	Mohr-Coulomb	22.00	c′ = 117.8	8.50°
Foundation sand	Mohr-Coulomb	20.00	c′ = 5.00	18.00°
Limestone	Mohr-Coulomb	23.00	c′ = 1000.0	45.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.974				2.973	+0.05 %

The source gives reliability indices on fixed circles; no factor of safety is published for the minimum-β circle.

Probabilistic analysis on the same surface: 4 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Taylor series)	–	3.353	3.354

3.103 V061: Clarence Cannon Dam (Wolff & Harr), specified surface

Wolff & Harr (1987).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Phase I and II fills with a chimney drain, a horizontal drain blanket (el. 548.2–552.8 ft) and a cutoff trench to rock, flat water table at el. 557 ft; the printed non-circular surface. Converted to SI.

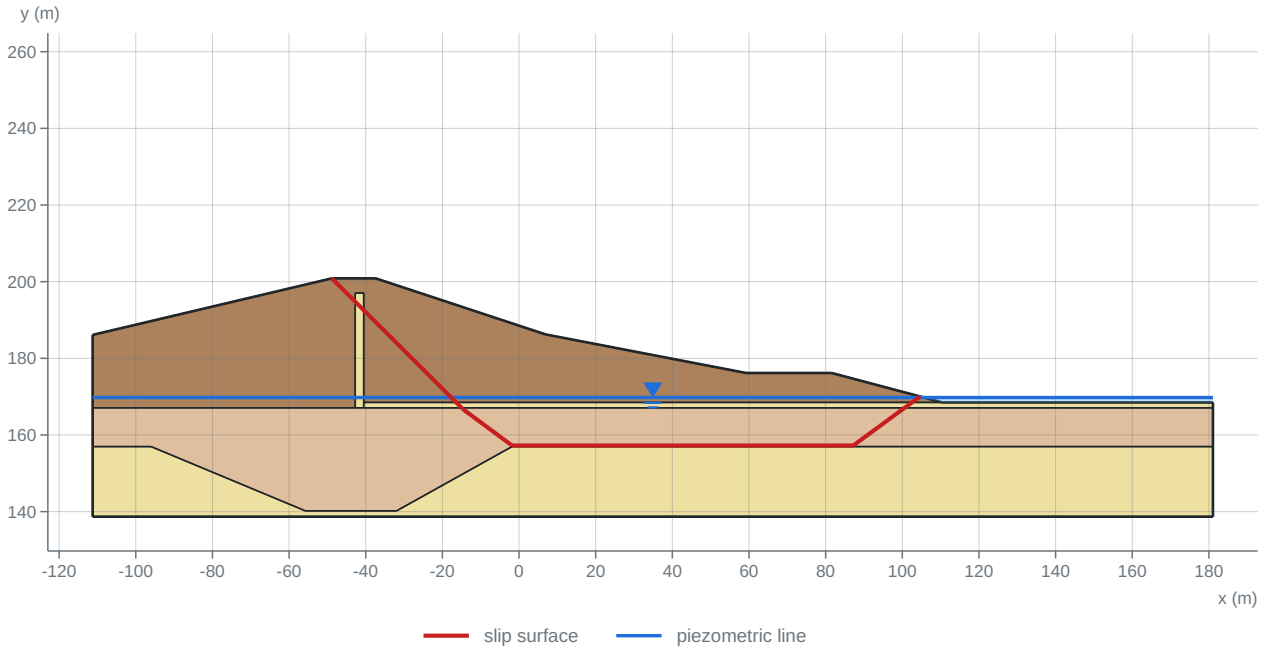


Figure 104. V061: analyzed surface (Morgenstern-Price (half-sine), F = 2.348).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	23.56	c′ = 138.9	14.80°
Sand drain	Mohr-Coulomb	18.85	c′ = 0.00	30.00°
Phase I fill	Mohr-Coulomb	23.56	c′ = 106.8	6.34°
Foundation sand	Mohr-Coulomb	18.85	c′ = 0.00	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	2.348	2.36	Wolff & Harr	−0.49 %	2.343	+0.23 %
Spencer	2.388				2.383	+0.21 %

3.104 V062a: Clarence Cannon Dam, surface 7A

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analyzed on the published circle 7A (center 320.175, 313.336; tangent elevation 145.347), Bishop.

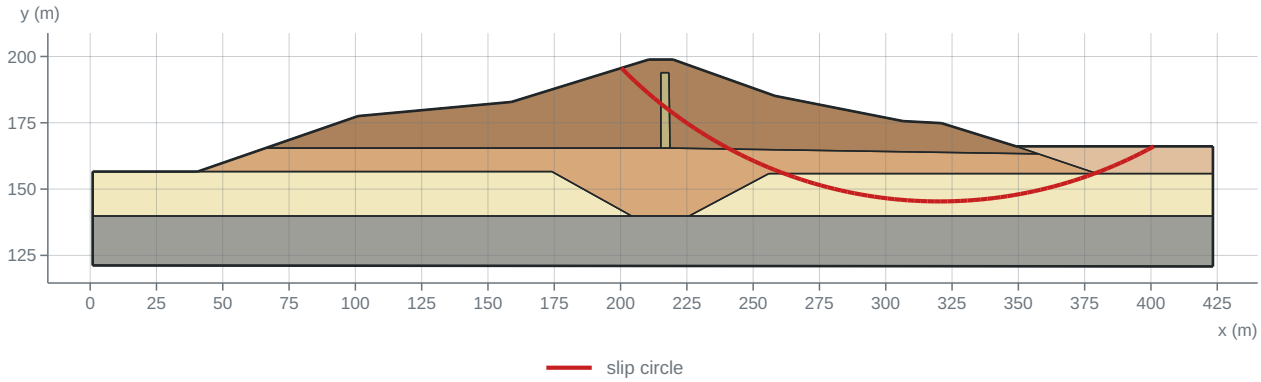


Figure 105. V062a: analyzed circle (Bishop simplified, F = 2.549).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c′ = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c′ = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c′ = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c′ = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c′ = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c′ = 100.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.549				2.551–2.560	−0.27 %
–	–	2.753	Hassan & Wolff (1999)			

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	10.953	11.033

3.105 V062b: Clarence Cannon Dam, surface 7B

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analyzed on the published circle 7B (center 295.235, 277.589; tangent elevation 156.298), Bishop.

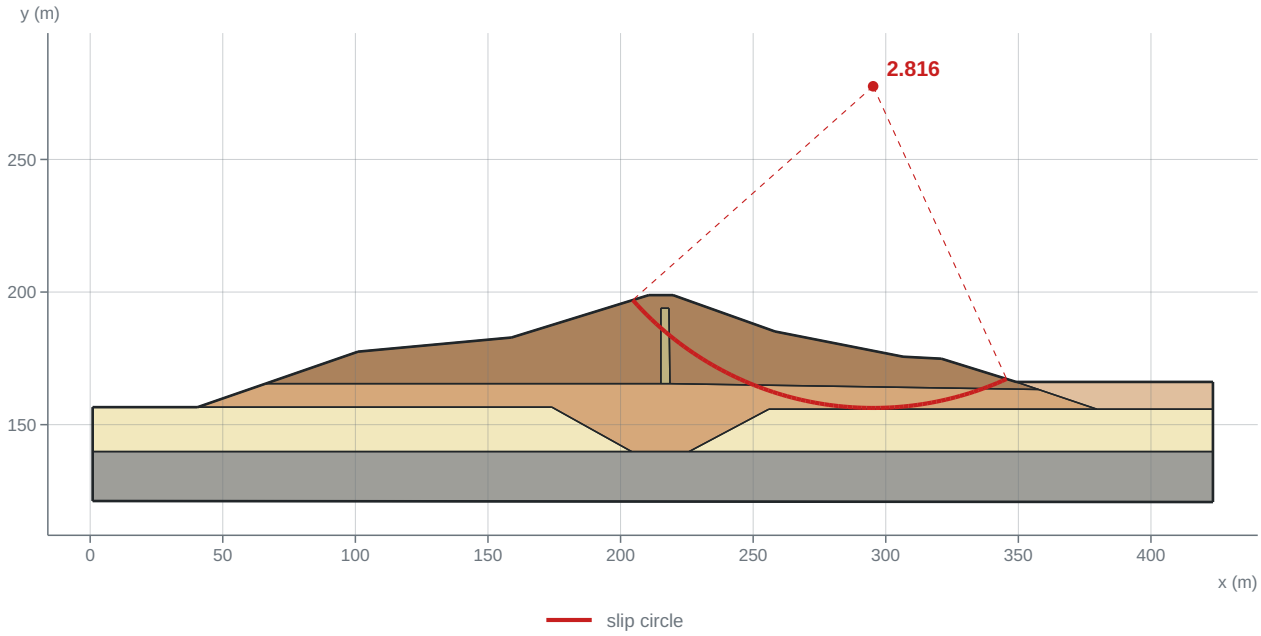


Figure 106. V062b: analyzed circle (Bishop simplified, F = 2.816).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c′ = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c′ = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c′ = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c′ = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c′ = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c′ = 100.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.816				2.806–2.820	+0.11 %
–	–	2.352	Hassan & Wolff (1999)			

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	4.351	4.402

3.106 V062c: Clarence Cannon Dam, surface 7C

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analyzed on the published circle 7C (center 303.683, 310.206; tangent elevation 155.928), Bishop.

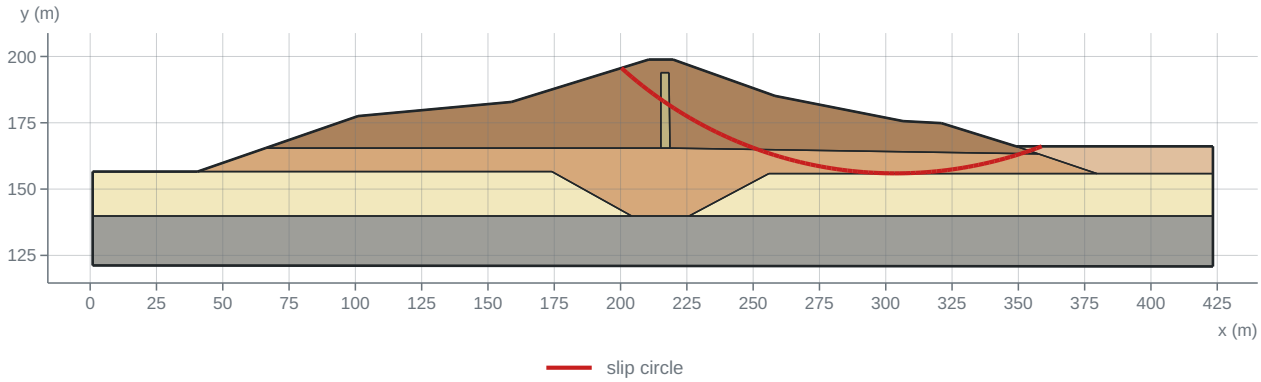


Figure 107. V062c: analyzed circle (Bishop simplified, F = 2.780).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c′ = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c′ = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c′ = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c′ = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c′ = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c′ = 100.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.780				2.771–2.777	+0.20 %
–	–	2.523	Hassan & Wolff (1999)			

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	4.263	4.325

3.107 V062d: Clarence Cannon Dam, surface 7D

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analyzed on the published circle 7D (center 322.152, 310.206; tangent elevation 145.36), Bishop.

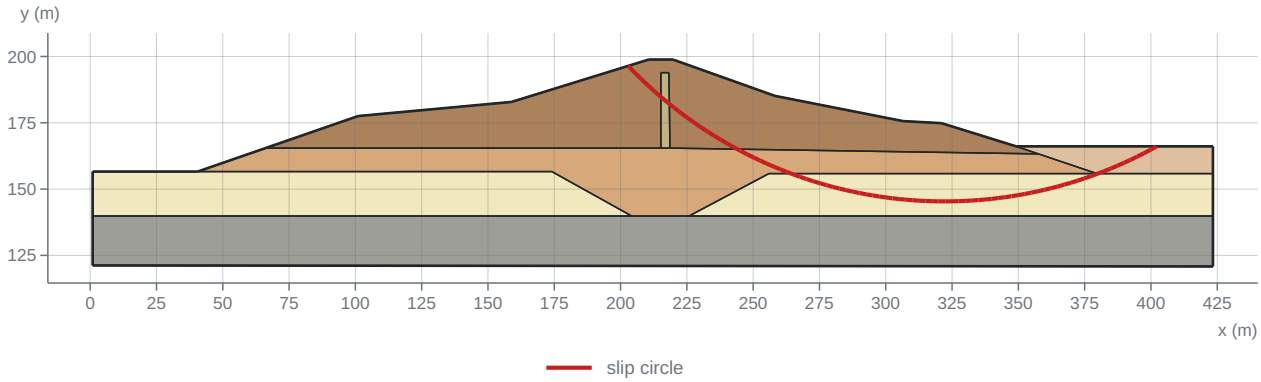


Figure 108. V062d: analyzed circle (Bishop simplified, F = 2.577).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c′ = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c′ = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c′ = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c′ = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c′ = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c′ = 100.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.577				2.583–2.589	−0.36 %
–	–	2.457	Hassan & Wolff (1999)			

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	11.092	11.074

3.108 V062e: Clarence Cannon Dam, surface 7E

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analyzed on the published circle 7E (center 327.693, 324.018; tangent elevation 147.163), Bishop.

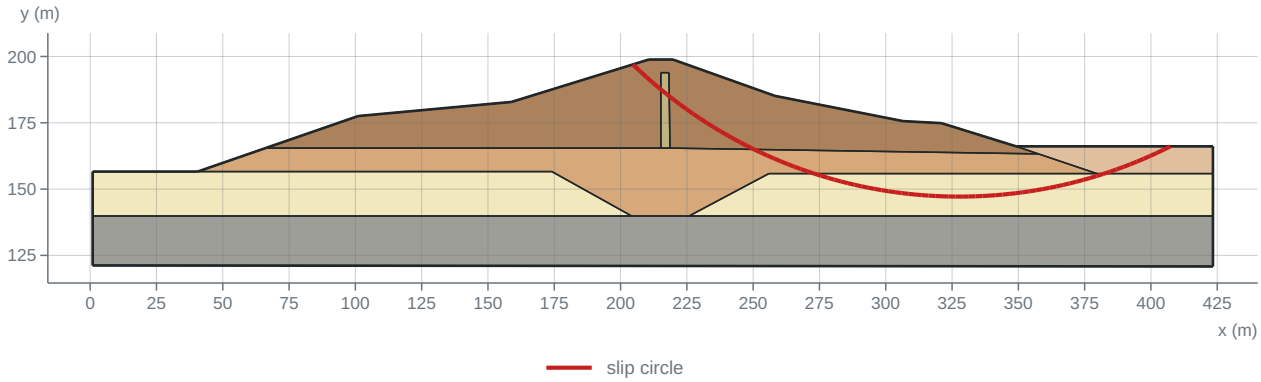


Figure 109. V062e: analyzed circle (Bishop simplified, F = 2.690).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c′ = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c′ = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c′ = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c′ = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c′ = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c′ = 100.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.690				2.692–2.703	−0.27 %
–	–	2.602	Hassan & Wolff (1999)			

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	10.281	10.328

3.109 V062f: Clarence Cannon Dam, surface 8B

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analyzed on the published circle 8B (center 304.239, 307.858; tangent elevation 155.311), Bishop.

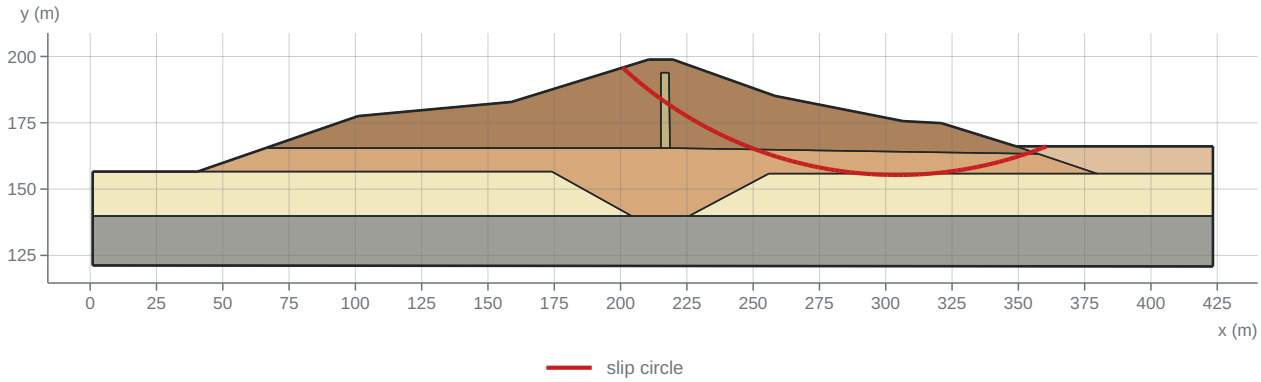


Figure 110. V062f: analyzed circle (Bishop simplified, F = 2.674).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c′ = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c′ = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c′ = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c′ = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c′ = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c′ = 100.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.674				2.672–2.673	+0.04 %
–	–	2.995	Hassan & Wolff (1999)			

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	4.858	4.933

3.110 V062g: Clarence Cannon Dam, surface 8F

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analyzed on the published circle 8F (center 313.114, 304.784; tangent elevation 160.368), Bishop.

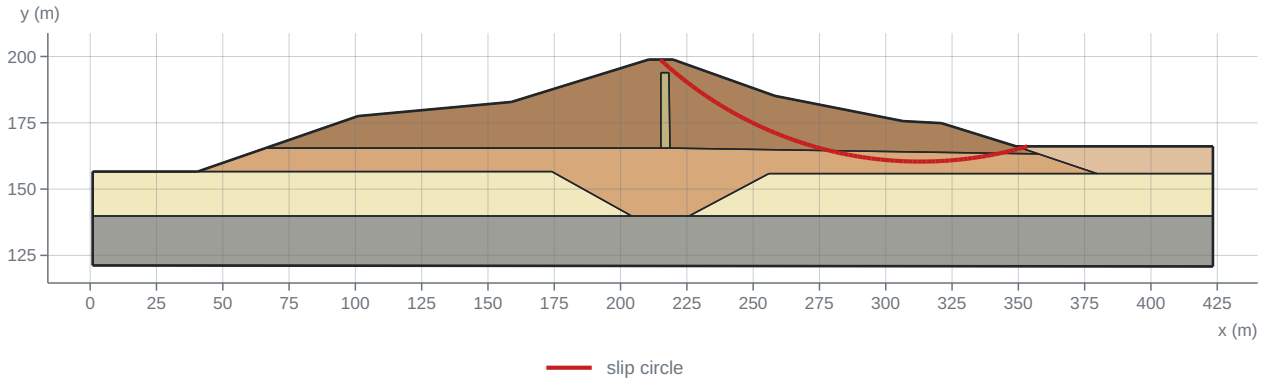


Figure 111. V062g: analyzed circle (Bishop simplified, F = 3.593).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c′ = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c′ = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c′ = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c′ = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c′ = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c′ = 100.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	3.593				3.586–3.598	+0.02 %
–	–	3.916	Hassan & Wolff (1999)			

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	5.485	5.528

3.111 V062h: Clarence Cannon Dam, surface 8G

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analyzed on the published circle 8G (center 314.856, 305.604; tangent elevation 165.756), Bishop.

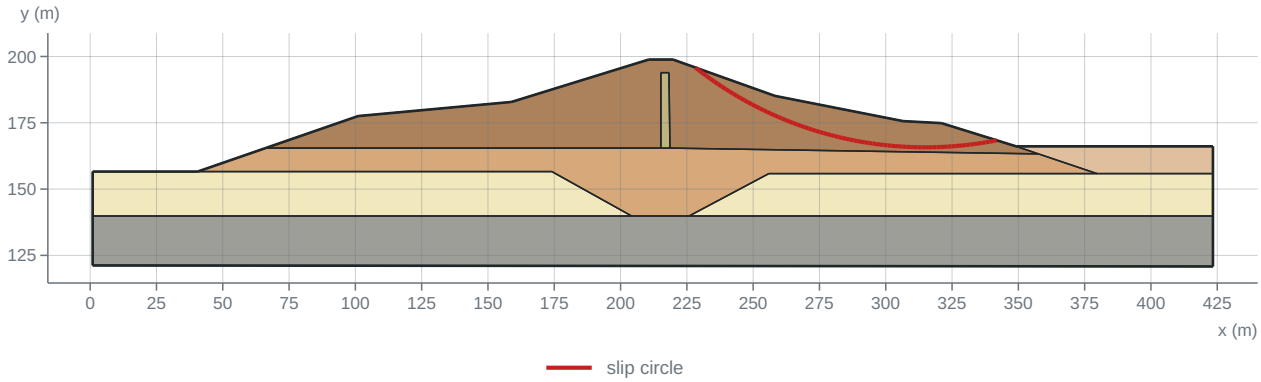


Figure 112. V062h: analyzed circle (Bishop simplified, F = 6.065).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c′ = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c′ = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c′ = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c′ = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c′ = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c′ = 100.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	6.065				6.069–6.074	−0.10 %
–	–	10.576	Hassan & Wolff (1999)			

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	5.563	5.491

3.112 V062i: Clarence Cannon Dam, surface 8H

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analyzed on the published circle 8H (center 316.198, 305.604; tangent elevation 170.537), Bishop.

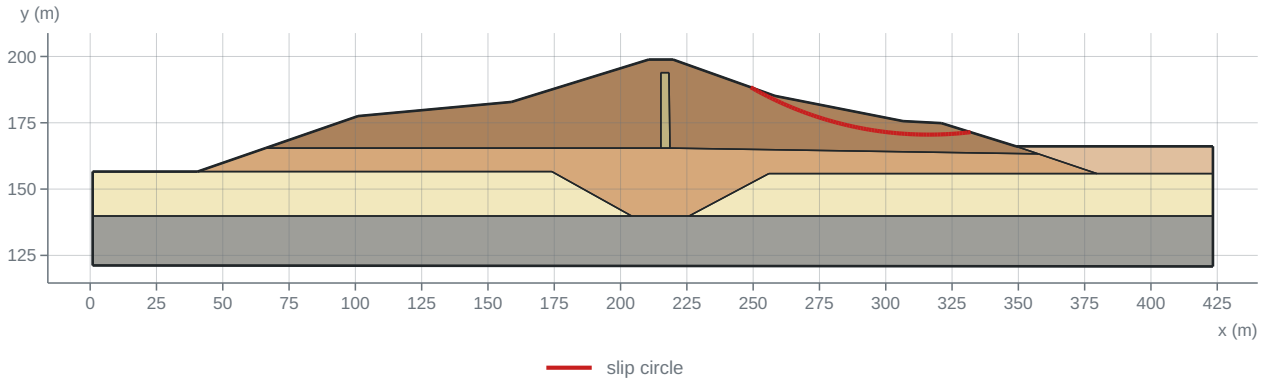


Figure 113. V062i: analyzed circle (Bishop simplified, F = 11.571).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c′ = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c′ = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c′ = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c′ = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c′ = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c′ = 100.0	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	11.571				11.583	−0.11 %
–	–	6.293	Hassan & Wolff (1999)			

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	6.394	6.432

3.113 V063: Huang & Jia earth dam, dry

Huang, M. & Jia, C.-Q. (2009). Strength reduction FEM in stability analysis of soil slopes subjected to transient unsaturated seepage. Computers and Geotechnics 36(1–2), 93–101.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous earth dam (c′ = 13.8 kPa, φ′ = 37°, γ = 18.2 kN/m³) before the reservoir is filled; circular search.

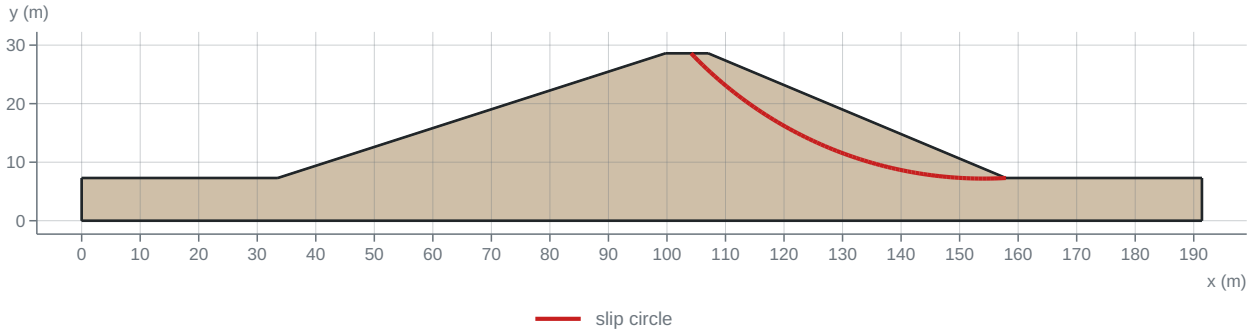


Figure 114. V063: critical circle (Spencer, F = 2.451).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	18.20	c′ = 13.80	37.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	2.451				2.455	−0.15 %
Bishop simplified	2.453	2.43	Huang & Jia	+0.95 %		
Strength reduction (FE)	2.403	2.50	Huang & Jia (2009): finite-element strength reduction	−3.88 %		

Huang & Jia's limit-equilibrium value 2.43 is by Bishop's method (their p. 95 and fig. 3a); their strength-reduction finite-element value is 2.50.

3.114 V100: Jiang, Baker & Yamagami (2003) example 1, power-law envelope, $r_u = 0.3$

Jiang, J.-C., Baker, R. & Yamagami, T. (2003). The effect of strength envelope nonlinearity on slope stability computations. Can. Geotech. J. 40(2), 308–325; after Charles & Soares (1984b).

Critical circle search · Units: SI (m, kPa, kN/m³)

20 m deep 1 in 4 cut in clay with the non-linear envelope τ = 1.4·σ'^0.8 (kPa), γ = 20 kN/m³ and r_u = 0.3. The envelope is a shear-normal function through the origin and 40 points of the curve from 1 to 1000 kPa (logarithmic spacing), straight lines between them; critical-surface searches.

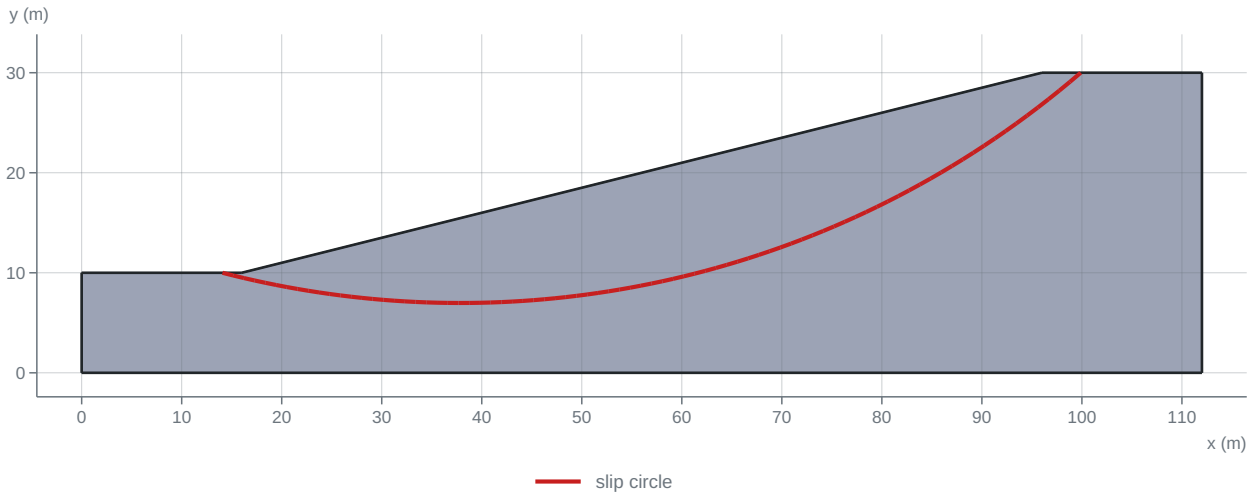


Figure 115. V100: critical circle (Bishop simplified, F = 1.668).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	shear-normal	20.00	see model	–

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.668	1.66	Charles & Soares	+0.48 %		
Janbu simplified, non-circular	1.559	1.56	Jiang, Baker & Yamagami: Janbu simplified (uncorrected), non-circular	−0.09 %	1.563	−0.28 %
Bishop simplified, non-circular	1.651				1.656	−0.28 %
–	–	1.67	Perry (1994): Janbu rigorous, on the critical Bishop circle			
–	–	1.60	Jiang, Baker & Yamagami (2003): simplified Janbu (uncorrected) on Charles & Soares's circle			

Charles & Soares's 1.66 (Bishop, critical circle), Perry's 1.67 (rigorous Janbu on that circle) and Jiang, Baker & Yamagami's 1.60 (simplified Janbu on that circle) and 1.56 (simplified Janbu, dynamic programming) are all quoted from Jiang, Baker & Yamagami (2003, p. 318 and Fig. 10). The values on that circle are shown without a comparison: cSlope has no rigorous Janbu, and its critical circle for simplified Janbu is a deeper one.

3.115 V064: Leshchinsky & Han MSE wall, five tiers

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

Five 1.8 m tiers (Ta = 10.1 kN/m).

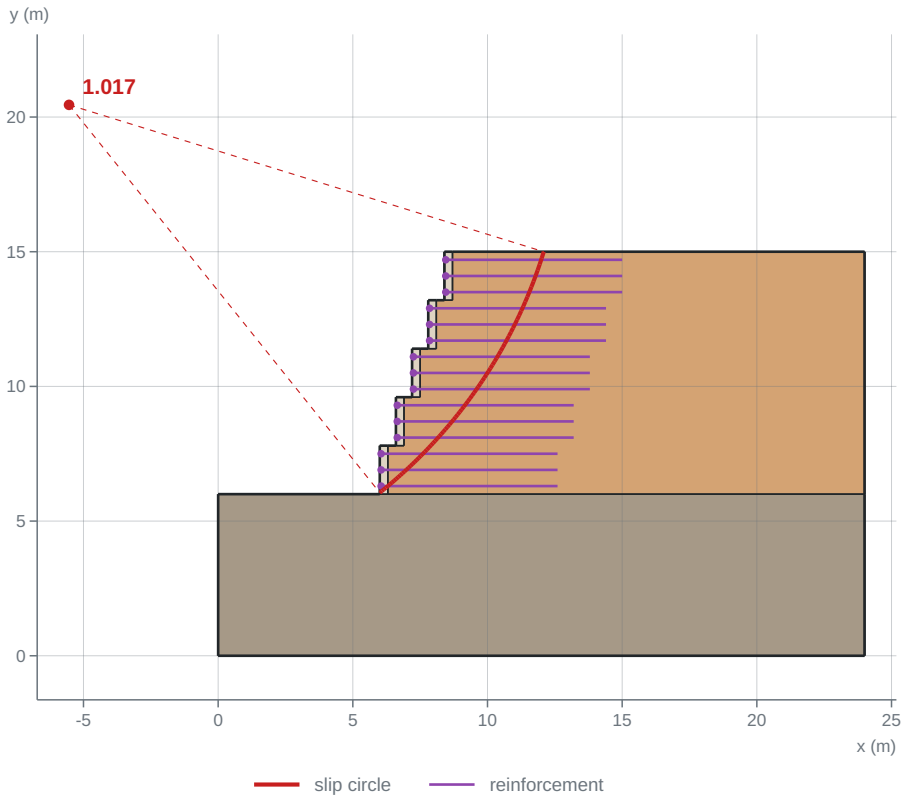


Figure 116. V064: analyzed circle (Bishop simplified, F = 1.017).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c′ = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c′ = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c′ = 2.50	34.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.017	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	+1.71 %	1.020	−0.28 %

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.116 V065: Leshchinsky & Han MSE wall, weak foundation

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a on a c = 0, φ = 18° foundation.

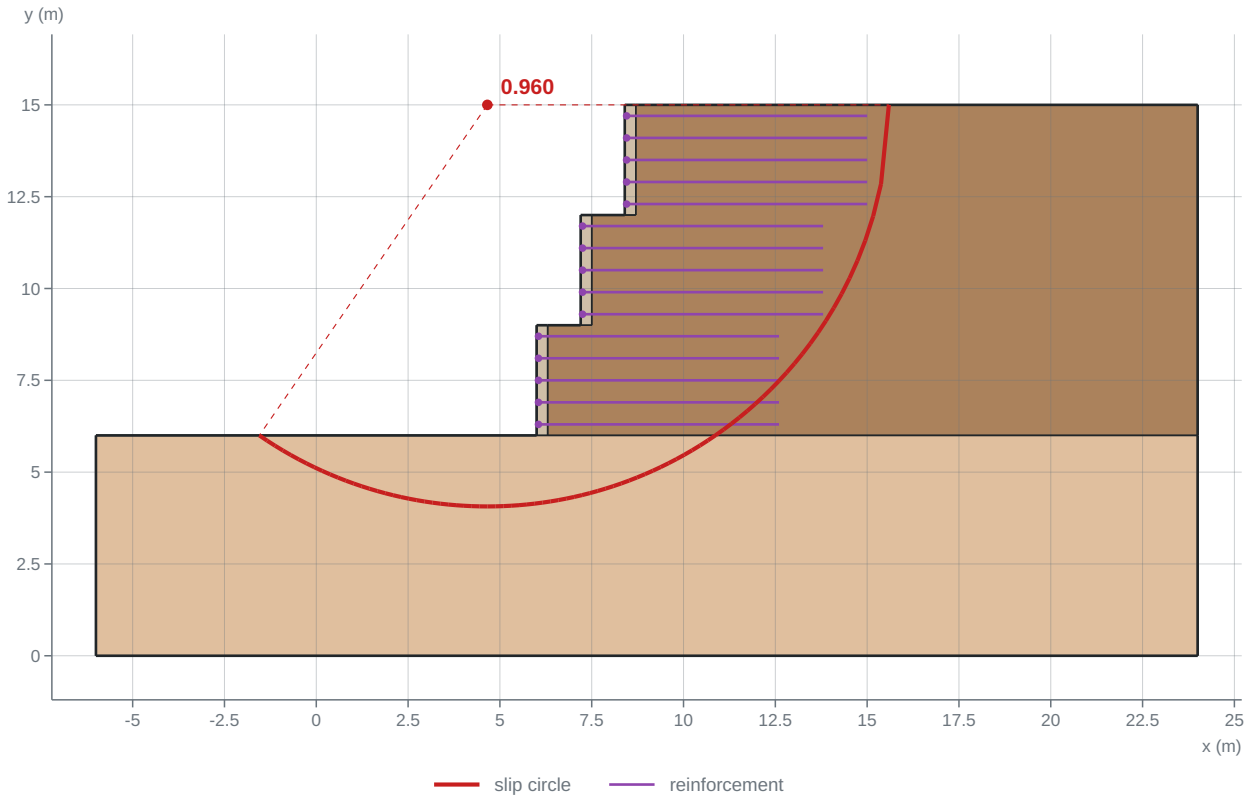


Figure 117. V065: analyzed circle (Spencer, F = 0.960).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c′ = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c′ = 0.00	18.00°
Blocks	Mohr-Coulomb	18.00	c′ = 2.50	34.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	0.960	0.92	Leshchinsky & Han (2004): Spencer, three-part wedge	+4.38 %	0.964	−0.38 %
Bishop simplified	0.979	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	−2.13 %		
		0.98	Leshchinsky & Han (2004): Bishop simplified, deep circle (read from fig. 7a)	−0.13 %		
–	–	0.86	Leshchinsky & Han (2004): finite differences			

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.117 V066a: Leshchinsky & Han three-tier MSE wall, baseline

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

Three 3 m tiers offset 1.2 m, block facing, geotextiles every 0.6 m (L = 6.3 m, Ta = 10 kN/m, applied horizontally, passive), a commercial program's critical circle.

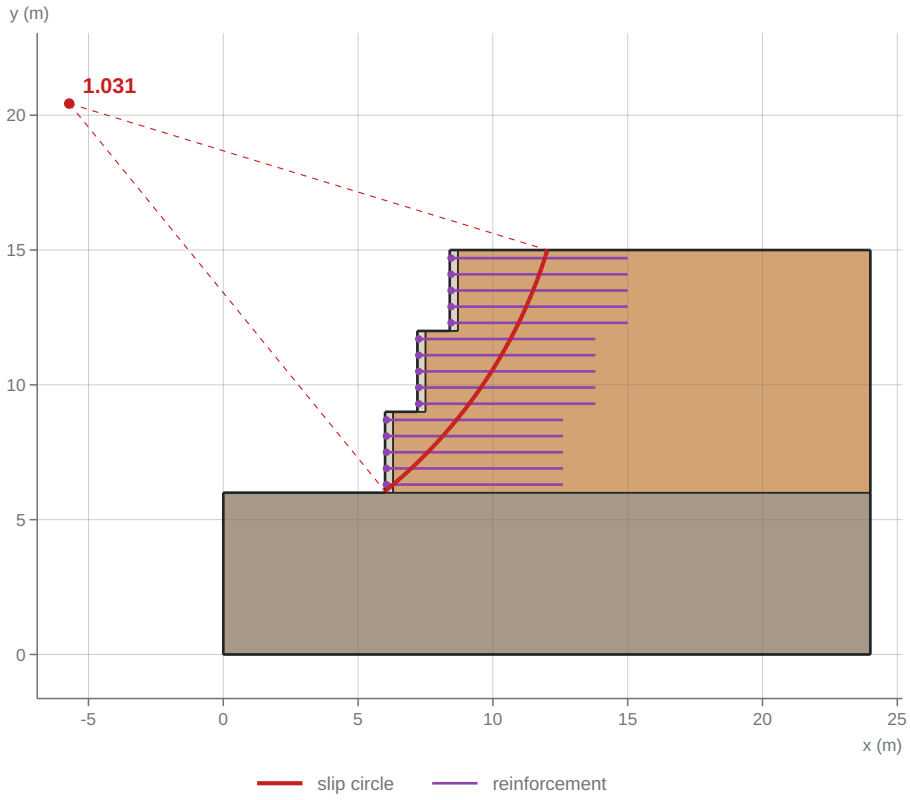


Figure 118. V066a: analyzed circle (Bishop simplified, F = 1.031).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c′ = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c′ = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c′ = 2.50	34.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.031	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	+3.06 %	1.040	−0.90 %
–	–	0.99	Leshchinsky & Han (2004): finite differences			

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.118 V066b: Leshchinsky & Han MSE wall, L = 4.2 m

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a with 4.2 m geotextiles (Ta = 11.4 kN/m).

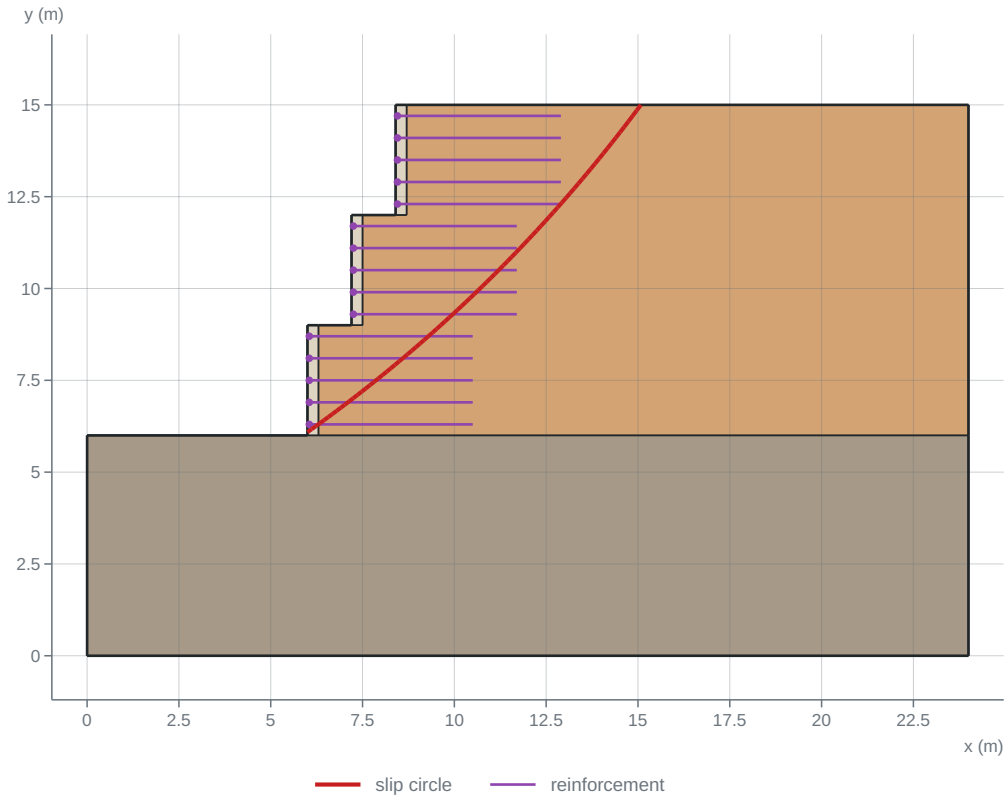


Figure 119. V066b: analyzed circle (Spencer, F = 1.011).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c′ = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c′ = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c′ = 2.50	34.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.011				1.011	+0.01 %
Bishop simplified	1.013	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	+1.32 %		
–	–	0.98	Leshchinsky & Han (2004): finite differences			

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.119 V066c: Leshchinsky & Han MSE wall, two reinforcement types

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a with two reinforcement strengths (7.5 and 11.0 kN/m).

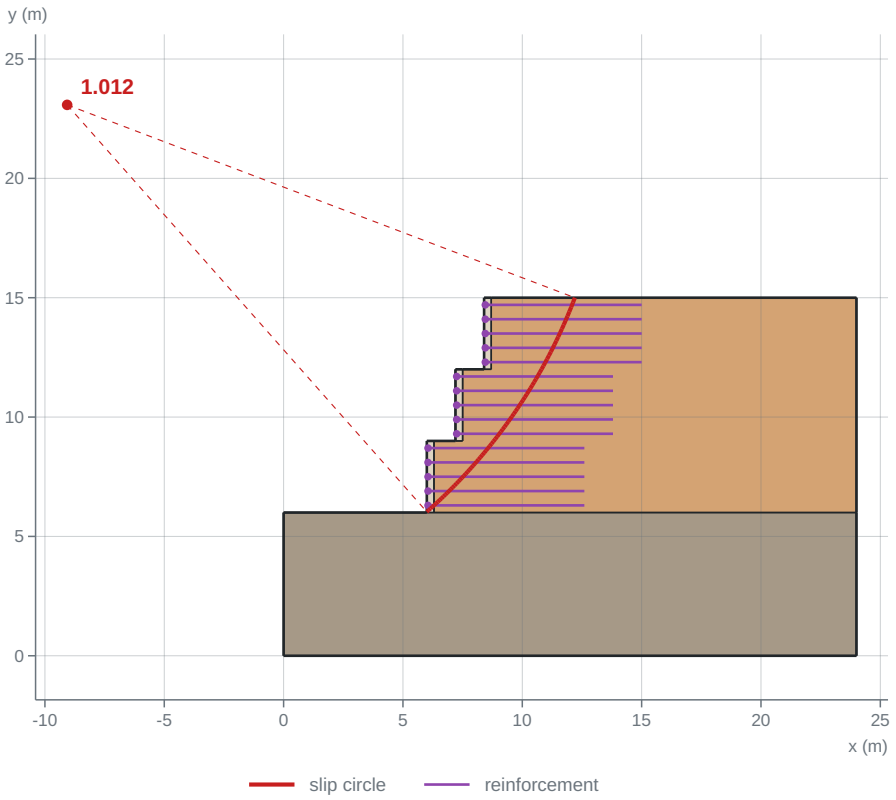


Figure 120. V066c: analyzed circle (Bishop simplified, F = 1.012).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c′ = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c′ = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c′ = 2.50	34.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.012	1.00	Leshchinsky & Han	+1.15 %		
–	–	1.01	Leshchinsky & Han (2004): finite differences			

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.120 V066d: Leshchinsky & Han MSE wall, water hw = 3 m

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a with the water table 3 m above the toe: pore pressure in the foundation only (drained fill); Ta = 9.25 kN/m. The pond in front of the wall is left out of this comparison: another program's water loads apply no pressure to the vertical facing, whereas cSlope does.

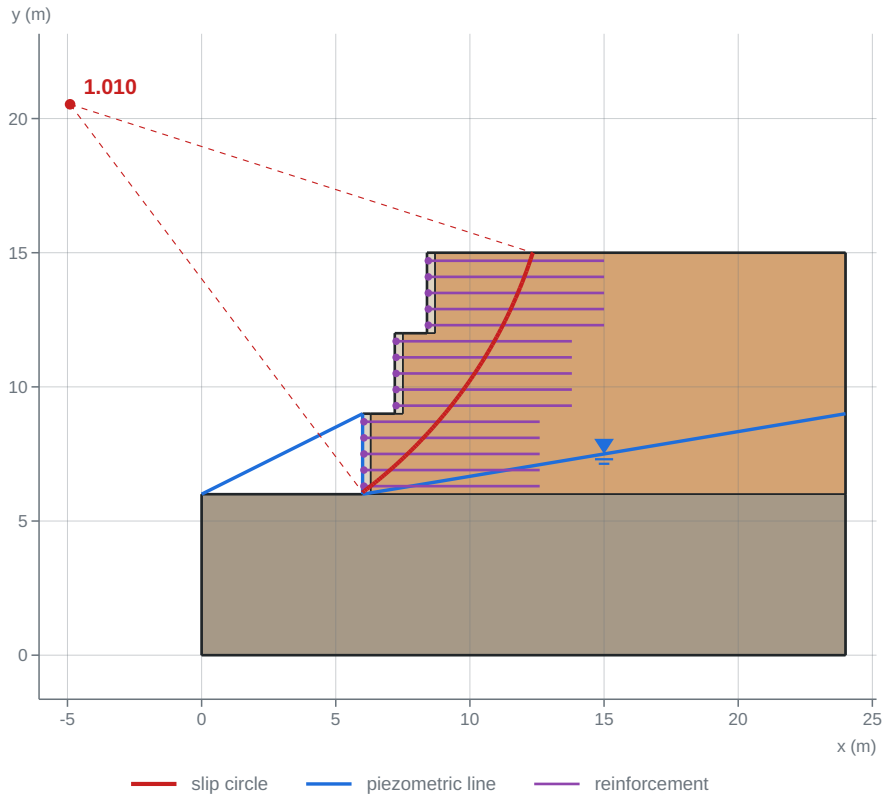


Figure 121. V066d: analyzed circle (Bishop simplified, F = 1.010).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c′ = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c′ = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c′ = 2.50	34.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.010	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	+1.00 %	1.010	0.00 %
–	–	1.01	Leshchinsky & Han (2004): finite differences			

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.121 V066e: Leshchinsky & Han MSE wall, surcharge

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a (Ta = 10 kN/m) with a 20 kPa surcharge on the top tier (x = 8.7 to 24).

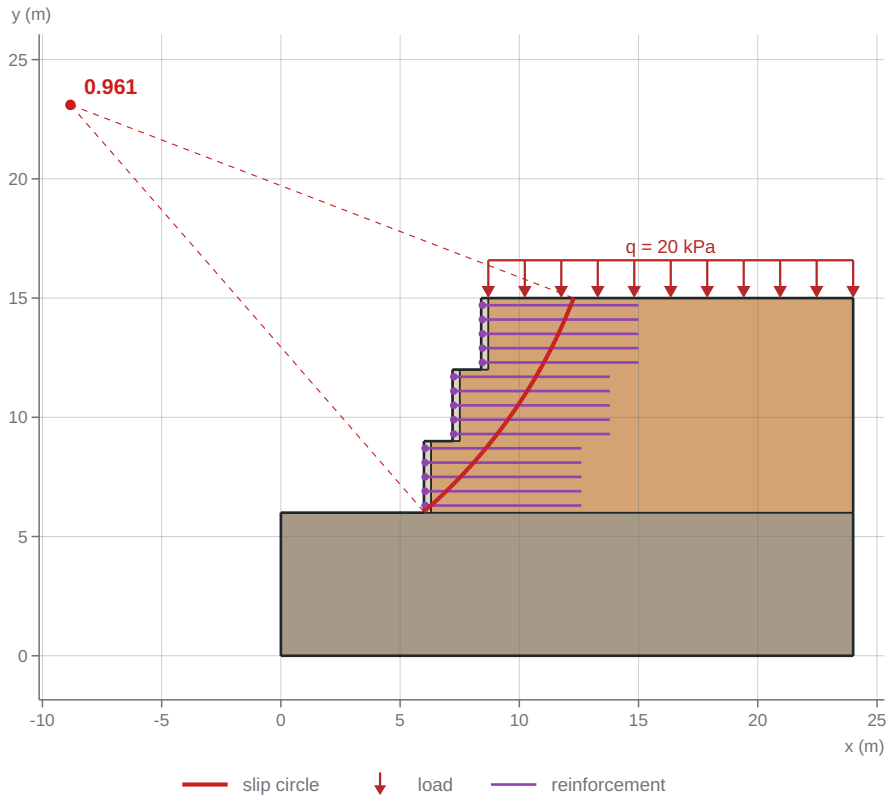


Figure 122. V066e: analyzed circle (Bishop simplified, F = 0.961).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c′ = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c′ = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c′ = 2.50	34.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	0.961	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	−3.90 %	0.958	+0.31 %
–	–	1.02	Leshchinsky & Han (2004): finite differences			

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value. Here the reinforcement keeps the baseline strength of 10 kN/m, as in the program compared with, not the 11.6 kN/m designed for the surcharge, so the published 1.00 is not directly comparable.

3.122 V101: Li & Lumb (1987) homogeneous slope, $r_u = 0.2$

Li, K.S. & Lumb, P. (1987). Probabilistic design of slopes. Can. Geotech. J. 24(4), 520–535; Hassan, A.M. & Wolff, T.F. (1999). J. Geotech. Geoenviron. Eng. 125(4), 301–308.

Critical circle search · Units: SI (m, kPa, kN/m³)

10 m high 1:1 slope (Li & Lumb's illustrative example, Fig. 5) with the means of their random variables: c′ = 18 kPa, tan φ′ = tan 30°, γ = 18 kN/m³, r_u = 0.2; critical-circle search. The ground 5 m below and beyond the toe and 5 m behind the crest is a program manual's model; the critical circle passes through the toe.

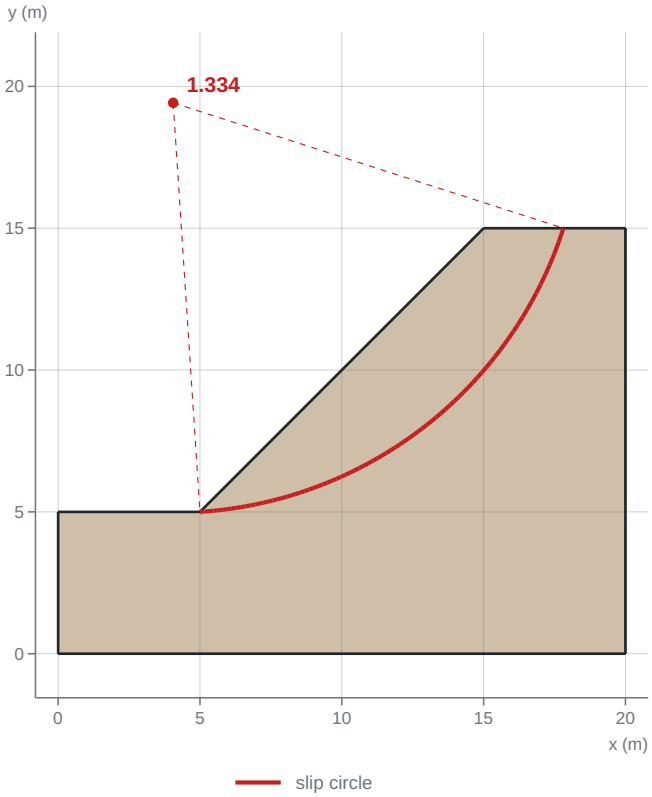


Figure 123. V101: critical circle (Bishop simplified, F = 1.334).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c′ = 18.00	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.334	1.334	Hassan & Wolff	0.00 %	1.340–1.343	−0.56 %

Probabilistic analysis on the same surface: 4 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1).

Quantity	Published	Other programs	cSlope
β lognormal (Taylor series)	2.336 Hassan & Wolff (1999)	–	2.328

3.123 V102a: Loukidis et al. (2003) slope at its critical seismic coefficient, dry

Loukidis, D., Bandini, P. & Salgado, R. (2003). Stability of seismically loaded slopes using limit analysis. Géotechnique 53(5), 463–479.

Critical circle search · Units: SI (m, kPa, kN/m³)

25 m high 3:1 slope, c′ = 25 kPa, φ′ = 30°, γ = 20 kN/m³, dry, with the horizontal seismic coefficient kh = 0.432, Loukidis, Bandini & Salgado's critical value; critical-surface searches.

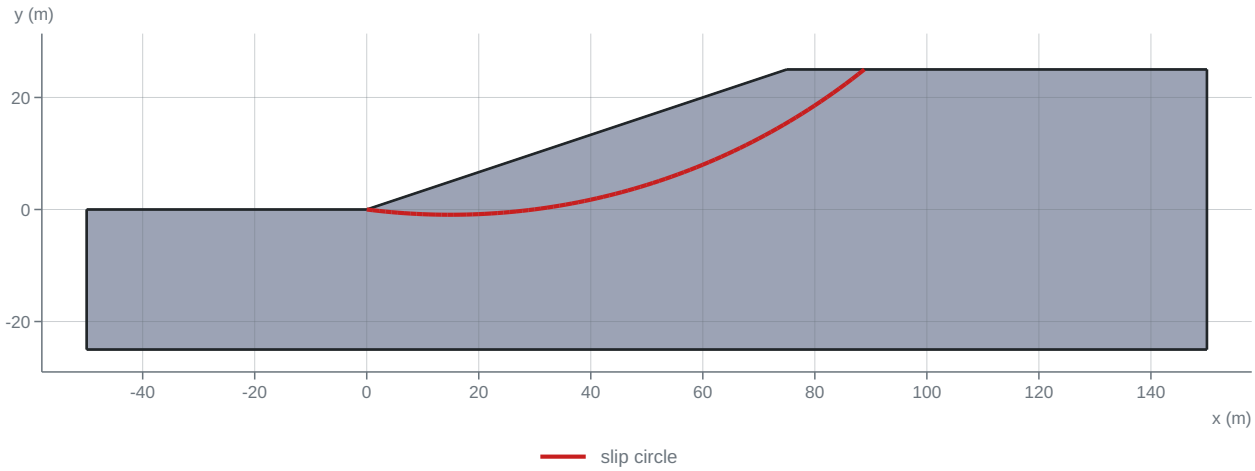


Figure 124. V102a: critical circle (Spencer, F = 1.001).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	Mohr-Coulomb	20.00	c′ = 25.00	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.001	1.00	Loukidis, Bandini & Salgado	+0.10 %	1.001	0.00 %
Bishop simplified	0.991				0.991–0.993	−0.07 %
Spencer, non-circular	0.998				0.999	−0.12 %

Loukidis, Bandini & Salgado (2003) give the critical seismic coefficient, at which the factor of safety is 1 by definition; the check is the factor of safety at their coefficient. Their coefficients are quoted from two program manuals, which agree.

3.124 V102b: Loukidis et al. (2003) slope at its critical seismic coefficient, $r_u = 0.5$

Loukidis, D., Bandini, P. & Salgado, R. (2003). Stability of seismically loaded slopes using limit analysis. Géotechnique 53(5), 463–479.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V102a with a pore pressure ratio r_u = 0.5 and kh = 0.132, Loukidis, Bandini & Salgado's critical value for it; critical-surface searches.

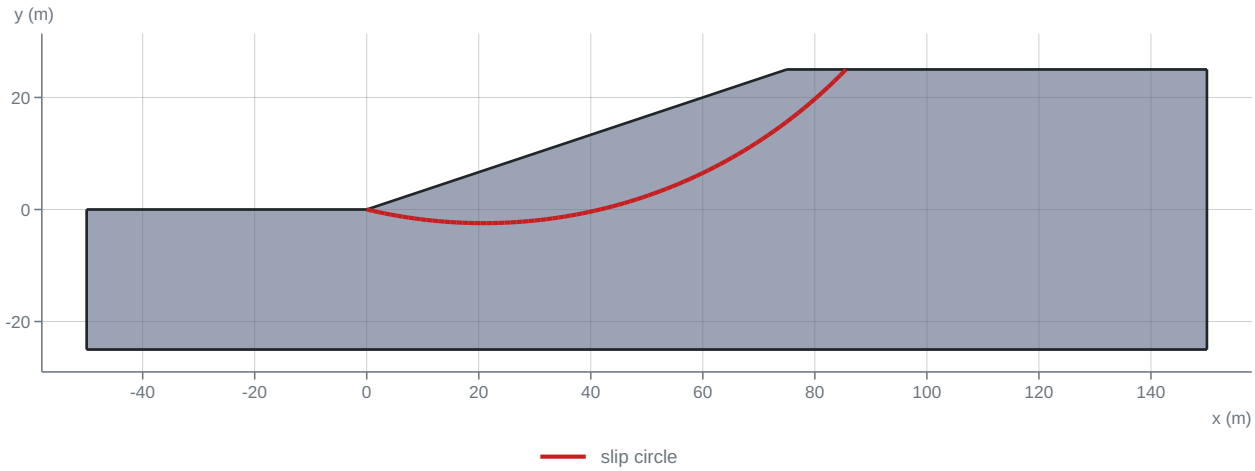


Figure 125. V102b: critical circle (Spencer, F = 1.001).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	Mohr-Coulomb	20.00	c′ = 25.00	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.001	1.00	Loukidis, Bandini & Salgado	+0.07 %	1.001	−0.03 %
Bishop simplified	0.987				0.987–0.988	−0.10 %
Spencer, non-circular	0.997				0.997	−0.05 %

Loukidis, Bandini & Salgado (2003) give the critical seismic coefficient, at which the factor of safety is 1 by definition; the check is the factor of safety at their coefficient. Their coefficients are quoted from two program manuals, which agree.

3.125 V067: Low (1989), layered undrained clay

Low (1989).

Critical circle search · Units: SI (m, kPa, kN/m³)

Embankment on three undrained clay layers of different strength; critical-circle search.

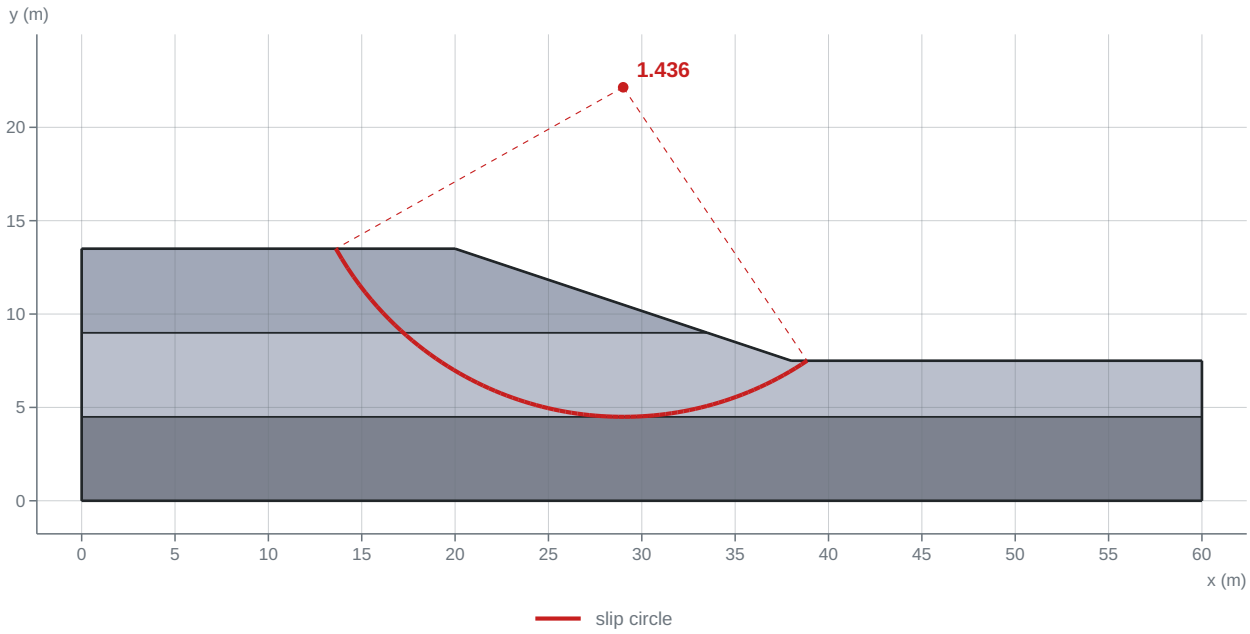


Figure 126. V067: critical circle (Bishop simplified, F = 1.436).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
A	Undrained (φ = 0)	18.00	Su = 30.00	0°
B	Undrained (φ = 0)	18.00	Su = 20.00	0°
C	Undrained (φ = 0)	18.00	Su = 150.0	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.436				1.439	−0.19 %
Strength reduction (FE)	1.439	1.41	another program's verification manual: finite-element strength reduction (verification manual, problem 19)	+2.08 %		
–	–	1.44	Low (1989): Ordinary			

3.126 V068: Low (1989), stiff crust over soft clays, published circle

Low, B.K. (1989). Stability analysis of embankments on soft ground. J. Geotech. Eng. 115(2), 211–227; example 2, fig. 11.

Fixed circle · Units: SI (m, kPa, kN/m³)

Stiff upper slope on two undrained clays, the lowest with Su increasing linearly with depth (S = f(datum)), on the critical circle a commercial program prints with its Bishop value (center (18.001, 16.000), R 15.556).

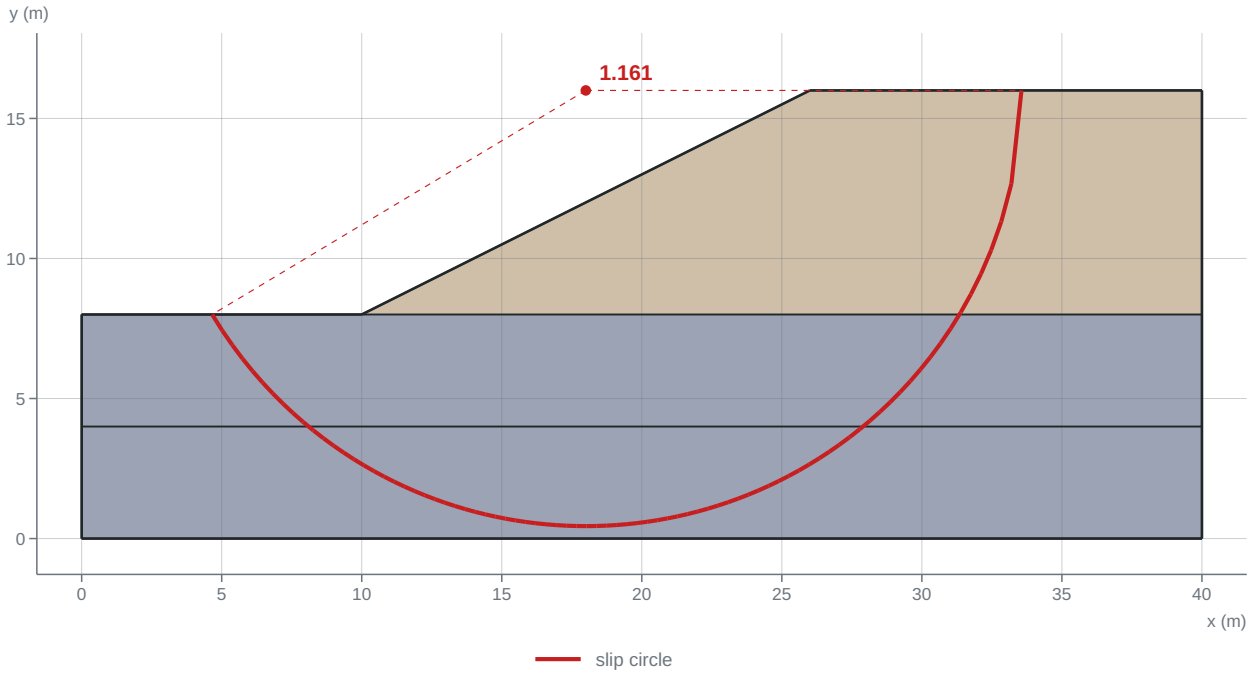


Figure 127. V068: analyzed circle (Bishop simplified, F = 1.161).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper	Mohr-Coulomb	20.00	c′ = 95.00	15.00°
Middle	Mohr-Coulomb	20.00	c′ = 15.00	0.00°
Lower	S = f(datum)	20.00	Su = 15.00 at y = 4.00, 3.75 kPa/m below	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.161	1.14	Low	+1.82 %		
Spencer, non-circular	–	1.17	Kim et al. (2002): Spencer, non-circular (random search)	–		
Strength reduction (FE)	1.177	1.13–1.27	Kim et al. (2002): limit analysis, lower and upper bound	in bounds		
–	–	1.36	Low (1989): Ordinary			
–	–	1.31	Low (1989): Ordinary and Bishop with φ = 0 in the bank			
–	–	1.38	Low (1989): hand calculation (Low's method)			

The comparison program's 1.19 is not reproduced on the circle it prints with that value: cSlope gives 1.16 there with 50 slices, against Low's Bishop value 1.14, which is the check. The circle's center lies at crest level, so its upper end is a vertical scarp, and the factor depends on how that steep part is divided into slices; the tolerance is 1 % wider for it. Low himself calls that Bishop value misleading, an artifact of negative effective normal stresses in the stiff bank with φ = 15°: with φ = 0 in the bank both Ordinary and Bishop give 1.31, and his own hand method 1.38. Kim et al. (2002) give 1.17 with Spencer's method on a non-circular surface and limit-analysis bounds of 1.13–1.27.

3.127 V069a: Morgenstern (1963) complete drawdown, 3:1

Morgenstern, N. (1963). Stability charts for earth slopes during rapid drawdown. Géotechnique 13(2).

Critical circle search · Units: SI (m, kPa, kN/m³)

3:1 slope after complete drawdown: piezometric line on the slope face, base at the toe (D = 1). Pore pressure from the vertical head. Converted to SI.

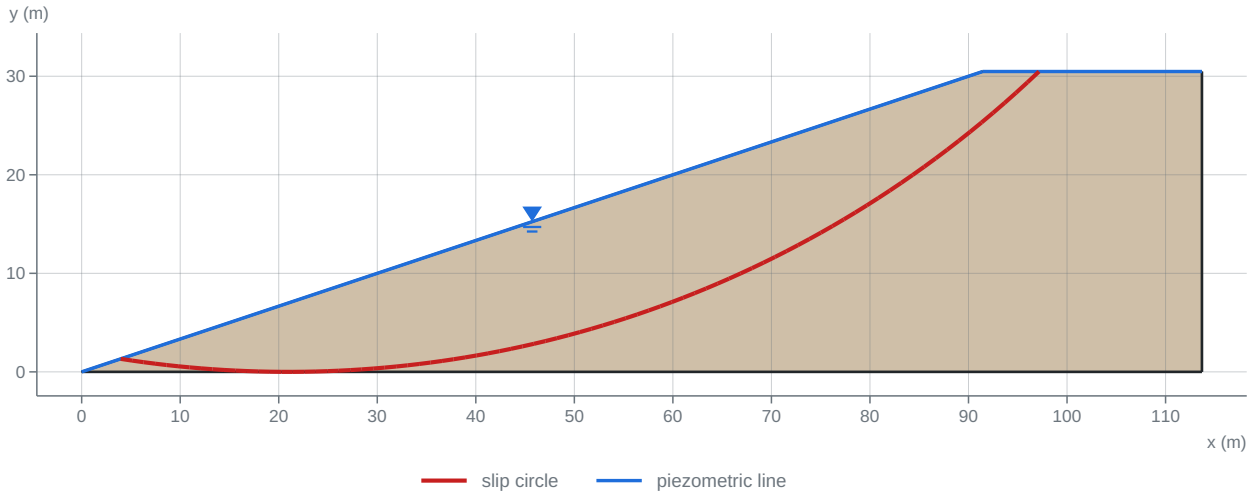


Figure 128. V069a: critical circle (Bishop simplified, F = 1.201).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	19.60	c′ = 14.94	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.201	1.20	Morgenstern: stability chart	+0.07 %		
Spencer	1.206					

3.128 V069b: Morgenstern complete drawdown, the reference circle

Morgenstern, N. (1963). Stability charts for earth slopes during rapid drawdown. Géotechnique 13(2). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V069a slope after complete drawdown on the reference critical circle (Fig. 100.2); pore pressure from the vertical head. Converted to SI.

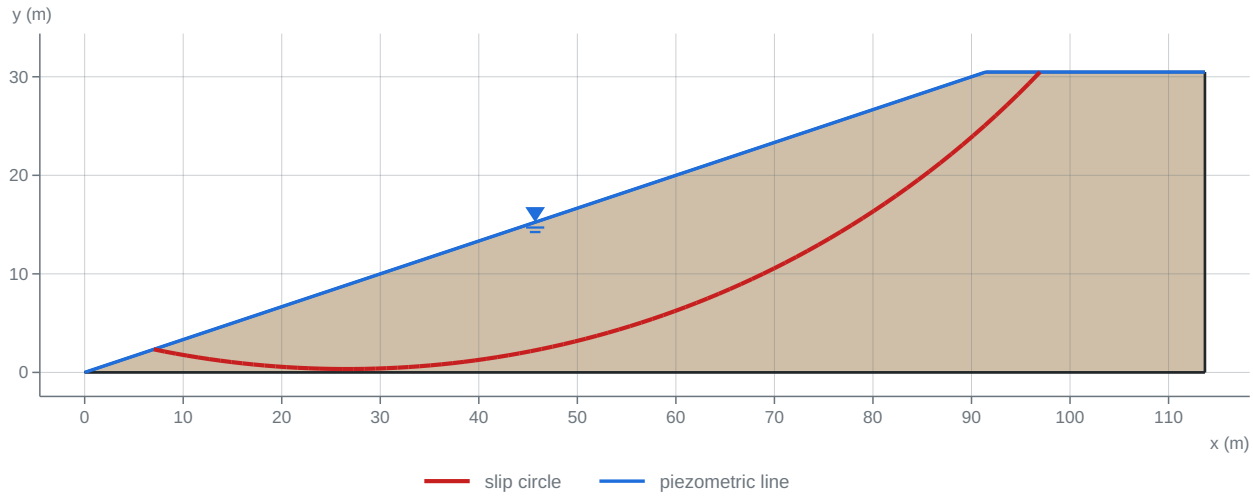


Figure 129. V069b: analyzed circle (Bishop simplified, F = 1.210).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	19.60	c′ = 14.94	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.210				1.212	−0.15 %
–	–	1.20	Morgenstern (1963): stability chart			

Morgenstern's 1.2 is read from a stability chart.

3.129 V070: Partial drawdown, $\bar{B} = 1$ (Morgenstern chart)

Morgenstern (1963).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous embankment after a partial drawdown from 100 to 50 ft with $\bar{B} = 1$: the piezometric line follows the face above the pool and stays at the pool below it, which loads the submerged face. Converted to SI.

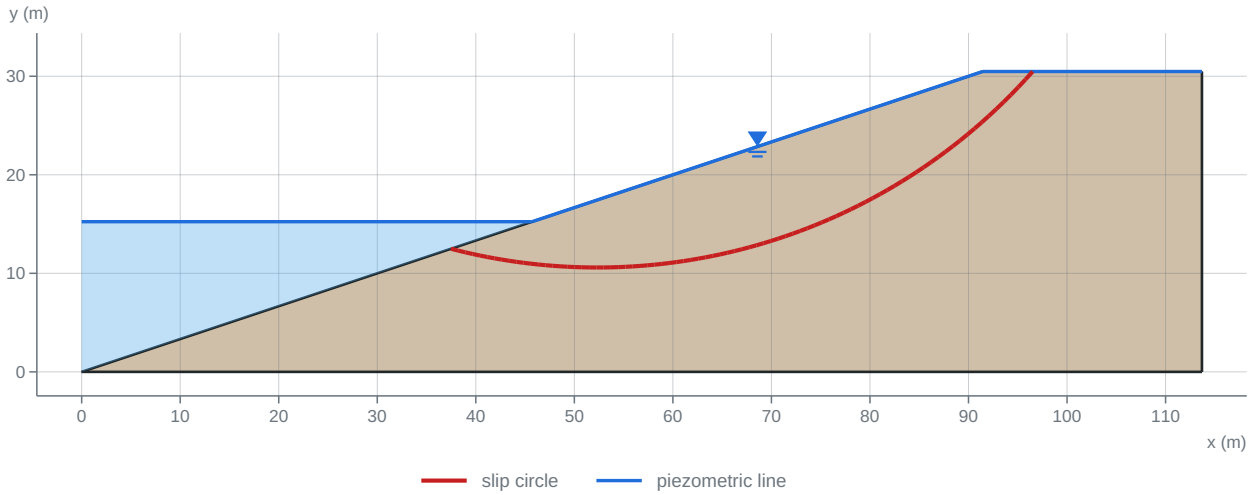


Figure 130. V070: critical circle (Bishop simplified, F = 1.416).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Morgenstern soil	Mohr-Coulomb	19.60	c′ = 14.94	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.415	1.41	Morgenstern: stability chart	+0.39 %	1.417	−0.11 %
Spencer	1.422				1.422	−0.01 %
–	–	1.52	Morgenstern (1963): stability chart, circles tangent to the base			
–	–	1.48	Morgenstern (1963): stability chart, circles tangent to H/2			

Morgenstern's 1.41 is the lowest of three chart readings for circles tangent to the base, to H/2 and to H/4 (1.52, 1.48 and 1.41, the last interpolated), with his remark that a slightly lower value could perhaps be found; a free search may therefore come in below it.

3.130 V071a: Pockoski & Duncan (2000) slope 1

Pockoski & Duncan (2000), slope 1.

Fixed circle · Units: SI (m, kPa, kN/m³)

2:1 slope with a water table, on the circle published by a commercial program; pore pressure from the vertical head below the piezometric line. Converted to SI.

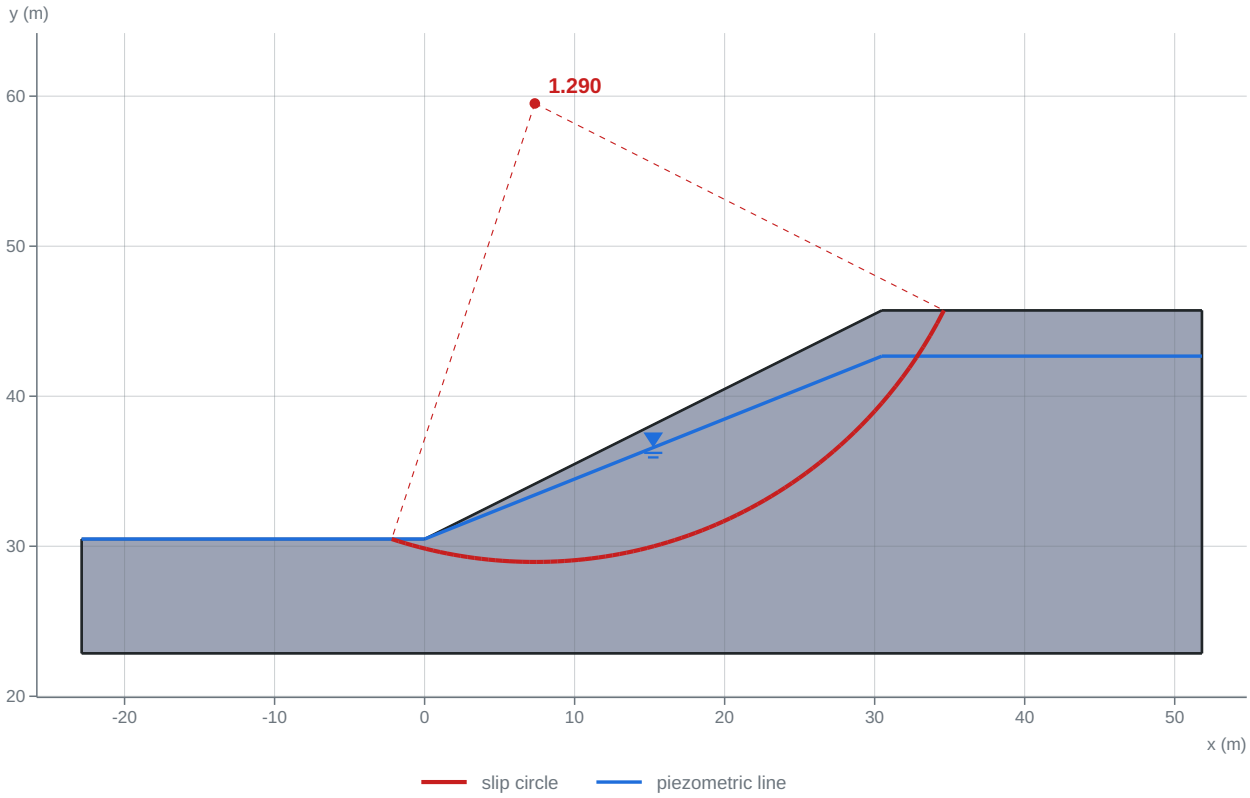


Figure 131. V071a: analyzed circle (Bishop simplified, F = 1.290).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c′ = 14.36	30.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.290				1.293	−0.19 %
Spencer	1.297				1.300	−0.22 %

Pockoski & Duncan compare computer programs and publish no independent value.

3.131 V071b: Pockoski & Duncan slope 1, circular search

Pockoski & Duncan (2000), slope 1. Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V071a slope (Pockoski & Duncan slope 1) with a circular search instead of the given circle; pore pressure from the vertical head. Converted to SI.

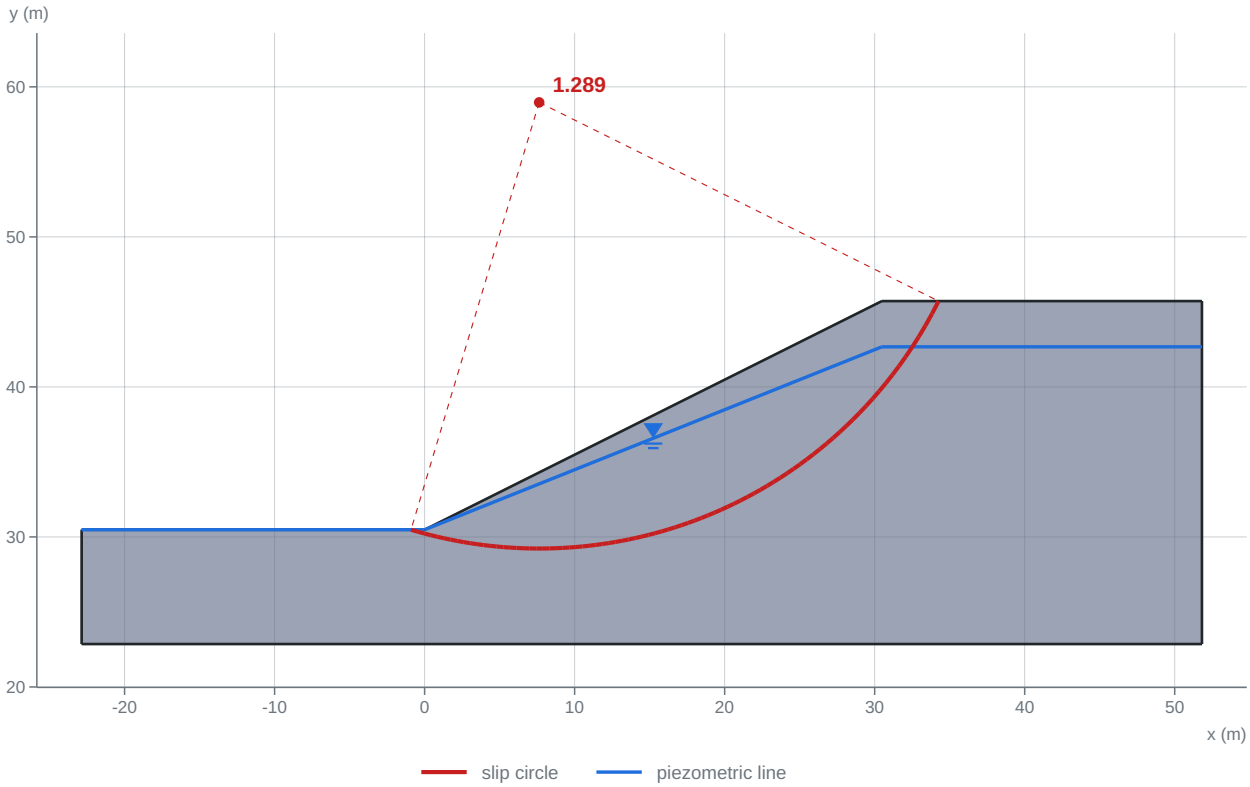


Figure 132. V071b: critical circle (Bishop simplified, F = 1.289).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c′ = 14.36	30.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.289				1.293	−0.31 %
Spencer	1.295				1.300	−0.38 %
Janbu simplified	1.149				1.151	−0.19 %

Pockoski & Duncan compare computer programs and publish no independent value.

3.132 V072a: Pockoski & Duncan test slope 2, dry tension crack

Pockoski & Duncan (2000).

Fixed circle · Units: SI (m, kPa, kN/m³)

The V071a slope with a dry 5.5 ft tension crack, on the reference critical circle. Converted to SI.

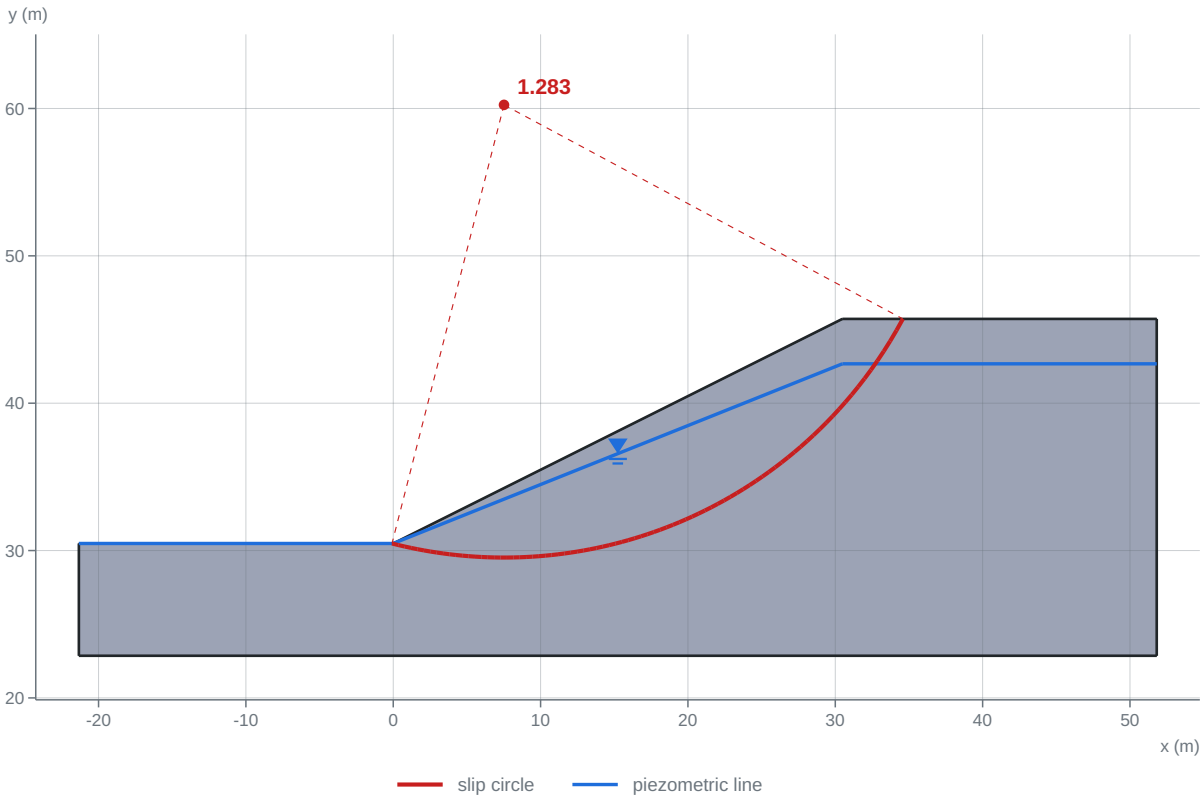


Figure 133. V072a: analyzed circle (Bishop simplified, F = 1.283).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c′ = 14.36	30.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.283				1.285	−0.17 %
Spencer	1.288				1.290	−0.15 %

Pockoski & Duncan compare computer programs and publish no independent value.

3.133 V072b: Pockoski & Duncan slope 2, circular search

Pockoski & Duncan (2000). Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V072a slope (dry 5.5 ft tension crack) with a circular search. Converted to SI.

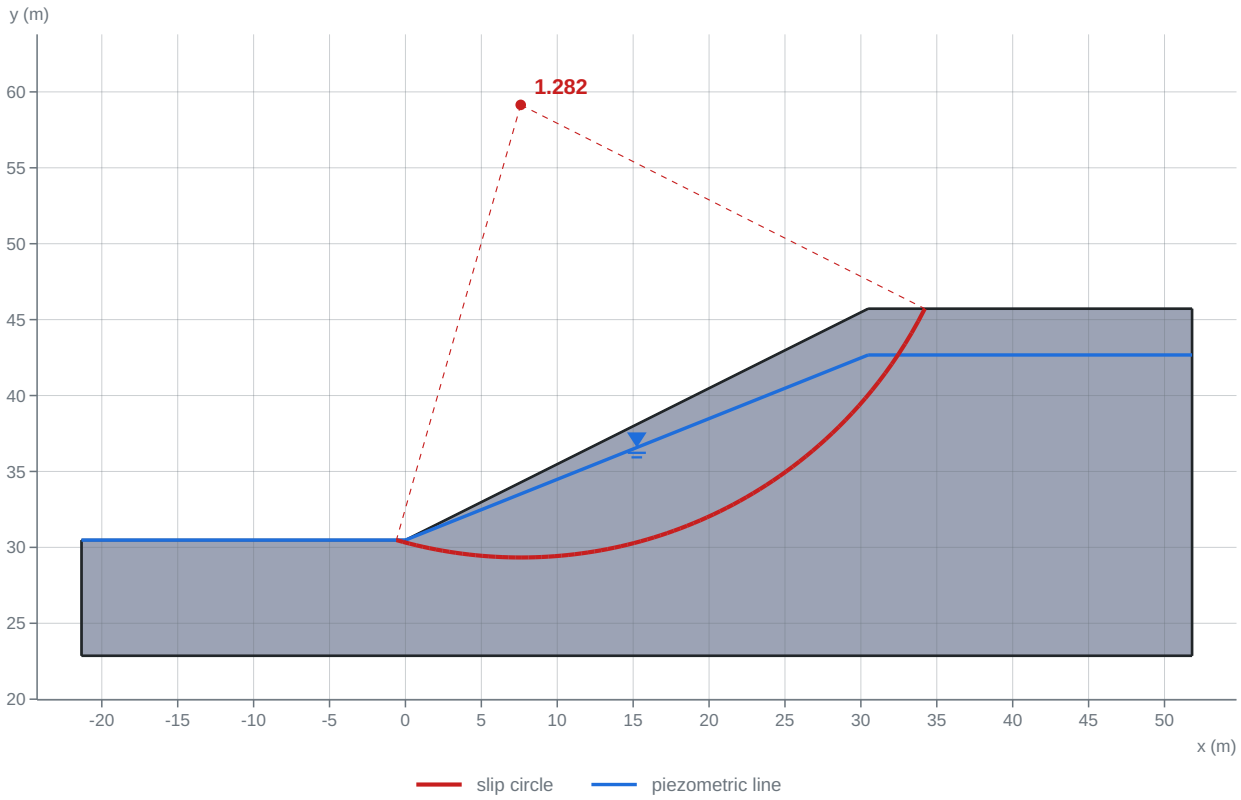


Figure 134. V072b: critical circle (Bishop simplified, F = 1.282).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c′ = 14.36	30.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.282				1.285	−0.22 %
Spencer	1.288				1.290	−0.18 %
Janbu simplified	1.136				1.141	−0.42 %

Pockoski & Duncan compare computer programs and publish no independent value.

3.134 V073: Pockoski & Duncan test slope 3, clay seam and dry crack

Pockoski & Duncan (2000).

Critical circle search · Units: SI (m, kPa, kN/m³)

Sandy clay over a 5 ft highly plastic clay seam on the model base, water table and a dry 6 ft tension crack; circular search. Converted to SI.

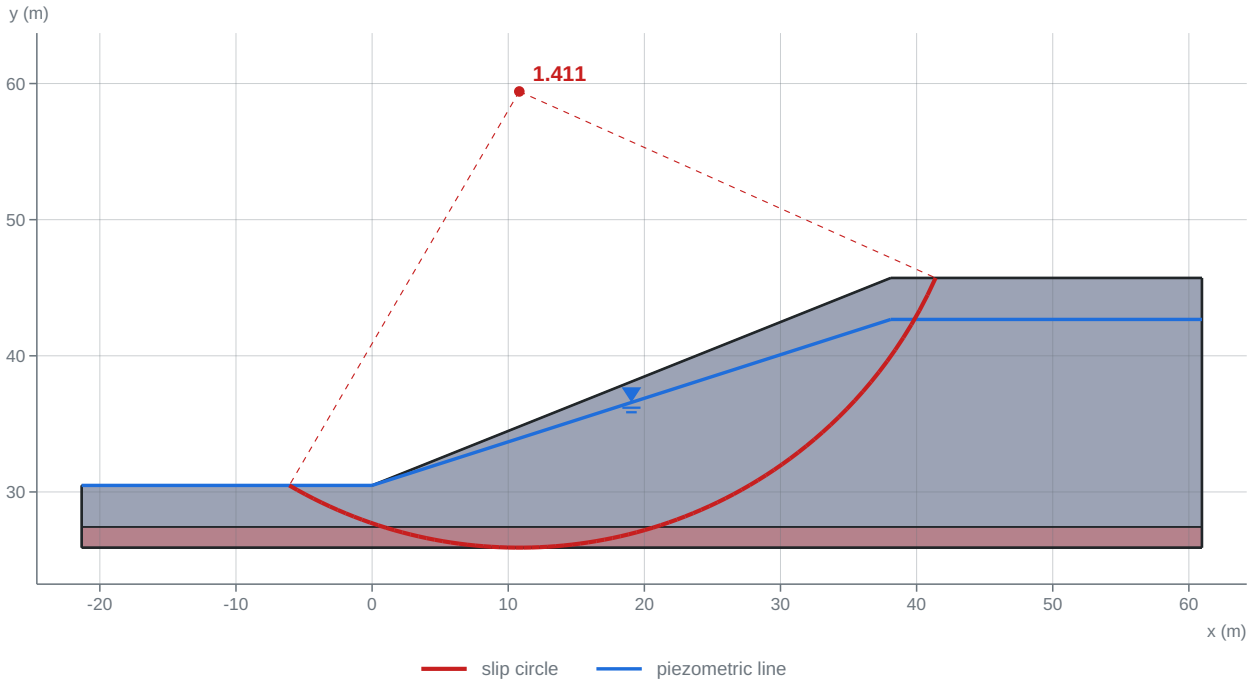


Figure 135. V073: critical circle (Bishop simplified, circles only, F = 1.411).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	20.42	c′ = 14.36	35.00°
Plastic clay seam	Mohr-Coulomb	20.42	c′ = 0.00	25.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified, circles only	1.411				1.417	−0.44 %
Spencer, circles only	1.415				1.422	−0.46 %
Bishop simplified	1.388				1.392	−0.28 %
Spencer	1.396				1.400	−0.29 %

Pockoski & Duncan compare computer programs and publish no independent value.

3.135 V074: Soil-nailed wall with tension crack and surcharges

Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

25 ft nailed wall in undrained clay (800 psf) with crest surcharges, a dry 7 ft crack and five passive nail rows at 15°; the reference circle truncated by the crack. Converted to SI.

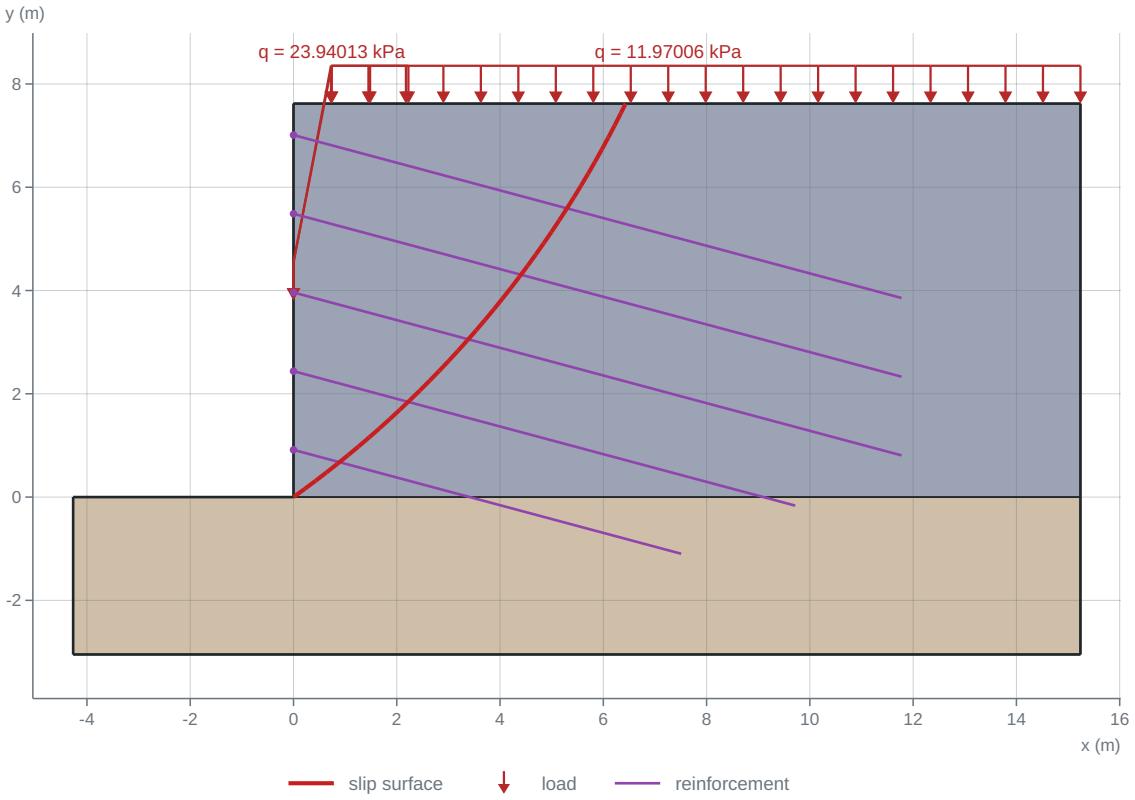


Figure 136. V074: analyzed surface (Spencer, F = 1.010).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Undrained (φ = 0)	18.85	Su = 38.30	0°
Firm soil	Mohr-Coulomb	20.42	c′ = 239.4	35.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.010				1.009	+0.10 %
Janbu corrected	1.075				1.073	+0.16 %

Pockoski & Duncan compare computer programs and publish no independent value.

3.136 V075: Tieback wall in sand, drawdown water table

Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Single-row tieback wall in sand (φ′ = 30°; the tieback from 12 ft up the wall, 15° down, 37 ft long of which 22 ft bonded) with a water table drawn down to the wall toe (digitized from the reference figure) and pore pressure from the vertical head; the printed surface from the wall toe. Converted to SI.

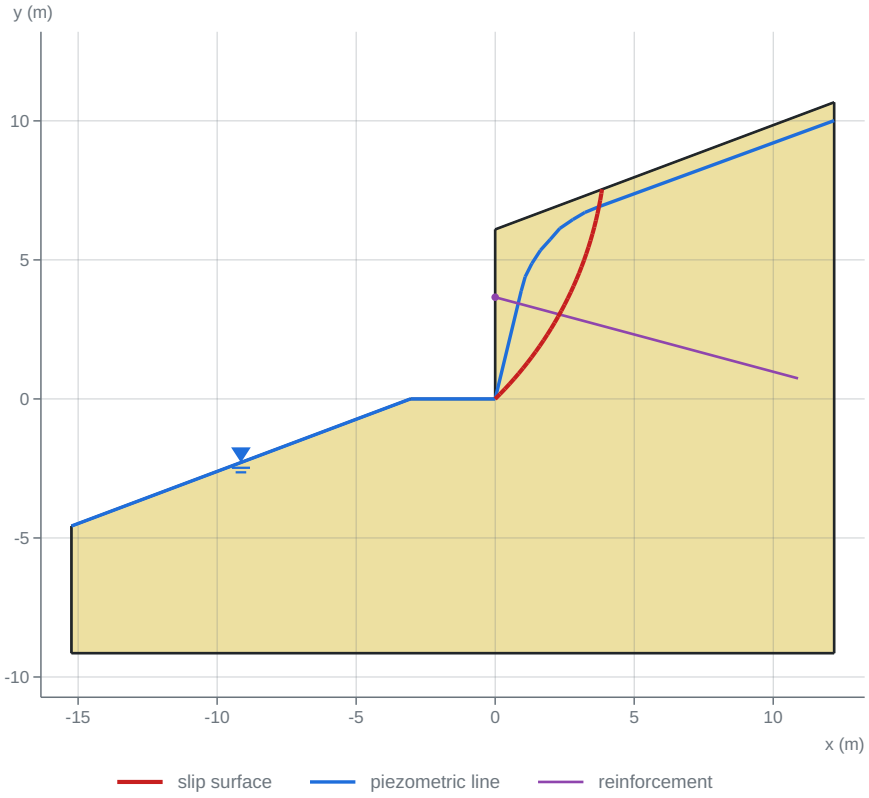


Figure 137. V075: analyzed surface (Spencer, F = 0.578).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Mohr-Coulomb	18.85	c′ = 0.00	30.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	0.578				0.570	+1.37 %

Pockoski & Duncan compare computer programs and publish no independent value.

3.137 V076: Tied-back wall in layered soil

Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.

Fixed circle · Units: SI (m, kPa, kN/m³)

44 ft tied-back excavation in eight layers, water table at grade in front of the wall and at el. 102.5 behind it (a vertical step at the wall), three tieback rows at 20° (88 ft, 40 ft bond); a commercial program's critical circle. Converted to SI.

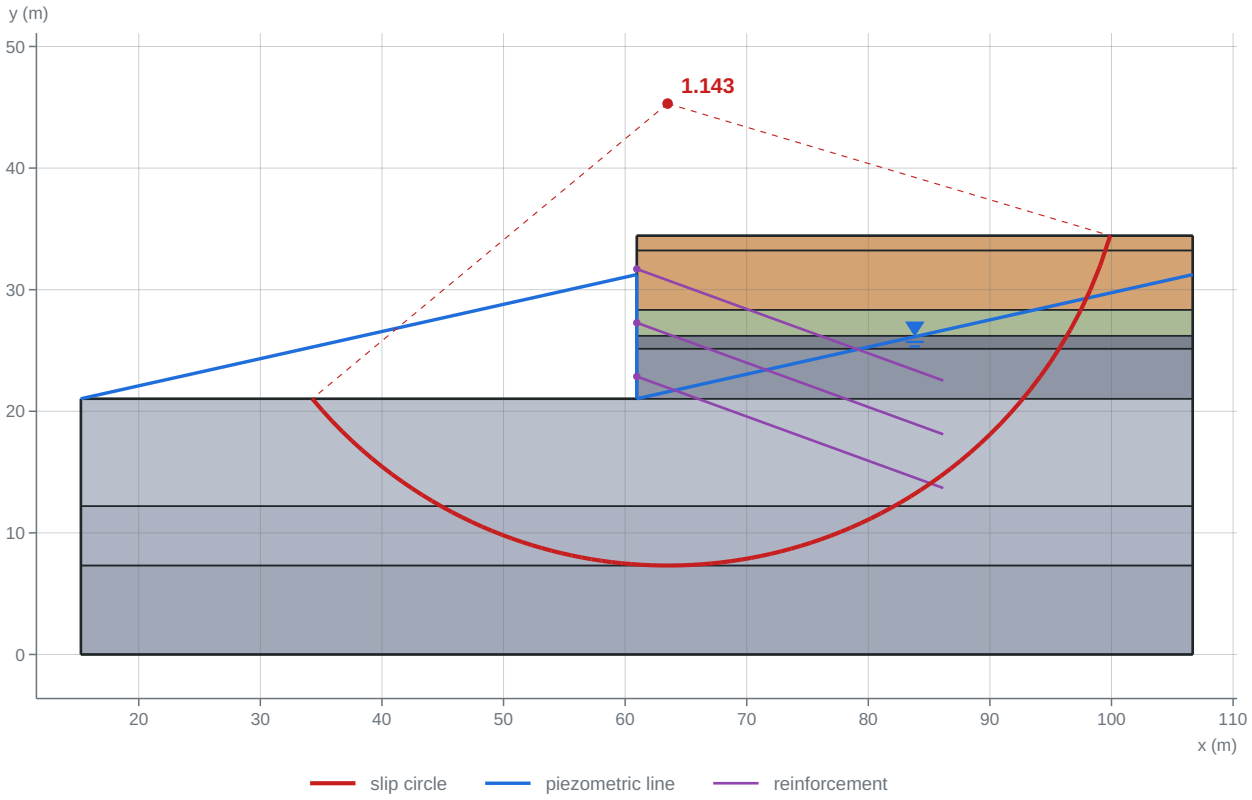


Figure 138. V076: analyzed circle (Bishop simplified, F = 1.143).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular Fill	Mohr-Coulomb	18.91	c′ = 0.00	30.00°
Cohesive Fill	Mohr-Coulomb	18.02	c′ = 0.00	30.00°
Organic Silt	Undrained (φ = 0)	17.31	Su = 43.09	0°
OC Crust	Undrained (φ = 0)	18.50	Su = 119.0	0°
Upper Marine Clay	Undrained (φ = 0)	18.50	Su = 79.96	0°
Middle Marine Clay	Undrained (φ = 0)	18.50	Su = 45.97	0°
Lower Marine Clay	Undrained (φ = 0)	18.50	Su = 51.95	0°
Glaciomarine Deposits	Undrained (φ = 0)	23.11	Su = 71.82	0°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.143				1.147	−0.35 %
Spencer	1.141				1.145	−0.36 %

Pockoski & Duncan compare computer programs and publish no independent value.

3.138 V077: Prandtl bearing capacity on level ground

Prandtl (1920). Über die Härte plastischer Körper. Nachr. Ges. Wiss. Göttingen, Math.-phys. Kl., 74–85.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Weightless (γ = 1e-6 kN/m³) c = 20 kPa soil (φ = 0) loaded by a strip load of 102.83 kPa = c·(2 + π) over 10 m, the exact bearing capacity. Analyzed on the Prandtl mechanism of Fig. 26.2: active wedge under the load, a 90° fan arc of radius 7.071 m about the load's edge (5, 10), lowest point 2.929, and the passive wedge.

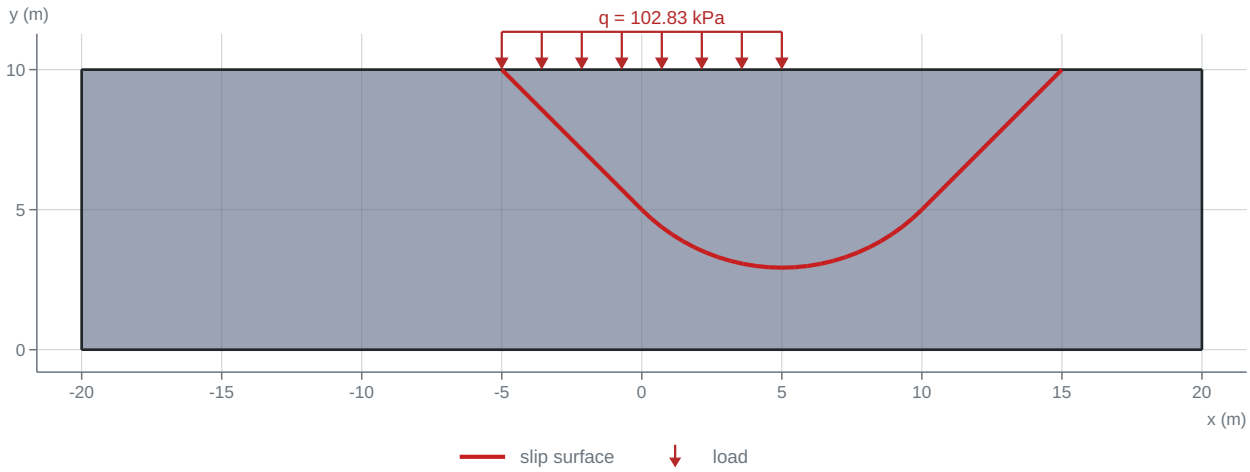


Figure 139. V077: analyzed surface (Spencer, F = 0.943).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Undrained (φ = 0)	0.00	Su = 20.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	0.943				0.940	+0.30 %
Strength reduction (FE)	1.001	1.00	Prandtl (1920): plasticity theory	+0.14 %		

The theoretical factor of safety is 1.0; limit equilibrium with slices gives about 6 % less on this mechanism.

3.139 V078: Prandtl bearing capacity, fully specified Prandtl surface (reference values)

Prandtl (1920). Über die Härte plastischer Körper. Nachr. Ges. Wiss. Göttingen, Math.-phys. Kl., 74–85.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Prandtl bearing capacity as a commercial program specifies it: weightless c = 20 kPa soil, 102.83 kPa over 10 m, the fully specified Prandtl surface (active wedge, 90° fan about the load edge, passive wedge).

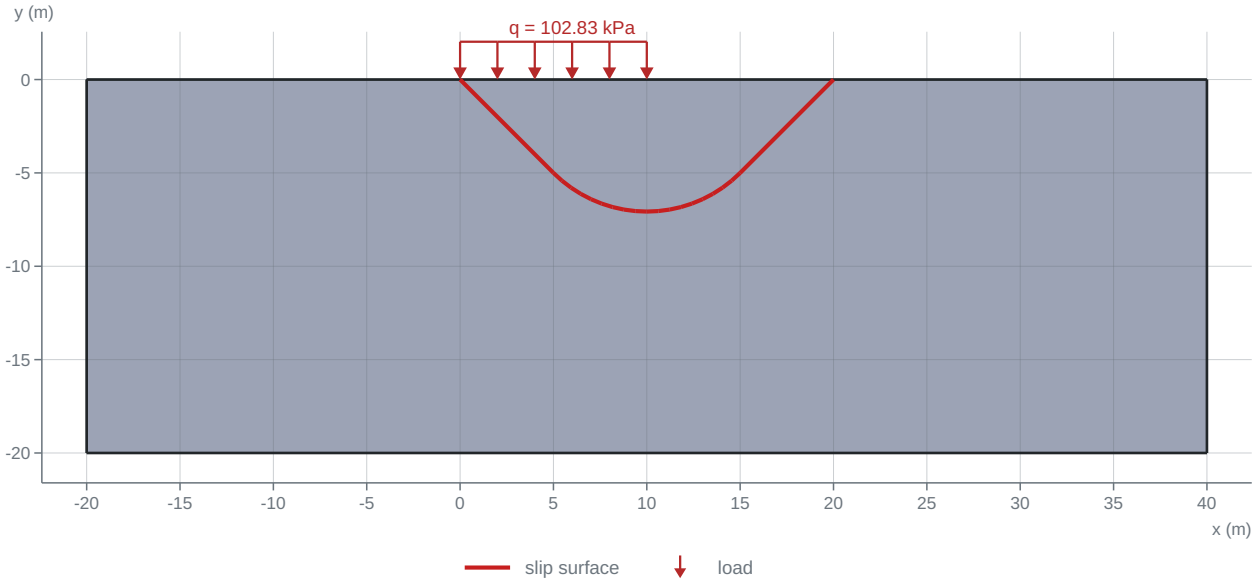


Figure 140. V078: analyzed surface (Morgenstern-Price (half-sine), F = 0.958).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Undrained (φ = 0)	0.00	Su = 20.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	0.958				0.960	−0.22 %
–	–	1.00	Prandtl (1920): plasticity theory			

The theoretical factor of safety is 1.0; limit equilibrium with slices gives about 4 % less on this mechanism.

3.140 V079: Priest (1993) rigid block on a plane, water-filled crack

Priest, S.D. (1993). Discontinuity Analysis for Rock Engineering. Chapman & Hall, London; example 8.1, pp. 224–225.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Priest's example 8.1 as two commercial programs model it: a 30 m face with the crest at x = 17.0 (60.46°, Priest's is 60° with the crest at 17.32, see V080), a 30° plane from the toe, a vertical 15 m tension crack holding 3.75 m of water, and a water table rising from the toe to 3.75 m above the plane at the crack, which gives Priest's linear pressure along the plane.

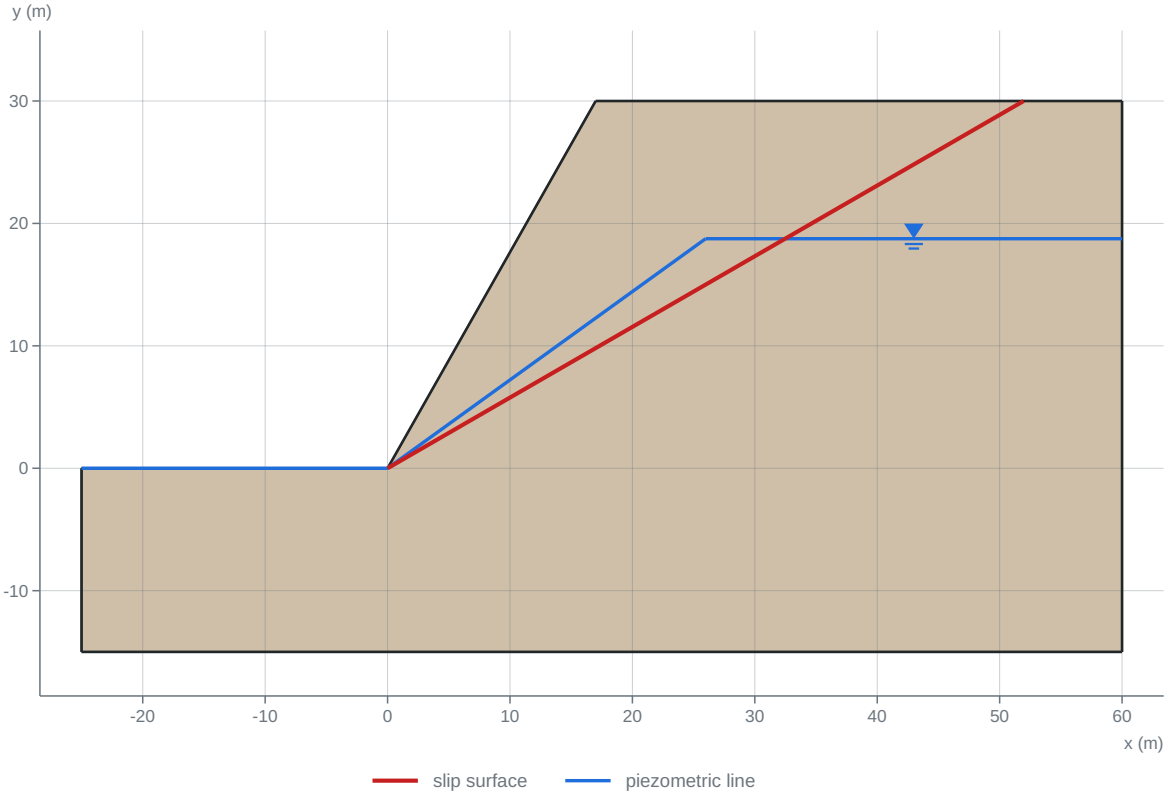


Figure 141. V079: analyzed surface (Janbu simplified, F = 1.048).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Block	Mohr-Coulomb	25.00	c′ = 20.00	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu simplified	1.048				1.049	−0.07 %
Spencer	1.048				1.049	−0.07 %
–	–	1.049	Priest (1993): rigid block, force equilibrium			

3.141 V080: Priest (1993) rigid block, reference geometry (60° face)

Priest, S.D. (1993). Discontinuity Analysis for Rock Engineering. Chapman & Hall, London; example 8.1, pp. 224–225.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Priest's example 8.1 with his geometry: a 60° face 30 m high (crest at x = 17.321), a 30° plane from the toe, a vertical 15 m tension crack 8.66 m behind the crest holding 3.75 m of water, and a water table rising from the toe to 3.75 m above the plane at the crack.

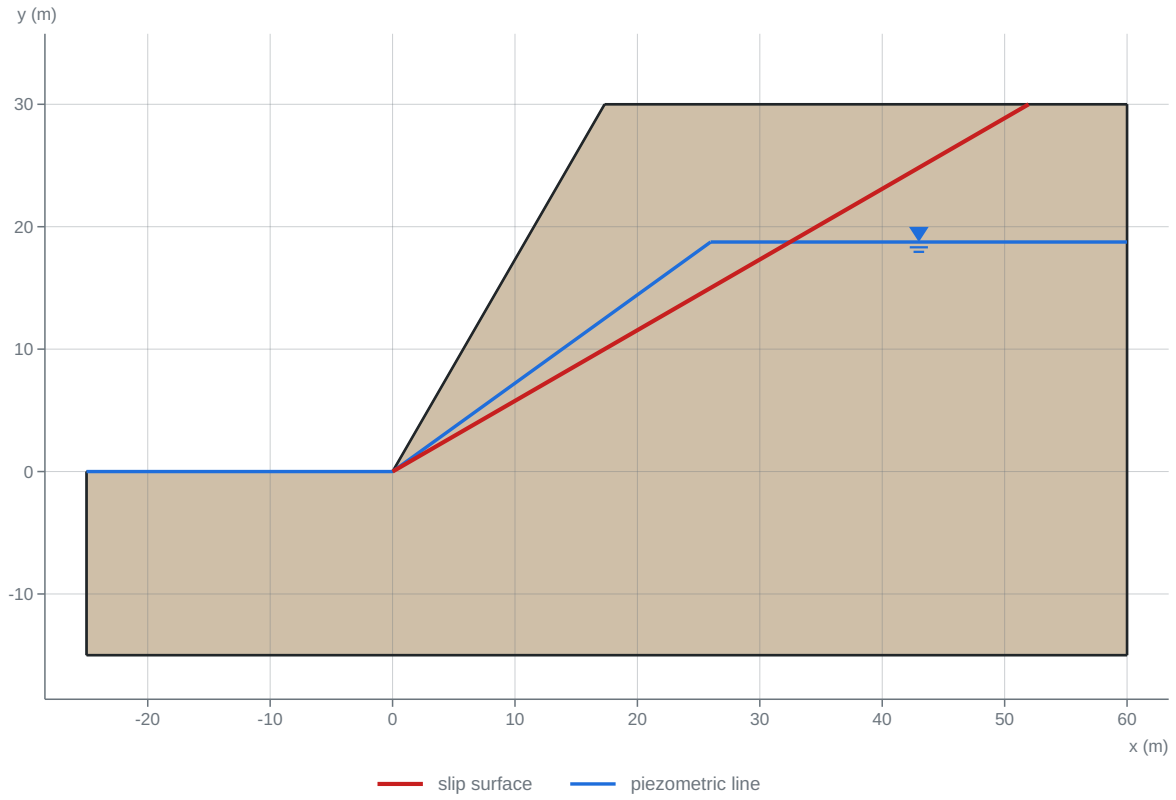


Figure 142. V080: analyzed surface (Morgenstern-Price (half-sine), F = 1.049).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Block	Mohr-Coulomb	25.00	c′ = 20.00	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Morgenstern-Price (half-sine)	1.049				1.049	0.00 %
Spencer	1.049	1.049	Priest	0.00 %		
–	–	1.049	Priest (1993): rigid block, force equilibrium			

3.142 V081a: Sharma (1996) reinforcement example, unreinforced

Sharma (1996).

Fixed circle · Units: SI (m, kPa, kN/m³)

45° cohesionless sand slope with a 40 kPa surcharge behind the crest, on the published toe circle.

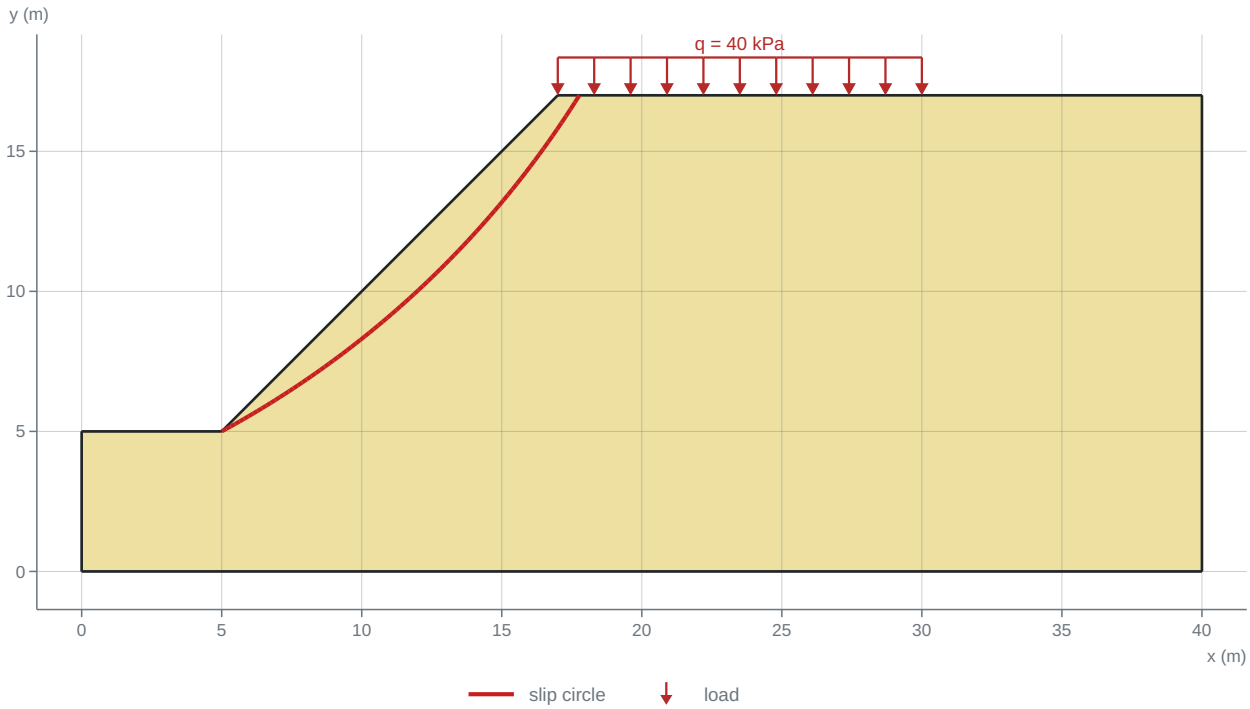


Figure 143. V081a: analyzed circle (Bishop simplified, F = 0.764).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Mohr-Coulomb	20.00	c′ = 0.00	36.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	0.764				0.764	+0.03 %

The source compares programs and publishes no independent value.

3.143 V081b: Sharma (1996) slope with a reinforced zone (φ = 54.93°)

Sharma (1996). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V081a sand slope (φ′ = 36°, 40 kPa surcharge) with a 7.6 m reinforced zone of φ = 54.93° (tan φ = 1.96 tan 36°) behind the face, on the reference critical toe circle (8.707, 18.307, R = 13.813).

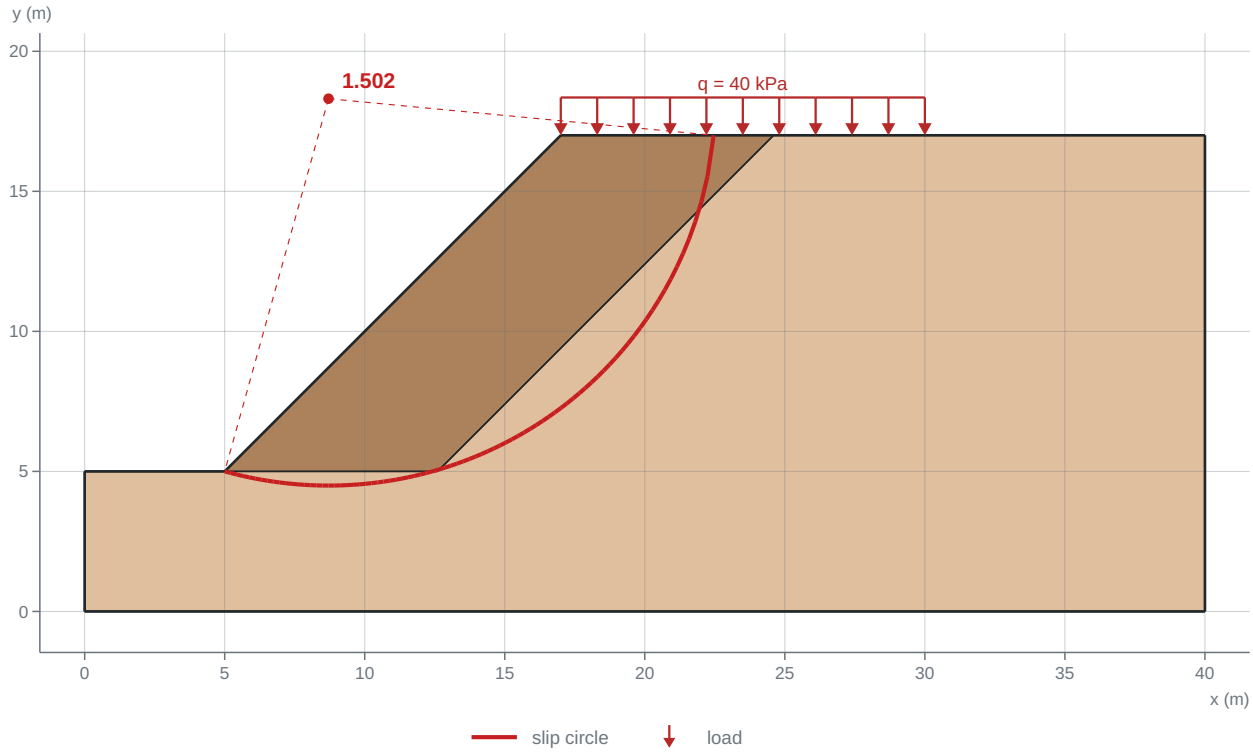


Figure 144. V081b: analyzed circle (Bishop simplified, F = 1.502).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	20.00	c′ = 0.00	36.00°
Reinforced zone	Mohr-Coulomb	20.00	c′ = 0.00	54.93°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.502				1.504	−0.12 %

The source compares programs and publishes no independent value.

3.144 V082a: Sharma (1996) slope, 11 ft dry tension crack instead of the zero-strength cap

Sharma (1996); Malkawi et al. (2001). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V082c slope with the zero-strength cap replaced by an 11 ft dry tension crack (Table 27.4, analysis 3), on the V082c circle; phreatic correction. Converted to SI.

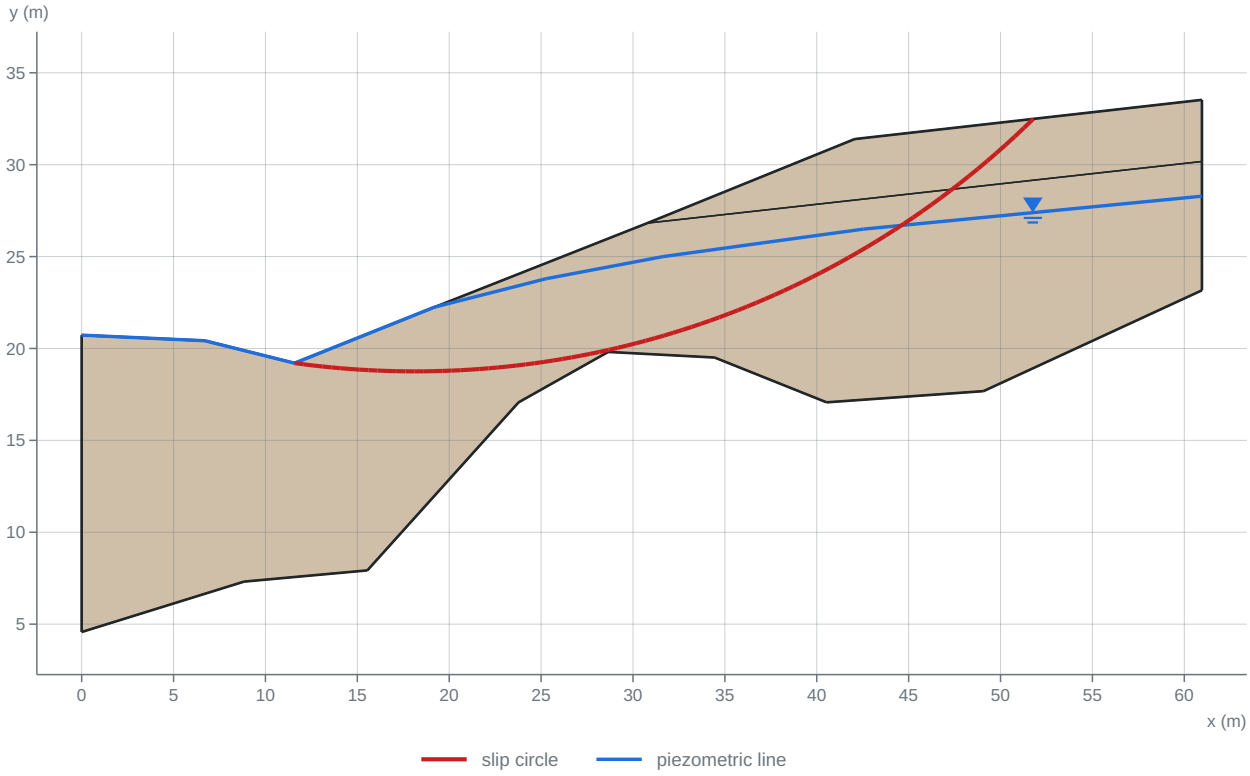


Figure 145. V082a: analyzed circle (Bishop simplified, F = 1.534).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.28	c′ = 23.94	14.00°
Soil 2 (zero strength)	Undrained (φ = 0)	18.28	Su = 0.00	0°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.534				1.532	+0.14 %
Spencer	1.533				1.532	+0.08 %
Morgenstern-Price (half-sine)	1.533				1.532	+0.08 %
Janbu corrected	1.546				1.544	+0.16 %

The source compares programs and publishes no independent value.

3.145 V082b: Sharma (1996) slope, 11 ft tension crack with 6 ft of water

Sharma (1996); Malkawi et al. (2001). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V082a with 6 ft of water in the 11 ft crack (Table 27.4, analysis 4).

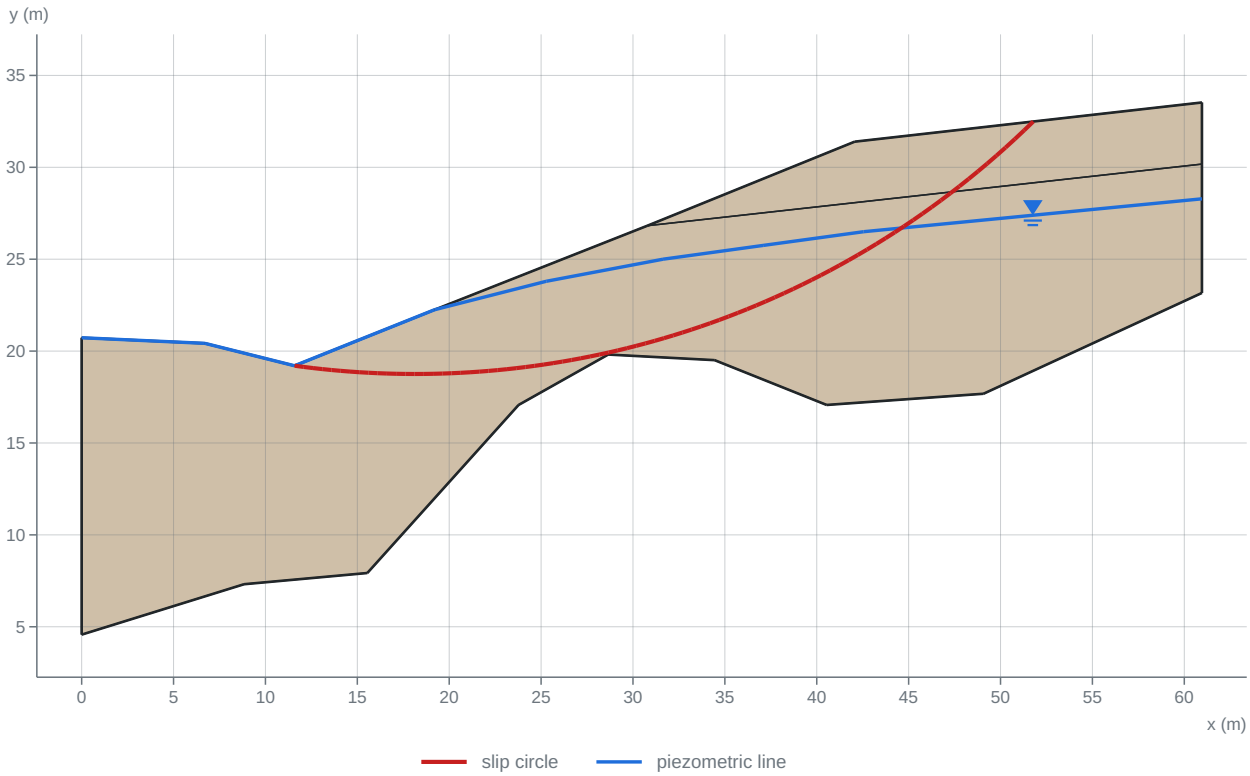


Figure 146. V082b: analyzed circle (Bishop simplified, F = 1.513).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.28	c′ = 23.94	14.00°
Soil 2 (zero strength)	Undrained (φ = 0)	18.28	Su = 0.00	0°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.513				1.511	+0.13 %
Spencer	1.512				1.510	+0.13 %
Morgenstern-Price (half-sine)	1.512				1.510	+0.14 %
Janbu corrected	1.522				1.520	+0.12 %

The source compares programs and publishes no independent value.

3.146 V082c: Sharma (1996) slope over undulating bedrock, zero-strength cap

Sharma (1996); Malkawi et al. (2001).

Fixed circle · Units: SI (m, kPa, kN/m³)

Two-material slope over undulating bedrock with a zero-strength cap layer and a water table (phreatic correction); soil 1 has moist/saturated unit weights 116.4/124.2 pcf. Specified circle (59.52, 219.21, R = 157.68). Converted to SI.

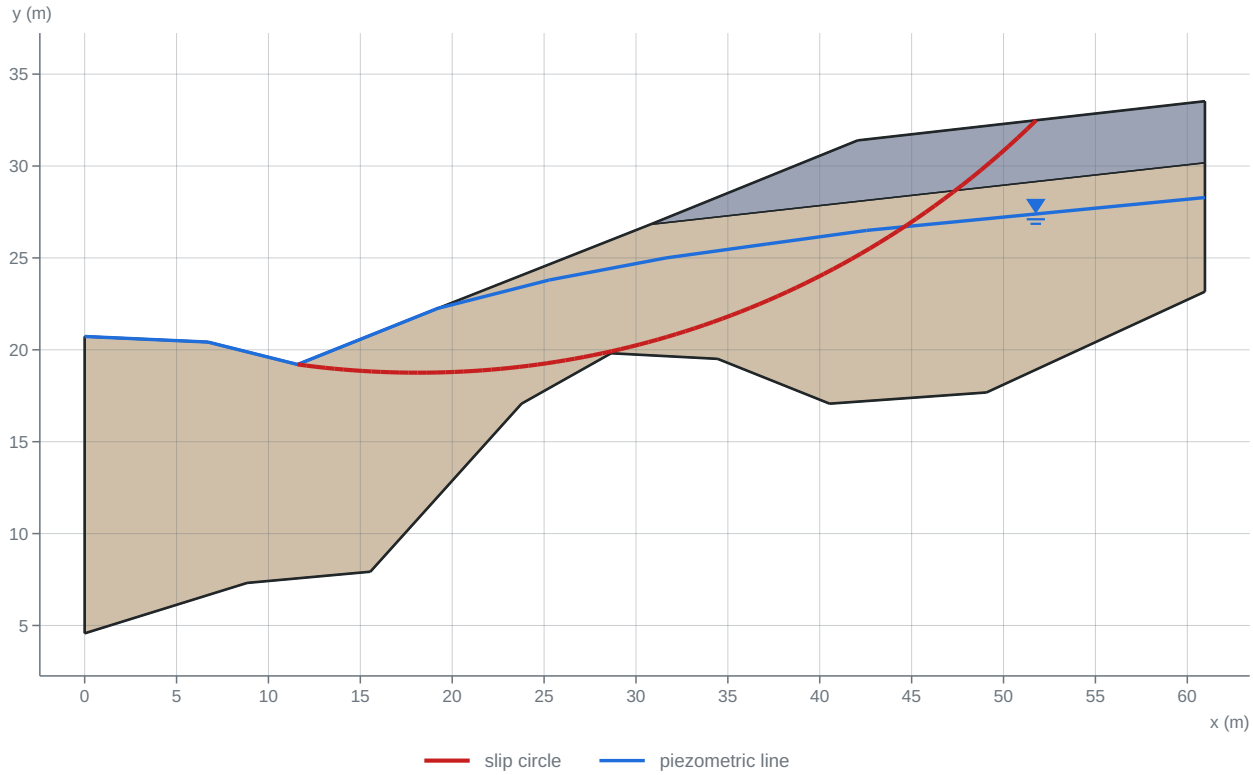


Figure 147. V082c: analyzed circle (Bishop simplified, F = 1.396).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.28	c′ = 23.94	14.00°
Soil 2 (zero strength)	Undrained (φ = 0)	18.28	Su = 0.00	0°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.396				1.396	+0.02 %
Spencer	1.403				1.402	+0.04 %
Janbu corrected	1.392				1.391	+0.09 %
Morgenstern-Price (half-sine)	1.399				1.398	+0.04 %

The source compares programs and publishes no independent value.

3.147 V082d: Sharma (1996) slope over undulating bedrock, circular search with exit limits

Sharma (1996); Malkawi et al. (2001).

Critical circle search · Units: SI (m, kPa, kN/m³)

The V082c slope (zero-strength cap, water table with phreatic correction) with a circular search whose circles must exit between x = 38 and 70 ft at the toe and 120 and 180 ft at the crest (Table 27.3). Converted to SI.

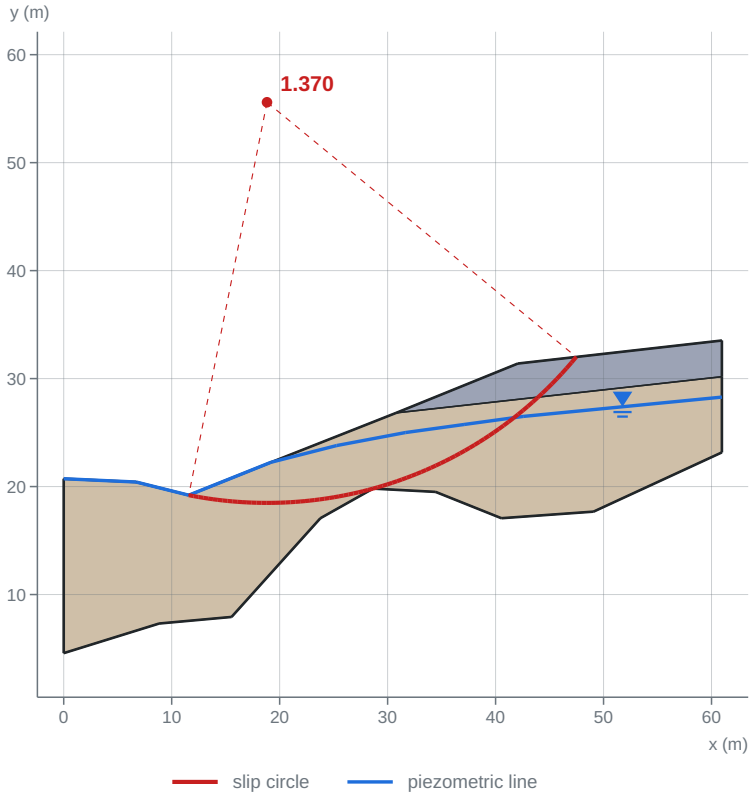


Figure 148. V082d: critical circle (Bishop simplified, F = 1.370).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.28	c′ = 23.94	14.00°
Soil 2 (zero strength)	Undrained (φ = 0)	18.28	Su = 0.00	0°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.370				1.376	−0.45 %
Spencer	1.376				1.382	−0.42 %
Morgenstern-Price (half-sine)	1.371				1.378	−0.48 %
Janbu corrected	1.344				1.345	−0.10 %

The source compares programs and publishes no independent value.

3.148 V083: Soil-nailed wall in clay (Amherst test wall)

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

6.1 m vertical cut in undrained clay (Su = 25 kPa) with two nail rows at 20° (tensile 118 kN, plate 86 kN, bond 15 kN/m, spacing 1.5 m; axial, passive) and the facing weight as a line load; the critical plane printed by a commercial program, from the toe (0, 0) to the crest (6.279, 6.1), 44.2°.

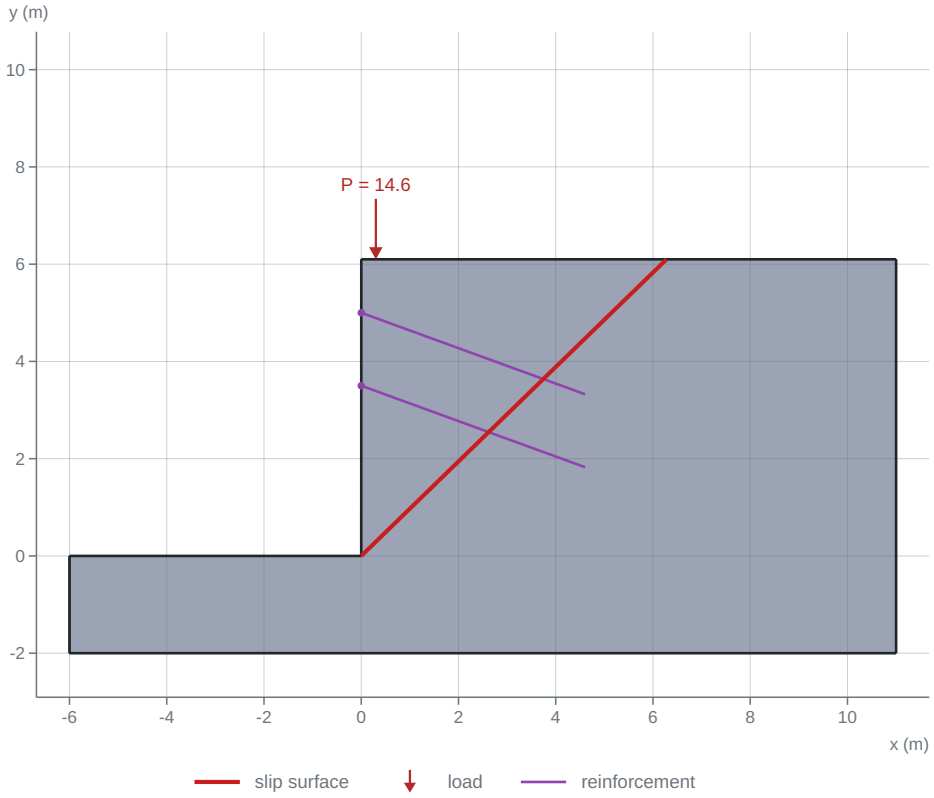


Figure 149. V083: analyzed surface (Janbu corrected, F = 0.885).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Amherst Clay	Undrained (φ = 0)	18.90	Su = 25.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	0.884				0.890	−0.62 %
–	–	0.887	Sheahan & Ho (2003): force-equilibrium trial wedge			

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.149 V084a: Soil-nailed wall in sand (Clouterre test wall)

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

7 m cut in sand (c′ = 3 kPa, φ′ = 38°) with seven passive nail rows at 10° (tensile 15 kN, plate 59 kN, bond 7.5 kN/m, i.e. the tensile capacity develops over 2 m; spacing 1.15 m) and the facing weight; 55° plane through the toe. V084b … V084f are the other planes of Table 48.3.

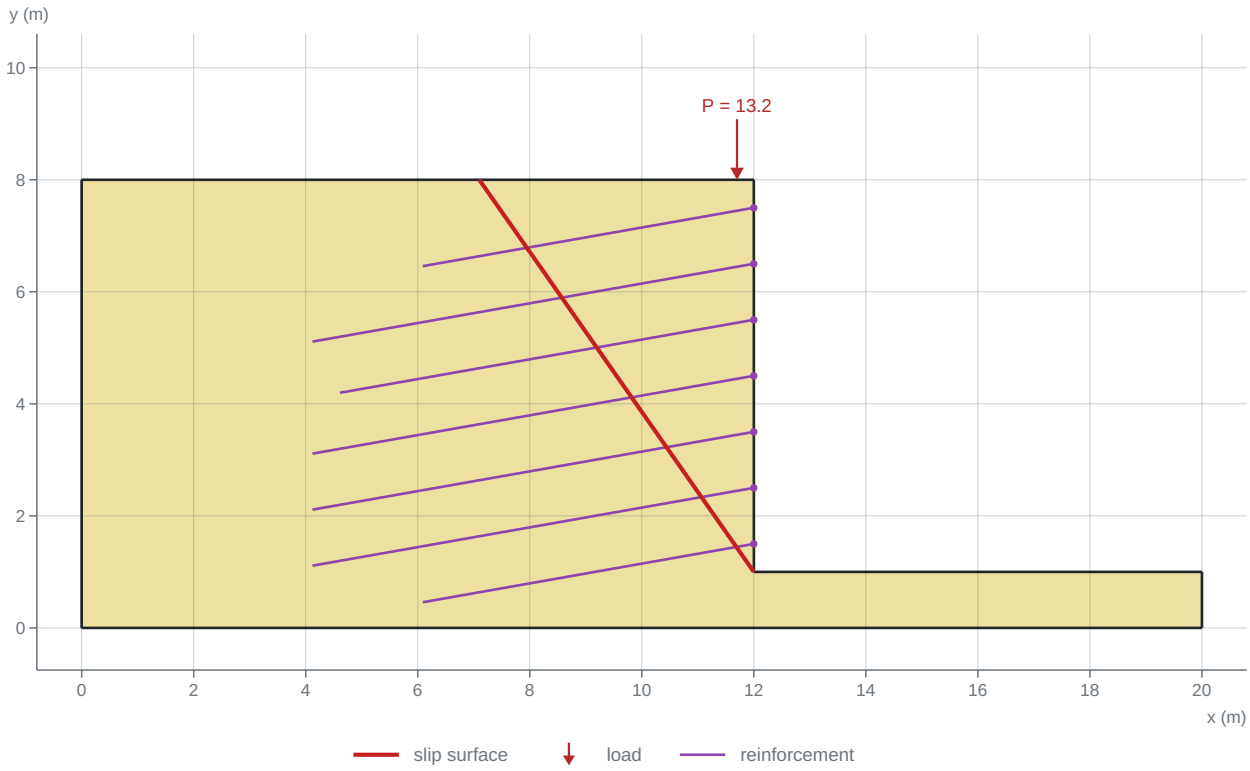


Figure 150. V084a: analyzed surface (Janbu corrected, F = 0.988).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c′ = 3.00	38.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	0.988				0.989	−0.06 %
–	–	0.989	Sheahan & Ho (2003): force-equilibrium trial wedge			

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.150 V084b: Soil-nailed wall in sand (Clouterre test wall), 45° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 45° plane through the toe.

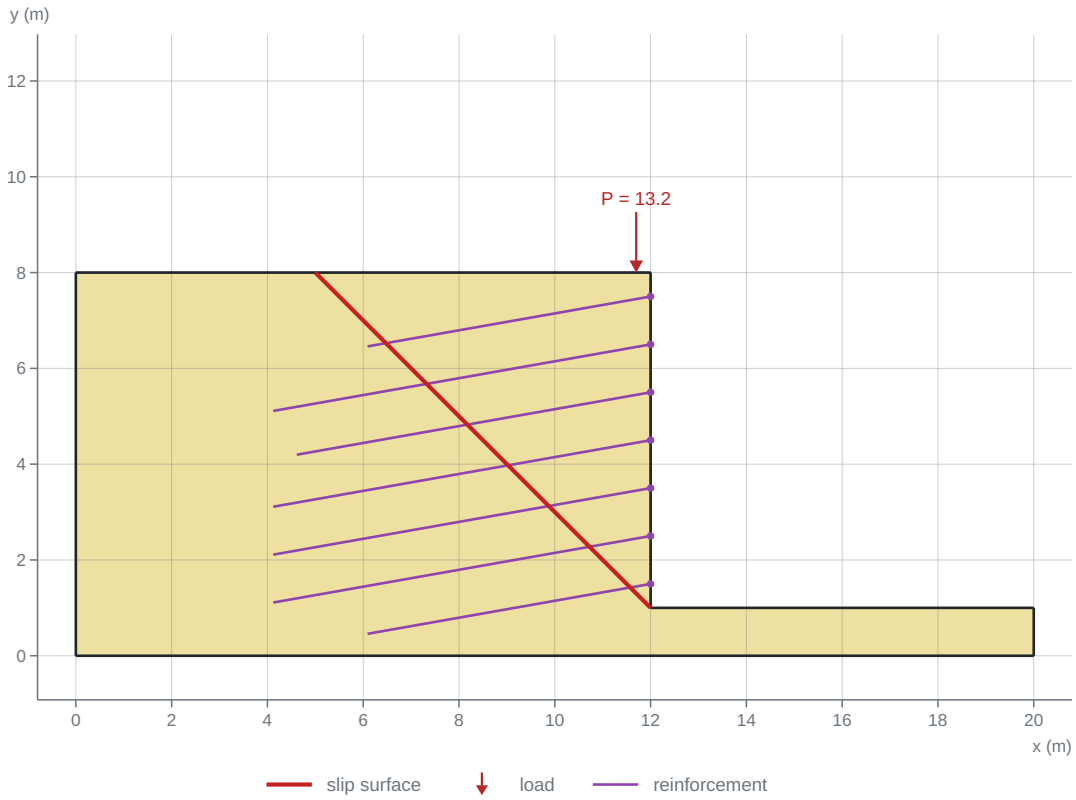


Figure 151. V084b: analyzed surface (Janbu corrected, F = 1.124).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c′ = 3.00	38.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	1.124				1.123	+0.12 %
–	–	1.176	Sheahan & Ho (2003): force-equilibrium trial wedge			

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 5 %.

3.151 V084c: Soil-nailed wall in sand (Clouterre test wall), 50° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 50° plane through the toe.

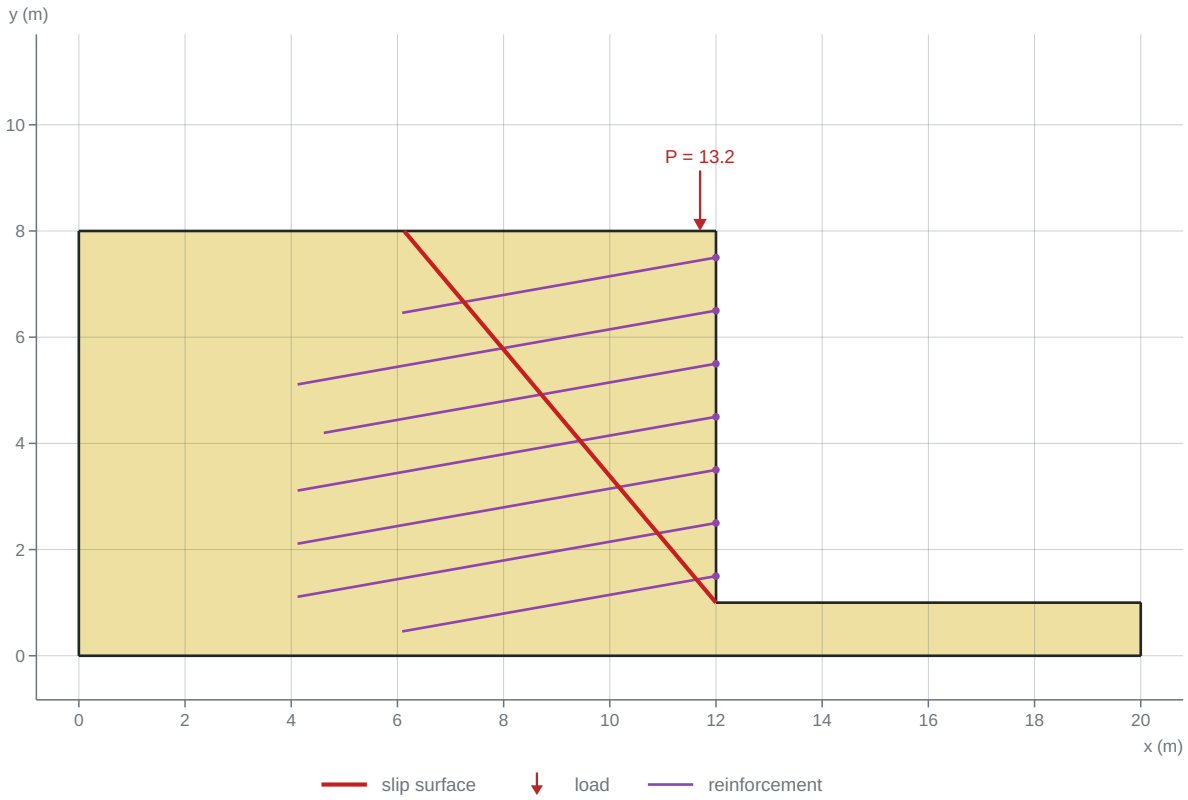


Figure 152. V084c: analyzed surface (Janbu corrected, F = 1.043).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c′ = 3.00	38.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	1.043				1.043	+0.03 %
–	–	1.07	Sheahan & Ho (2003): force-equilibrium trial wedge			

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.152 V084d: Soil-nailed wall in sand (Clouterre test wall), 60° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 60° plane through the toe.

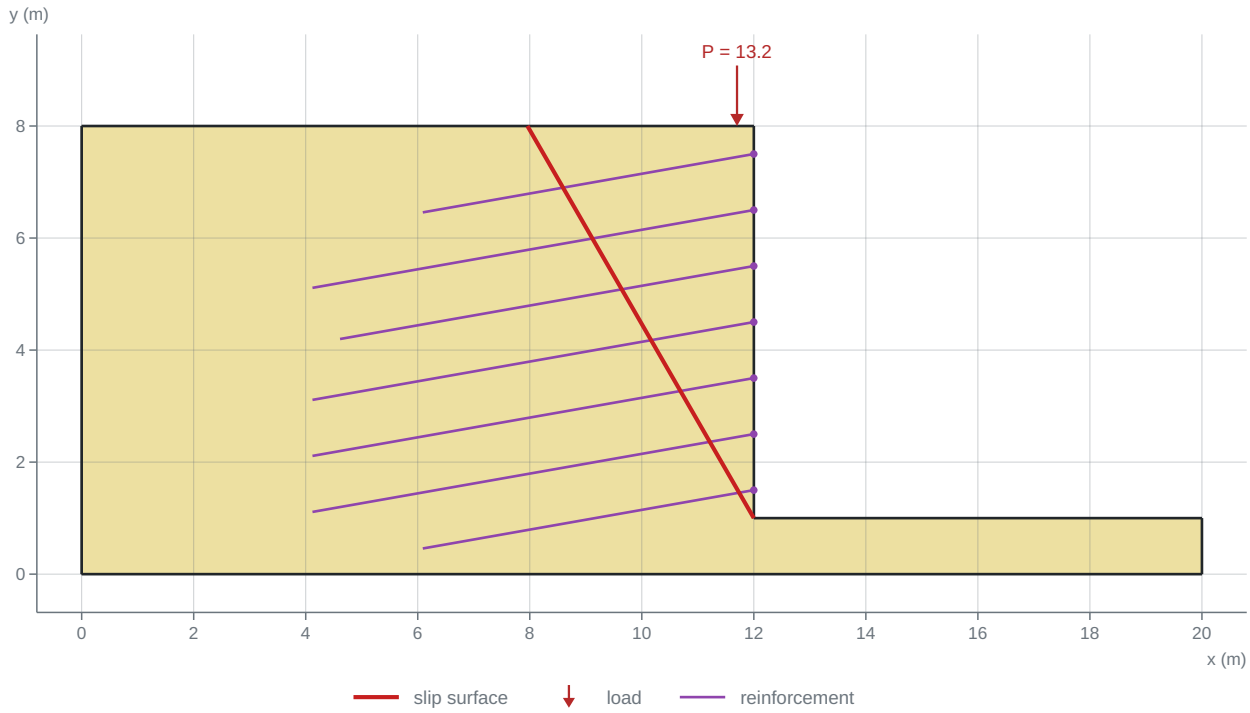


Figure 153. V084d: analyzed surface (Janbu corrected, F = 0.944).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c′ = 3.00	38.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	0.944				0.945	−0.07 %
–	–	0.929	Sheahan & Ho (2003): force-equilibrium trial wedge			

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.153 V084e: Soil-nailed wall in sand (Clouterre test wall), 65° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 65° plane through the toe.

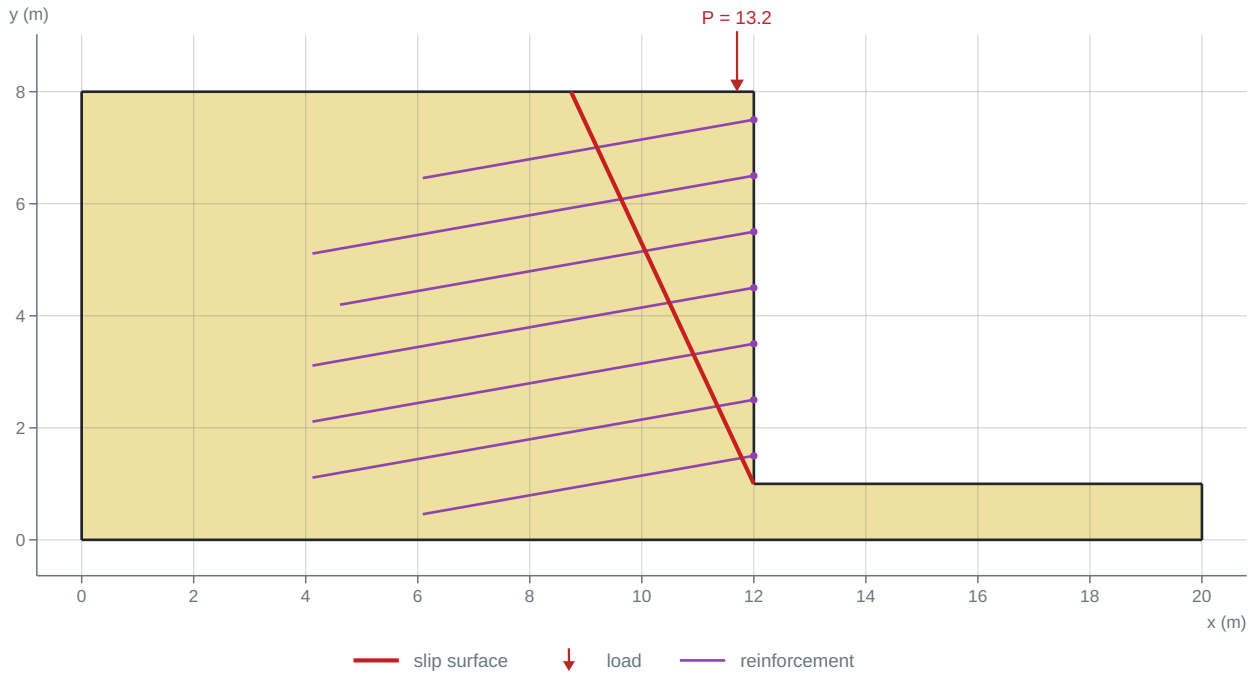


Figure 154. V084e: analyzed surface (Janbu corrected, F = 0.920).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c′ = 3.00	38.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	0.920				0.922	−0.23 %
–	–	0.893	Sheahan & Ho (2003): force-equilibrium trial wedge			

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.154 V084f: Soil-nailed wall in sand (Clouterre test wall), 70° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 70° plane through the toe.

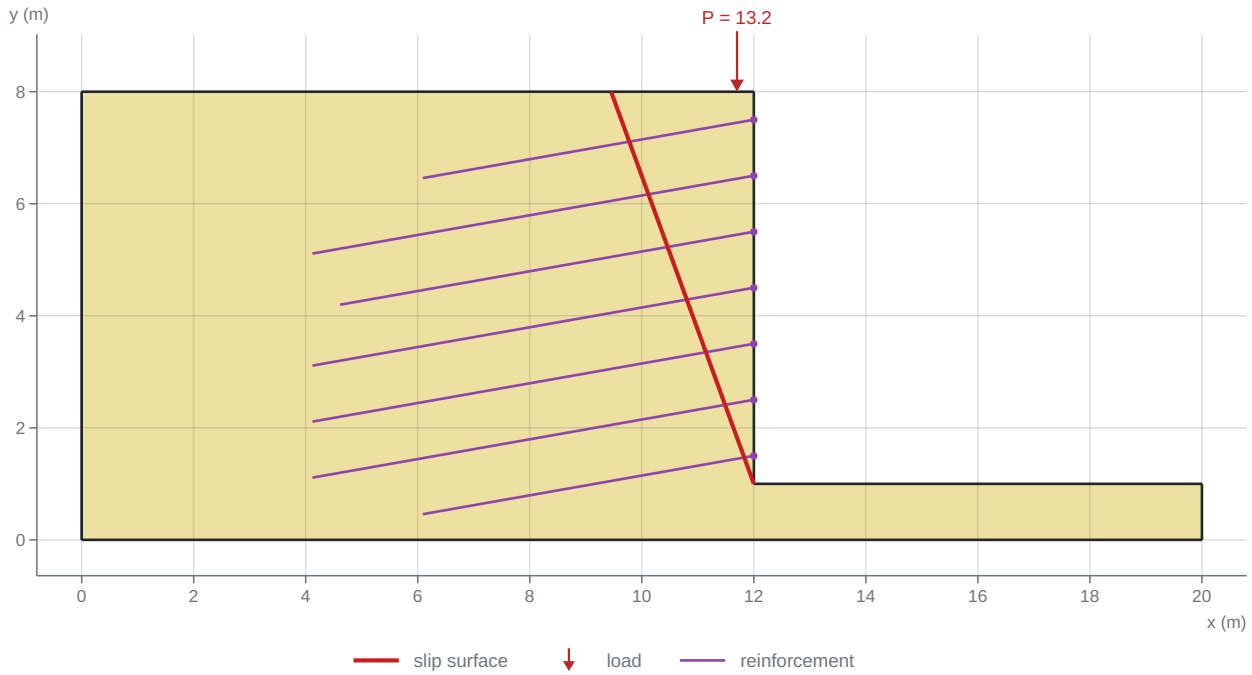


Figure 155. V084f: analyzed surface (Janbu corrected, F = 0.920).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c′ = 3.00	38.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	0.920				0.923	−0.27 %
–	–	0.887	Sheahan & Ho (2003): force-equilibrium trial wedge			

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.155 V085: Bradwell excavated slope, stiff-fissured London clay

Skempton, A.W. & La Rochelle, P. (1965). The Bradwell slip: a short-term failure in London clay. Géotechnique 15(3), 221–242; Duncan, Wright & Brandon (2014), Soil Strength and Slope Stability, 2nd ed., §7.7.3.

Critical circle search · Units: imperial (ft, psf, pcf)

Excavation in London clay (six sublayers with Su increasing with depth, S = f(datum)) under Marsh clay and a clay fill cracked to its full depth (11.4 ft); search for the deep failure, entering on the crest and leaving at the excavation toe.

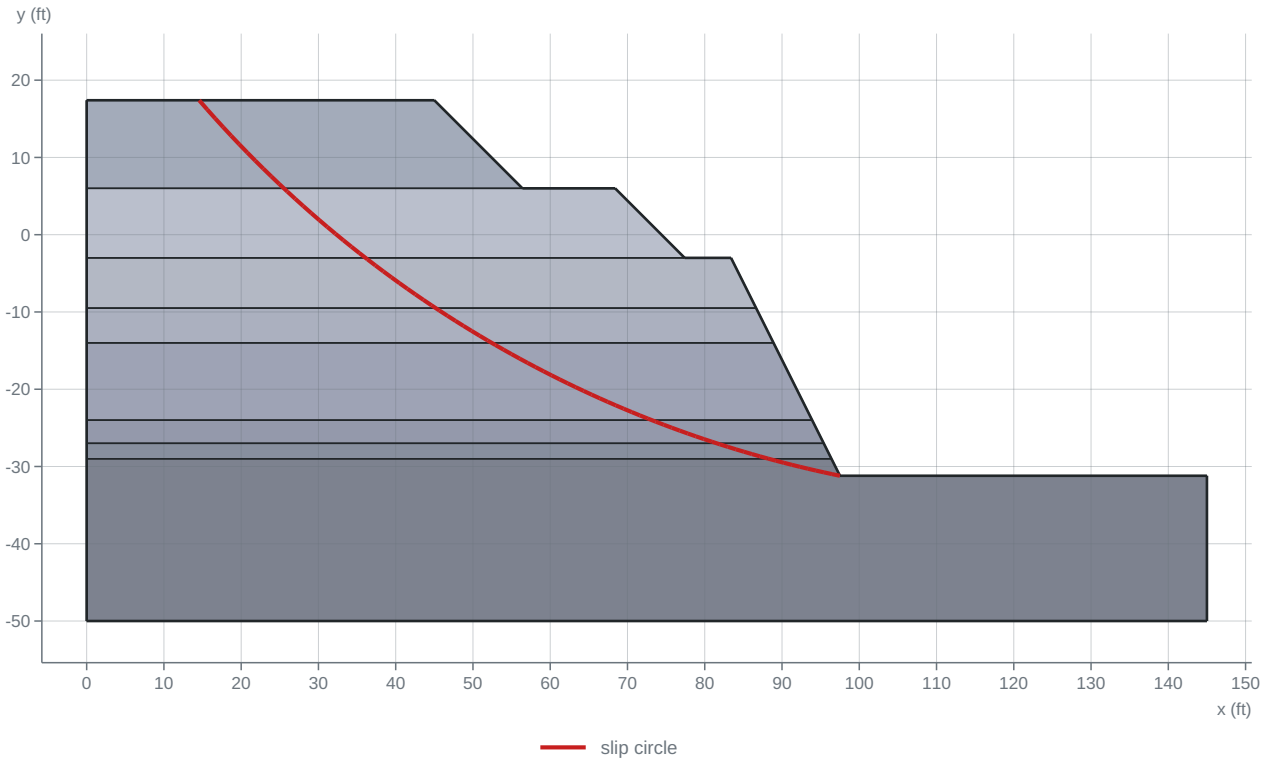


Figure 156. V085: critical circle (Bishop simplified, F = 1.759).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay fill	Mohr-Coulomb	110.0	c′ = 1.00	35.00°
Marsh clay	Mohr-Coulomb	105.0	c′ = 300.0	0.00°
Brown London clay 1	S = f(datum)	120.0	Su = 750.0 at y = -3.00, 90.00 psf/ft below	0°
Brown London clay 2	S = f(datum)	120.0	Su = 1335.0 at y = -9.50, 82.00 psf/ft below	0°
Brown London clay 3	S = f(datum)	120.0	Su = 1704.0 at y = -14.00, 53.00 psf/ft below	0°
Brown London clay 4	S = f(datum)	120.0	Su = 2234.0 at y = -24.00, 47.00 psf/ft below	0°
Blue London clay 1	S = f(datum)	120.0	Su = 2375.0 at y = -27.00, 47.00 psf/ft below	0°
Blue London clay 2	S = f(datum)	120.0	Su = 2469.0 at y = -29.00, 39.00 psf/ft below	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.759	1.76	Duncan & Wright	−0.07 %	1.762	−0.18 %
Spencer	1.759				1.758	+0.05 %
Janbu corrected	1.735				1.736	−0.06 %
Janbu simplified	1.627	1.63	Duncan & Wright	−0.20 %	1.628	−0.08 %

3.156 V103: Spencer's slope, r_u = 0.5, non-circular critical surface

Fredlund, D.G. & Krahn, J. (1977), Can. Geotech. J. 14(3), Figs. 3 and 7; Baker, R. (1980). Determination of the critical slip surface in slope stability computations. Int. J. Numer. Anal. Meth. Geomech. 4, 333–359; Spencer (1969).

Circle search with non-circular optimization · Units: SI (m, kPa, kN/m³)

30 m high 2:1 (26.5°) homogeneous slope with c′/γH = 0.02 (c′ = 10.8 kPa, γ = 18 kN/m³), φ′ = 40° and r_u = 0.5; non-circular Spencer search. The crest and the ground beyond the toe are 10 m long and the base is 10 m below the toe, as in a program manual's model.

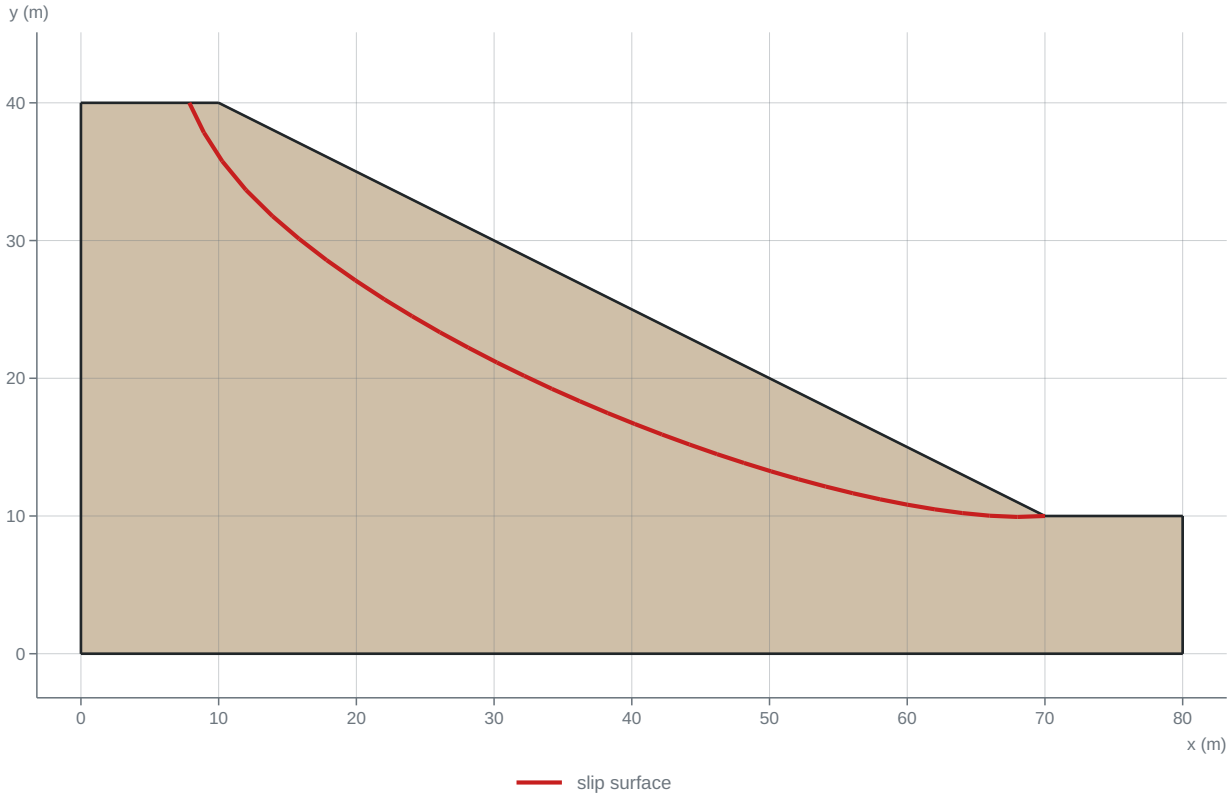


Figure 157. V103: critical surface (Spencer, non-circular, F = 1.008).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c′ = 10.80	40.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer, non-circular	1.008	1.02	Baker	−1.20 %	1.010	−0.22 %
–	–	1.08	Spencer (1969)			

Spencer's own factor of safety (1.08, quoted from a program manual) is for his own trial surface, not a critical non-circular surface, and is shown without a comparison: cSlope's critical circle gives 1.015 with Spencer's method, and the non-circular searches 1.01–1.02.

3.157 V086a: Tandjiria embankment, clay fill, geosynthetic T = 169 kN/m

Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V086b clay-fill embankment with a 169 kN/m geosynthetic at its base (axial, active, fully anchored) and the water-filled 2 m crack, on the critical circle.

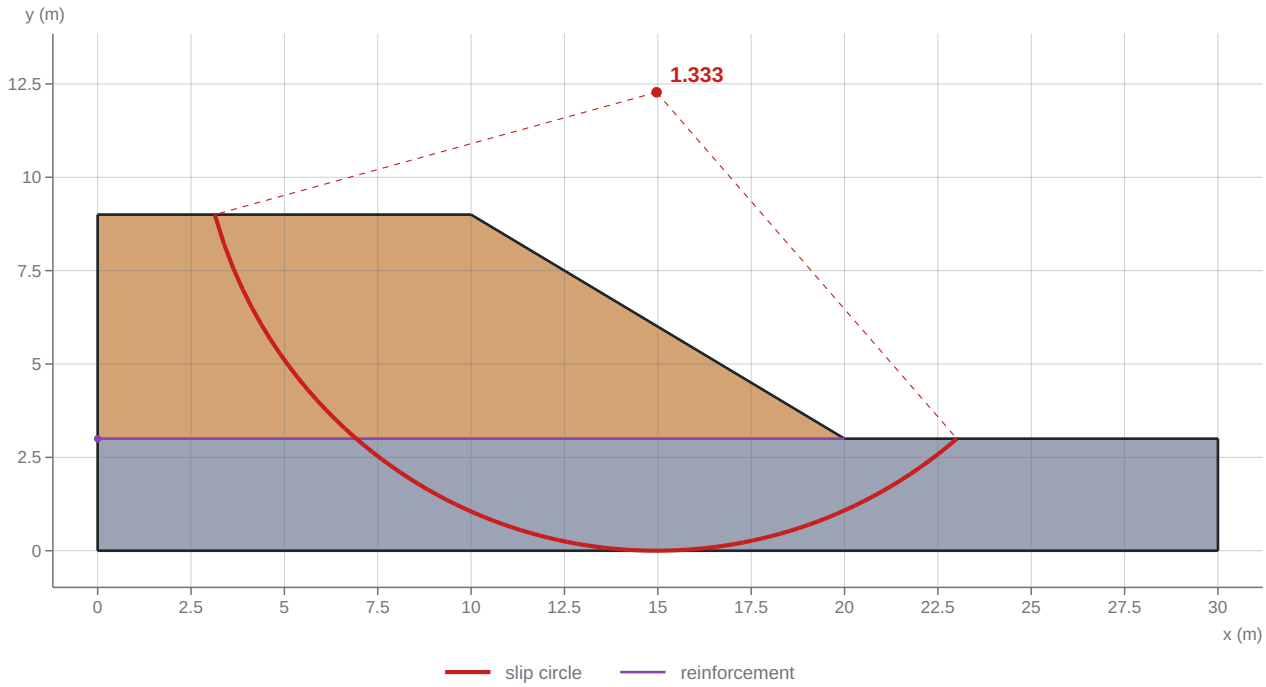


Figure 158. V086a: analyzed circle (Spencer, F = 1.333).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Undrained (φ = 0)	19.40	Su = 20.00	0°
Soft Clay	Undrained (φ = 0)	19.40	Su = 20.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.333				1.332–1.350	−0.57 %
–	–	1.35	Tandjiria, Low & Teh (2002): generalized method of slices, at T = 170.5 kN/m			

3.158 V086b: Tandjiria embankment, clay fill, water-filled crack

Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443.

Critical circle search · Units: SI (m, kPa, kN/m³)

Half-embankment (9 m crest, 10 m wide crest, 1V:1.67H face) of clay fill (Su = 20 kPa) on soft clay (Su = 20 kPa), γ = 19.4; 2 m water-filled tension crack, unreinforced; circular search. Also run with the automatic (Rankine) crack depth.

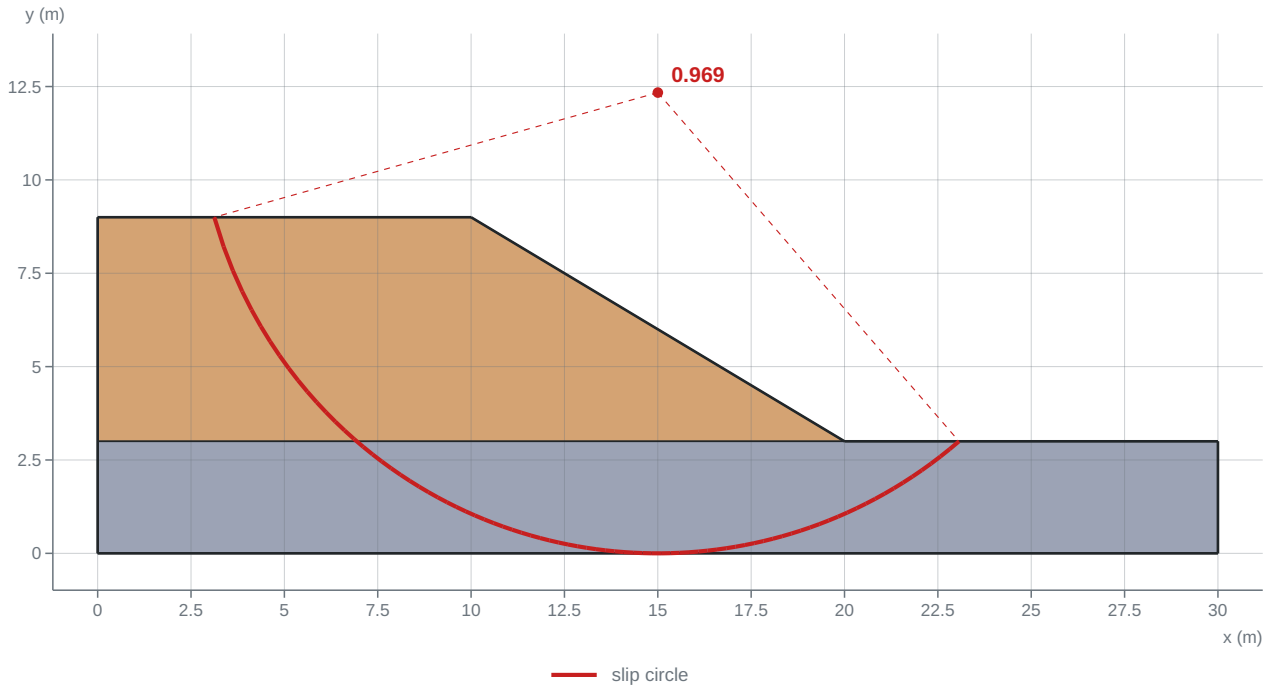


Figure 159. V086b: critical circle (Spencer (2 m tension crack), F = 0.969).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Undrained (φ = 0)	19.40	Su = 20.00	0°
Soft clay	Undrained (φ = 0)	19.40	Su = 20.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer (2 m tension crack)	0.969	0.981	Tandjiria, Low & Teh (2002): Spencer	−1.27 %	0.975	−0.67 %
Spencer (tension crack of automatic depth)	0.967				0.975	−0.86 %
Morgenstern-Price (half-sine)	0.969				0.975	−0.67 %

3.159 V086c: Tandjiria embankment, sand fill, dry crack

Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V086 embankment as a sand fill (φ′ = 37°, γ = 17) on soft clay (Su = 20 kPa, γ = 20) with a dry 2 m tension crack, unreinforced; circular search.

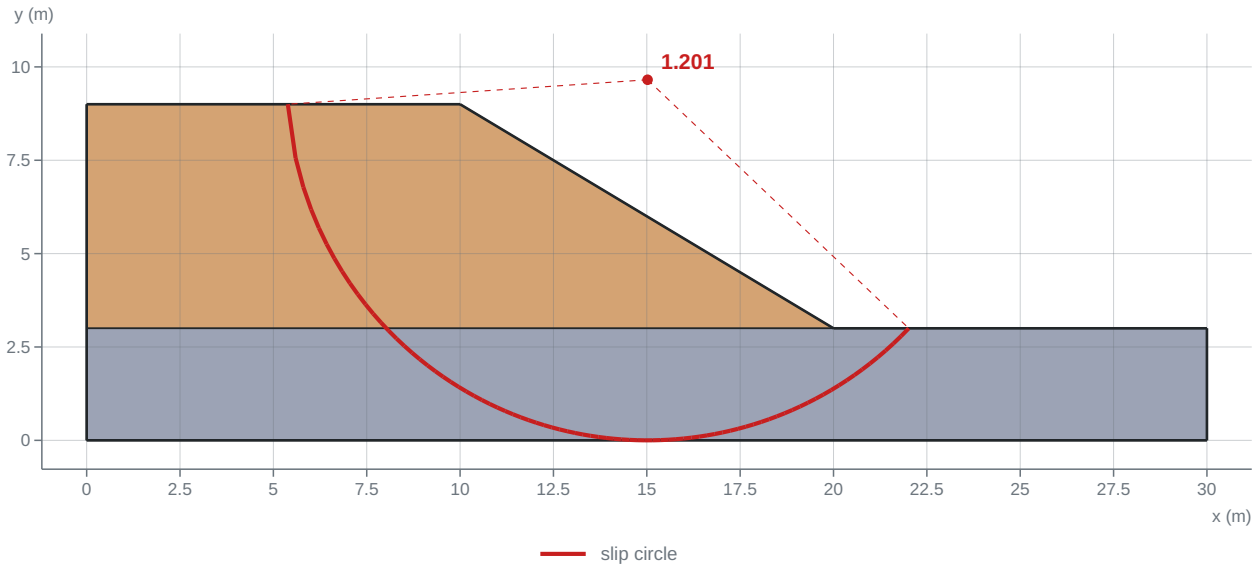


Figure 160. V086c: critical circle (Spencer, F = 1.201).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Mohr-Coulomb	17.00	c′ = 0.00	37.00°
Soft clay	Undrained (φ = 0)	20.00	Su = 20.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.200	1.219	Tandjiria, Low & Teh (2002)	−1.52 %	1.209	−0.70 %
Morgenstern-Price (half-sine)	1.211				1.218	−0.58 %

3.160 V086d: Tandjiria embankment, clay fill, the paper's geometry

Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443; problem 1, Table 2.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V086b clay-fill embankment as the paper gives it: a 30° face (p. 431, Fig. 4) and a water-filled tension crack 2c/γ = 2.06 m deep (Table 2 note), unreinforced; circular search.

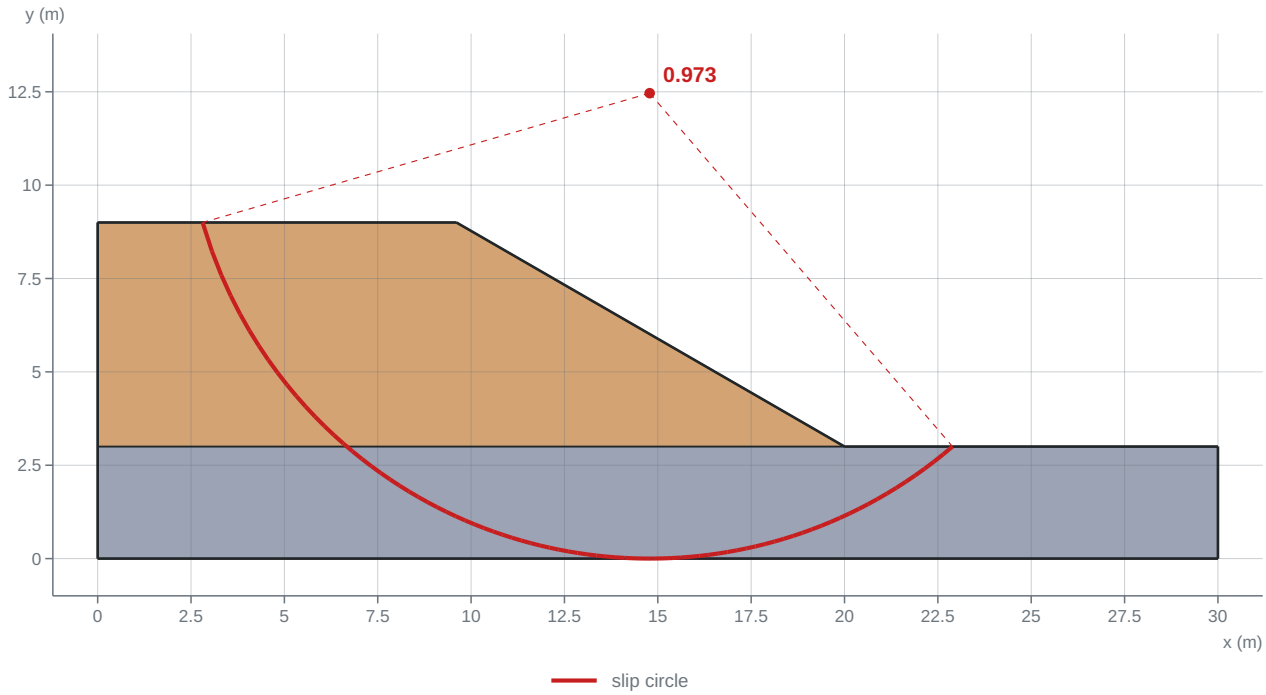


Figure 161. V086d: critical circle (Spencer, F = 0.973).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Undrained (φ = 0)	19.40	Su = 20.00	0°
Soft clay	Undrained (φ = 0)	19.40	Su = 20.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	0.973	0.981	Tandjiria, Low & Teh	−0.80 %		

The paper uses the generalized method of slices without naming its side-force function; Spencer's method is cSlope's nearest. The soft clay's unit weight is not printed and is taken as the fill's (19.4).

3.161 V086e: Tandjiria embankment, sand fill, the paper's geometry

Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443; problem 2.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V086c sand-fill embankment as the paper gives it: a 30° face and no tension crack (the paper mentions one only for the clay fill), unreinforced; circular search.

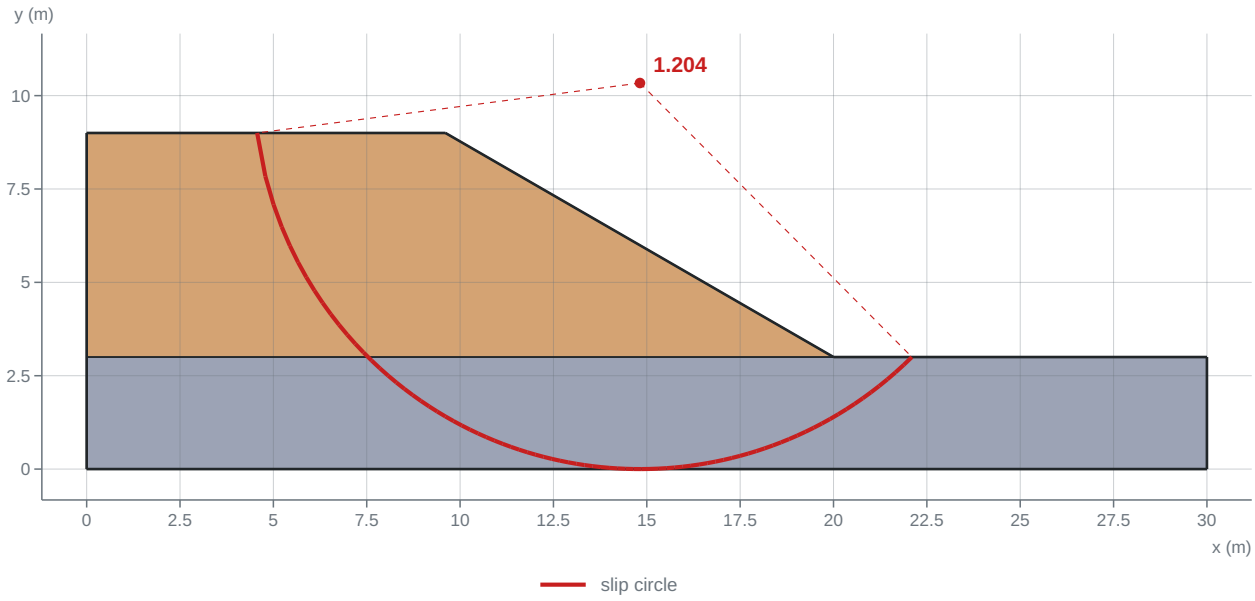


Figure 162. V086e: critical circle (Spencer, F = 1.204).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Mohr-Coulomb	17.00	c′ = 0.00	37.00°
Soft clay	Undrained (φ = 0)	20.00	Su = 20.00	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.204	1.219	Tandjiria, Low & Teh	−1.19 %		

The paper uses the generalized method of slices without naming its side-force function; Spencer's method is cSlope's nearest. The soft clay's unit weight is not printed; 20 kN/m³ is assumed (17 gives 0.1 % less).

3.162 V087: USACE dam, chart-check properties (EM 1110-2-1902 fig. 4-3)

USACE EM 1110-2-1902 (2003), fig. 4-3.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V088 dam family with single unit weights (embankment c = 200 psf, φ = 25°, γ = 115; sand 0/35/130; clay 0/27/115), slightly flatter faces and ponded water at el. 20 on both faces; the reference circle (−135, 169, R = 169.0), ending at the crest (−15, 50). Converted to SI.

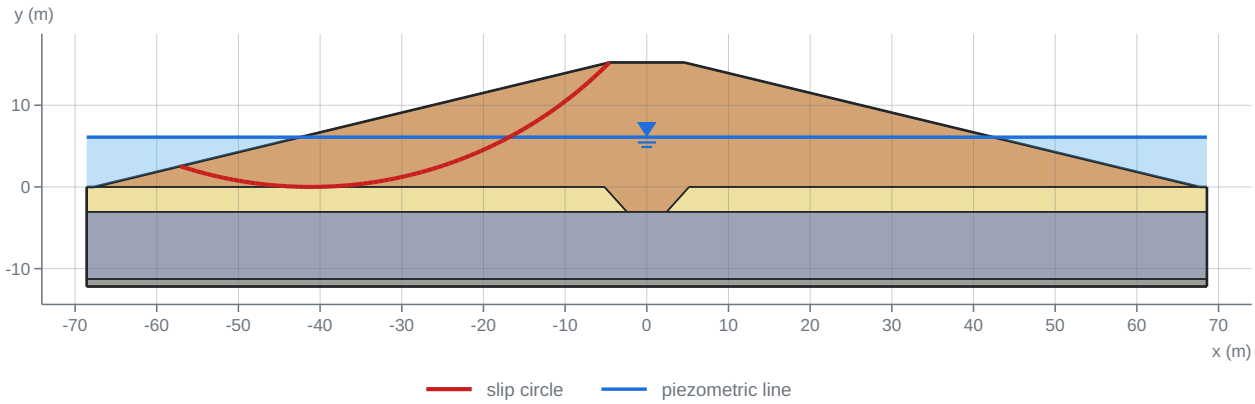


Figure 163. V087: analyzed circle (Janbu corrected, F = 2.249).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.07	c′ = 9.58	25.00°
Sand	Mohr-Coulomb	20.42	c′ = 0.00	35.00°
Foundation clay	Mohr-Coulomb	18.07	c′ = 0.00	27.00°
Rock	Mohr-Coulomb	25.13	c′ = 0.00	45.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Janbu corrected	2.248				2.290	−1.81 %
–	–	2.30	USACE EM 1110-2-1902			

3.163 V088: USACE dam, end of construction (EM 1110-2-1902 fig. 4-1)

USACE EM 1110-2-1902 (2003), fig. 4-1.

Fixed circle · Units: SI (m, kPa, kN/m³)

Symmetric 50 ft embankment (crest ±12 ft, toes ±206.18 ft) with c = 1000 psf, φ = 5°, a core trench through a 10 ft sand blanket over foundation clay (Su = 3000 psf) and rock; groundwater at the sand top, 7 ft dry crest crack; moist/saturated unit weights. Specified circle (102, 163, R = 163). Converted to SI.

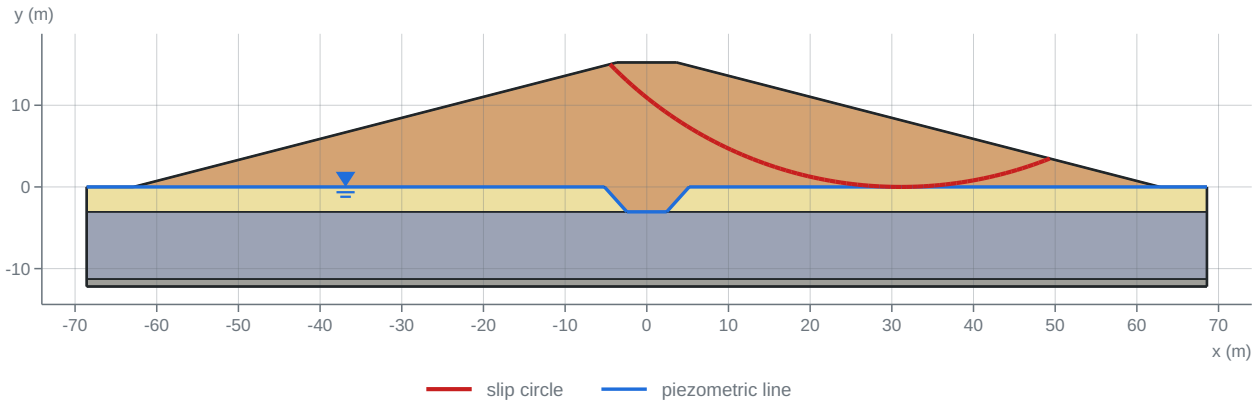


Figure 164. V088: analyzed circle (Spencer, F = 2.442).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.07	c′ = 47.88	5.00°
Sand	Mohr-Coulomb	19.64	c′ = 0.00	35.00°
Foundation clay	Undrained (φ = 0)	17.28	Su = 143.6	0°
Rock	Mohr-Coulomb	25.13	c′ = 0.00	45.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	2.442	2.44	USACE EM 1110-2-1902	+0.08 %	2.445	−0.12 %
Bishop simplified	2.443				2.447	−0.18 %
Janbu corrected	2.444				2.430	+0.59 %
Morgenstern-Price (half-sine)	2.442				2.447	−0.20 %

3.164 V089: USACE dam, upstream low pool (EM 1110-2-1902 fig. 4-2)

USACE EM 1110-2-1902 (2003), fig. 4-2.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V088 dam family, 45 ft high (crest ±15 ft, toes ±200 ft), with drained strengths (embankment c = 100 psf, φ = 25°; sand 0/35; clay 0/28; rock 0/45; moist/saturated unit weights) and the pool at el. 20 standing on the upstream face. USACE's circle (−102, 163, R = 173), tangent to the clay top. Converted to SI.

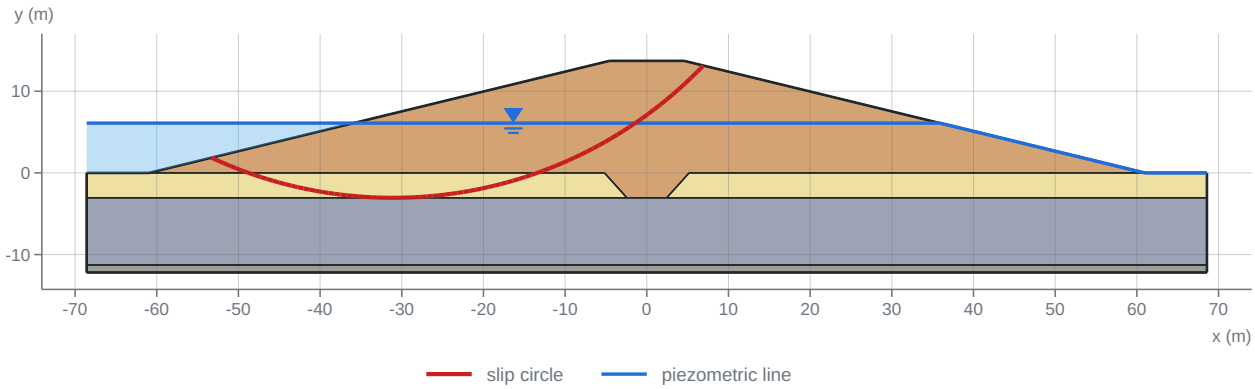


Figure 165. V089: analyzed circle (Bishop simplified, F = 2.699).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.07	c′ = 4.79	25.00°
Sand	Mohr-Coulomb	19.64	c′ = 0.00	35.00°
Foundation clay	Mohr-Coulomb	17.28	c′ = 0.00	28.00°
Rock	Mohr-Coulomb	25.13	c′ = 0.00	45.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.699	2.71	USACE EM 1110-2-1902	−0.40 %	2.716	−0.62 %
Spencer	2.722				2.736	−0.52 %
Janbu corrected	2.648				2.650	−0.06 %
Morgenstern-Price (half-sine)	2.728				2.744	−0.58 %

3.165 V090: USACE EM 1110-2-1902, example F-5

US Army Corps of Engineers (2003). Slope Stability, EM 1110-2-1902, Appendix F.

Fixed circle · Units: imperial (ft, psf, pcf)

Embankment (c = 1780 psf, φ = 5°) on a foundation (c = 1600 psf, φ = 2°), analyzed with the circle given in the Corps of Engineers example.

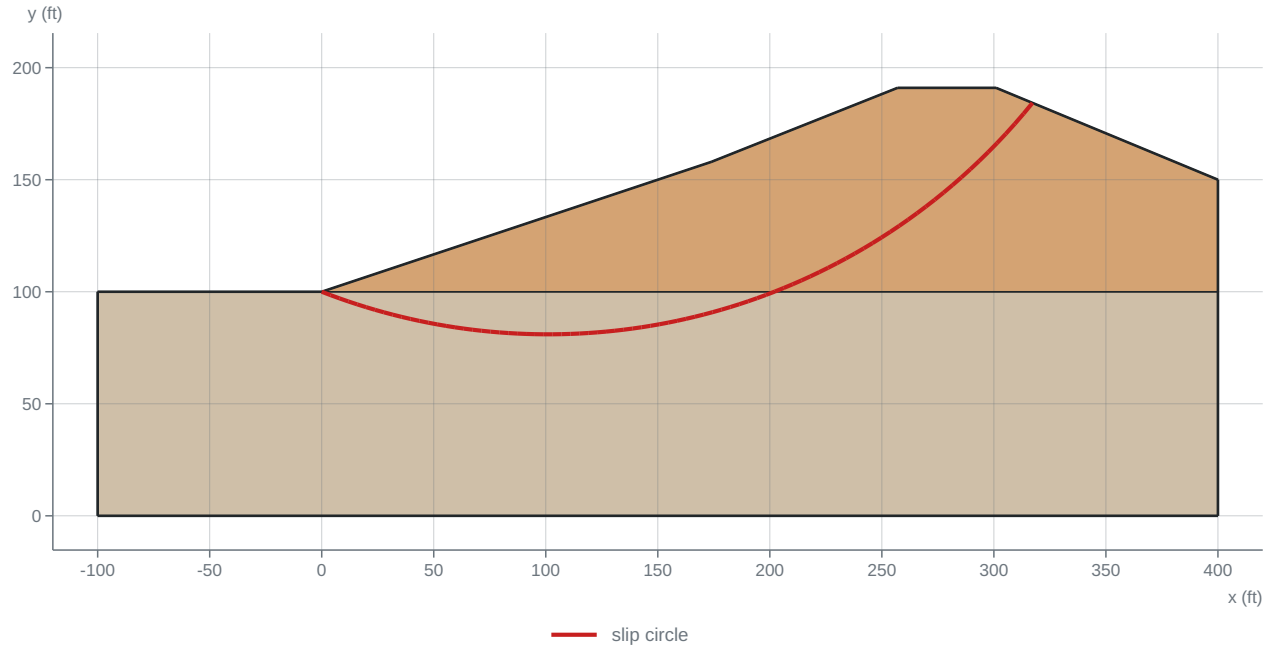


Figure 166. V090: analyzed circle (Bishop simplified, F = 1.321).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	135.0	c′ = 1780.0	5.00°
Foundation	Mohr-Coulomb	127.0	c′ = 1600.0	2.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.321				1.332	−0.85 %
Spencer	1.317				1.328	−0.84 %
Janbu corrected	1.341				1.345	−0.31 %
Morgenstern-Price (half-sine)	1.317				1.327	−0.78 %
–	–	1.33	USACE EM 1110-2-1902			

3.166 V091: USACE example E-2 (fig. E-10), ponded water

USACE EM 1110-2-1902 (2003), appendix E, example E-2, fig. E-10.

Fixed circle · Units: SI (m, kPa, kN/m³)

Undrained three-layer slope with 8 ft of water ponded against its toe (pool el. 0), on a commercial program's circle (48.262, 28, R = 47.683 ft; Fig. 68.2). Converted to SI.

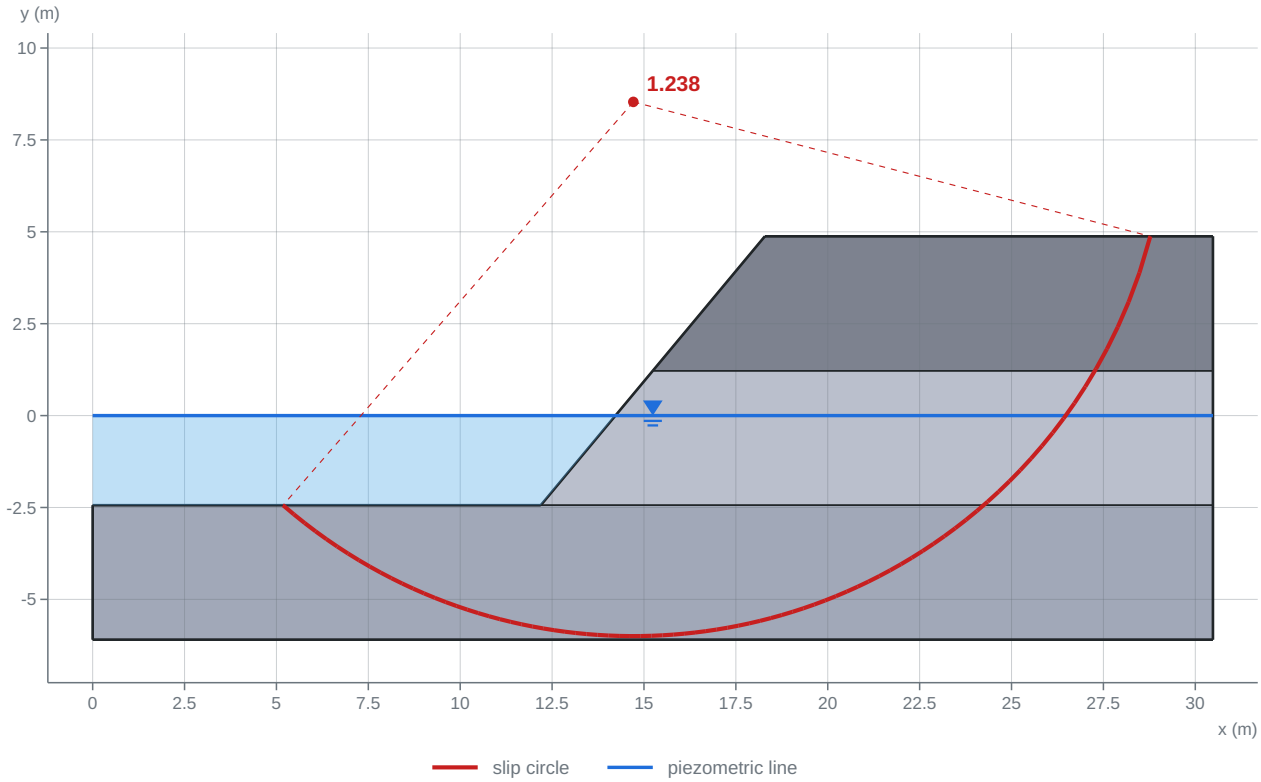


Figure 167. V091: analyzed circle (Bishop simplified, F = 1.238).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Undrained (φ = 0)	18.85	Su = 28.73	0°
Soil 2	Undrained (φ = 0)	15.71	Su = 19.15	0°
Soil 3	Undrained (φ = 0)	16.49	Su = 23.94	0°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.238				1.241	−0.27 %
Spencer	1.238				1.241	−0.27 %
Morgenstern-Price (half-sine)	1.238				1.244	−0.51 %
Janbu corrected	1.389				1.385	+0.31 %
–	–	1.24	USACE EM 1110-2-1902: stability chart (example E-2)			

USACE's chart solution for this circle (example E-2, p. E-20) gives 1.24; the programs and cSlope give 1.235–1.244 on it. The manual's shallower circle tangent to el. −8 ft (example E-1) gives 1.14 and is the more critical of the two.

3.167 V092: USACE example F-6, steady-seepage dam

USACE EM 1110-2-1902, example F-6.

Fixed circle · Units: SI (m, kPa, kN/m³)

112 ft embankment on a granular foundation under steady seepage: piezometric line from the pool through the chimney drain to the tailwater, which stands on the downstream toe (free water). The upstream face rises from (30.2, 100) ft to the crest (60.8, 112). USACE's circle. Converted to SI.

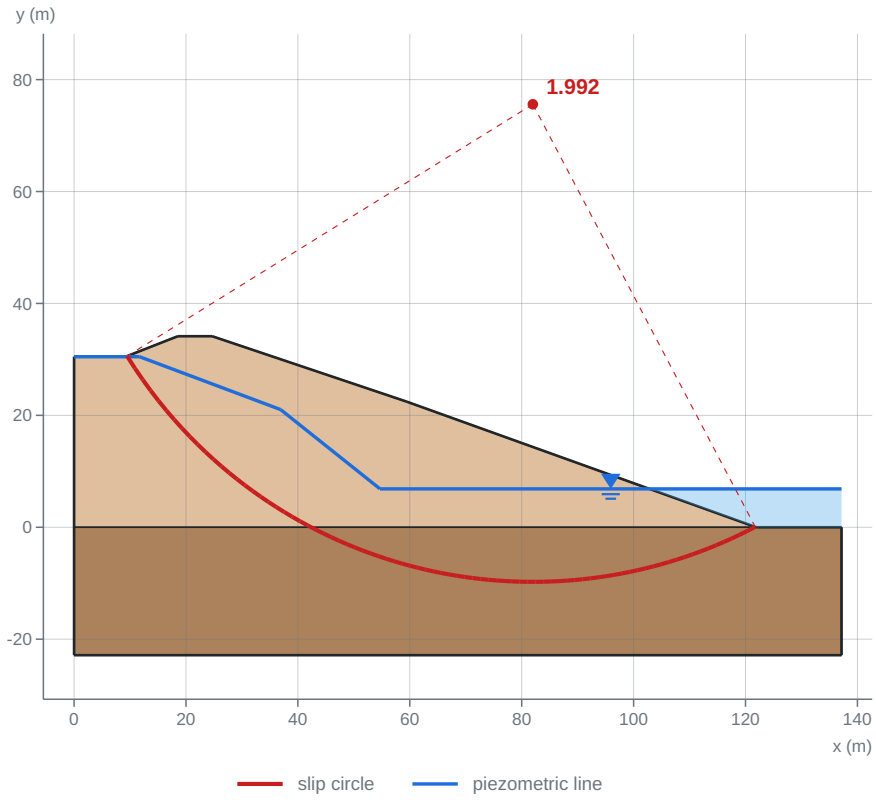


Figure 168. V092: analyzed circle (Bishop simplified, F = 1.992).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	20.42	c′ = 0.00	34.00°
Foundation	Mohr-Coulomb	19.64	c′ = 0.00	35.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.992				2.011	−0.97 %
Spencer	2.006				2.026	−1.00 %
Janbu corrected	1.825				1.830	−0.29 %
Morgenstern-Price (half-sine)	2.006				2.027	−1.04 %
–	–	2.01	USACE EM 1110-2-1902: Simplified Bishop, nine-slice hand calculation on the manual's own piezometric line (example F-6)			
–	–	2.07	USACE EM 1110-2-1902: Modified Swedish (θ = 19.7°) on the manual's own piezometric line (example F-6)			

USACE's values are for the manual's own piezometric line, which its slice table (fig. F-17a) places lower than the line of the program version analyzed here: the kink at about el. 53–59 ft instead of 69 ft, and the tailwater level reached about 18 ft further

upstream. On the manual's line cSlope gives Bishop 2.05–2.06 and Spencer 2.06–2.08, depending on where the table's slices are placed: within 0.5 % of USACE's Modified Swedish value 2.07, and 2–2.6 % above its nine-slice Bishop hand calculation. The manual's values are shown for reference and not compared with cSlope's analyses on the program's line.

3.168 V093: Yamagami & Ueta (1988)

Yamagami & Ueta (1988).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 2:1 slope, 5 m high; critical-circle search.

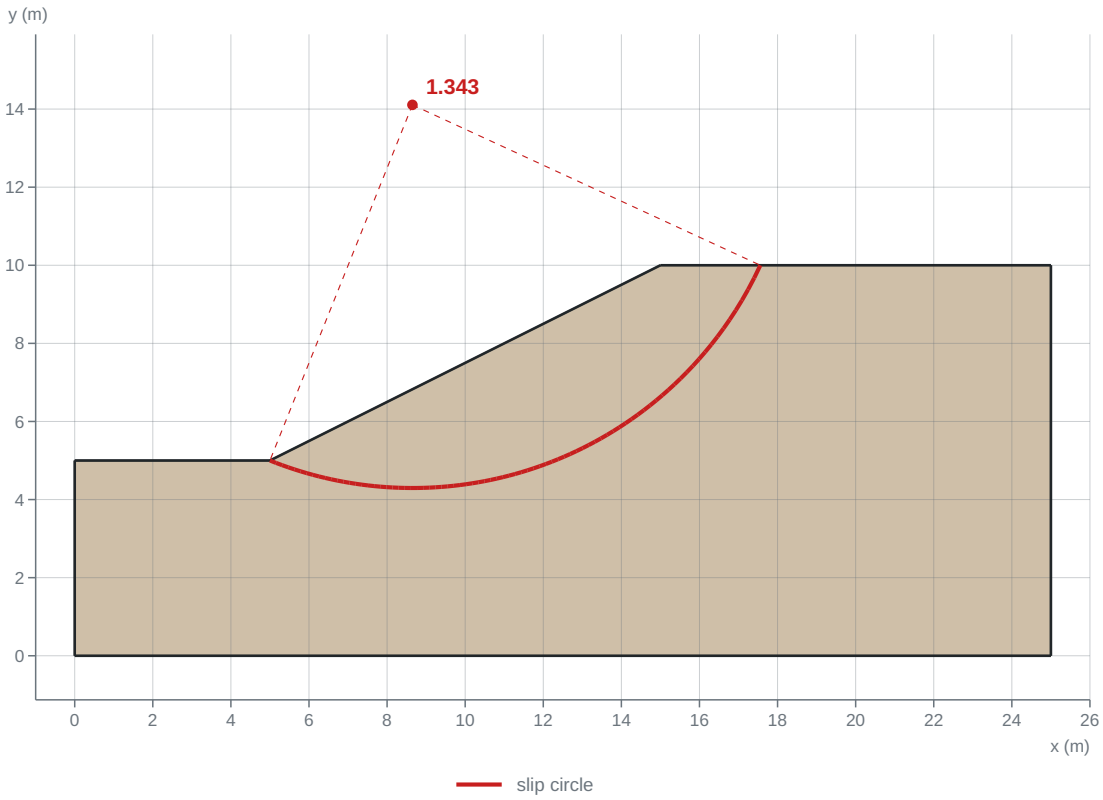


Figure 169. V093: critical circle (Bishop simplified, F = 1.343).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	17.64	c′ = 9.80	10.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.343	1.348	Yamagami & Ueta (1988)	−0.37 %	1.344	−0.07 %
Spencer, non-circular	1.323	1.339	Yamagami & Ueta (1988)	−1.23 %	1.325	−0.19 %
		1.33	Greco (1996)	−0.56 %		
Janbu simplified, non-circular	1.176	1.185	Yamagami & Ueta (1988)	−0.73 %	1.178	−0.14 %
Strength reduction (FE)	1.339	1.33	another program's verification manual: finite-element strength reduction (verification manual, problem 13)	+0.70 %		
–	–	1.282	Yamagami & Ueta (1988): Ordinary			

3.169 V094: Yamagami & Ueta (1988) four-layer slope, non-circular

Greco (1996) example 4; Yamagami & Ueta (1988).

Circle search with non-circular optimization · Units: SI (m, kPa, kN/m³)

Four-layer slope without water whose critical surface is non-circular; circular search followed by non-circular optimization.

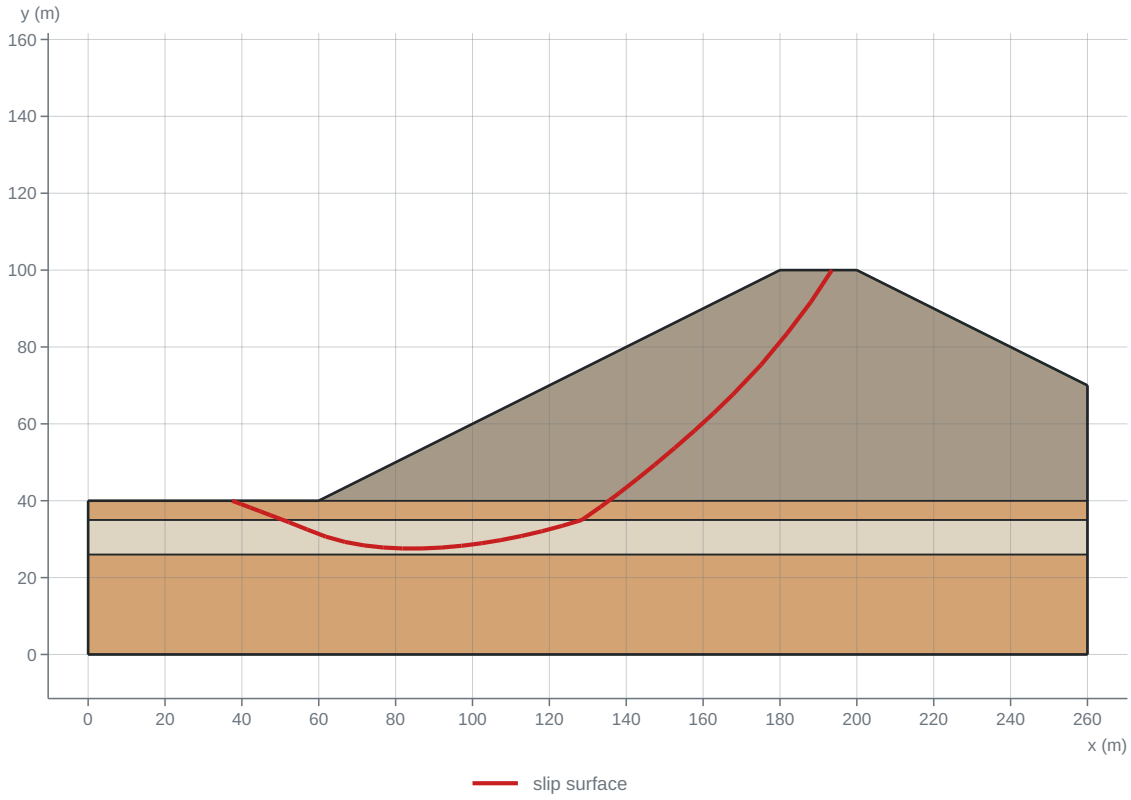


Figure 170. V094: critical surface (Spencer, non-circular, F = 1.399).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper layer	Mohr-Coulomb	20.38	c′ = 49.00	29.00°
Layer 2	Mohr-Coulomb	17.64	c′ = 0.00	30.00°
Layer 3	Mohr-Coulomb	20.38	c′ = 7.84	20.00°
Bottom layer	Mohr-Coulomb	17.64	c′ = 0.00	30.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer, non-circular	1.399	1.40–1.42	Greco (1996)	−0.07 %	1.398	+0.07 %
Morgenstern-Price (half-sine), non-circular	1.388				1.389	−0.06 %
Strength reduction (FE)	1.351	1.39	another program's verification manual: finite-element strength reduction (verification manual, problem 15)	−2.83 %		

3.170 V095a: Yamagami micro-pile slope, no pile

Yamagami et al. (2000).

Fixed circle · Units: SI (m, kPa, kN/m³)

Homogeneous 2:1 slope, 4 m high, without piles, on the published circle.

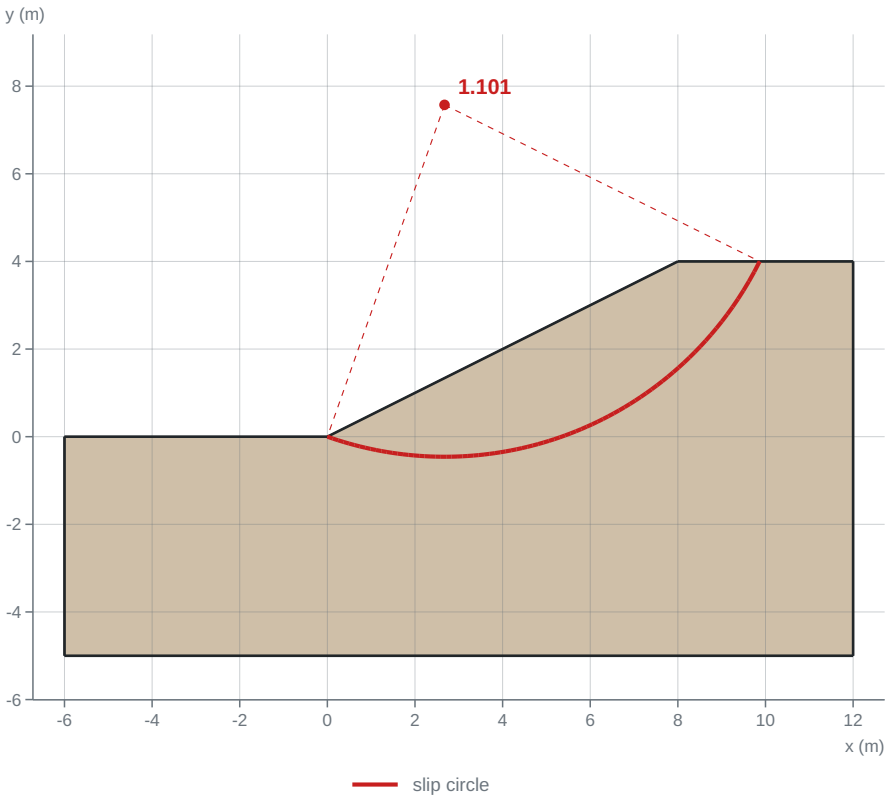


Figure 171. V095a: analyzed circle (Bishop simplified, F = 1.101).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	15.68	c′ = 4.90	10.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.101				1.102	−0.09 %
–	–	1.10	Yamagami et al. (2000)			

3.171 V095c: Homogeneous slope with a micro-pile row

Yamagami et al. (2000).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V095a with a micro-pile row (1 m spacing) whose shear strength of 10.7 kN acts as a passive force tangent to the slip surface, on the reference with-pile circle (2.674, 7.376, R = 8.102).

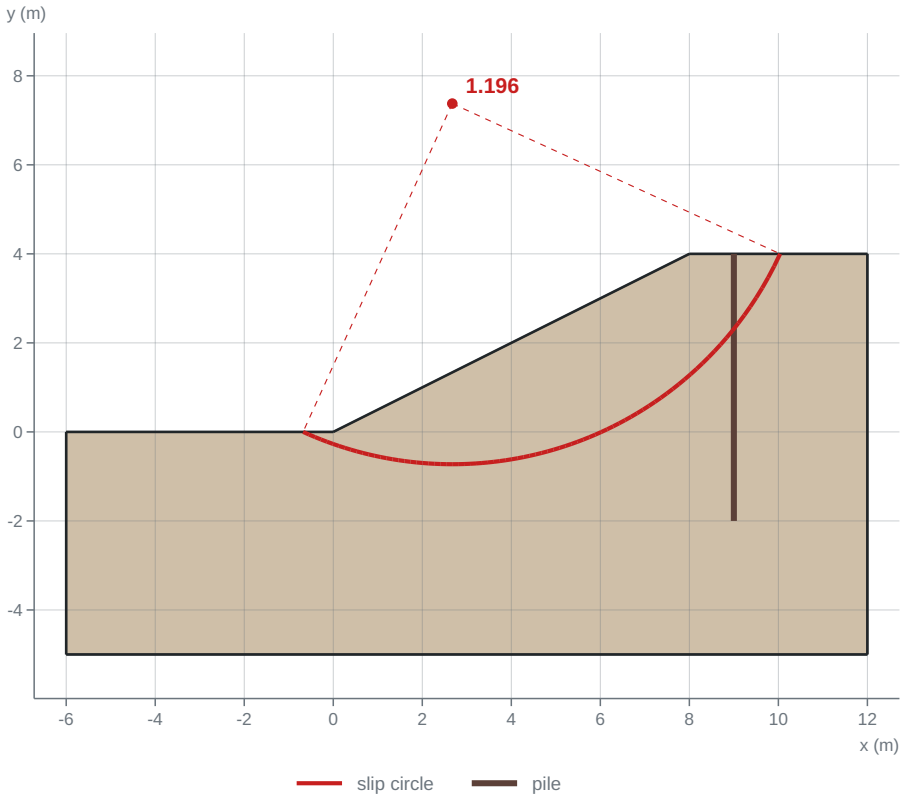


Figure 172. V095c: analyzed circle (Bishop simplified, F = 1.196).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	15.68	c′ = 4.90	10.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.196				1.193	+0.25 %
–	–	1.20	Yamagami et al. (2000)			

3.172 V096a: Zhu & Lee heterogeneous slope, dry, deep circle

Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.

Critical circle search · Units: SI (m, kPa, kN/m³)

Four horizontal layers under a two-bench slope with a dry tension crack 1.015 m deep at the crest (Fig. 52.1; the depth of a commercial program's surface end below the crest), dry; the circular search lands in the deep family (surface 3).

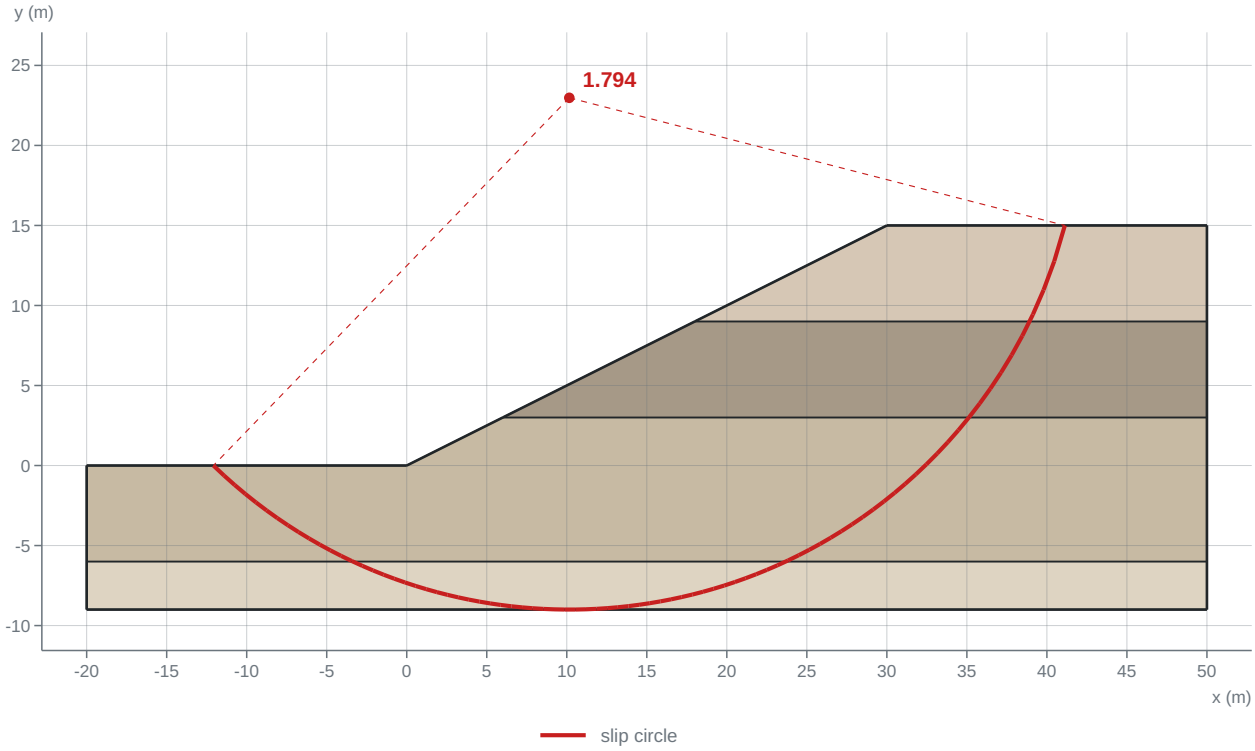


Figure 173. V096a: critical circle (Bishop simplified, F = 1.794).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1 (top)	Mohr-Coulomb	18.80	c′ = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c′ = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c′ = 25.00	26.00°
Layer 4 (bottom)	Mohr-Coulomb	18.00	c′ = 10.00	12.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.794				1.804	−0.56 %
Spencer	1.795				1.804	−0.52 %
Morgenstern-Price (half-sine)	1.780				1.790	−0.54 %

The circle and the 1.015 m crack follow a program's version of the problem. The paper gives its surface 3 only in a figure, and it enters the crest about 3 m nearer the slope than the program's circle; the paper's values for it are checked on the paper's own circle in V096d (dry) and V096e (with the water table).

3.173 V096b: Zhu & Lee heterogeneous slope with a water table, deep circle

Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V096a (dry 1.015 m crack) with the water table of Table 52.2.

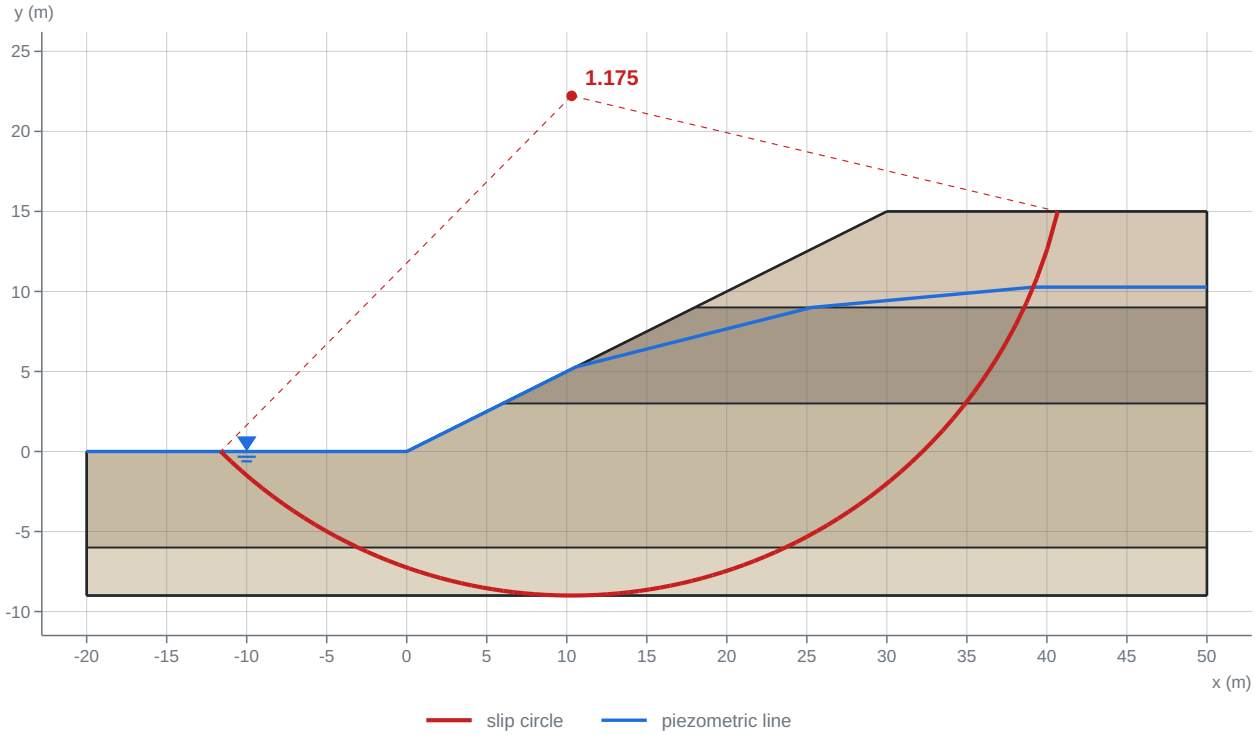


Figure 174. V096b: critical circle (Bishop simplified, F = 1.175).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1 (top)	Mohr-Coulomb	18.80	c′ = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c′ = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c′ = 25.00	26.00°
Layer 4 (bottom)	Mohr-Coulomb	18.00	c′ = 10.00	12.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.175				1.176	−0.12 %
Spencer	1.188				1.189	−0.10 %
Morgenstern-Price (half-sine)	1.173				1.174	−0.07 %

As V096a: the paper's values for its surface 3 with the water table are checked on the paper's own circle in V096e.

3.174 V096c: Zhu & Lee slope, surface 3 (the reference circle), dry, 1 m crack

Zhu & Lee (2002), Int. J. Numer. Anal. Meth. Geomech. 26(15). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V096a slope with the dry 1.015 m crack on the reference surface 3 circle (10.911, 23.208, R = 32.106; Fig. 52.4).

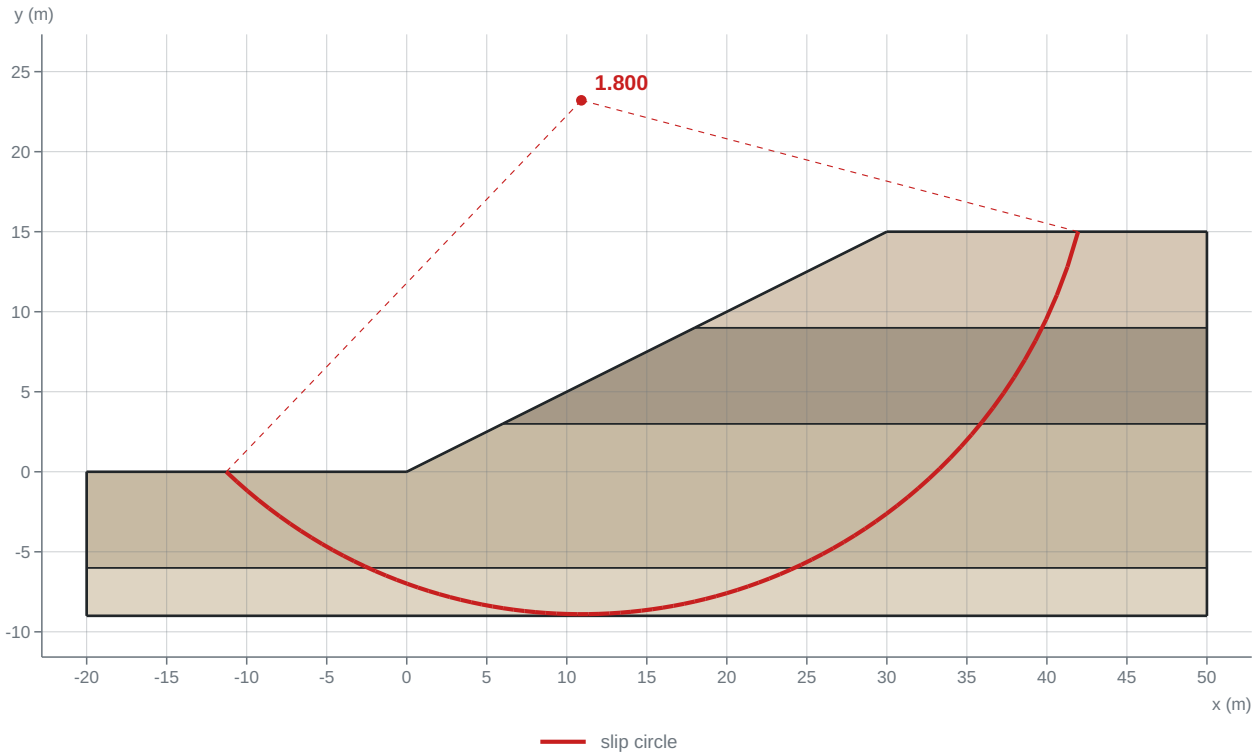


Figure 175. V096c: analyzed circle (Spencer, F = 1.800).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	18.80	c′ = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c′ = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c′ = 25.00	26.00°
Layer 4	Mohr-Coulomb	18.00	c′ = 10.00	12.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.800				1.804	−0.23 %
Bishop simplified	1.800				1.804	−0.24 %

The program's reference circle enters the crest at x = 41.9 m, about 3 m further from the slope than the paper's surface 3 as drawn in its fig. 3; the paper's values are checked on the paper's circle in V096d and V096e.

3.175 V096d: Zhu & Lee slope, the paper's surface 3, dry

Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V096a slope, dry, no tension crack, on the paper's surface 3 digitized from its fig. 3: least-squares circle (9.7, 21.3, R = 29.8) through the drawn surface, leaving the lower bench at x = −11.1 and entering the crest at x = 38.8 (±0.3 m).

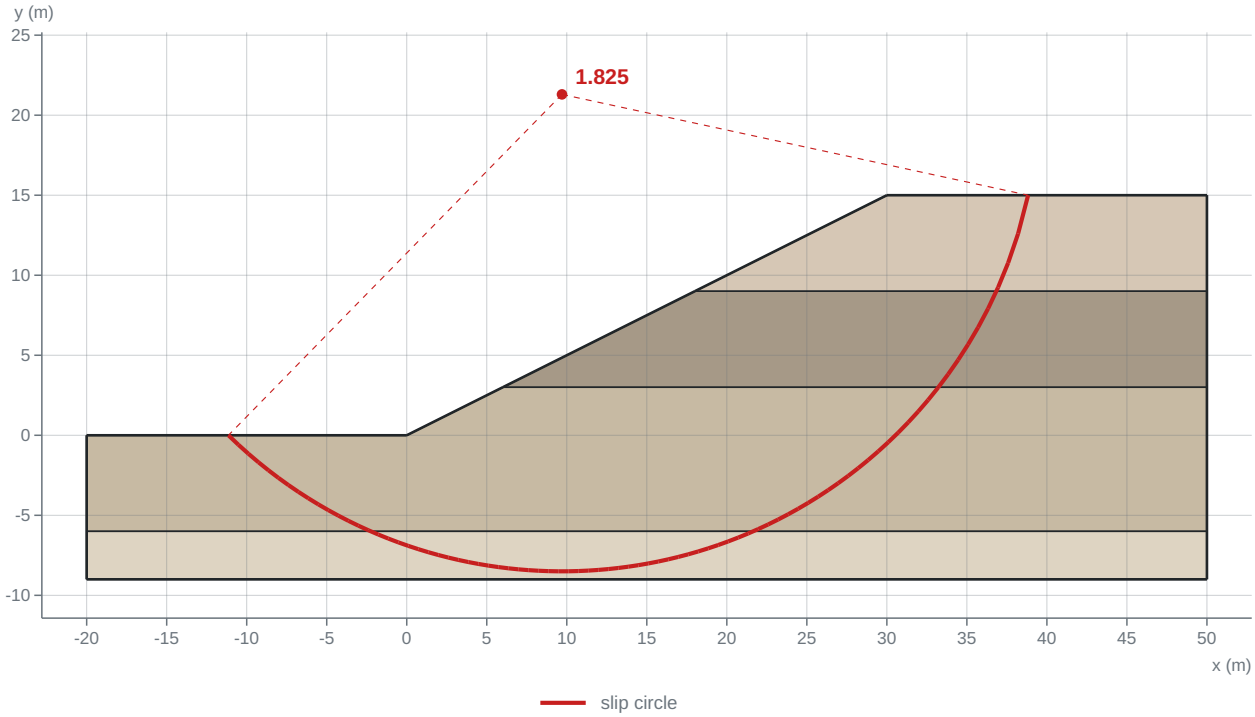


Figure 176. V096d: analyzed circle (Spencer, F = 1.825).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	18.80	c′ = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c′ = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c′ = 25.00	26.00°
Layer 4	Mohr-Coulomb	18.00	c′ = 10.00	12.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.825	1.836	Zhu & Lee	−0.59 %		
Morgenstern-Price (half-sine)	1.810	1.823	Zhu & Lee	−0.73 %		
Bishop simplified	1.824	1.429	Zhu & Lee (2002)	*+27.66 % **		
–	–	1.744	Zhu & Lee (2002): explicit solution (the paper's method)			
–	–	1.229	Zhu & Lee (2002): Ordinary			

** Disputed, not counted in section 2: Zhu & Lee (2002)'s 1.429. Confirmed in Zhu & Lee (2002), Table II, which also gives Spencer 1.836 and Morgenstern-Price 1.823 for the same circle; the authors read the gap as simplified Bishop underestimating deep circles by up to 18 %. On a circle simplified Bishop satisfies moment equilibrium about the center as the rigorous methods do and normally comes within a few per cent of them; cSlope gives 1.823 on the paper's circle. Not counted in the error statistics.*

The paper gives surface 3 only in a figure. On the circle fitted to it cSlope comes within 1 % of the paper's Spencer and Morgenstern-Price values (1.836, 1.823), against 2 % on the program's circle of V096c; moving the fitted center or radius by 0.3 m changes the result by up to 1 %. The paper's simplified Bishop value 1.429 is 22 % below its own rigorous values for the same circle, where cSlope's Bishop gives 1.823 as expected on a circle; it is shown but not counted in the error statistics.

3.176 V096e: Zhu & Lee slope, the paper's surface 3, with the water table

Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V096d with the water table of the paper's fig. 3 (the points of V096b, which match the drawn line within 0.3 m).

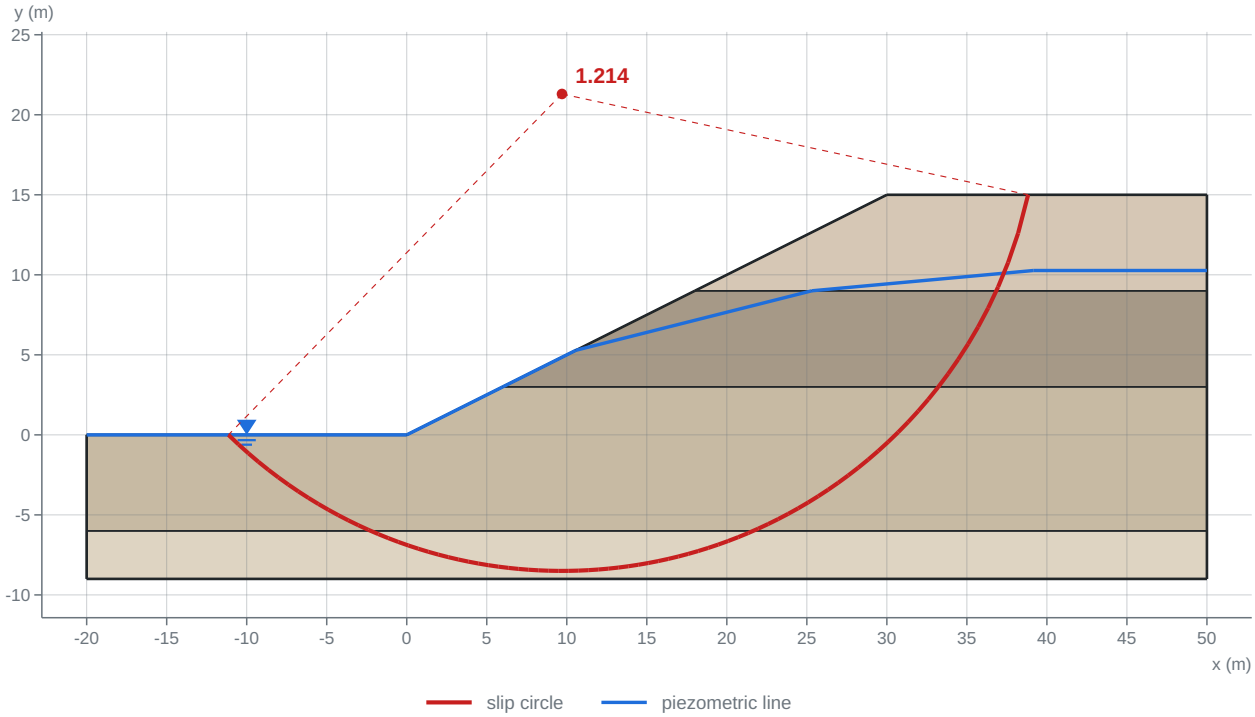


Figure 177. V096e: analyzed circle (Spencer, F = 1.214).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	18.80	c′ = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c′ = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c′ = 25.00	26.00°
Layer 4	Mohr-Coulomb	18.00	c′ = 10.00	12.00°

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.214	1.211	Zhu & Lee	+0.27 %		
Morgenstern-Price (half-sine)	1.199	1.197	Zhu & Lee	+0.19 %		
Bishop simplified	1.201	1.079	Zhu & Lee (2002)	*+11.32 % **		
–	–	1.166	Zhu & Lee (2002): explicit solution (the paper's method)			
–	–	0.922	Zhu & Lee (2002): Ordinary			

** Disputed, not counted in section 2: Zhu & Lee (2002)'s 1.079. Zhu & Lee (2002), Table II: 11 % below their own Spencer (1.211) and Morgenstern-Price (1.197) for the same circle, as their dry Bishop value is; cSlope gives 1.199 on the paper's circle with the water table. Not counted in the error statistics.*

On the circle fitted to the paper's figure cSlope reproduces the paper's Spencer and Morgenstern-Price values (1.211, 1.197) within 0.1 %. The paper's simplified Bishop value 1.079 lies 11 % below its own rigorous values for the same circle, as its dry value does; cSlope's Bishop gives 1.199. It is shown but not counted in the error statistics.

3.177 V097a: Eurocode 7 cutting with surcharge

Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

Cutting in clay with an inclined piezometric line and a 35 kPa surcharge on the crest, analyzed on the published circle.

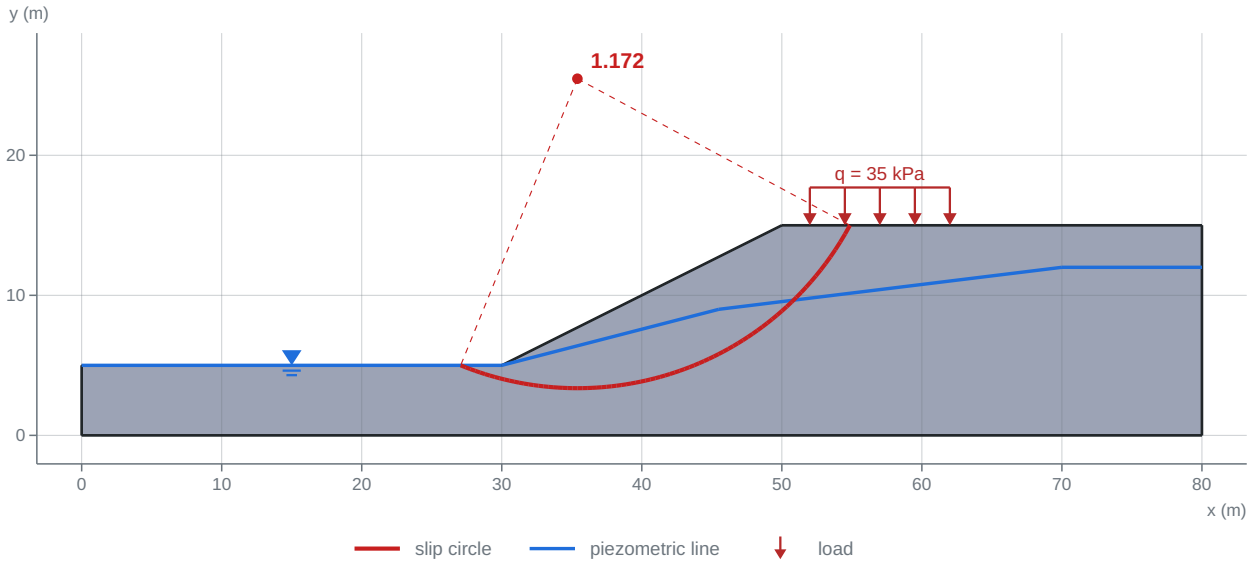


Figure 178. V097a: analyzed circle (Bishop simplified, F = 1.172).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	Mohr-Coulomb	20.00	c′ = 8.00	23.04°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.173				1.173	−0.04 %

The problem comes from a program manual only. The manual cites 1.193 (Bishop) from the textbook example it is based on, but its water table is approximate because the book gives no coordinates, so that value is not used as a check.

3.178 V097b: Eurocode 7 cutting, partial factors applied by cSlope

Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V097a, but with the characteristic strength c′ = 10 kPa, φ′ = 28° and Design Approach 3 material factors (set M2: γc′ = γφ′ = 1.25) applied by the analysis; the surcharge is permanent (factor 1.0). F is the over-design factor on design values.

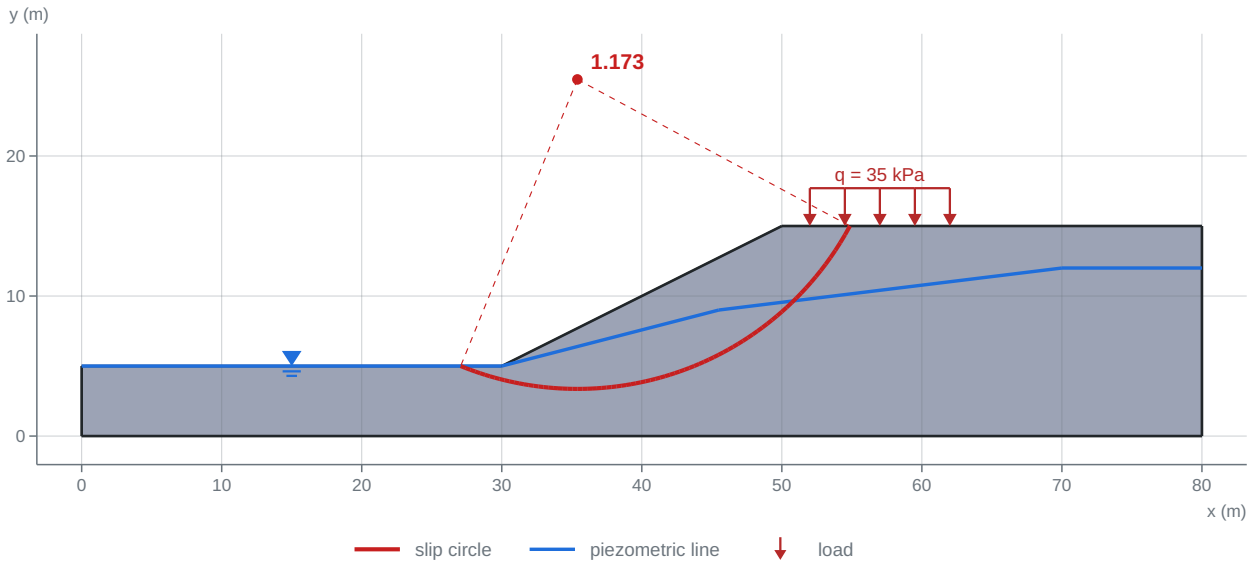


Figure 179. V097b: analyzed circle (Bishop simplified, F = 1.173).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	Mohr-Coulomb	20.00	c′ = 10.00	28.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.173				1.173	−0.03 %
Spencer	1.174				1.174	−0.01 %

The problem comes from a program manual only. The manual cites 1.193 (Bishop) from the textbook example it is based on, but its water table is approximate because the book gives no coordinates, so that value is not used as a check.

3.179 V098: Example slope without seismic load

Verification manual of a commercial slope stability program.

Circle search with non-circular optimization · Units: SI (m, kPa, kN/m³)

10 m, 2:1 slope in three layers over a horizontal base, dry, no seismic load; Spencer search, circular and non-circular.

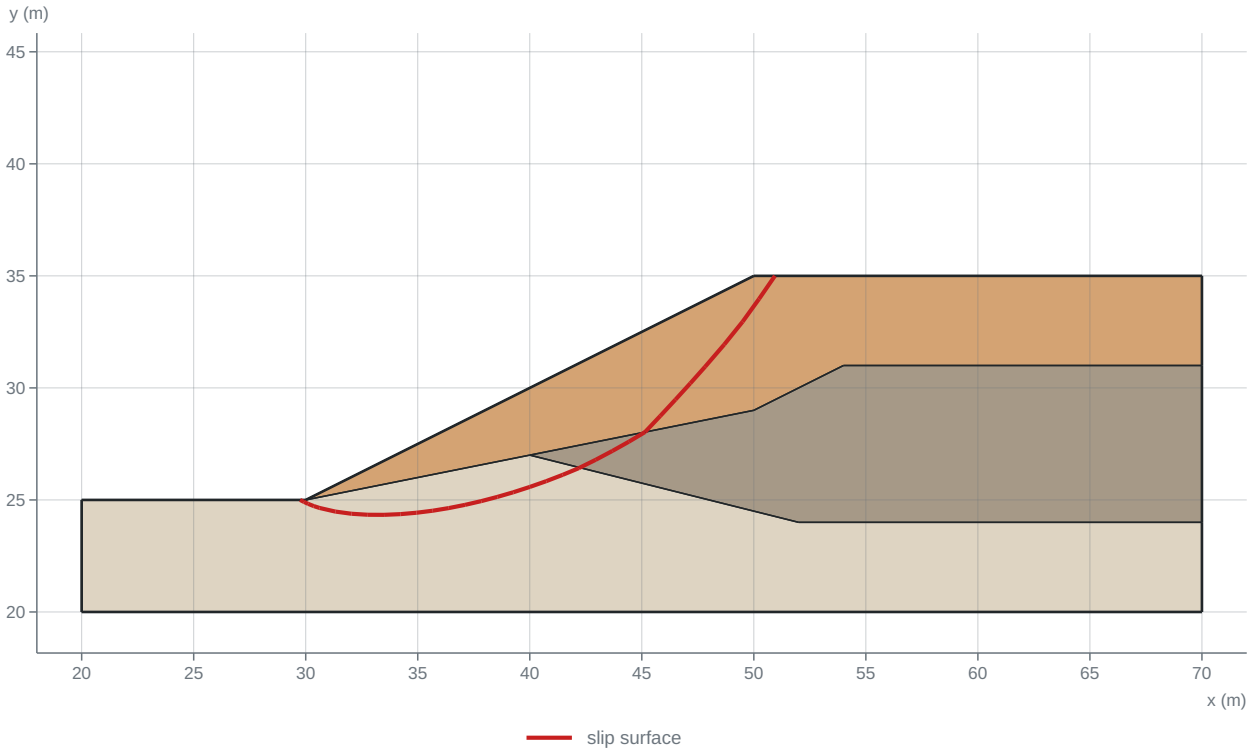


Figure 180. V098: critical surface (Spencer, non-circular, F = 1.357).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil1	Mohr-Coulomb	19.50	c′ = 0.00	38.00°
soil2	Mohr-Coulomb	19.50	c′ = 5.30	23.00°
soil3	Mohr-Coulomb	19.50	c′ = 7.20	20.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer, non-circular	1.357				1.360	−0.22 %
Spencer	1.372				1.372	+0.03 %

The problem comes from a program manual only; no independent value is published.

3.180 V104a: Embankment on clay, no excess pore pressure (B-bar tutorial, step 1)

Tutorial of a commercial slope stability program: undrained loading by the B-bar method (its tutorial pages and the tutorial's project file).

Critical circle search · Units: SI (m, kPa, kN/m³)

A 2 m embankment (crest x = 13–20.6 at y = 7, 1.5:1 faces to y = 5 at x = 10 and 23.6; γ = 20 kN/m³, c′ = 5 kPa, φ′ = 32°) on 1 m of upper clay (c′ = 10 kPa, φ′ = 20°) over 4 m of lower clay (c′ = 15 kPa, φ′ = 20°; both γ = 17 kN/m³), water table at the clay surface (y = 5), dry embankment; failure towards −x; circular search, Bishop simplified. Step 1: no excess pore pressure; the critical circle stays in the embankment.

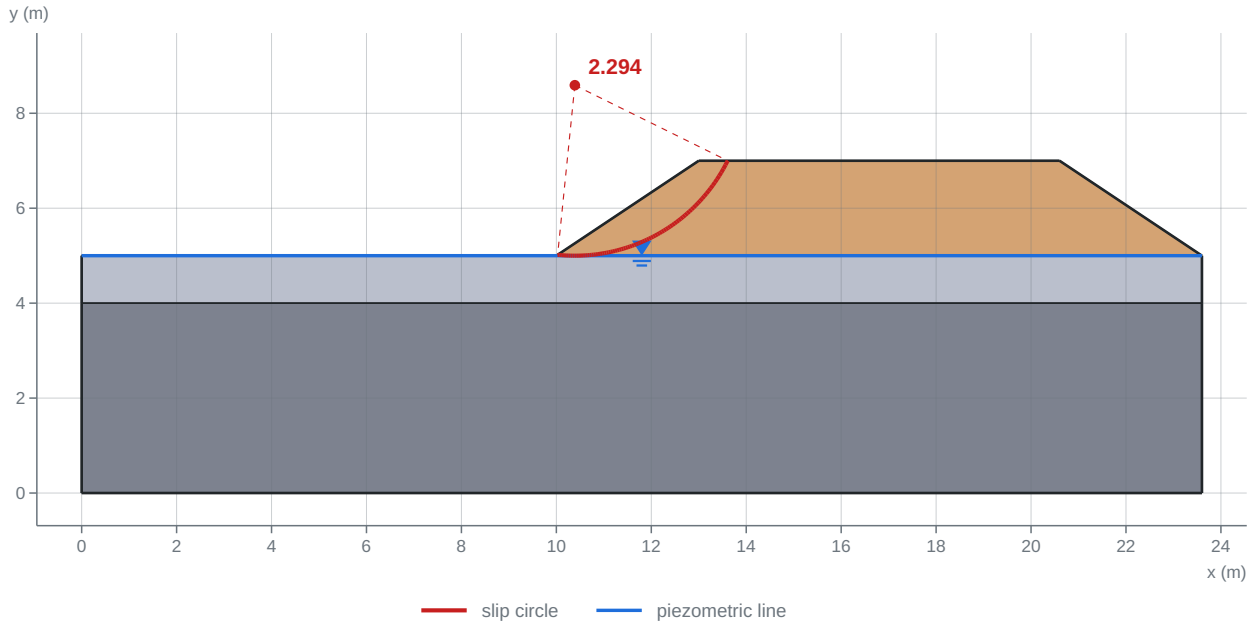


Figure 181. V104a: critical circle (Bishop simplified, F = 2.294).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
embankment	Mohr-Coulomb	20.00	c′ = 5.00	32.00°
upper clay	Mohr-Coulomb	17.00	c′ = 10.00	20.00°
lower clay	Mohr-Coulomb	17.00	c′ = 15.00	20.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	2.294				2.300	−0.26 %

The problem comes from a program's tutorial only; no independent value is published. The tutorial does not name the method; its project file runs Bishop simplified.

3.181 V104b: Embankment on clay, $\bar{B}$ = 1 from the fill weight (B-bar tutorial, step 2)

Tutorial of a commercial slope stability program: undrained loading by the B-bar method (its tutorial pages and the tutorial's project file).

Critical circle search · Units: SI (m, kPa, kN/m³)

A 2 m embankment (crest x = 13–20.6 at y = 7, 1.5:1 faces to y = 5 at x = 10 and 23.6; γ = 20 kN/m³, c′ = 5 kPa, φ′ = 32°) on 1 m of upper clay (c′ = 10 kPa, φ′ = 20°) over 4 m of lower clay (c′ = 15 kPa, φ′ = 20°; both γ = 17 kN/m³), water table at the clay surface (y = 5), dry embankment; failure towards −x; circular search, Bishop simplified. Step 2: $\bar{B}$ = 1 in both clays; the weight of the embankment above a slice base raises the pore pressure there by $\bar{B}$ × its vertical stress (the embankment creates excess pore pressure, $\bar{B}$ = 0 in itself).

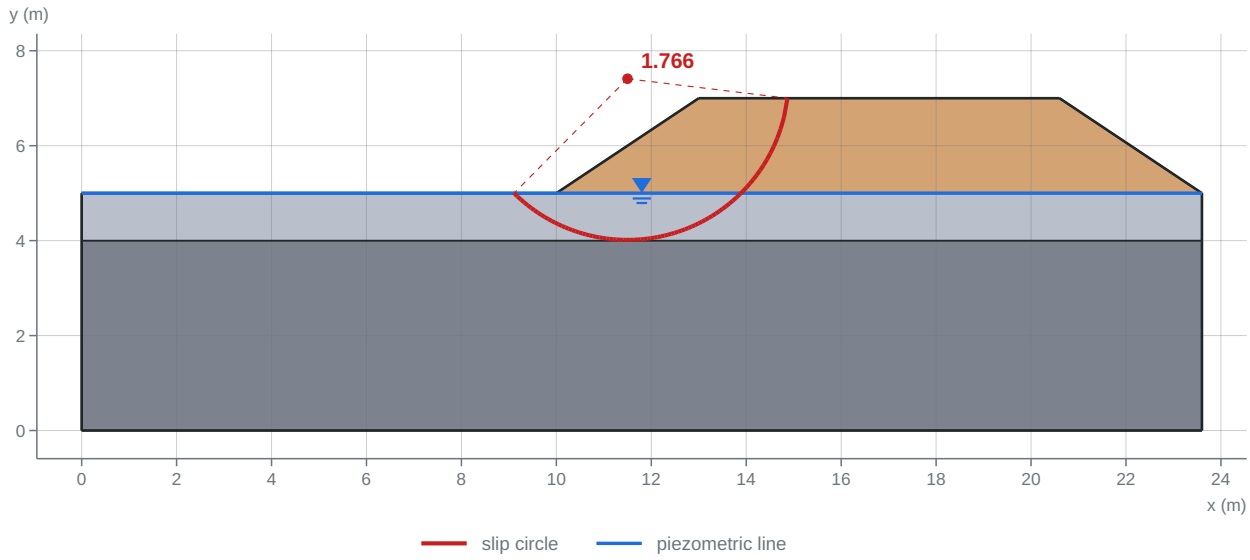


Figure 182. V104b: critical circle (Bishop simplified, F = 1.766).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
embankment	Mohr-Coulomb	20.00	c′ = 5.00	32.00°
upper clay	Mohr-Coulomb	17.00	c′ = 10.00	20.00°
lower clay	Mohr-Coulomb	17.00	c′ = 15.00	20.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.766				1.770	−0.23 %

The problem comes from a program's tutorial only; no independent value is published. The tutorial does not name the method; its project file runs Bishop simplified.

3.182 V104c: Embankment on clay, $\bar{B} = 1$ with a 10 kPa crest load (B-bar tutorial, step 3)

Tutorial of a commercial slope stability program: undrained loading by the B-bar method (its tutorial pages and the tutorial's project file).

Critical circle search · Units: SI (m, kPa, kN/m³)

A 2 m embankment (crest x = 13–20.6 at y = 7, 1.5:1 faces to y = 5 at x = 10 and 23.6; γ = 20 kN/m³, c′ = 5 kPa, φ′ = 32°) on 1 m of upper clay (c′ = 10 kPa, φ′ = 20°) over 4 m of lower clay (c′ = 15 kPa, φ′ = 20°; both γ = 17 kN/m³), water table at the clay surface (y = 5), dry embankment; failure towards −x; circular search, Bishop simplified. Step 3: as V104b with a 10 kPa vertical load on the crest (x = 13–20.6) that also creates excess pore pressure ($\bar{B}$ × the load, straight down) in the clays.

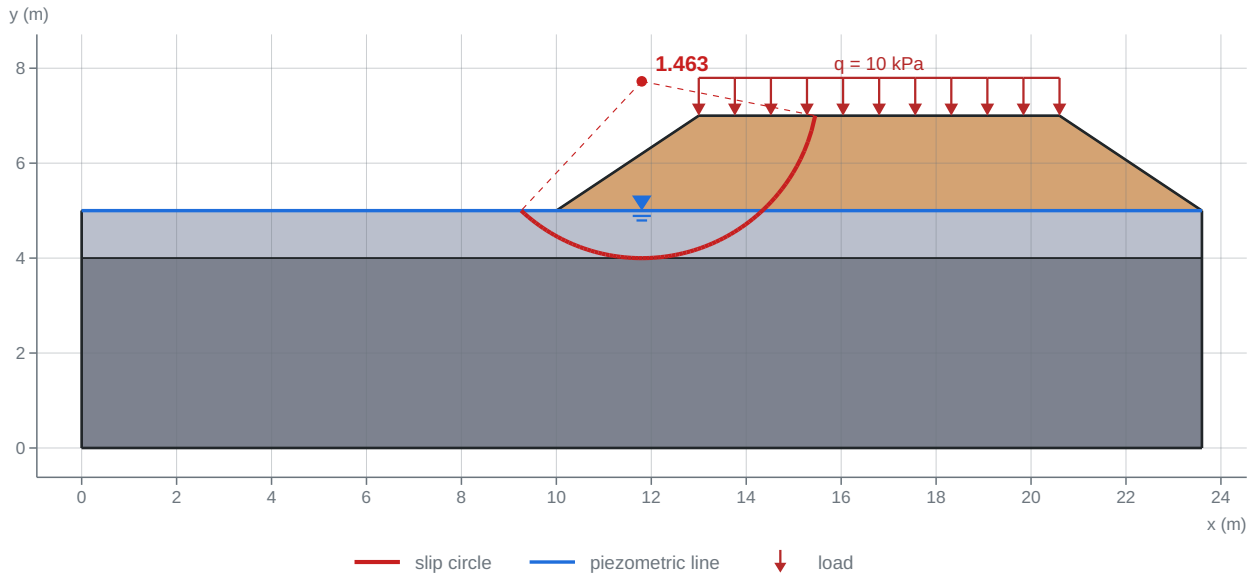


Figure 183. V104c: critical circle (Bishop simplified, F = 1.463).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
embankment	Mohr-Coulomb	20.00	c′ = 5.00	32.00°
upper clay	Mohr-Coulomb	17.00	c′ = 10.00	20.00°
lower clay	Mohr-Coulomb	17.00	c′ = 15.00	20.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.462				1.470	−0.51 %

The problem comes from a program's tutorial only; no independent value is published. The tutorial does not name the method; its project file runs Bishop simplified.

3.183 V104d: Embankment on clay, $\bar{B}$ = 1, crest load and r_u = 0.4 in the fill (B-bar tutorial, step 4)

Tutorial of a commercial slope stability program: undrained loading by the B-bar method (its tutorial pages and the tutorial's project file).

Critical circle search · Units: SI (m, kPa, kN/m³)

A 2 m embankment (crest x = 13–20.6 at y = 7, 1.5:1 faces to y = 5 at x = 10 and 23.6; γ = 20 kN/m³, c′ = 5 kPa, φ′ = 32°) on 1 m of upper clay (c′ = 10 kPa, φ′ = 20°) over 4 m of lower clay (c′ = 15 kPa, φ′ = 20°; both γ = 17 kN/m³), water table at the clay surface (y = 5), dry embankment; failure towards −x; circular search, Bishop simplified. Step 4: as V104c with r_u = 0.4 in the embankment (its own pore pressure; it still creates excess pore pressure in the clays and takes none from them).

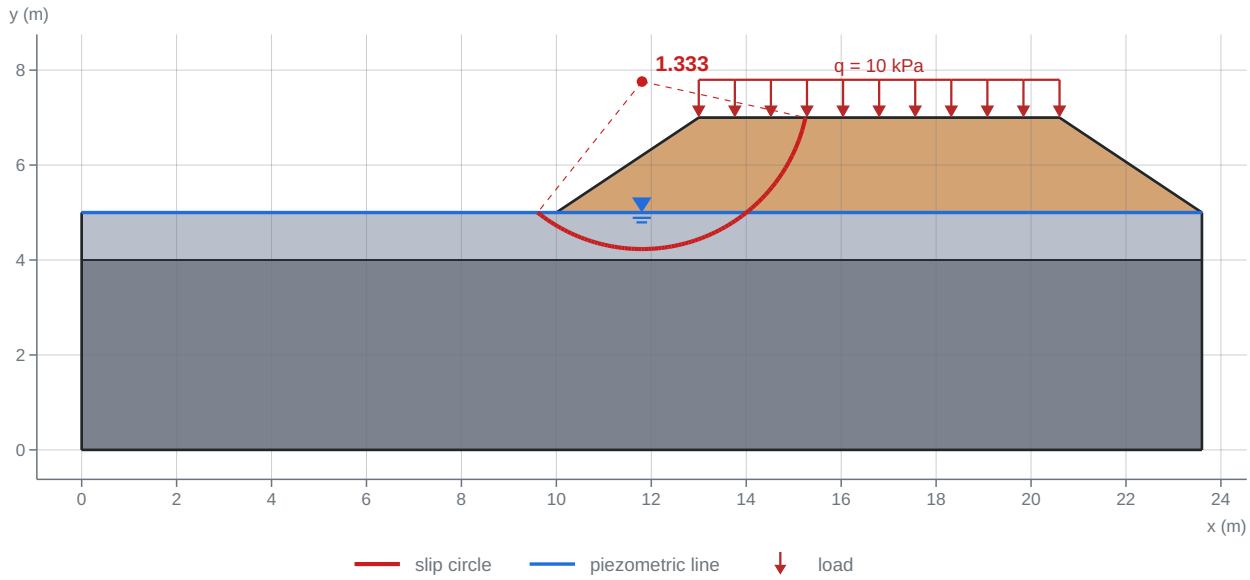


Figure 184. V104d: critical circle (Bishop simplified, F = 1.333).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
upper clay	Mohr-Coulomb	17.00	c′ = 10.00	20.00°
lower clay	Mohr-Coulomb	17.00	c′ = 15.00	20.00°
embankment modified	Mohr-Coulomb	20.00	c′ = 5.00	32.00°

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.333				1.330	+0.24 %

The problem comes from a program's tutorial only; no independent value is published. The tutorial does not name the method; its project file runs Bishop simplified.

3.184 V105a: Dike on soft soil, SHANSEP with POP and yield stress, daily water level

Tutorial of a commercial slope stability program (its user manual): a dike on soft soil, stage "Daily conditions" and stage "High Water". The project file with the program's results is distributed with the program and in the test data of an open-source library.

Fixed circle · Units: SI (m, kPa, kN/m³)

A dike on soft soil (x = −100 to 150, crest at y ≈ 12.7 near x = 3.5–4.5, polder level y ≈ 7–5.5): dike body "Sand with clay" (γ = 18 kN/m³) over shallow clay, organic clay, deep peat and an aquifer sand (γ = 18/20 kN/m³, φ′ = 30° with ψ = 0°, entered as the program's reduced φ = 26.57°, tan φ = sin φ′ cos ψ / (1 − sin φ′ sin ψ)). The soft layers are SHANSEP (S = 0.27/0.24/0.23/0.22, m = 0.9/0.85/0.9/0.9 for peat, organic clay, shallow clay and dike body) with POP 5, 20, 10 kPa and, in the dike body, a yield stress of 86 kPa at (5, 8.7) (POP 20.18 kPa there, which the program keeps constant in the layer); above the phreatic line the dike body takes c′ = 0, φ′ = 33° (the strength above the water line). Daily conditions: a phreatic line at 5.6 (outer side) rising to 7.5 under the dike and 5.5 in the polder for all layers; Bishop on the tutorial's given circle (20, 20, R 20).

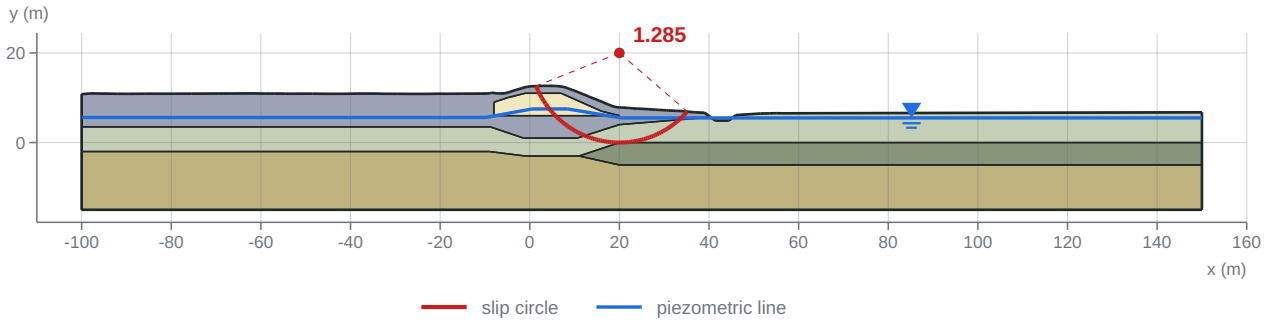


Figure 185. V105a: analyzed circle (Bishop simplified, F = 1.285).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Mohr-Coulomb	18.00	c′ = 0.00	26.57°
Peat, deep	shansep	11.00	see model	–
Organic clay	shansep	13.90	see model	–
Clay, shallow	shansep	14.80	see model	–
Sand with clay	shansep	18.00	see model	–

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.285				1.285	+0.01 %

The problem comes from a program's tutorial only; no independent value is published. The checks are on the program's own surfaces, stored with its results in the tutorial's project file.

3.185 V105b: Dike on soft soil at high water: waternet, excess pore pressure under load, yield stress from the earlier stage

Tutorial of a commercial slope stability program (its user manual): a dike on soft soil, stage "Daily conditions" and stage "High Water". The project file with the program's results is distributed with the program and in the test data of an open-source library.

Fixed circle · Units: SI (m, kPa, kN/m³)

A dike on soft soil (x = −100 to 150, crest at y ≈ 12.7 near x = 3.5–4.5, polder level y ≈ 7–5.5): dike body "Sand with clay" (γ = 18 kN/m³) over shallow clay, organic clay, deep peat and an aquifer sand (γ = 18/20 kN/m³, φ′ = 30° with ψ = 0°, entered as the program's reduced φ = 26.57°, tan φ = sin φ′ cos ψ / (1 − sin φ′ sin ψ)). The soft layers are SHANSEP (S = 0.27/0.24/0.23/0.22, m = 0.9/0.85/0.9/0.9 for peat, organic clay, shallow clay and dike body) with POP 5, 20, 10 kPa and, in the dike body, a yield stress of 86 kPa at (5, 8.7) (POP 20.18 kPa there, which the program keeps constant in the layer); above the phreatic line the dike body takes c′ = 0, φ′ = 33° (the strength above the water line). High water: a waternet with the phreatic line (12 at the outer toe, 10.9 on the crest, 5.7 at x = 40) as the head above the top of the soft layers and a head of 10 → 8.4 → 8 in the aquifer below the top of the sand (the head varies linearly between the two reference lines in the soft layers); a 13 kPa traffic load at x = 3.5–6 of which 90 % is carried by excess pore pressure in the soft layers (10 % consolidation); the SHANSEP yield stress is the daily-conditions stage's effective vertical stress plus the POP (state correlated with the earlier stage), so that the higher water lowers σ′v but not the yield stress. Bishop on the program's critical circle of its grid search (15.7, 18.6, R 13.7).

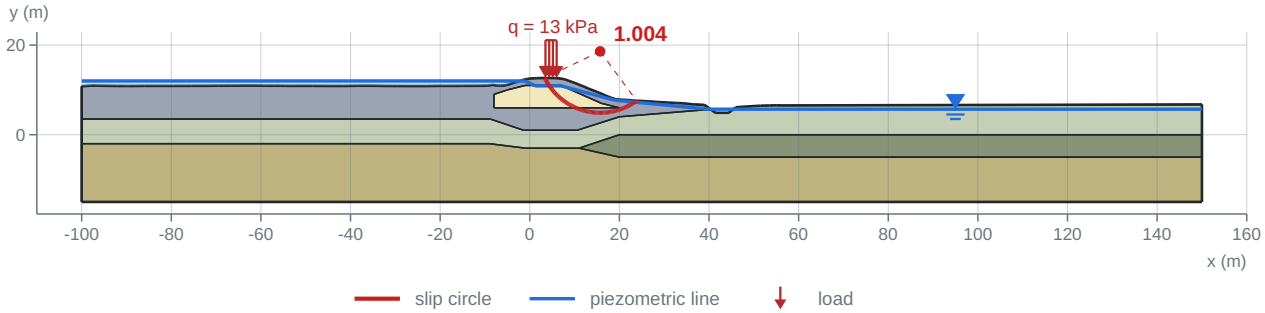


Figure 186. V105b: analyzed circle (Bishop simplified, F = 1.004).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Mohr-Coulomb	18.00	c′ = 0.00	26.57°
Peat, deep	shansep	11.00	see model	–
Organic clay	shansep	13.90	see model	–
Clay, shallow	shansep	14.80	see model	–
Sand with clay	shansep	18.00	see model	–

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Bishop simplified	1.004				1.004	+0.01 %

The problem comes from a program's tutorial only; no independent value is published. The checks are on the program's own surfaces, stored with its results in the tutorial's project file.

3.186 V105c: Dike on soft soil at high water, the program's non-circular critical surface

Tutorial of a commercial slope stability program (its user manual): a dike on soft soil, stage "Daily conditions" and stage "High Water". The project file with the program's results is distributed with the program and in the test data of an open-source library.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

A dike on soft soil (x = −100 to 150, crest at y ≈ 12.7 near x = 3.5–4.5, polder level y ≈ 7–5.5): dike body "Sand with clay" (γ = 18 kN/m³) over shallow clay, organic clay, deep peat and an aquifer sand (γ = 18/20 kN/m³, φ′ = 30° with ψ = 0°, entered as the program's reduced φ = 26.57°, tan φ = sin φ′ cos ψ / (1 − sin φ′ sin ψ)). The soft layers are SHANSEP (S = 0.27/0.24/0.23/0.22, m = 0.9/0.85/0.9/0.9 for peat, organic clay, shallow clay and dike body) with POP 5, 20, 10 kPa and, in the dike body, a yield stress of 86 kPa at (5, 8.7) (POP 20.18 kPa there, which the program keeps constant in the layer); above the phreatic line the dike body takes c′ = 0, φ′ = 33° (the strength above the water line). High water: a waternet with the phreatic line (12 at the outer toe, 10.9 on the crest, 5.7 at x = 40) as the head above the top of the soft layers and a head of 10 → 8.4 → 8 in the aquifer below the top of the sand (the head varies linearly between the two reference lines in the soft layers); a 13 kPa traffic load at x = 3.5–6 of which 90 % is carried by excess pore pressure in the soft layers (10 % consolidation); the SHANSEP yield stress is the daily-conditions stage's effective vertical stress plus the POP (state correlated with the earlier stage), so that the higher water lowers σ′v but not the yield stress. Spencer on the program's critical non-circular surface of its genetic search (x = 0.5 to 35, lowest point y = 2.19).

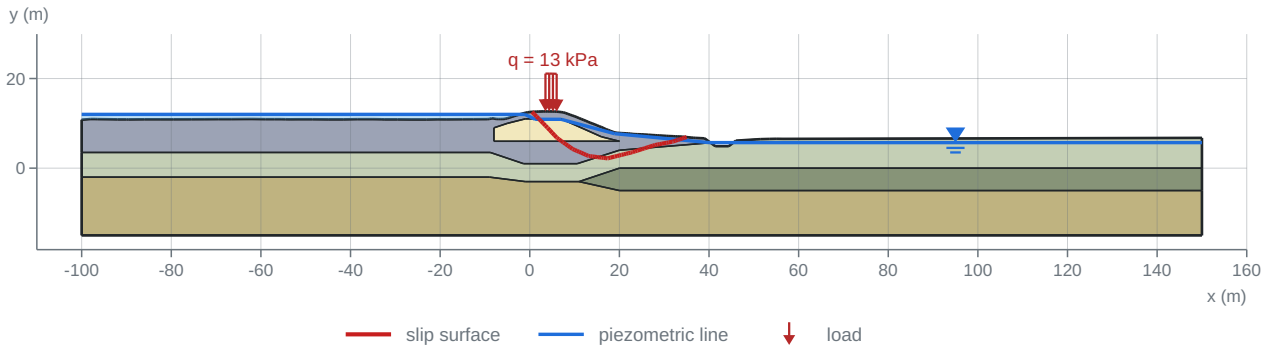


Figure 187. V105c: analyzed surface (Spencer, F = 1.085).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Mohr-Coulomb	18.00	c′ = 0.00	26.57°
Peat, deep	shansep	11.00	see model	–
Organic clay	shansep	13.90	see model	–
Clay, shallow	shansep	14.80	see model	–
Sand with clay	shansep	18.00	see model	–

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.085				1.085	−0.01 %

The problem comes from a program's tutorial only; no independent value is published. The checks are on the program's own surfaces, stored with its results in the tutorial's project file.

3.187 V105d: Dike on soft soil at high water, deep surface into the aquifer head zone

Tutorial of a commercial slope stability program (its user manual): a dike on soft soil, stage "Daily conditions" and stage "High Water". The project file with the program's results is distributed with the program and in the test data of an open-source library.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

A dike on soft soil (x = −100 to 150, crest at y ≈ 12.7 near x = 3.5–4.5, polder level y ≈ 7–5.5): dike body "Sand with clay" (γ = 18 kN/m³) over shallow clay, organic clay, deep peat and an aquifer sand (γ = 18/20 kN/m³, φ′ = 30° with ψ = 0°, entered as the program's reduced φ = 26.57°, tan φ = sin φ′ cos ψ / (1 − sin φ′ sin ψ)). The soft layers are SHANSEP (S = 0.27/0.24/0.23/0.22, m = 0.9/0.85/0.9/0.9 for peat, organic clay, shallow clay and dike body) with POP 5, 20, 10 kPa and, in the dike body, a yield stress of 86 kPa at (5, 8.7) (POP 20.18 kPa there, which the program keeps constant in the layer); above the phreatic line the dike body takes c′ = 0, φ′ = 33° (the strength above the water line). High water: a waternet with the phreatic line (12 at the outer toe, 10.9 on the crest, 5.7 at x = 40) as the head above the top of the soft layers and a head of 10 → 8.4 → 8 in the aquifer below the top of the sand (the head varies linearly between the two reference lines in the soft layers); a 13 kPa traffic load at x = 3.5–6 of which 90 % is carried by excess pore pressure in the soft layers (10 % consolidation); the SHANSEP yield stress is the daily-conditions stage's effective vertical stress plus the POP (state correlated with the earlier stage), so that the higher water lowers σ′v but not the yield stress. Spencer on the tutorial's given non-circular surface (x = −12.5 to 37, down to y = −2.13 in the soft layers above the aquifer, where the waternet's head varies between the phreatic line and the aquifer's).

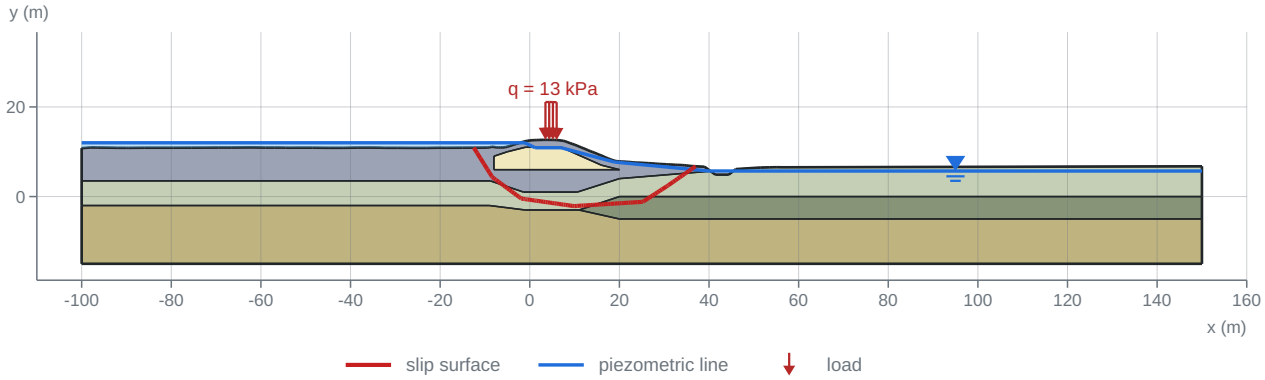


Figure 188. V105d: analyzed surface (Spencer, F = 1.761).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Mohr-Coulomb	18.00	c′ = 0.00	26.57°
Peat, deep	shansep	11.00	see model	–
Organic clay	shansep	13.90	see model	–
Clay, shallow	shansep	14.80	see model	–
Sand with clay	shansep	18.00	see model	–

The sources publish no factor of safety for this problem.

Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.
Spencer	1.761				1.761	+0.01 %

The problem comes from a program's tutorial only; no independent value is published. The checks are on the program's own surfaces, stored with its results in the tutorial's project file.

4 Strength reduction by finite elements

cSlope's finite element analysis reduces c and tan φ by a factor F until the elastic, perfectly plastic slope can no longer be brought into equilibrium. The mesh is of six-node triangles. It is refined adaptively, at most twice, where the plastic dissipation at failure is largest, until F changes by less than 0.01. Each problem is run with its source's flow rule (Davis B, associated, or non-associated with ψ = 0); the source's elastic parameters do not change F appreciably. The finite element factor is compared with the source's finite element, finite difference, limit analysis or theoretical value. It agrees within 2 % of a theoretical value or of a pair of bounds, and within 5 % of a published numerical value, which depends on the mesh, the elements and the failure criterion. 34 of the 34 references agree; the disputed one is not counted.

Table 6. Finite element factors of safety against the published values; the mesh refinement as number of elements: F.

Problem	Published by	Published	Flow rule	Elements: F	cSlope	Dev.
V001	finite-element strength reduction (verification manual, problem 10), another program's verification manual	1.40	non-associated	2811: 1.406, 4033: 1.402	1.402	+0.17 %
V002	limit analysis, lower and upper bound, Kim et al. (2002)	0.40–0.45	associated	2491: 0.425, 2573: 0.425	0.425	in bounds
V003	finite-element strength reduction (verification manual, problem 12), another program's verification manual	1.09	non-associated	1403: 1.107, 1846: 1.107	1.107	+1.55 %
V012	finite-element strength reduction (verification manual, problem 24), another program's verification manual	0.95	non-associated	6162: 0.910, 7014: 0.928, 8724: 0.928	0.928	−2.35 %
V013a	finite-element strength reduction (3D, without piles), Cai & Ugai (2000)	1.14	non-associated	1366: 1.141, 2149: 1.130, 3035: 1.133	1.133	−0.64 %
V018	plasticity theory (Prandtl mechanism), Chen & Shao (1988)	1.00	Davis B	1543: 1.000, 1751: 1.000	1.000	0.00 %
V019	finite-element strength reduction, Cheng et al. (2007)	1.47	associated	1832: 1.426, 1946: 1.408, 2157: 1.404	1.404	−4.46 %
V020	finite-element strength reduction, Cheng et al. (2007)	1.327	associated	1476: 1.377, 1532: 1.356, 1643: 1.353	1.353	+1.95 %
V026	finite-element strength reduction, Duncan, Wright & Brandon (2014)	1.19	non-associated	3026: 1.189, 3566: 1.186	1.186	−0.30 %
V038b	finite-element strength reduction, piezometric line, Duncan, Wright & Brandon (2014)	1.07	non-associated	2978: 1.084, 3424: 1.074, 4196: 1.074	1.073	+0.33 %
V039	finite-element strength reduction, Duncan, Wright & Brandon (2014)	1.26	non-associated	5324: 1.240, 6292: 1.240	1.240	−1.56 %
V044	finite-element strength reduction (verification manual, problem 17), another program's verification manual	2.00	non-associated	2133: 1.985, 3190: 1.980	1.980	−0.99 %
V045	finite-element strength reduction (verification manual, problem 17), another program's verification manual	1.78	non-associated	2140: 1.785, 3064: 1.781	1.781	+0.05 %
V046a	finite-element strength reduction (verification manual, problem 18), another program's verification manual	1.34	non-associated	2802: 1.340, 3113: 1.337	1.337	−0.25 %
V046b	finite-element strength reduction (verification manual, problem 18), another program's verification manual	1.13	non-associated	2922: 1.183, 3425: 1.186	1.186	+4.99 %
V047a	finite-element strength reduction (verification manual, problem 1), another program's verification manual	0.99	non-associated	2555: 0.971, 3807: 0.951, 5401: 0.956	0.956	−3.44 %
V049	finite-element strength reduction (verification manual, problem 2), another program's verification manual	1.36	non-associated	4522: 1.335, 6232: 1.335	1.335	−1.82 %

Problem	Published by	Published	Flow rule	Elements: F	cSlope	Dev.
V050b	finite-element strength reduction (verification manual, problem 5), another program's verification manual	1.26	non-associated	2530: 1.283, 2855: 1.283	1.283	+1.84 %
V052a	finite-element strength reduction (verification manual, problem 4), another program's verification manual	1.88	non-associated	3092: 1.780, 3417: 1.789	1.789	−4.85 %
V099	finite-element strength reduction with finite element seepage (verification manual, problem 7), another program's verification manual	1.48	non-associated	4354: 1.551, 4959: 1.543	1.543	+4.28 %
V053	finite-element strength reduction (verification manual, problem 16), another program's verification manual	1.02	non-associated	3380: 0.979, 3688: 0.969	0.969	−4.97 %
V055	finite elements, Griffiths & Lane (1999)	1.40	non-associated	2200: 1.356, 3288: 1.346, 5077: 1.342	1.342	−4.11 %
V056	finite elements (read from fig. 7), Griffiths & Lane (1999) (disputed: see below)	0.60	non-associated	2663: 0.538, 2845: 0.517, 3293: 0.509	0.509	−15.10 %
V057a	finite elements, Griffiths & Lane (1999)	1.45	non-associated	2296: 1.455, 2836: 1.451	1.451	+0.10 %
V057b	finite elements, Griffiths & Lane (1999)	2.03	non-associated	2296: 2.094, 2444: 2.071, 2912: 2.042	2.042	+0.60 %
V058	finite elements, Griffiths & Lane (1999)	1.90	non-associated	3566: 1.863, 4066: 1.854	1.854	−2.44 %
V059a	finite elements (read from fig. 2), Guo & Griffiths (2020)	1.28	non-associated	2046: 1.284, 2603: 1.281	1.281	+0.08 %
V059b	finite elements, Guo & Griffiths (2020)	1.35	non-associated	2046: 1.353, 2695: 1.353	1.353	+0.19 %
V059c	finite elements (read from fig. 2), Guo & Griffiths (2020)	1.36	non-associated	2046: 1.396, 2260: 1.369, 2713: 1.362	1.362	+0.12 %
V063	finite-element strength reduction, Huang & Jia (2009)	2.50	non-associated	3530: 2.427, 4872: 2.403, 6770: 2.403	2.403	−3.88 %
V067	finite-element strength reduction (verification manual, problem 19), another program's verification manual	1.41	non-associated	3620: 1.465, 3896: 1.447, 4469: 1.439	1.439	+2.08 %
V068	limit analysis, lower and upper bound, Kim et al. (2002)	1.13–1.27	associated	2024: 1.180, 2445: 1.177	1.177	in bounds
V077	plasticity theory, Prandtl (1920)	1.00	Davis B	4162: 1.004, 5303: 1.001	1.001	+0.14 %
V093	finite-element strength reduction (verification manual, problem 13), another program's verification manual	1.33	non-associated	2115: 1.339, 3399: 1.339	1.339	+0.70 %
V094	finite-element strength reduction (verification manual, problem 15), another program's verification manual	1.39	non-associated	2134: 1.371, 2626: 1.354, 3700: 1.351	1.351	−2.83 %

V056: Griffiths & Lane's 0.60 is read from fig. 7 on their mesh of 60 × 20 eight-node elements of 0.1H (fig. 8), one element across the inclined part of the weak layer, with failure as non-convergence in 1000 viscoplastic iterations. It disagrees with the paper's own Janbu wedge for the same mechanism (0.485) and with the limit analysis bounds, 0.488–0.498 (exact for φu = 0; cSlope's lower and upper bound programs on refined meshes): cSlope's strength reduction falls with the mesh, 0.64 on a uniform 0.2H mesh, 0.59 on 0.16H, 0.545 on 0.1H, 0.51 refined and 0.504 on a 0.5 m mesh, so the paper's value measures its mesh rather than cSlope. Not counted in the agreement or the error statistics.

Appendix A Results of all problems

One row per cSlope analysis of every problem, with the same columns as the tables of section 3 and the page of the problem. The problem id links to its section.

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V001	Bishop simplified	1.404	1.451	Arai & Tagyo (1985)	−3.21 %	1.409–1.417	−0.61 %	7
	Morgenstern-Price (half-sine)	1.401				1.414	−0.90 %	
	Spencer, non-circular	1.384				1.386	−0.17 %	
	Janbu simplified, non-circular	1.252	1.265	Arai & Tagyo (1985)	−1.06 %			
	Janbu corrected, non-circular	1.345	1.357	Arai & Tagyo (1985)	−0.91 %			
	Strength reduction (FE)	1.402	1.40	another program's verification manual: finite-element strength reduction (verification manual, problem 10)	+0.17 %			
V002	Bishop simplified	0.419	0.417	Arai & Tagyo (1985)	+0.55 %	0.420	−0.17 %	8
			0.43	Kim et al. (2002)	−2.49 %			
	Spencer, non-circular	0.411	0.44	Kim et al. (2002)	−6.59 %	0.414	−0.72 %	
			0.39	Greco (1996)	+5.38 %			
	Janbu simplified, non-circular	0.392	0.405	Arai & Tagyo (1985)	−3.23 %			
	Janbu corrected, non-circular	0.413	0.43	Arai & Tagyo (1985)	−3.95 %			
	Strength reduction (FE)	0.425	0.40–0.45	Kim et al. (2002): limit analysis, lower and upper bound	in bounds			
V003	Bishop simplified	1.114	1.138	Arai & Tagyo (1985)	−2.11 %	1.112–1.118	−0.09 %	9
	Spencer	1.115				1.118	−0.30 %	
	Janbu corrected	1.130				1.131	−0.05 %	
	Janbu simplified, non-circular	0.966	0.995	Arai & Tagyo (1985)	−2.91 %			
	Janbu corrected, non-circular	1.047	1.071	Arai & Tagyo (1985)	−2.23 %			
	Strength reduction (FE)	1.107	1.09	another program's verification manual: finite-element strength reduction (verification manual, problem 12)	+1.55 %			
V004a	Spencer	2.802				2.794	+0.30 %	10
	Janbu simplified	2.677				2.662	+0.55 %	
	–	–	2.66	Baker (2003): Mohr-Coulomb strength				
V004b	Janbu simplified	2.609				2.610	−0.04 %	11
	Spencer	2.667				2.696	−1.09 %	
	–	–	2.64	Baker (2003): non-linear strength envelope				
	–	–	2.66	Baker (2003): Mohr-Coulomb strength				
V005	Bishop simplified	2.502	2.50	Theory: infinite slope	+0.07 %			12
	Spencer	2.502						
	Spencer, non-circular	2.500	2.41	Baker, Frydman & Talesnick (1993): Spencer, non-circular (dynamic programming)	+3.75 %			
V006	Spencer	1.352				1.351	+0.07 %	13
	Janbu simplified	1.352				1.351	+0.07 %	
	–	–	1.35	Baker & Leshchinsky (2001): Culmann (planar wedge)				

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V007a	Janbu simplified	1.469				1.469	−0.03 %	14
	–	–	1.50	Baker (2003): Mohr-Coulomb strength				
V007b	Janbu simplified	0.955				0.957	−0.17 %	15
	Spencer	0.978				0.981	−0.35 %	
	–	–	0.97	Baker (2003): non-linear strength envelope				
	–	–	1.50	Baker (2003): Mohr-Coulomb strength				
V007c	Janbu simplified	1.290				1.291	−0.05 %	16
	–	–	1.35	Baker (2003): Mohr-Coulomb strength				
V008a	Spencer	1.519				1.536	−1.11 %	17
	–	–	1.50	Baker (2003): Mohr-Coulomb strength				
V008b	Spencer	1.365				1.366	−0.09 %	18
	–	–	1.35	Baker (2003): Mohr-Coulomb strength				
V009a	Spencer	1.927				1.925	+0.08 %	19
V009b	Spencer	1.882	1.91	Baker & Leshchinsky	−1.47 %			20
V009c	Spencer, non-circular	1.867	1.91	Baker & Leshchinsky (2001)	−2.25 %	1.868	−0.05 %	21
V010	Morgenstern-Price (half-sine)	1.164				1.171	−0.56 %	22
	Bishop simplified	1.166				1.170	−0.36 %	
V010b	Bishop simplified	1.159	1.15	Borges & Cardoso	+0.79 %			23
	–	–	1.19	Borges & Cardoso (2002): authors' moment method, circle A				
	–	–	1.22	Borges & Cardoso (2002): finite elements, circle A				
V011a	Bishop simplified	1.221				1.230	−0.72 %	24
	–	–	1.25	Borges & Cardoso (2002)				
V011b	Bishop simplified	1.217				1.220	−0.27 %	25
	–	–	1.19	Borges & Cardoso (2002)				
V012	Bishop simplified	0.982				0.980	+0.19 %	26
	Strength reduction (FE)	0.928	0.95	another program's verification manual: finite-element strength reduction (verification manual, problem 24)	−2.35 %			
	–	–	0.99	Borges & Cardoso (2002)				
V013a	Bishop simplified	1.143	1.13	Cai & Ugai	+1.15 %	1.140	+0.26 %	27
	Strength reduction (FE)	1.133	1.14	Cai & Ugai (2000): finite-element strength reduction (3D, without piles)	−0.64 %			
V013b	Bishop simplified	1.541				1.540	+0.07 %	28
	–	–	1.54	Cai & Ugai (2000)				
V013c	Bishop simplified	1.437				1.430	+0.51 %	29
	–	–	1.37	Cai & Ugai (2000)				
V013d	Bishop simplified	1.332				1.330	+0.12 %	30
	–	–	1.31	Cai & Ugai (2000)				
V013e	Bishop simplified	1.253				1.250	+0.24 %	31
	–	–	1.25	Cai & Ugai (2000)				

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V014	Janbu corrected	1.448				1.448–1.460	−0.40 %	32
	Spencer	1.575				1.576	−0.03 %	
V015	Janbu corrected	1.476				1.479	−0.18 %	33
	Janbu simplified	1.442				1.446	−0.29 %	
V016a	Bishop simplified	1.791				1.787–1.790	+0.15 %	34
	Spencer	1.798				1.791	+0.40 %	
V016b	Bishop simplified	1.831				1.830–1.835	−0.09 %	35
	Spencer	1.836				1.839	−0.17 %	
V016c	Bishop simplified	1.791				1.790	+0.08 %	36
V017a	Bishop simplified	1.384				1.373	+0.82 %	37
	Spencer	1.400				1.386	+1.02 %	
V017b	Bishop simplified	1.384				1.378	+0.46 %	38
V018	Spencer	1.052	1.05	Chen & Shao (1988)	+0.19 %	1.051	+0.10 %	39
	Morgenstern-Price (specified f(x))	1.008				1.009–1.017	−0.51 %	
	Strength reduction (FE)	1.000	1.00	Chen & Shao (1988): plasticity theory (Prandtl mechanism)	0.00 %			
V019	Spencer (40 slices, vertical pore-pressure head)	1.399				1.380–1.398	+0.71 %	40
	Spencer (vertical pore-pressure head)	1.399	1.383	Cheng et al.: Spencer	+1.13 %			
	Bishop simplified	1.401				1.401	+0.01 %	
	Strength reduction (FE)	1.404	1.47	Cheng et al. (2007): finite-element strength reduction	−4.46 %			
V020	Spencer	1.333	1.33	Cheng et al.	+0.23 %	1.336–1.338	−0.29 %	41
	Bishop simplified	1.337				1.342	−0.34 %	
	Strength reduction (FE)	1.353	1.327	Cheng et al. (2007): finite-element strength reduction	+1.95 %			
V021a	Morgenstern-Price (half-sine)	1.129				1.132	−0.29 %	42
	Bishop simplified	1.129	1.1178	Chowdhury & Xu	+0.98 %			
	Bishop simplified (60 slices)	1.129				1.128	+0.04 %	
V021b	Morgenstern-Price (half-sine)	1.110				1.116	−0.52 %	43
	Bishop simplified	1.110	1.1088	Chowdhury & Xu	+0.13 %			
	Bishop simplified (60 slices)	1.110				1.109	+0.10 %	
V021c	Morgenstern-Price (half-sine)	1.109				1.110	−0.11 %	44
	Bishop simplified	1.109	1.1096	Chowdhury & Xu	−0.07 %	1.108	+0.07 %	
V021d	Morgenstern-Price (half-sine)	1.058				1.063	−0.43 %	45
	Bishop simplified	1.058	1.0639	Chowdhury & Xu	−0.52 %	1.058	+0.04 %	
V021e	Morgenstern-Price (half-sine)	2.246				2.248	−0.10 %	46
	Bishop simplified	2.246	2.2343	Chowdhury & Xu	+0.51 %	2.245	+0.03 %	
V021f	Morgenstern-Price (half-sine)	2.129				2.138	−0.43 %	47
	Bishop simplified	2.129	2.1396	Chowdhury & Xu	−0.50 %	2.128	+0.04 %	
V021g	Morgenstern-Price (half-sine)	1.429				1.422	+0.47 %	48
	Bishop simplified	1.429	1.4239	Chowdhury & Xu	+0.33 %			
V022a	Morgenstern-Price (half-sine)	1.157				1.158	−0.06 %	49
	Bishop simplified	1.159	1.1625	Chowdhury & Xu	−0.33 %			
	Bishop simplified (60 slices)	1.159				1.160	−0.12 %	

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V022b	Morgenstern-Price (half-sine)	1.176				1.178	−0.20 %	51
	Bishop simplified	1.178	1.1479	Chowdhury & Xu (1995)	+2.62 %	1.185	−0.59 %	
V023	Spencer	1.138				1.145	−0.65 %	52
	–	–	1.17	Duncan (2000)				
V024	Bishop simplified	1.572	1.57	Duncan & Wright	+0.15 %	1.563–1.572	+0.31 %	53
	Spencer	1.563				1.557	+0.37 %	
V025	Bishop simplified	1.133				1.142	−0.81 %	54
	Spencer	1.133				1.142	−0.80 %	
	–	–	1.141	Duncan & Wright				
V026	Bishop simplified	1.220	1.22	Duncan & Wright	+0.01 %	1.228	−0.64 %	55
	Spencer	1.195	1.19	Duncan & Wright	+0.45 %			
	Spencer, non-circular	1.170				1.190	−1.66 %	
	Strength reduction (FE)	1.186	1.19	Duncan, Wright & Brandon (2014): finite-element strength reduction	−0.30 %			
V027	Bishop simplified	1.123				1.126	−0.24 %	56
	Spencer	1.136	1.124	Duncan & Wright	+1.03 %			
V028	Bishop simplified	1.125				1.126	−0.12 %	57
	Spencer	1.124	1.124	Duncan & Wright	+0.03 %			
V029	Bishop simplified	1.119				1.125	−0.50 %	58
	Spencer	1.119	1.119	Duncan & Wright	+0.04 %			
V030	Bishop simplified	1.409				1.412	−0.21 %	59
	Spencer	1.400	1.40	Duncan & Wright	−0.02 %			
V031a	Bishop simplified	2.534				2.549	−0.59 %	60
	Spencer	2.531	2.56	Duncan & Wright	−1.14 %	2.545	−0.55 %	
V031b	Bishop simplified	1.389				1.398	−0.61 %	61
	Spencer	1.352	1.35	Duncan & Wright	+0.16 %	1.359	−0.51 %	
V032a	Bishop simplified	1.224				1.230	−0.46 %	62
	Spencer	1.206	1.21	Duncan & Wright	−0.36 %	1.209	−0.27 %	
V032b	Bishop simplified	1.155				1.155	−0.03 %	63
	Spencer	1.155	1.15	Duncan & Wright	+0.41 %	1.155	−0.03 %	
V033	Bishop simplified	1.522				1.533	−0.73 %	64
	Spencer	1.534	1.529	Duncan & Wright	+0.30 %			
	Spencer, non-circular	1.522				1.534	−0.78 %	
V034a	Bishop simplified	1.306				1.313	−0.51 %	65
	Spencer	1.277	1.276	Duncan & Wright	+0.06 %			
	Spencer, non-circular	1.261				1.262	−0.09 %	
V034b	Bishop simplified	1.330				1.335	−0.35 %	66
	Spencer	1.328	1.328	Duncan & Wright	+0.03 %			
	Spencer, non-circular	1.168				1.182	−1.18 %	
V035a	Morgenstern-Price (half-sine)	1.569				1.575	−0.40 %	67
	–	–	1.51	Duncan & Wright				
V035b	Bishop simplified	1.321	1.32	Duncan & Wright	+0.07 %	1.324	−0.23 %	68
V036a	Bishop simplified	0.756				0.761	−0.60 %	69
	Spencer	0.751	0.75	Duncan & Wright	+0.20 %	0.756	−0.60 %	
	Spencer, non-circular	0.737				0.740	−0.43 %	

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V036b	Bishop simplified	0.906				0.909	−0.35 %	70
	Spencer	0.898	0.90	Duncan & Wright	−0.23 %	0.898	−0.01 %	
V036c	Bishop simplified	1.042				1.045	−0.25 %	71
	Spencer	1.029	1.03	Duncan & Wright	−0.13 %	1.032	−0.32 %	
	Spencer, non-circular	1.016				1.018	−0.15 %	
V036d	Bishop simplified	1.152				1.154	−0.17 %	72
	Spencer	1.132	1.13	Duncan & Wright	+0.14 %	1.134	−0.21 %	
	Spencer, non-circular	1.119				1.116	+0.30 %	
V037	Bishop simplified	1.614				1.629	−0.95 %	73
	Spencer	1.607	1.62	Duncan, Wright & Brandon (2014)	−0.77 %	1.620	−0.77 %	
V038	Bishop simplified, circles only	1.071				1.070–1.090	−0.87 %	74
	Spencer, circles only	1.078				1.100	−2.01 %	
	–	–	1.16	Duncan & Wright				
V038b	Spencer	1.158	1.16	Duncan & Wright	−0.17 %			75
			1.19	Duncan, Wright & Brandon (2014): Spencer, finite-element pore pressures	−2.69 %			
	Strength reduction (FE)	1.073	1.07	Duncan, Wright & Brandon (2014): finite-element strength reduction, piezometric line	+0.33 %			
V039	Spencer, circles only	1.447	1.45	Duncan & Wright: Spencer	−0.19 %			76
	Spencer	1.166	1.17	Duncan & Wright: Spencer, composite circle, composite surface	−0.32 %			
	Spencer, non-circular	1.160	1.16	Duncan & Wright	+0.03 %	1.167	−0.57 %	
	Strength reduction (FE)	1.240	1.26	Duncan, Wright & Brandon (2014): finite-element strength reduction	−1.56 %			
V040	Bishop simplified, circles only	1.460	1.46	El-Ramly et al. (2002): Bishop simplified	−0.03 %	1.461	−0.10 %	77
V041a	Bishop simplified	1.598				1.603	−0.29 %	78
	Spencer	1.596	1.60	Duncan & Wright	−0.28 %	1.599	−0.21 %	
	Spencer, non-circular	1.591				1.592	−0.04 %	
V041b	Bishop simplified	1.598				1.603	−0.29 %	79
	Spencer	1.595	1.60	Duncan & Wright	−0.31 %	1.599	−0.24 %	
	Spencer, non-circular	1.588				1.590	−0.13 %	
V042	Bishop simplified	1.582				1.584	−0.11 %	80
	Spencer	1.649				1.648	+0.08 %	
	–	–	1.67	Duncan & Wright				
V043	Bishop simplified	1.320	1.31	El-Ramly et al.	+0.78 %	1.320	+0.02 %	81
V044	Bishop simplified	2.077	2.08	Fredlund & Krahn	−0.16 %			82
	Morgenstern-Price (half-sine)	2.072	2.076	Fredlund & Krahn	−0.17 %			
	Spencer	2.073	2.073	Fredlund & Krahn	−0.00 %			
	Strength reduction (FE)	1.980	2.00	another program's verification manual: finite-element strength reduction (verification manual, problem 17)	−0.99 %			
V044b	Bishop simplified	1.760	1.766	Fredlund & Krahn	−0.33 %			83
	Morgenstern-Price (half-sine)	1.758	1.764	Fredlund & Krahn	−0.36 %			
	Spencer	1.758	1.761	Fredlund & Krahn	−0.15 %			

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V045	Bishop simplified	1.830	1.834	Fredlund & Krahn	−0.22 %			84
	Morgenstern-Price (half-sine)	1.828	1.832	Fredlund & Krahn	−0.23 %			
	Spencer	1.828	1.83	Fredlund & Krahn	−0.09 %			
	Strength reduction (FE)	1.781	1.78	another program's verification manual: finite-element strength reduction (verification manual, problem 17)	+0.05 %			
V046a	Bishop simplified	1.380	1.377	Fredlund & Krahn	+0.25 %	1.382	−0.12 %	85
	Spencer	1.379	1.373	Fredlund & Krahn	+0.47 %	1.382	−0.19 %	
	Morgenstern-Price (half-sine)	1.370	1.37	Fredlund & Krahn	−0.01 %			
	Strength reduction (FE)	1.337	1.34	another program's verification manual: finite-element strength reduction (verification manual, problem 18)	−0.25 %			
V046b	Bishop simplified	1.241	1.248	Fredlund & Krahn	−0.57 %			86
	Spencer	1.242	1.245	Fredlund & Krahn	−0.27 %			
	Morgenstern-Price (half-sine)	1.235						
	Strength reduction (FE)	1.186	1.13	another program's verification manual: finite-element strength reduction (verification manual, problem 18)	+4.99 %			
V046c	Bishop simplified	1.122	1.124	Fredlund & Krahn	−0.19 %	1.124	−0.19 %	87
	Spencer	1.122				1.124	−0.16 %	
	Morgenstern-Price (half-sine)	1.113	1.118	Fredlund & Krahn	−0.49 %			
V047a	Bishop simplified	0.985				0.987–0.992	−0.44 %	88
	Spencer	0.984	1.00	ACADS	−1.59 %	0.986	−0.19 %	
	Janbu corrected	0.986				0.990	−0.44 %	
	Morgenstern-Price (half-sine)	0.984				0.991	−0.70 %	
	Strength reduction (FE)	0.956	0.99	another program's verification manual: finite-element strength reduction (verification manual, problem 1)	−3.44 %			
V047b	Bishop simplified	1.659				1.664	−0.30 %	89
	Morgenstern-Price (half-sine)	1.656				1.660	−0.25 %	
	–	–	1.65–1.70	Giam & Donald (1989), ACADS				
V048	Bishop simplified	1.587				1.596	−0.57 %	90
	Spencer	1.583				1.592	−0.60 %	
	Janbu corrected	1.475				1.489	−0.93 %	
	–	–	1.65	Giam & Donald (1989), ACADS				
V049	Bishop simplified	1.403				1.403–1.405	−0.06 %	91
	Spencer	1.372				1.375	−0.19 %	
	Janbu corrected	1.356				1.357	−0.10 %	
	Morgenstern-Price (half-sine)	1.372				1.382	−0.73 %	
	Strength reduction (FE)	1.335	1.36	another program's verification manual: finite-element strength reduction (verification manual, problem 2)	−1.82 %			
	–	–	1.39	Giam & Donald (1989), ACADS				

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V050a	Janbu corrected	1.294				1.294	−0.02 %	92
	Spencer	1.276				1.277	−0.06 %	
	Morgenstern-Price (half-sine)	1.260				1.261	−0.04 %	
	–	–	1.34	Giam & Donald (1989), ACADS				
V050b	Spencer, non-circular	1.225				1.246	−1.65 %	93
	Strength reduction (FE)	1.283	1.26	another program's verification manual: finite-element strength reduction (verification manual, problem 5)	+1.84 %			
	–	–	1.24–1.27	Giam & Donald (1989), ACADS				
V051	Spencer, non-circular	0.714				0.707	+0.99 %	94
	–	–	0.78	Giam & Donald (1989), ACADS				
V052a	Bishop simplified	1.948				1.948–1.951	−0.09 %	95
	Spencer	1.948				1.948	−0.02 %	
	Janbu corrected	1.950				1.949	+0.06 %	
	Strength reduction (FE)	1.789	1.88	another program's verification manual: finite-element strength reduction (verification manual, problem 4)	−4.85 %			
	–	–	1.95	Giam & Donald (1989), ACADS				
V052b	Bishop simplified	2.206				2.208	−0.07 %	96
	Spencer	2.290				2.292	−0.08 %	
	Janbu corrected	2.073				2.073	−0.00 %	
	Morgenstern-Price (half-sine)	2.299				2.299	+0.01 %	
	–	–	2.29	Giam & Donald (1989), ACADS				
V099	Strength reduction (FE)	1.543	1.48	another program's verification manual: finite-element strength reduction with finite element seepage (verification manual, problem 7)	+4.28 %			97
V053	Spencer, non-circular	1.006	1.009–1.025	Chen & Shao (1988): Spencer, non-circular (test problem 3)	−0.28 %	1.010	−0.38 %	98
			0.97	Greco (1996): Spencer, non-circular (global minimum along the weak layer)	+3.73 %			
	Spencer	1.092	1.08	Greco (1996): Spencer, local minimum (nearly circular surface from the toe)	+1.08 %			
	Strength reduction (FE)	0.969	1.02	another program's verification manual: finite-element strength reduction (verification manual, problem 16)	−4.97 %			
V054	Bishop simplified	1.086				1.087	−0.09 %	99
	Spencer	1.092	1.08	Greco (1996): Spencer, local minimum (nearly circular surface from the toe)	+1.08 %	1.093	−0.12 %	
V055	Spencer	1.367	1.40	Griffiths & Lane: finite elements	−2.39 %	1.366	+0.04 %	100
	Strength reduction (FE)	1.342	1.40	Griffiths & Lane (1999): finite elements	−4.11 %			
V056	Janbu corrected	0.499	0.485	Griffiths & Lane: Janbu wedge	+2.97 %			101
	Strength reduction (FE)	0.509	0.60	Griffiths & Lane (1999): finite elements (read from fig. 7)	*−15.10 %* *			

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V057a	Spencer	1.474	1.46	Griffiths & Lane: limit-equilibrium chart	+0.93 %	1.468	+0.38 %	102
	Strength reduction (FE)	1.451	1.45	Griffiths & Lane (1999): finite elements	+0.10 %			
	–	–	1.47	Griffiths & Lane (1999): stability chart (Taylor)				
V057b	Spencer	2.023	2.04	Griffiths & Lane: limit-equilibrium chart	−0.84 %	2.022	+0.04 %	103
	Strength reduction (FE)	2.042	2.03	Griffiths & Lane (1999): finite elements	+0.60 %			
	–	–	2.10	Griffiths & Lane (1999): stability chart (Taylor)				
V058	Spencer	1.917	1.90	Griffiths & Lane: finite elements	+0.87 %	1.915	+0.08 %	104
	Strength reduction (FE)	1.854	1.90	Griffiths & Lane (1999): finite elements	−2.44 %			
V059a	Spencer, non-circular	1.214				1.215	−0.11 %	105
	Strength reduction (FE)	1.281	1.28	Guo & Griffiths (2020): finite elements (read from fig. 2)	+0.08 %			
V059b	Spencer, non-circular	1.312				1.290	+1.69 %	106
	Strength reduction (FE)	1.353	1.35	Guo & Griffiths (2020): finite elements	+0.19 %			
V059c	Spencer, non-circular	1.312				1.315	−0.24 %	107
	Strength reduction (FE)	1.362	1.36	Guo & Griffiths (2020): finite elements (read from fig. 2)	+0.12 %			
V060a	Bishop simplified	2.530				2.529	+0.05 %	108
V060b	Bishop simplified	2.974				2.973	+0.05 %	109
V061	Morgenstern-Price (half-sine)	2.348	2.36	Wolff & Harr	−0.49 %	2.343	+0.23 %	110
	Spencer	2.388				2.383	+0.21 %	
V062a	Bishop simplified	2.549				2.551–2.560	−0.27 %	111
	–	–	2.753	Hassan & Wolff (1999)				
V062b	Bishop simplified	2.816				2.806–2.820	+0.11 %	112
	–	–	2.352	Hassan & Wolff (1999)				
V062c	Bishop simplified	2.780				2.771–2.777	+0.20 %	113
	–	–	2.523	Hassan & Wolff (1999)				
V062d	Bishop simplified	2.577				2.583–2.589	−0.36 %	114
	–	–	2.457	Hassan & Wolff (1999)				
V062e	Bishop simplified	2.690				2.692–2.703	−0.27 %	115
	–	–	2.602	Hassan & Wolff (1999)				
V062f	Bishop simplified	2.674				2.672–2.673	+0.04 %	116
	–	–	2.995	Hassan & Wolff (1999)				
V062g	Bishop simplified	3.593				3.586–3.598	+0.02 %	117
	–	–	3.916	Hassan & Wolff (1999)				
V062h	Bishop simplified	6.065				6.069–6.074	−0.10 %	118
	–	–	10.576	Hassan & Wolff (1999)				
V062i	Bishop simplified	11.571				11.583	−0.11 %	119
	–	–	6.293	Hassan & Wolff (1999)				

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V063	Spencer	2.451				2.455	−0.15 %	120
	Bishop simplified	2.453	2.43	Huang & Jia	+0.95 %			
	Strength reduction (FE)	2.403	2.50	Huang & Jia (2009): finite-element strength reduction	−3.88 %			
V100	Bishop simplified	1.668	1.66	Charles & Soares	+0.48 %			121
	Janbu simplified, non-circular	1.559	1.56	Jiang, Baker & Yamagami: Janbu simplified (uncorrected), non-circular	−0.09 %	1.563	−0.28 %	
	Bishop simplified, non-circular	1.651				1.656	−0.28 %	
	–	–	1.67	Perry (1994): Janbu rigorous, on the critical Bishop circle				
	–	–	1.60	Jiang, Baker & Yamagami (2003): simplified Janbu (uncorrected) on Charles & Soares's circle				
V064	Bishop simplified	1.017	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	+1.71 %	1.020	−0.28 %	122
V065	Spencer	0.960	0.92	Leshchinsky & Han (2004): Spencer, three-part wedge	+4.38 %	0.964	−0.38 %	123
	Bishop simplified	0.979	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	−2.13 %			
			0.98	Leshchinsky & Han (2004): Bishop simplified, deep circle (read from fig. 7a)	−0.13 %			
	–	–	0.86	Leshchinsky & Han (2004): finite differences				
V066a	Bishop simplified	1.031	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	+3.06 %	1.040	−0.90 %	124
	–	–	0.99	Leshchinsky & Han (2004): finite differences				
V066b	Spencer	1.011				1.011	+0.01 %	125
	Bishop simplified	1.013	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	+1.32 %			
	–	–	0.98	Leshchinsky & Han (2004): finite differences				
V066c	Bishop simplified	1.012	1.00	Leshchinsky & Han	+1.15 %			126
	–	–	1.01	Leshchinsky & Han (2004): finite differences				
V066d	Bishop simplified	1.010	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	+1.00 %	1.010	0.00 %	127
	–	–	1.01	Leshchinsky & Han (2004): finite differences				
V066e	Bishop simplified	0.961	1.00	Leshchinsky & Han (2004): Bishop simplified, circular	−3.90 %	0.958	+0.31 %	128
	–	–	1.02	Leshchinsky & Han (2004): finite differences				
V101	Bishop simplified	1.334	1.334	Hassan & Wolff	0.00 %	1.340–1.343	−0.56 %	129
V102a	Spencer	1.001	1.00	Loukidis, Bandini & Salgado	+0.10 %	1.001	0.00 %	130
	Bishop simplified	0.991				0.991–0.993	−0.07 %	
	Spencer, non-circular	0.998				0.999	−0.12 %	
V102b	Spencer	1.001	1.00	Loukidis, Bandini & Salgado	+0.07 %	1.001	−0.03 %	131
	Bishop simplified	0.987				0.987–0.988	−0.10 %	
	Spencer, non-circular	0.997				0.997	−0.05 %	

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V067	Bishop simplified	1.436				1.439	−0.19 %	132
	Strength reduction (FE)	1.439	1.41	another program's verification manual: finite-element strength reduction (verification manual, problem 19)	+2.08 %			
	–	–	1.44	Low (1989): Ordinary				
V068	Bishop simplified	1.161	1.14	Low	+1.82 %			133
	Spencer, non-circular	–	1.17	Kim et al. (2002): Spencer, non-circular (random search)	–			
	Strength reduction (FE)	1.177	1.13–1.27	Kim et al. (2002): limit analysis, lower and upper bound	in bounds			
	–	–	1.36	Low (1989): Ordinary				
	–	–	1.31	Low (1989): Ordinary and Bishop with φ = 0 in the bank				
	–	–	1.38	Low (1989): hand calculation (Low's method)				
V069a	Bishop simplified	1.201	1.20	Morgenstern: stability chart	+0.07 %			134
	Spencer	1.206						
V069b	Bishop simplified	1.210				1.212	−0.15 %	135
	–	–	1.20	Morgenstern (1963): stability chart				
V070	Bishop simplified	1.415	1.41	Morgenstern: stability chart	+0.39 %	1.417	−0.11 %	136
	Spencer	1.422				1.422	−0.01 %	
	–	–	1.52	Morgenstern (1963): stability chart, circles tangent to the base				
	–	–	1.48	Morgenstern (1963): stability chart, circles tangent to H/2				
V071a	Bishop simplified	1.290				1.293	−0.19 %	137
	Spencer	1.297				1.300	−0.22 %	
V071b	Bishop simplified	1.289				1.293	−0.31 %	138
	Spencer	1.295				1.300	−0.38 %	
	Janbu simplified	1.149				1.151	−0.19 %	
V072a	Bishop simplified	1.283				1.285	−0.17 %	139
	Spencer	1.288				1.290	−0.15 %	
V072b	Bishop simplified	1.282				1.285	−0.22 %	140
	Spencer	1.288				1.290	−0.18 %	
	Janbu simplified	1.136				1.141	−0.42 %	
V073	Bishop simplified, circles only	1.411				1.417	−0.44 %	141
	Spencer, circles only	1.415				1.422	−0.46 %	
	Bishop simplified	1.388				1.392	−0.28 %	
	Spencer	1.396				1.400	−0.29 %	
V074	Spencer	1.010				1.009	+0.10 %	142
	Janbu corrected	1.075				1.073	+0.16 %	
V075	Spencer	0.578				0.570	+1.37 %	143
V076	Bishop simplified	1.143				1.147	−0.35 %	144
	Spencer	1.141				1.145	−0.36 %	
V077	Spencer	0.943				0.940	+0.30 %	145
	Strength reduction (FE)	1.001	1.00	Prandtl (1920): plasticity theory	+0.14 %			

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V078	Morgenstern-Price (half-sine)	0.958				0.960	−0.22 %	146
	–	–	1.00	Prandtl (1920): plasticity theory				
V079	Janbu simplified	1.048				1.049	−0.07 %	147
	Spencer	1.048				1.049	−0.07 %	
	–	–	1.049	Priest (1993): rigid block, force equilibrium				
V080	Morgenstern-Price (half-sine)	1.049				1.049	0.00 %	148
	Spencer	1.049	1.049	Priest	0.00 %			
	–	–	1.049	Priest (1993): rigid block, force equilibrium				
V081a	Bishop simplified	0.764				0.764	+0.03 %	149
V081b	Bishop simplified	1.502				1.504	−0.12 %	150
V082a	Bishop simplified	1.534				1.532	+0.14 %	151
	Spencer	1.533				1.532	+0.08 %	
	Morgenstern-Price (half-sine)	1.533				1.532	+0.08 %	
	Janbu corrected	1.546				1.544	+0.16 %	
V082b	Bishop simplified	1.513				1.511	+0.13 %	152
	Spencer	1.512				1.510	+0.13 %	
	Morgenstern-Price (half-sine)	1.512				1.510	+0.14 %	
	Janbu corrected	1.522				1.520	+0.12 %	
V082c	Bishop simplified	1.396				1.396	+0.02 %	153
	Spencer	1.403				1.402	+0.04 %	
	Janbu corrected	1.392				1.391	+0.09 %	
	Morgenstern-Price (half-sine)	1.399				1.398	+0.04 %	
V082d	Bishop simplified	1.370				1.376	−0.45 %	154
	Spencer	1.376				1.382	−0.42 %	
	Morgenstern-Price (half-sine)	1.371				1.378	−0.48 %	
	Janbu corrected	1.344				1.345	−0.10 %	
V083	Janbu corrected	0.884				0.890	−0.62 %	155
	–	–	0.887	Sheahan & Ho (2003): force-equilibrium trial wedge				
V084a	Janbu corrected	0.988				0.989	−0.06 %	156
	–	–	0.989	Sheahan & Ho (2003): force-equilibrium trial wedge				
V084b	Janbu corrected	1.124				1.123	+0.12 %	157
	–	–	1.176	Sheahan & Ho (2003): force-equilibrium trial wedge				
V084c	Janbu corrected	1.043				1.043	+0.03 %	158
	–	–	1.07	Sheahan & Ho (2003): force-equilibrium trial wedge				
V084d	Janbu corrected	0.944				0.945	−0.07 %	159
	–	–	0.929	Sheahan & Ho (2003): force-equilibrium trial wedge				
V084e	Janbu corrected	0.920				0.922	−0.23 %	160
	–	–	0.893	Sheahan & Ho (2003): force-equilibrium trial wedge				

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V084f	Janbu corrected	0.920				0.923	−0.27 %	161
	–	–	0.887	Sheahan & Ho (2003): force-equilibrium trial wedge				
V085	Bishop simplified	1.759	1.76	Duncan & Wright	−0.07 %	1.762	−0.18 %	162
	Spencer	1.759				1.758	+0.05 %	
	Janbu corrected	1.735				1.736	−0.06 %	
	Janbu simplified	1.627	1.63	Duncan & Wright	−0.20 %	1.628	−0.08 %	
V103	Spencer, non-circular	1.008	1.02	Baker	−1.20 %	1.010	−0.22 %	163
	–	–	1.08	Spencer (1969)				
V086a	Spencer	1.333				1.332–1.350	−0.57 %	164
	–	–	1.35	Tandjiria, Low & Teh (2002): generalized method of slices, at T = 170.5 kN/m				
V086b	Spencer (2 m tension crack)	0.969	0.981	Tandjiria, Low & Teh (2002): Spencer	−1.27 %	0.975	−0.67 %	165
	Spencer (tension crack of automatic depth)	0.967				0.975	−0.86 %	
	Morgenstern-Price (half-sine)	0.969				0.975	−0.67 %	
V086c	Spencer	1.200	1.219	Tandjiria, Low & Teh (2002)	−1.52 %	1.209	−0.70 %	166
	Morgenstern-Price (half-sine)	1.211				1.218	−0.58 %	
V086d	Spencer	0.973	0.981	Tandjiria, Low & Teh	−0.80 %			167
V086e	Spencer	1.204	1.219	Tandjiria, Low & Teh	−1.19 %			168
V087	Janbu corrected	2.248				2.290	−1.81 %	169
	–	–	2.30	USACE EM 1110-2-1902				
V088	Spencer	2.442	2.44	USACE EM 1110-2-1902	+0.08 %	2.445	−0.12 %	170
	Bishop simplified	2.443				2.447	−0.18 %	
	Janbu corrected	2.444				2.430	+0.59 %	
	Morgenstern-Price (half-sine)	2.442				2.447	−0.20 %	
V089	Bishop simplified	2.699	2.71	USACE EM 1110-2-1902	−0.40 %	2.716	−0.62 %	171
	Spencer	2.722				2.736	−0.52 %	
	Janbu corrected	2.648				2.650	−0.06 %	
	Morgenstern-Price (half-sine)	2.728				2.744	−0.58 %	
V090	Bishop simplified	1.321				1.332	−0.85 %	172
	Spencer	1.317				1.328	−0.84 %	
	Janbu corrected	1.341				1.345	−0.31 %	
	Morgenstern-Price (half-sine)	1.317				1.327	−0.78 %	
	–	–	1.33	USACE EM 1110-2-1902				
V091	Bishop simplified	1.238				1.241	−0.27 %	173
	Spencer	1.238				1.241	−0.27 %	
	Morgenstern-Price (half-sine)	1.238				1.244	−0.51 %	
	Janbu corrected	1.389				1.385	+0.31 %	
	–	–	1.24	USACE EM 1110-2-1902: stability chart (example E-2)				

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V092	Bishop simplified	1.992				2.011	−0.97 %	174
	Spencer	2.006				2.026	−1.00 %	
	Janbu corrected	1.825				1.830	−0.29 %	
	Morgenstern-Price (half-sine)	2.006				2.027	−1.04 %	
	–	–	2.01	USACE EM 1110-2-1902: Simplified Bishop, nine-slice hand calculation on the manual's own piezometric line (example F-6)				
	–	–	2.07	USACE EM 1110-2-1902: Modified Swedish (θ = 19.7°) on the manual's own piezometric line (example F-6)				
V093	Bishop simplified	1.343	1.348	Yamagami & Ueta (1988)	−0.37 %	1.344	−0.07 %	176
	Spencer, non-circular	1.323	1.339	Yamagami & Ueta (1988)	−1.23 %	1.325	−0.19 %	
			1.33	Greco (1996)	−0.56 %			
	Janbu simplified, non-circular	1.176	1.185	Yamagami & Ueta (1988)	−0.73 %	1.178	−0.14 %	
	Strength reduction (FE)	1.339	1.33	another program's verification manual: finite-element strength reduction (verification manual, problem 13)	+0.70 %			
	–	–	1.282	Yamagami & Ueta (1988): Ordinary				
V094	Spencer, non-circular	1.399	1.40–1.42	Greco (1996)	−0.07 %	1.398	+0.07 %	177
	Morgenstern-Price (half-sine), non-circular	1.388				1.389	−0.06 %	
	Strength reduction (FE)	1.351	1.39	another program's verification manual: finite-element strength reduction (verification manual, problem 15)	−2.83 %			
V095a	Bishop simplified	1.101				1.102	−0.09 %	178
	–	–	1.10	Yamagami et al. (2000)				
V095c	Bishop simplified	1.196				1.193	+0.25 %	179
	–	–	1.20	Yamagami et al. (2000)				
V096a	Bishop simplified	1.794				1.804	−0.56 %	180
	Spencer	1.795				1.804	−0.52 %	
	Morgenstern-Price (half-sine)	1.780				1.790	−0.54 %	
V096b	Bishop simplified	1.175				1.176	−0.12 %	181
	Spencer	1.188				1.189	−0.10 %	
	Morgenstern-Price (half-sine)	1.173				1.174	−0.07 %	
V096c	Spencer	1.800				1.804	−0.23 %	182
	Bishop simplified	1.800				1.804	−0.24 %	
V096d	Spencer	1.825	1.836	Zhu & Lee	−0.59 %			183
	Morgenstern-Price (half-sine)	1.810	1.823	Zhu & Lee	−0.73 %			
	Bishop simplified	1.824	1.429	Zhu & Lee (2002)	*+27.66 %* *			
	–	–	1.744	Zhu & Lee (2002): explicit solution (the paper's method)				
	–	–	1.229	Zhu & Lee (2002): Ordinary				

Problem	Analysis	cSlope	Published	Source	Dev.	Other programs	Dev.	Page
V096e	Spencer	1.214	1.211	Zhu & Lee	+0.27 %			184
	Morgenstern-Price (half-sine)	1.199	1.197	Zhu & Lee	+0.19 %			
	Bishop simplified	1.201	1.079	Zhu & Lee (2002)	*+11.32 %* *			
	–	–	1.166	Zhu & Lee (2002): explicit solution (the paper's method)				
	–	–	0.922	Zhu & Lee (2002): Ordinary				
V097a	Bishop simplified	1.173				1.173	−0.04 %	185
V097b	Bishop simplified	1.173				1.173	−0.03 %	186
	Spencer	1.174				1.174	−0.01 %	
V098	Spencer, non-circular	1.357				1.360	−0.22 %	187
	Spencer	1.372				1.372	+0.03 %	
V104a	Bishop simplified	2.294				2.300	−0.26 %	188
V104b	Bishop simplified	1.766				1.770	−0.23 %	189
V104c	Bishop simplified	1.462				1.470	−0.51 %	190
V104d	Bishop simplified	1.333				1.330	+0.24 %	191
V105a	Bishop simplified	1.285				1.285	+0.01 %	192
V105b	Bishop simplified	1.004				1.004	+0.01 %	193
V105c	Spencer	1.085				1.085	−0.01 %	194
V105d	Spencer	1.761				1.761	+0.01 %	195

Probabilistic problems

Problem	Quantity	Published	Other programs	cSlope
V021a	Pf, share F < 1 (Monte Carlo)	–	0.246	0.239
V021a	β lognormal (Monte Carlo)	–	0.650	0.673
V021a	Pf, share F < 1 (Monte Carlo)	–	0.253	0.245
V021b	Pf, share F < 1 (Monte Carlo)	–	0.279	0.281
V021b	β lognormal (Monte Carlo)	–	0.537	0.555
V022a	Pf, share F < 1 (Monte Carlo)	–	0.206–0.212	0.206
V022a	β lognormal (Monte Carlo)	–	0.799–0.800	0.812
V022a	β lognormal (Taylor series)	–	0.787	0.789
V022a	Pf, share F < 1 (Monte Carlo)	–	0.210	0.208
V022b	Pf, share F < 1 (Monte Carlo)	–	0.199–0.209	0.211
V022b	β lognormal (Monte Carlo)	–	0.789–0.820	0.793
V022b	β lognormal (Taylor series)	–	0.798	0.806
V022b	Pf, share F < 1 (Monte Carlo)	–	0.214	0.215
V060b	β lognormal (Taylor series)	–	3.353	3.354
V023	β lognormal (Taylor series)	0.900 Duncan (2000)	–	0.889
V023	Pf lognormal (Taylor series)	0.180 Duncan (2000)	–	0.187
V062a	β lognormal (Monte Carlo)	–	10.953	11.033
V062b	β lognormal (Monte Carlo)	–	4.351	4.402
V062c	β lognormal (Monte Carlo)	–	4.263	4.325
V062d	β lognormal (Monte Carlo)	–	11.092	11.074
V062e	β lognormal (Monte Carlo)	–	10.281	10.328
V062f	β lognormal (Monte Carlo)	–	4.858	4.933

Problem	Quantity	Published	Other programs	cSlope
V062g	β lognormal (Monte Carlo)	–	5.485	5.528
V062h	β lognormal (Monte Carlo)	–	5.563	5.491
V062i	β lognormal (Monte Carlo)	–	6.394	6.432
V101	β lognormal (Taylor series)	2.336 Hassan & Wolff (1999)	–	2.328